

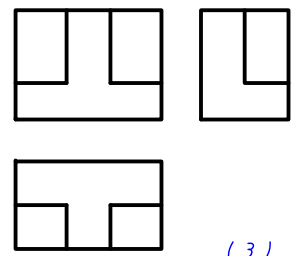
一、绪论

班级

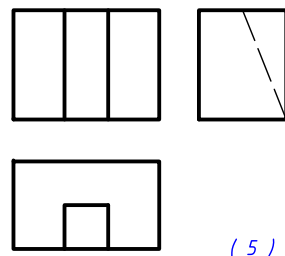
姓名

学号

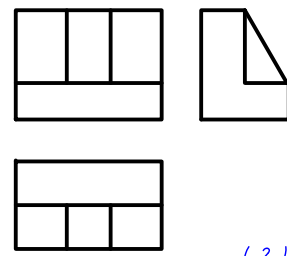
根据物体的三视图，找出对应的立体图，并将序号填进括号内。



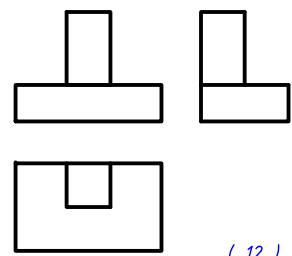
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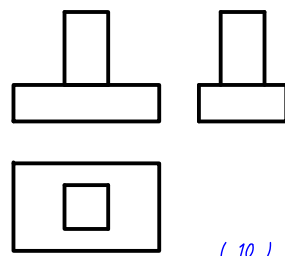
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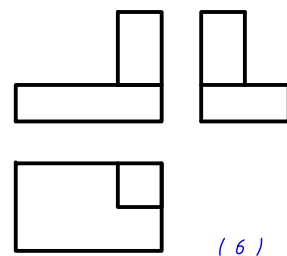
(2)



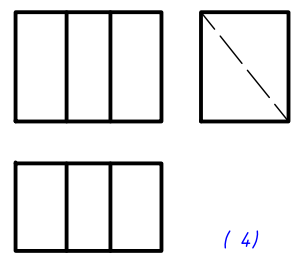
(12)



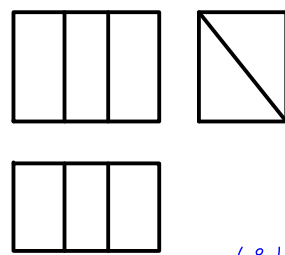
(10)



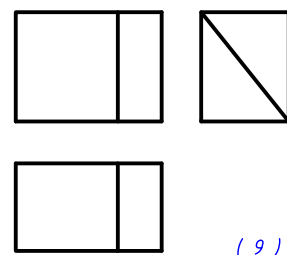
(6)



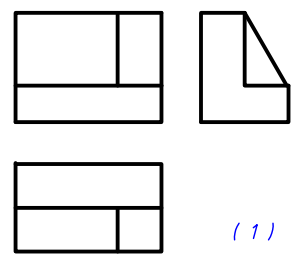
(4)



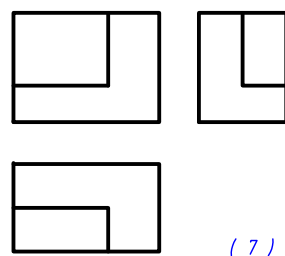
(8)



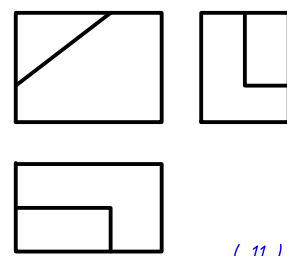
(9)



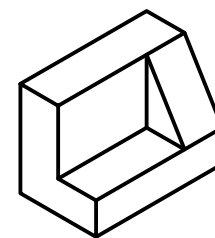
(1)



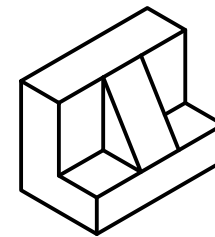
(7)



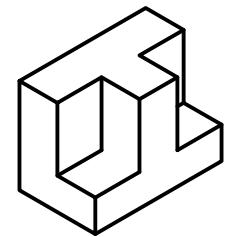
(11)



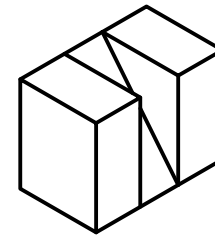
(1)



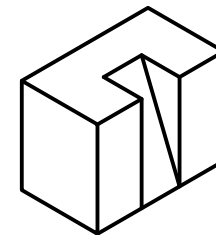
(2)



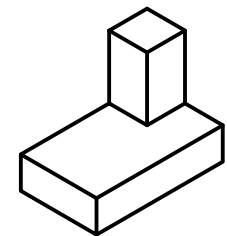
(3)



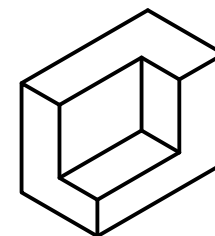
(4)



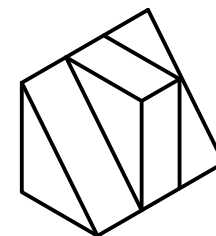
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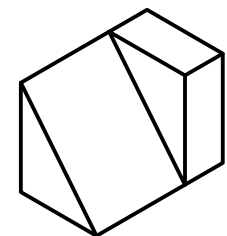
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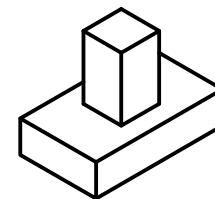
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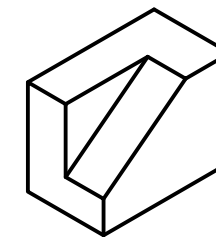
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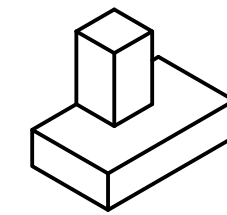
(9)



(10)



(11)



(12)

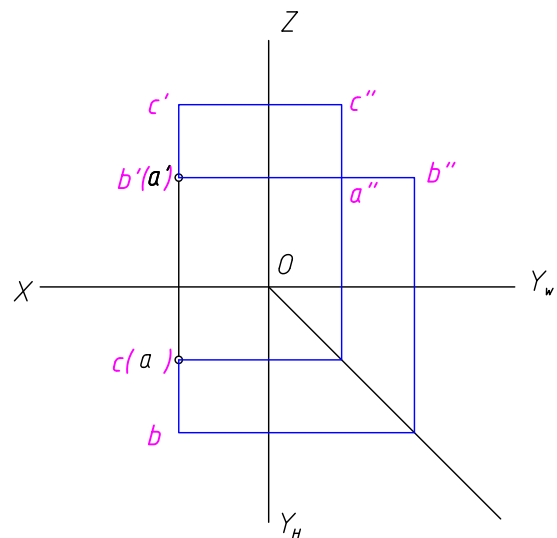
二、点

班级

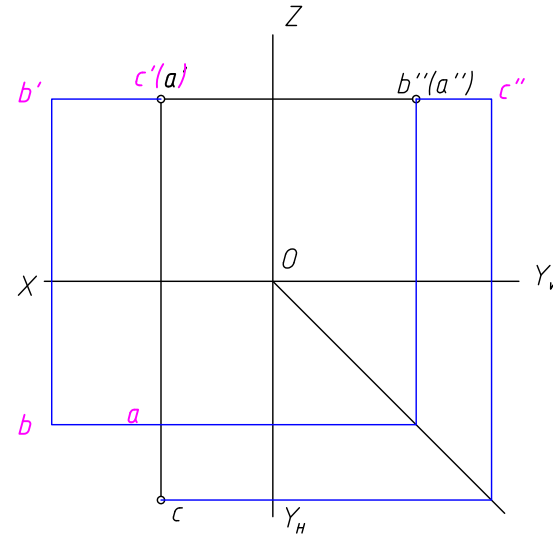
姓名

学号

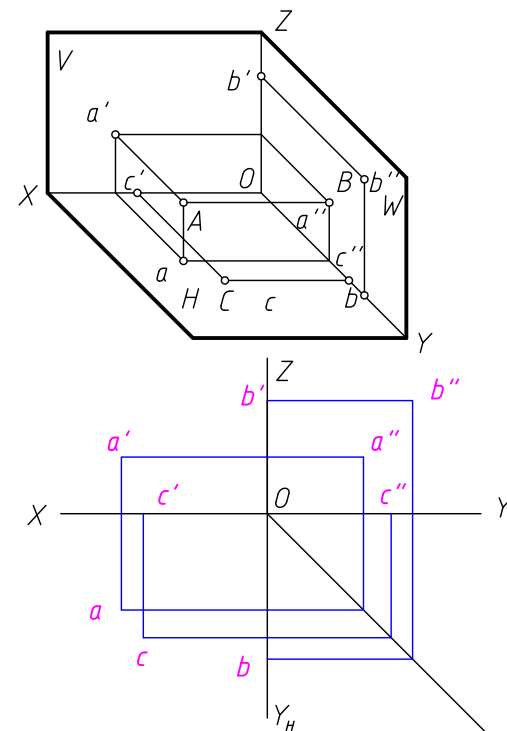
1. 已知点A，点B位于点A的正前方10mm，点C位于点A的正上方10mm。求作各点的三面投影图，并标明可见性。



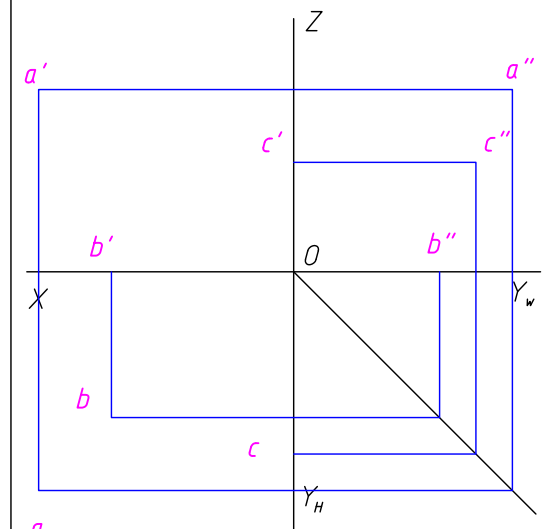
2. 已知点A的两面投影，点B距点A15mm，点C与点A为对V面的重影点。补全各点的三面投影，并标明可见性。



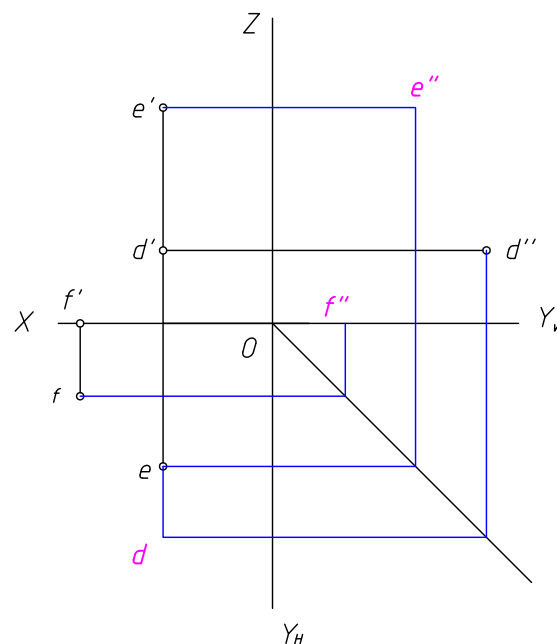
3. 按立体图作点A、B、C的三面投影图。



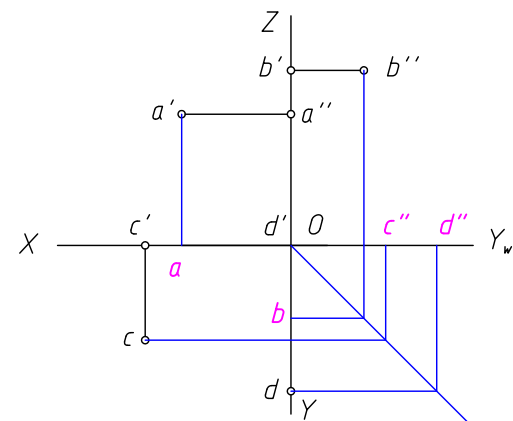
4. 已知点A (35, 30, 25)、点B (25, 20, 0)、点C (0, 25, 15)，作各点的三面投影图。



5. 求点D、E、F的第三面投影。

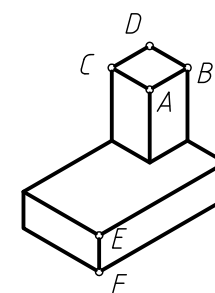
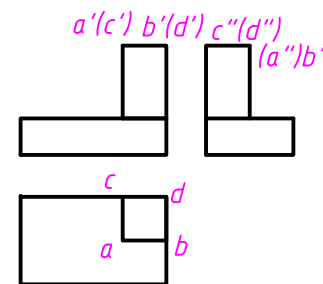


6. 已知点的两面投影，求作第三面投影，并说明各点的空间位置。



A 在V面上 B 在W面上
C 在H面上 D 在Y轴上

7. 根据立体图和三视图，按要求作图和填空。



- 在三视图中标出点A、B、C、D的三面投影。
- 在这些点中，对V面的重影点有A, C; B, D对W面的重影点有A, B; C, D对H面的重影点有E, F。
- 点A在点B的 正左方。（左方？正左方？）
点E在点F的 正上方。（上方？正上方？）
点A在点C的 正前方（前方？正前方？）
点A在点F的 左（左？右？），后（前？后？），上（上？下？）方。

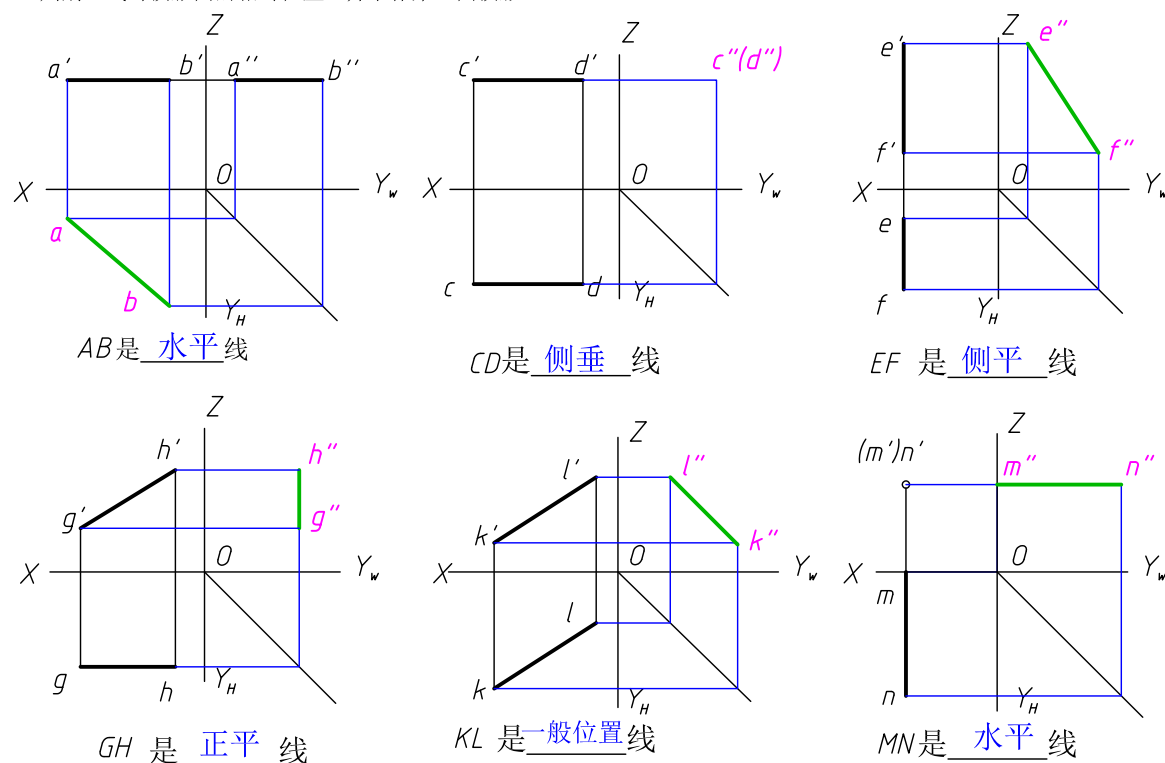
三、直线（一）

班级

姓名

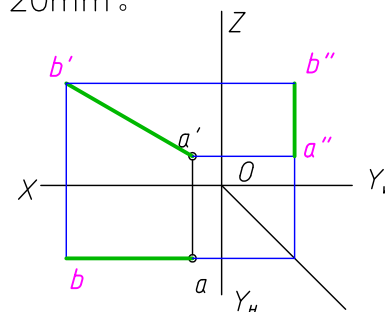
学号

1. 判别直线对投影面的相对位置，并求作第三面投影。

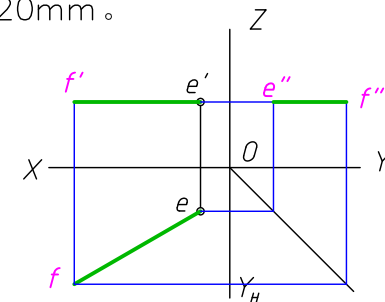


2. 过点作直线的三面投影（只作一个解）。

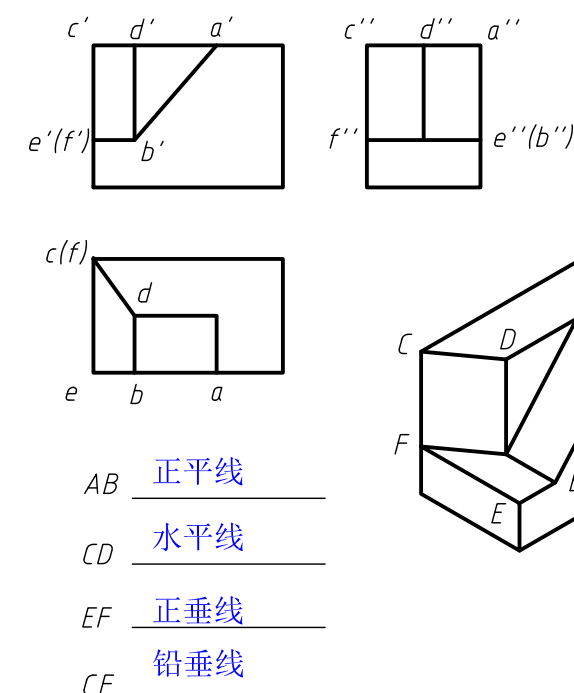
(1) 正平线AB， $\alpha=30^\circ$ ，长20mm。



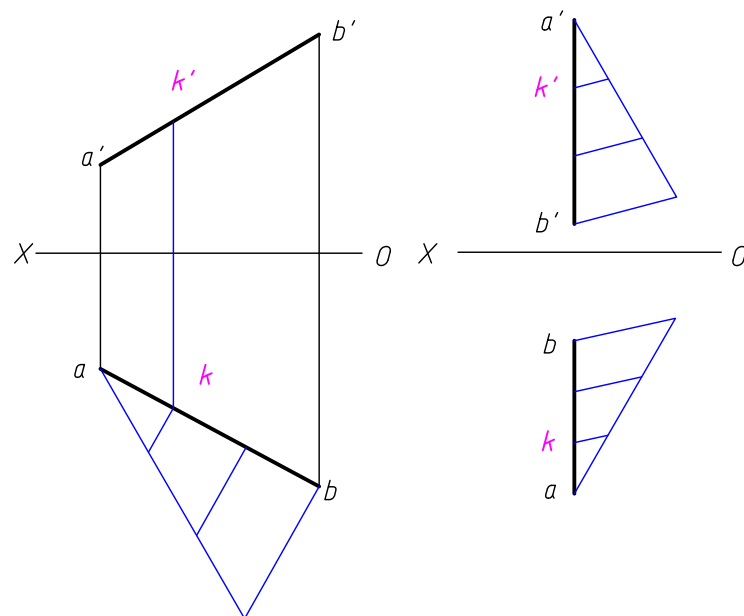
(2) 水平线EF， $\beta=30^\circ$ ，长20mm。



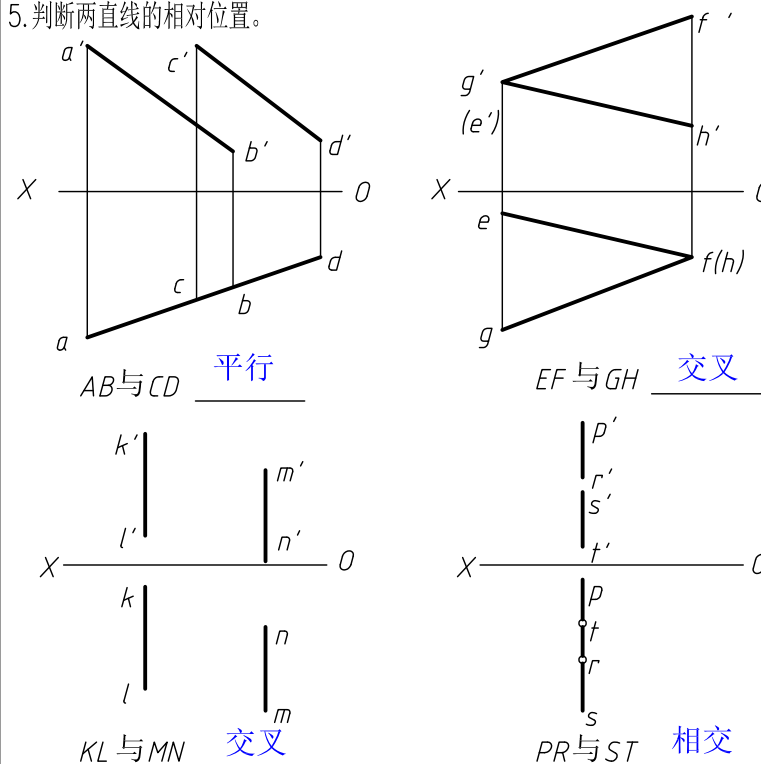
3. 判断立体上各直线对投影面的相对位置。



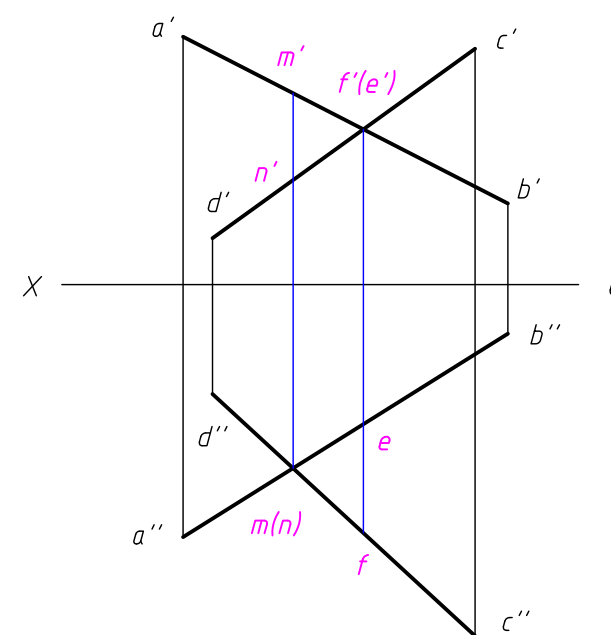
4. 在直线AB上取点K，使AK:KB=1:2。



5. 判断两直线的相对位置。



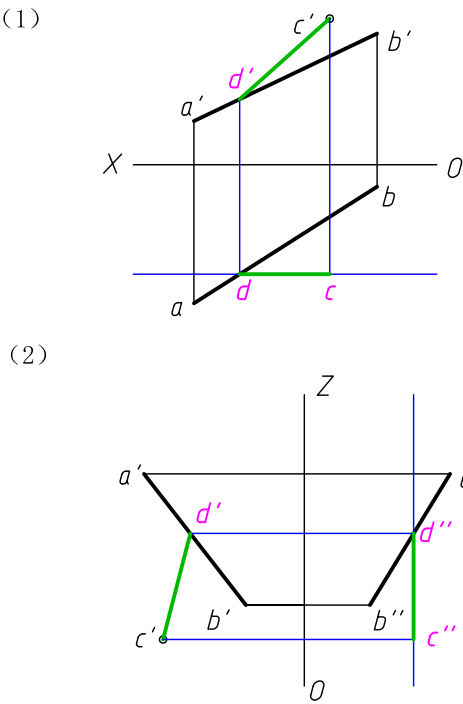
6. 作直线AB、CD对V面的重影点E、F和对H面的重影点M、N的两面投影，并表明其可见性。



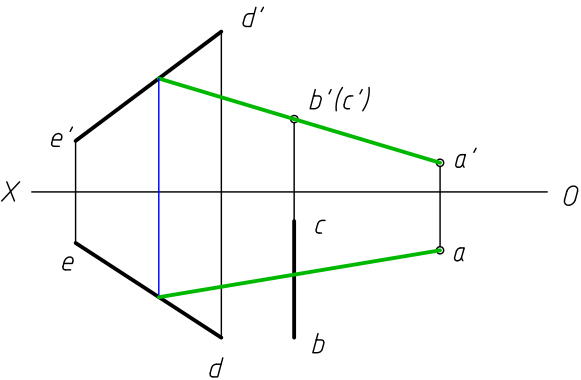
三、直线（二）

班级 姓名 学号

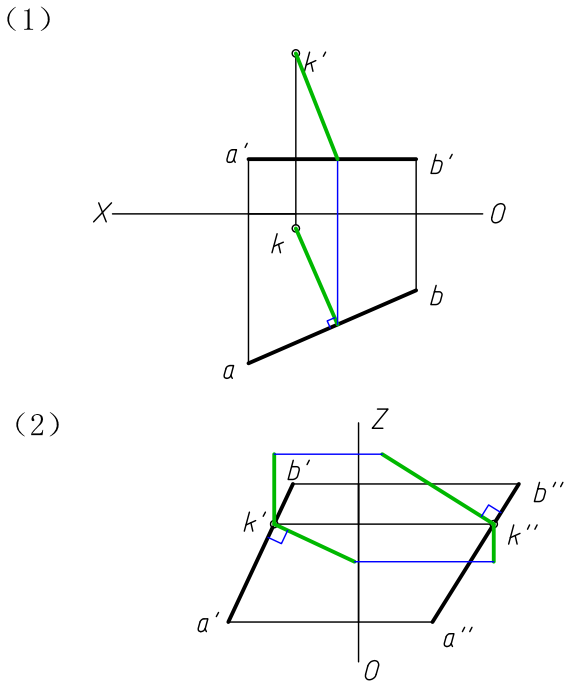
7. 作正平线CD与AB相交，且距V面15mm。



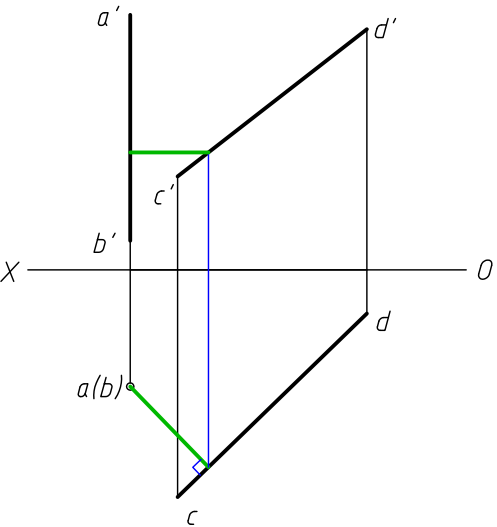
8. 过点A作直线与BC和DE相交。



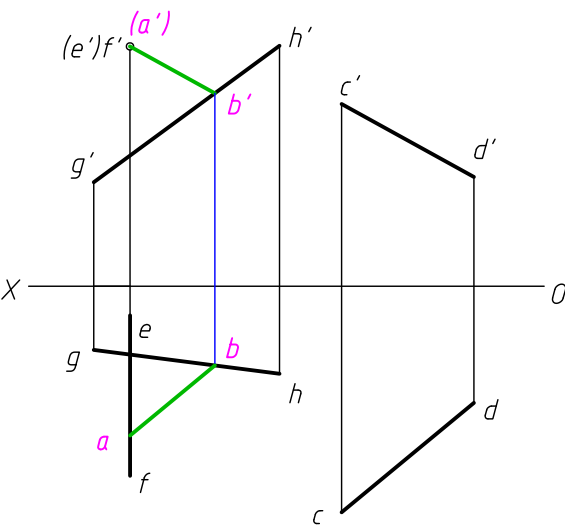
9. 过点K作直线与AB正交。



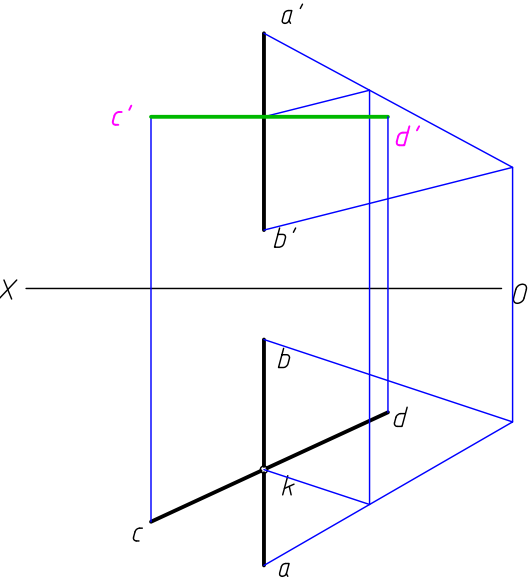
10. 作交叉两直线AB与CD的公垂线。



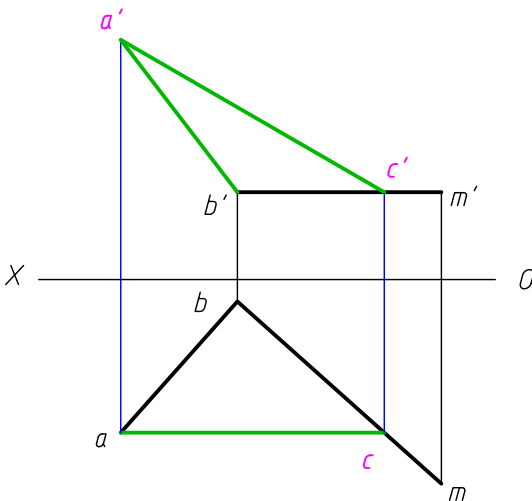
11. 作直线AB与CD平行，并与EF、GH分别相交于点A和B。



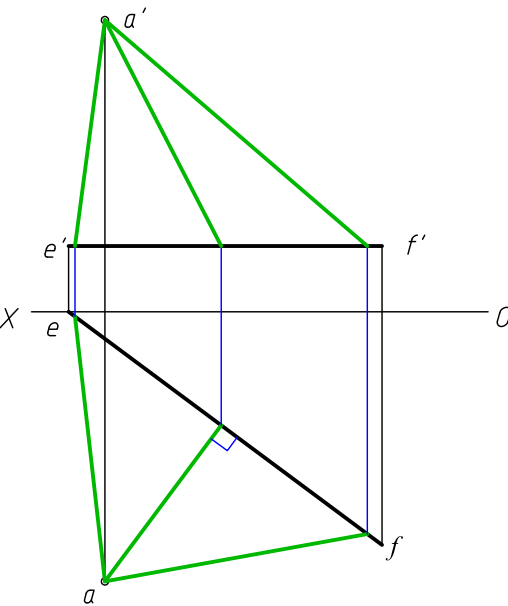
12. 水平线CD与AB相交于点K，求c'd'。



13. 求作直角三角形ABC的两面投影。已知一直角边BC在BM上，斜边AC为正平线，且 $\alpha = 30^\circ$ 。



14. 过点A作直角三角形ABC。已知直角边BC位于EF上，BC=25mm，另一直角边为AB。



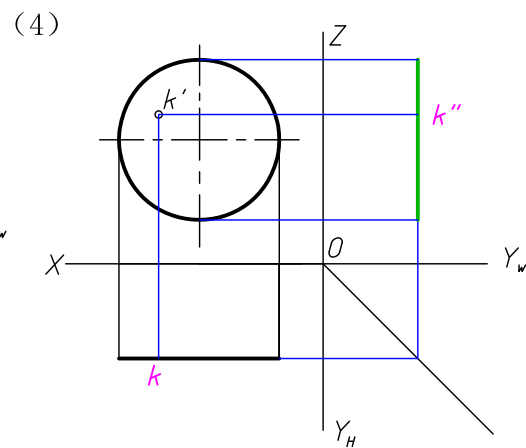
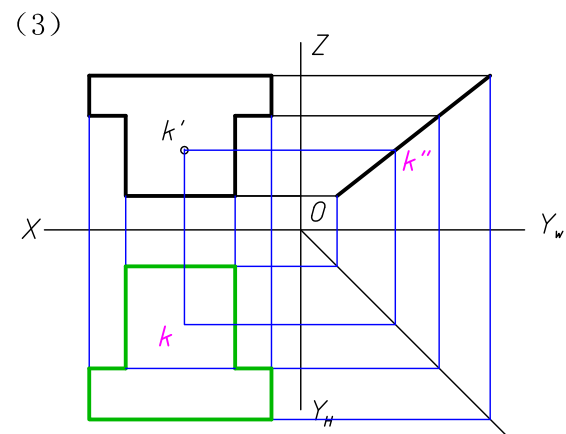
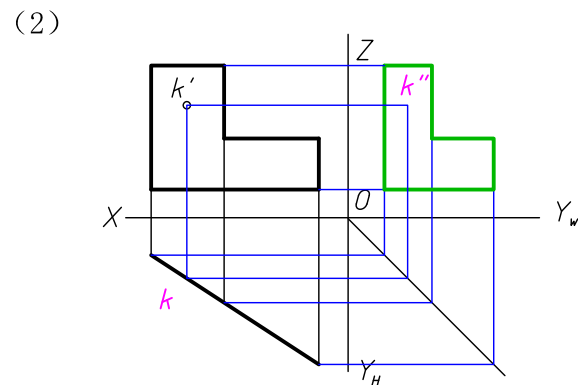
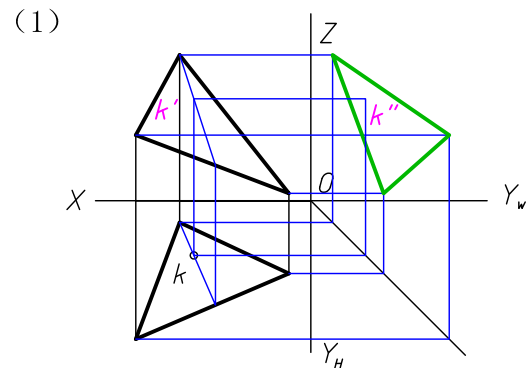
四、平面

班级

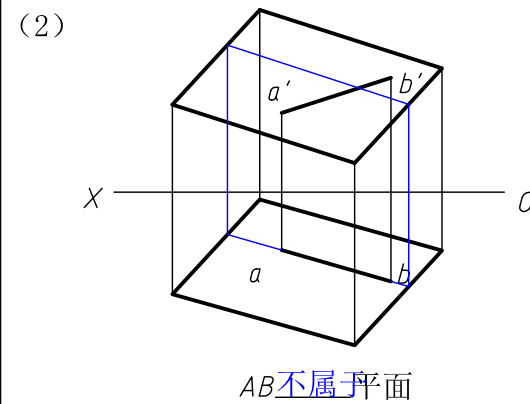
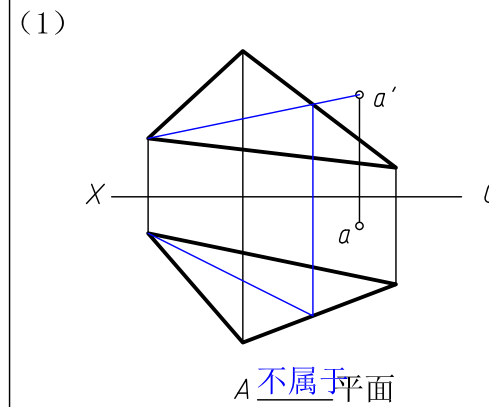
姓名

学号

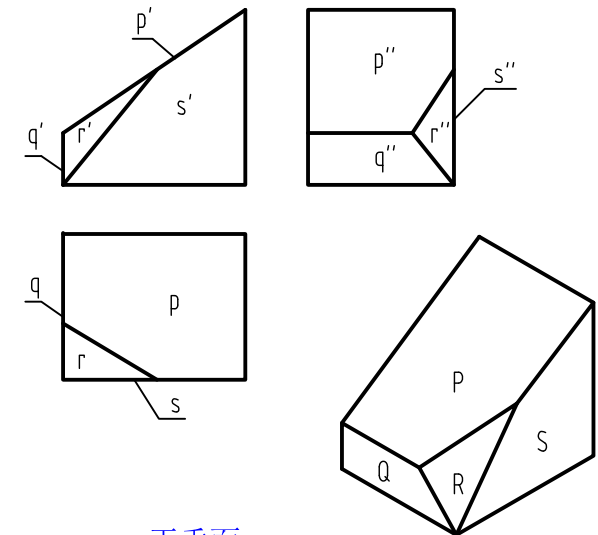
1. 求平面的第三投影，并作出平面上点K的其他两投影。



2. 判别点A和直线AB是否属于给定的平面。

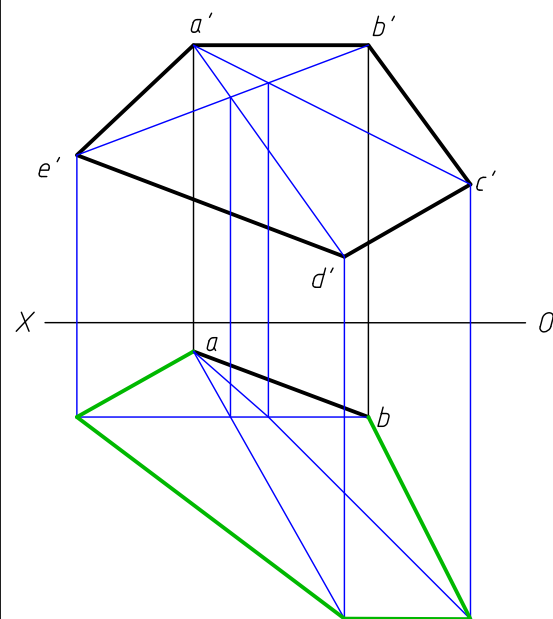


3. 判断立体上各表面对投影面的相对位置。

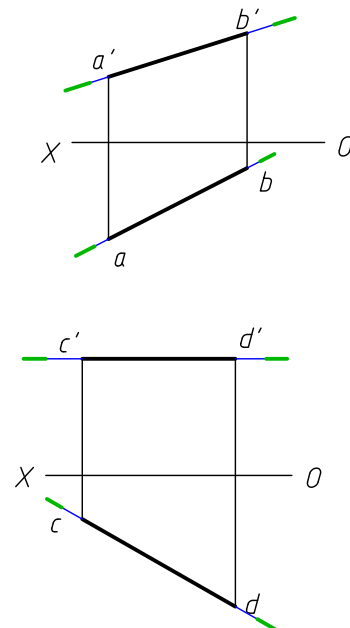


平面 P 正垂面
平面 Q 侧平面
平面 R 一般位置平面
平面 S 正平面

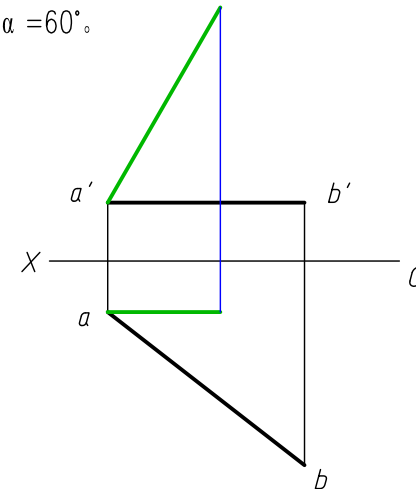
4. 已知直线BE为正平线，求五边形平面的水平投影。



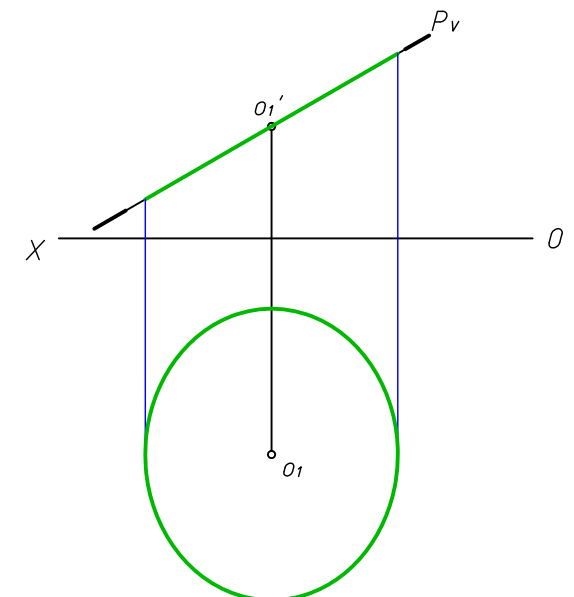
5. 过直线AB和CD各作两个用迹线表示的特殊位置平面。



6. 作平面P的两面投影，已知直线AB为平面P上的水平线，其正平线 $\alpha = 60^\circ$ 。



7. 作出P面上以O₁为圆心，直径为40mm的圆的投影。



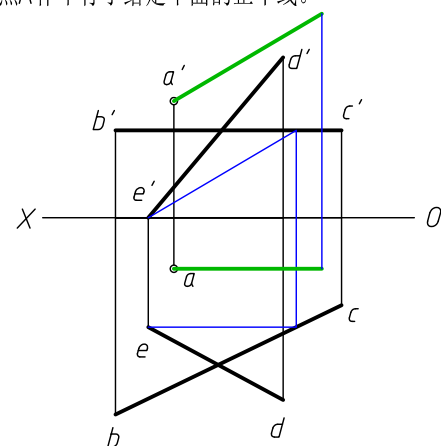
五、直线与平面、平面与平面的相对位置（一）

班级

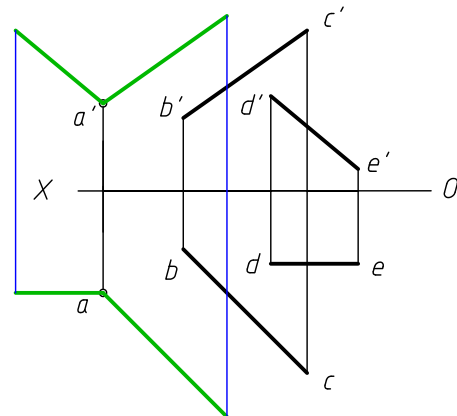
姓名

学号

1. 过点A作平行于给定平面的正平线。

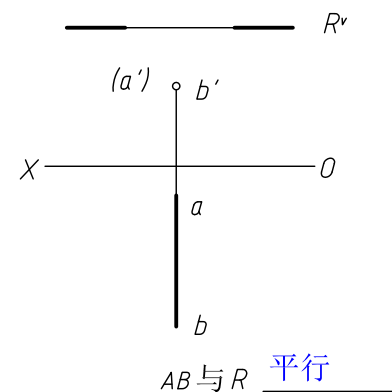


2. 过点A作平面平行于给定的两条直线。

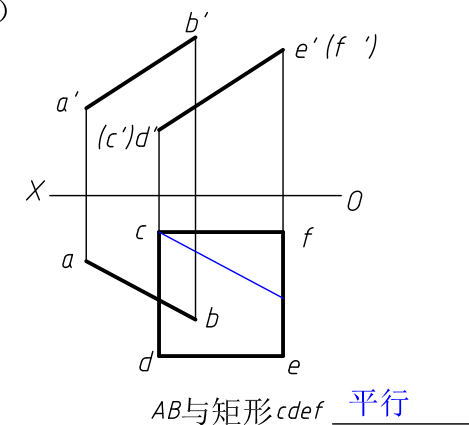


3. 判断直线与平面是否平行。

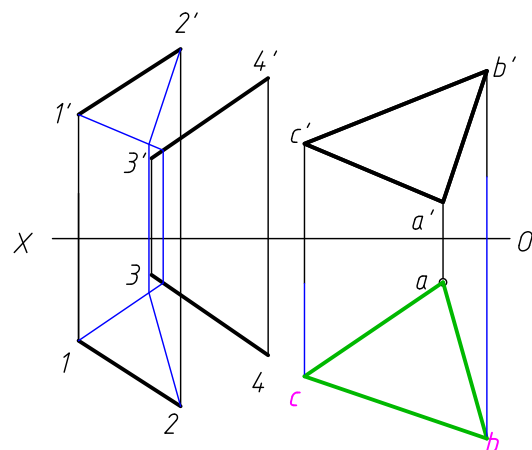
(1)



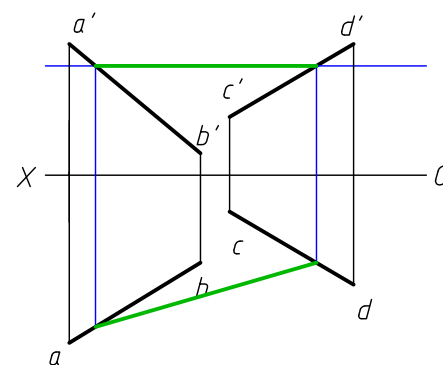
(2)



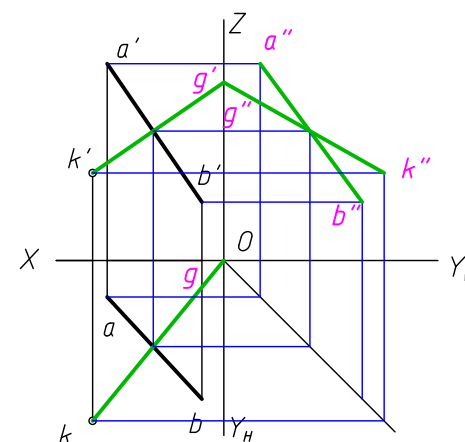
4. 已知两平面互相平行，求三角形ABC的水平投影。



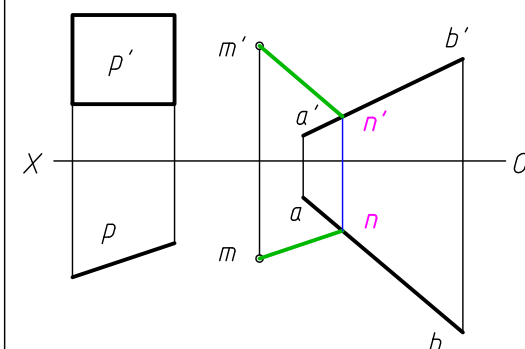
5. 作一水平线EF，距H面15mm，并与直线AB、CD相交。



6. 过一点K作一直线KG与AB相交，G点在Z轴上。

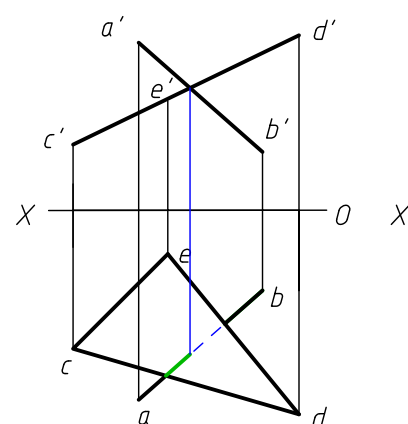


7. 作直线MN平行于P平面且与直线AB相交。

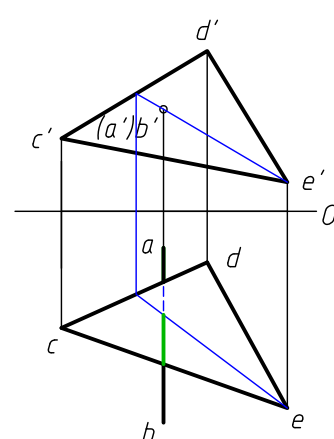


8. 求直线与平面的交点，并表明其可见性。

(1)

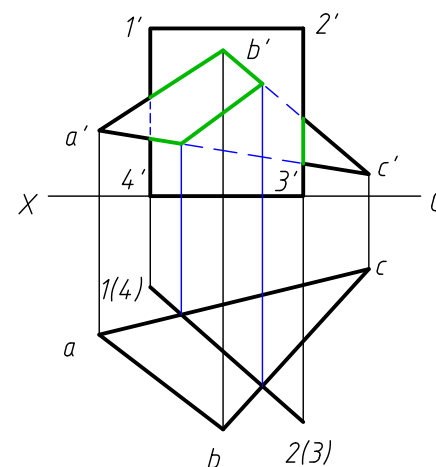


(2)

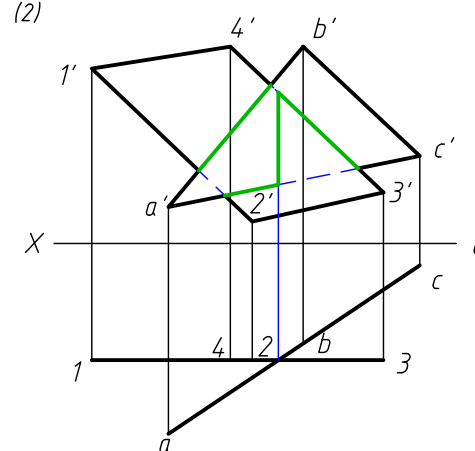


9. 求两平面的交线，并表明其可见性。

(1)



(2)



五、直线与平面、平面与平面的相对位置（二）

班级

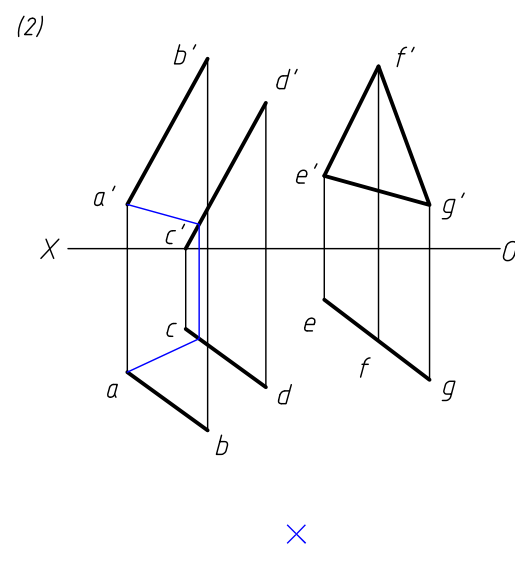
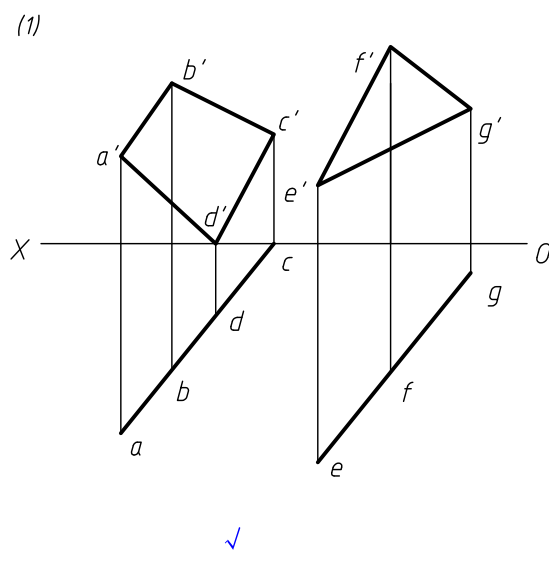
姓名

学号

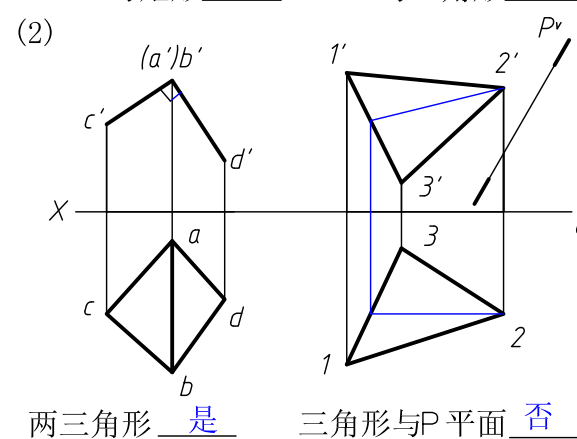
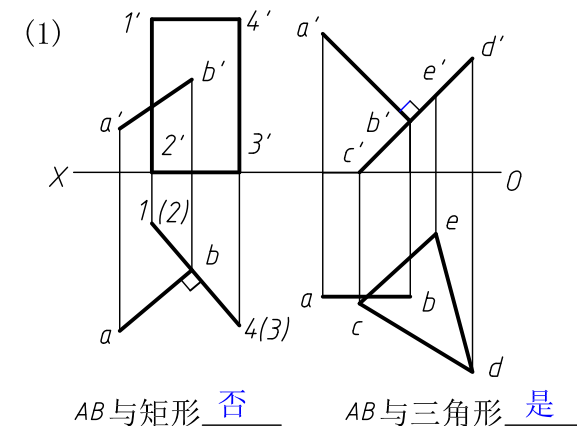
10. 判断下列平面是否平行。是打“√”， 否打“×”。

(1) $abdc // efg$

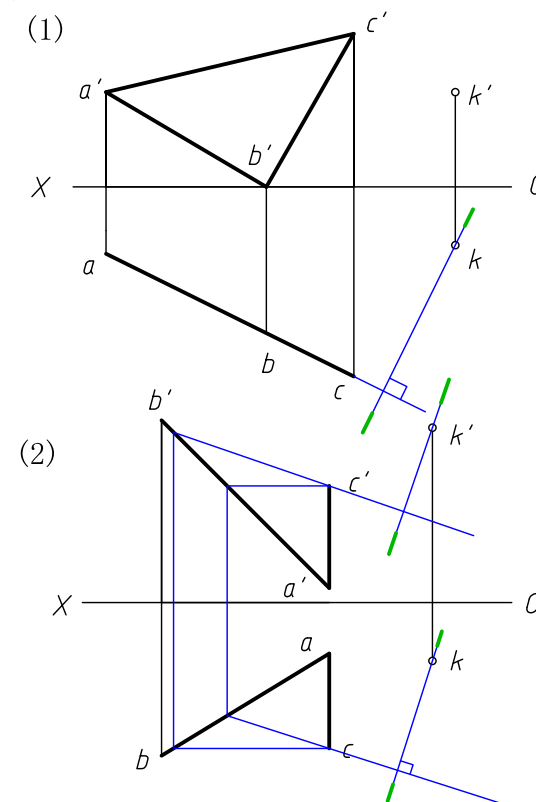
(2) $ab // cd // efg, a'b' // c'd' // e'f'$



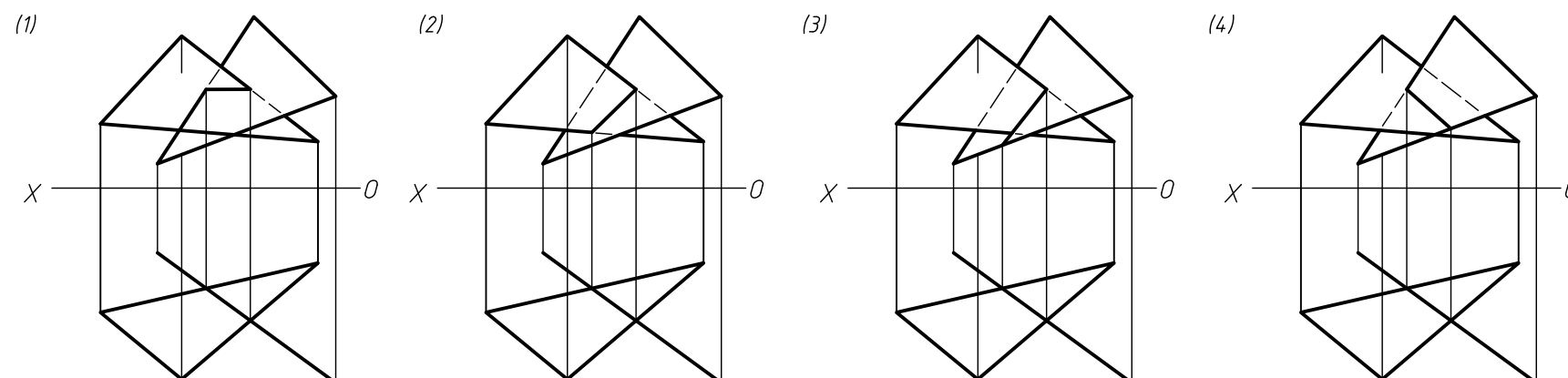
11. 判断直线与平面或两平面是否垂直。



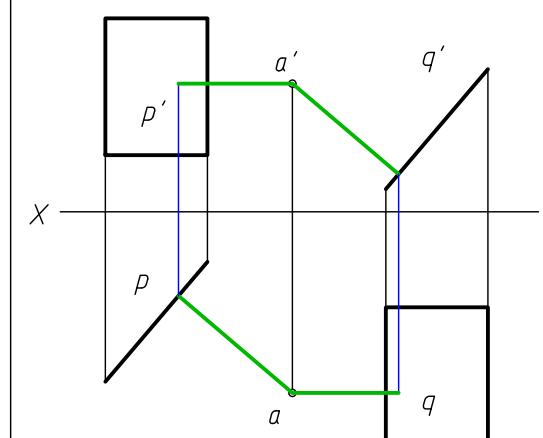
12. 过点K作特殊位置平面（用迹线表示），垂直于给定的平面。



13. 两平面相交的正确投影是（ 2 ）。



14. 过点A作平面R(AB, AC)与两平面垂直。



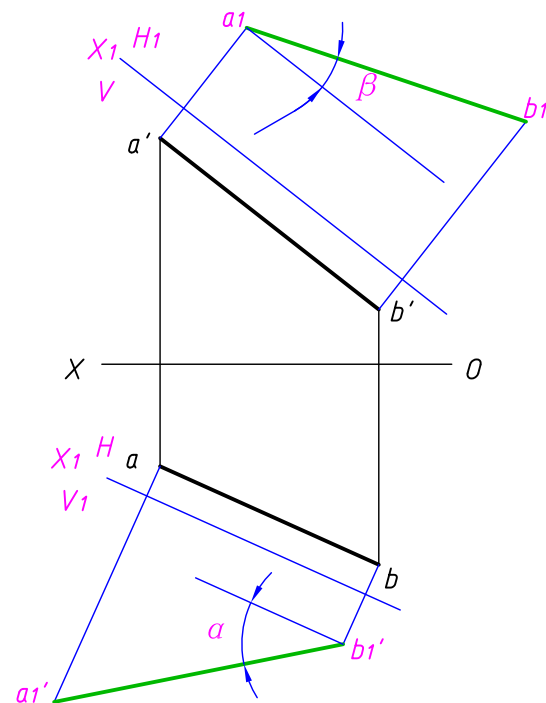
六、换面法

班级

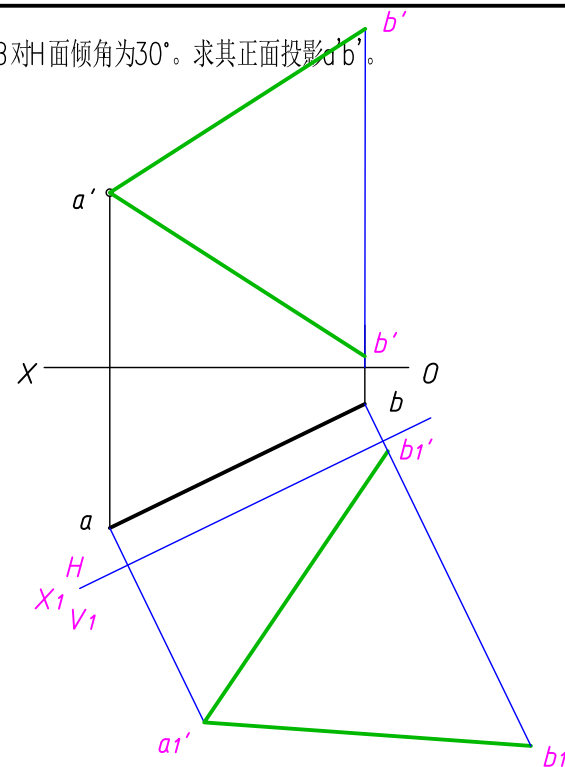
姓名

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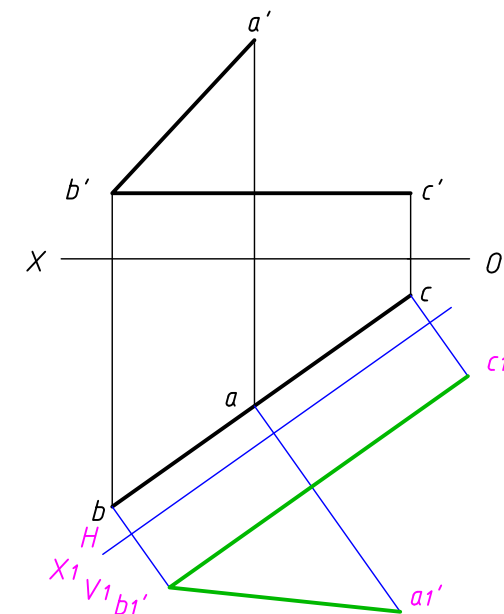
1.求直线AB的实长及其对H、V面的倾角 α 、 β 。



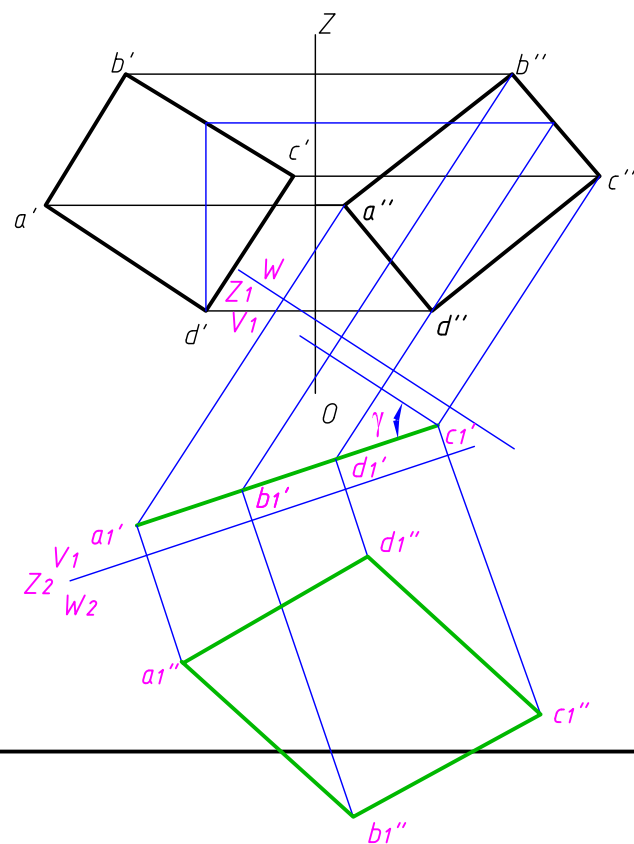
2.已知直线AB对H面倾角为 30° 。求其正面投影 $a'b'$ 。



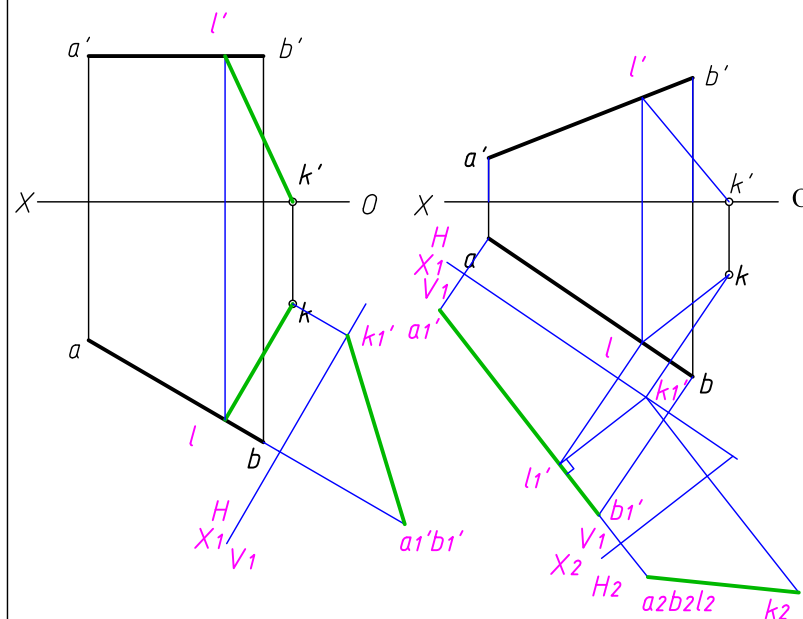
3.求 $\angle ABC$ 的实际大小。



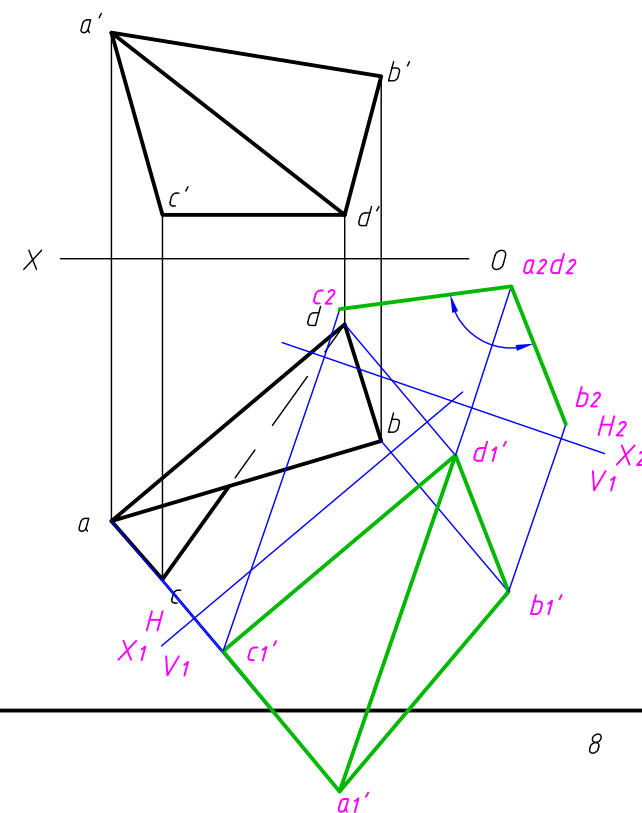
4.求平行四边形ABCD平面对W面的倾角 γ ，并求其实形。



5.求点K至直线AB的距离，并作出表示距离的线段在H、V面上的投影。



6.求两个三角形平面的夹角。



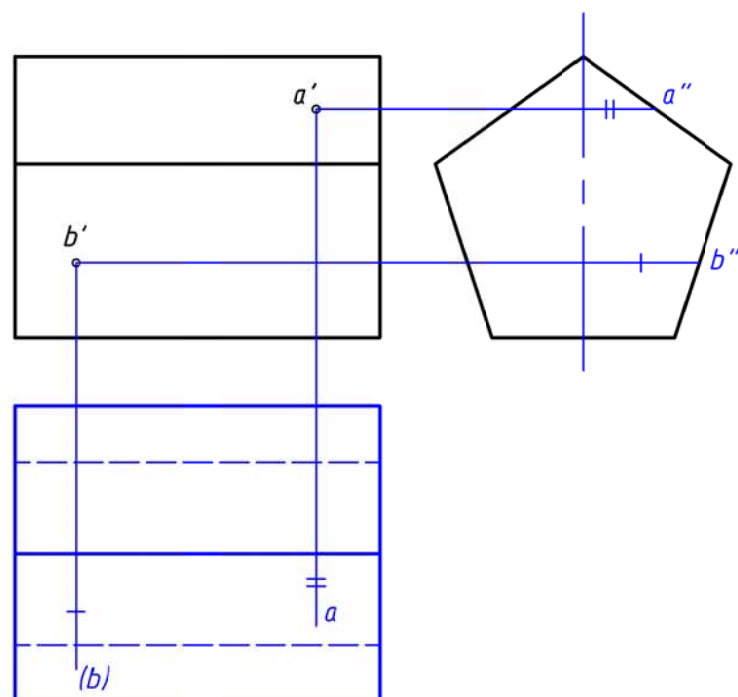
七、平面立体的投影

班级

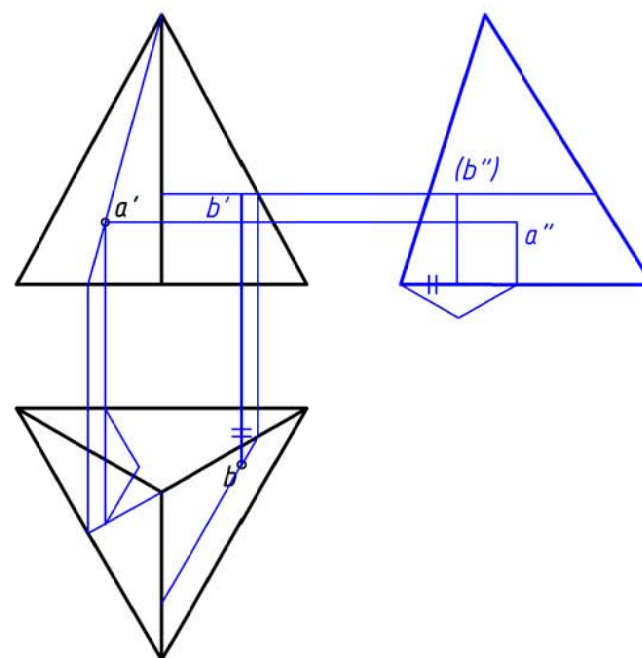
姓名

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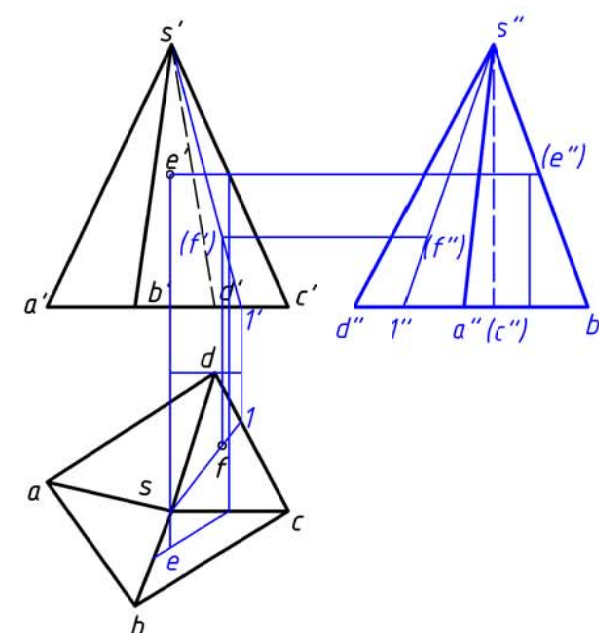
1. 完成五棱柱及其表面上点A和B的三面投影。



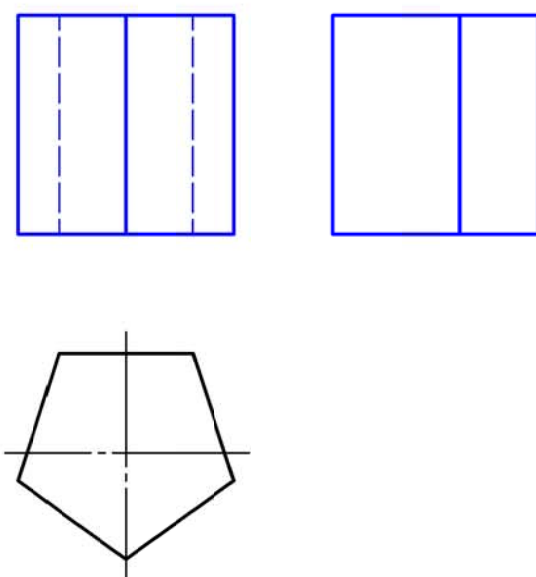
2. 完成三棱锥及其表面上点A和B的三面投影。



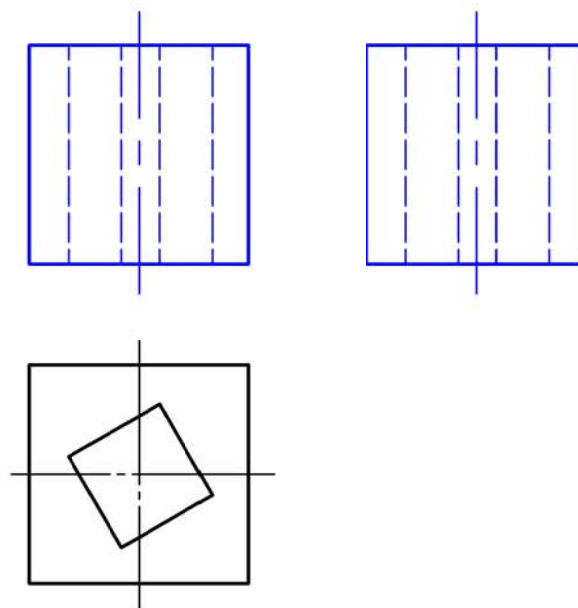
3. 完成四棱锥及其表面上点E和F的三面投影。



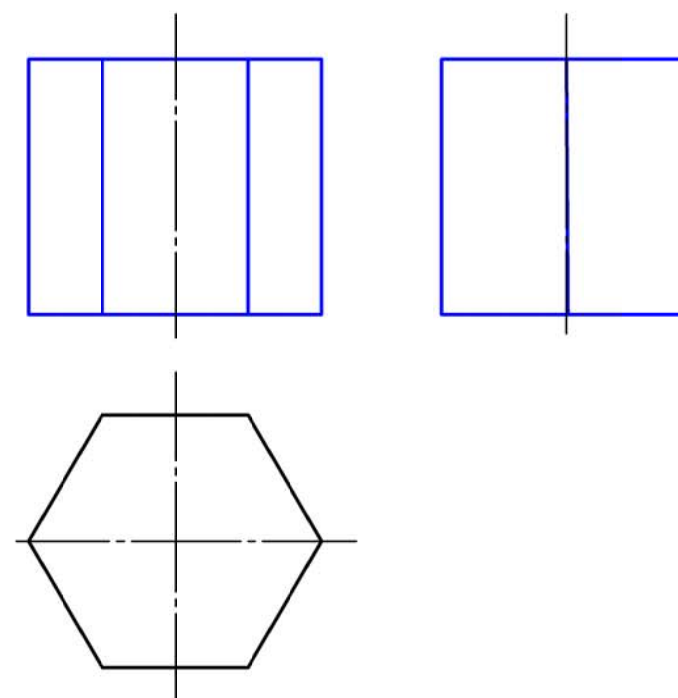
4. 正五棱柱高30mm。已知其水平投影，求正面及侧面投影。



5. 四棱柱高30mm，中间有一方通孔。已知其水平投影，求正面及侧面投影。



6. 完成上、下底为水平面，高为35mm的正六棱柱的正面投影和侧面投影。



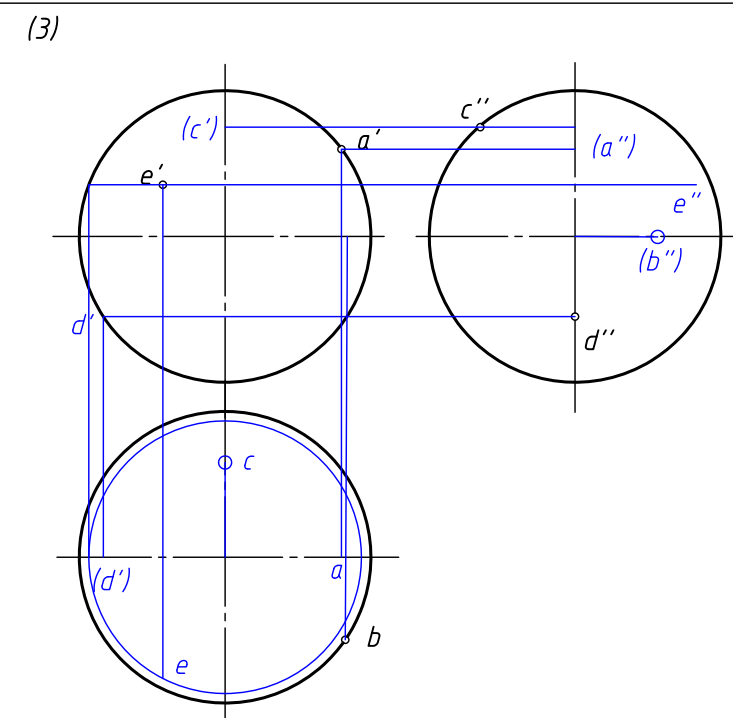
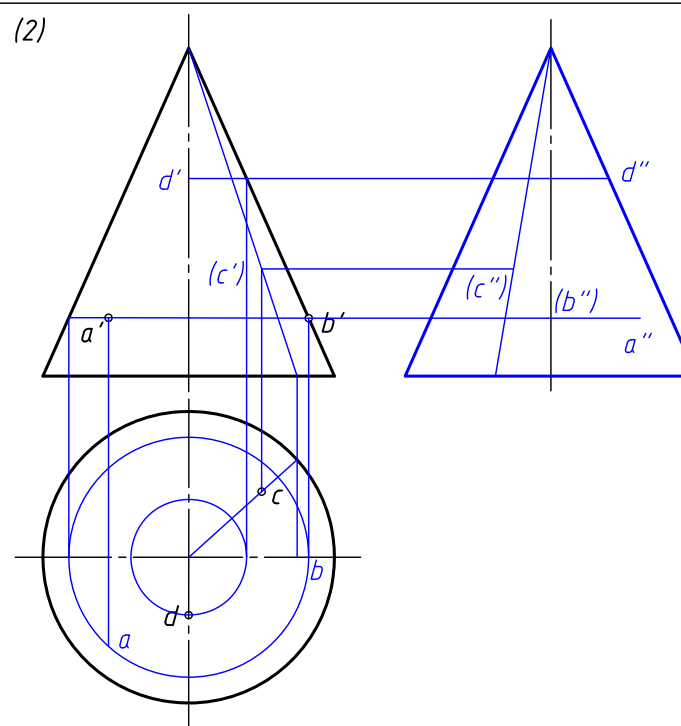
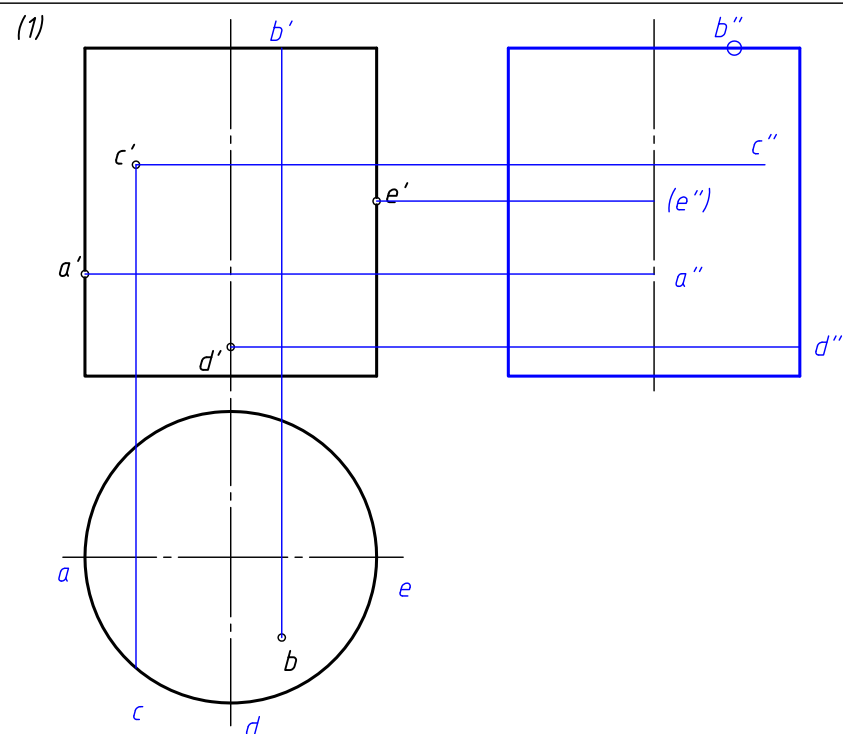
八、曲面立体的投影

班级

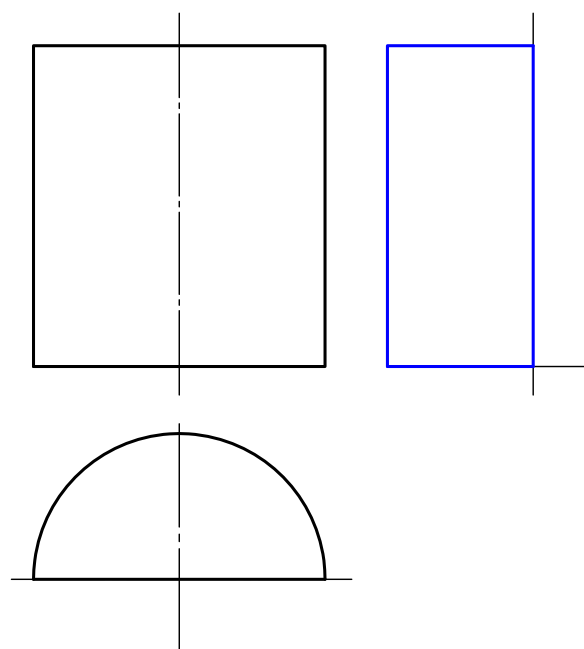
姓名

学号

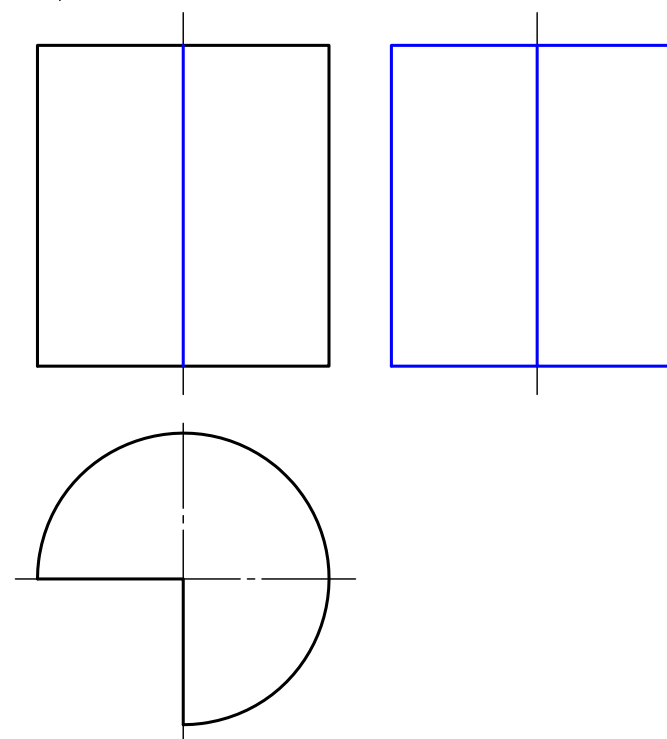
1. 作出下列各立体的第三面投影，并补全立体表面上各点的其余投影。



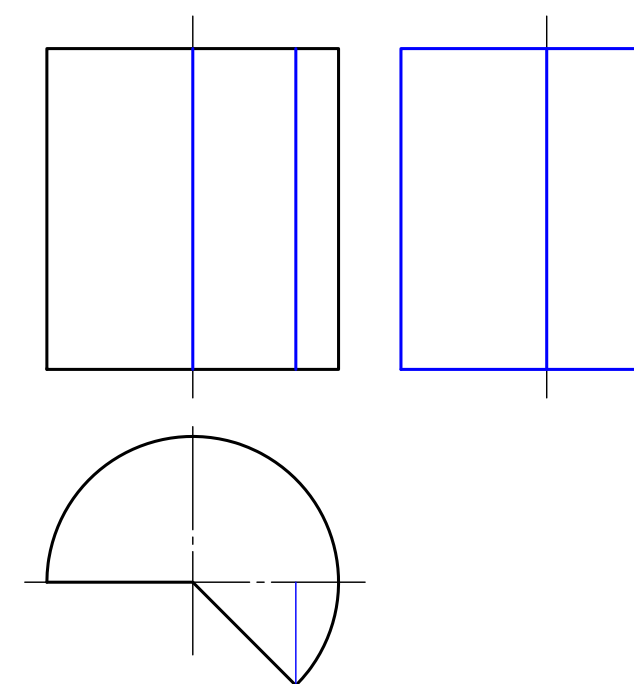
2. 完成1/2圆的柱三面投影。



3. 完成3/4圆柱的三面投影。



4. 完成不完整圆柱的三面投影。



八、曲面立体的投影

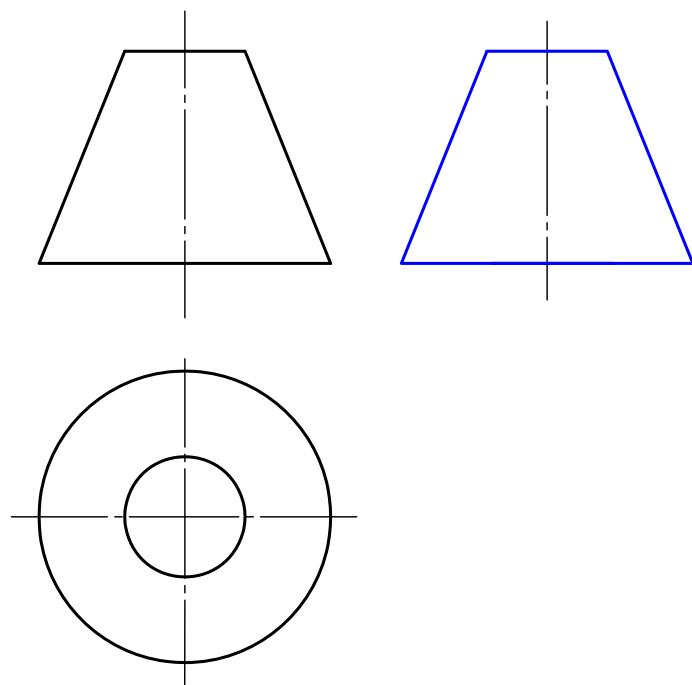
班级

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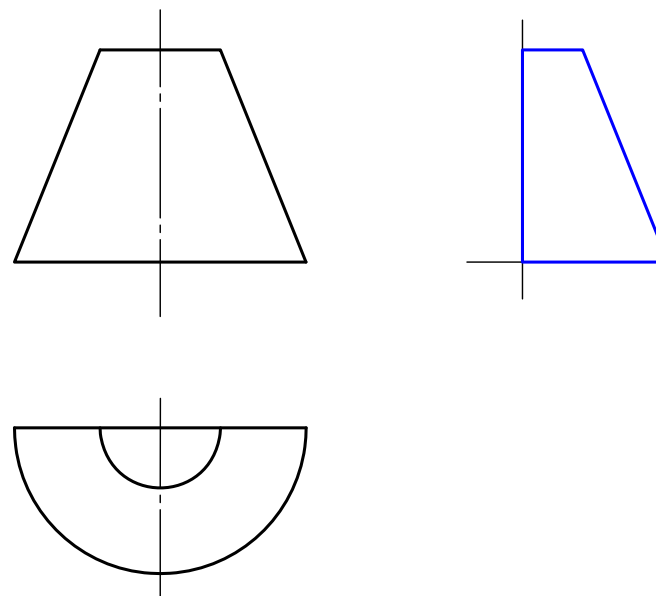
学号

5. 完成下列各立体的三面投影。

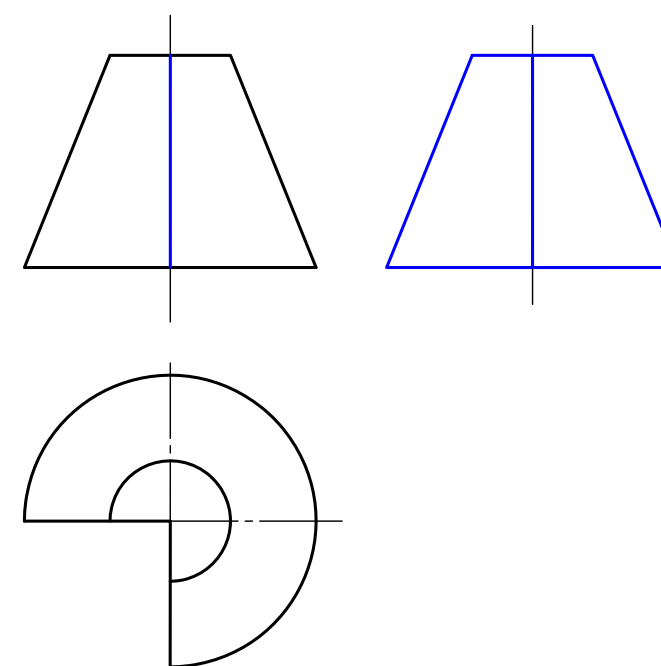
(1)



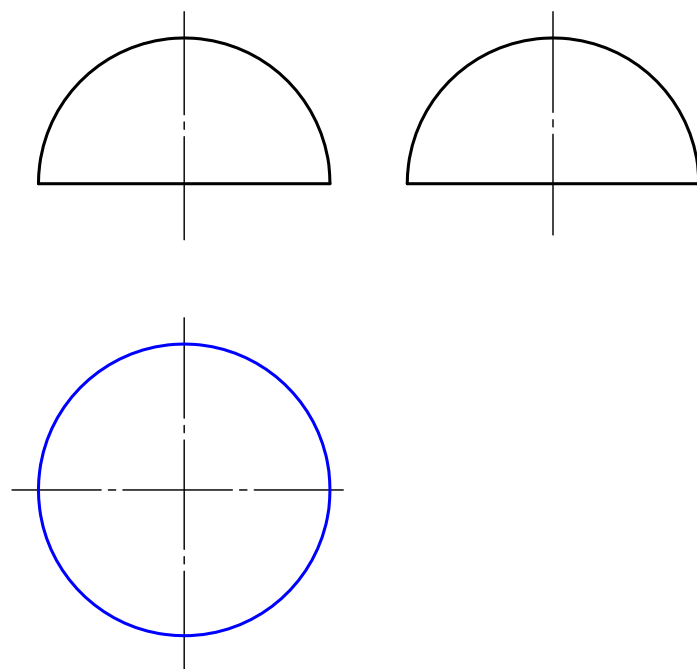
(2)



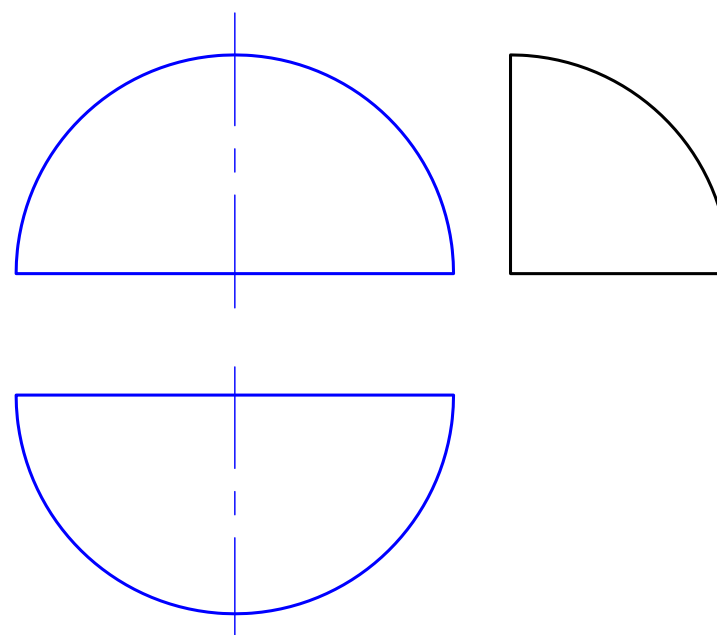
(3)



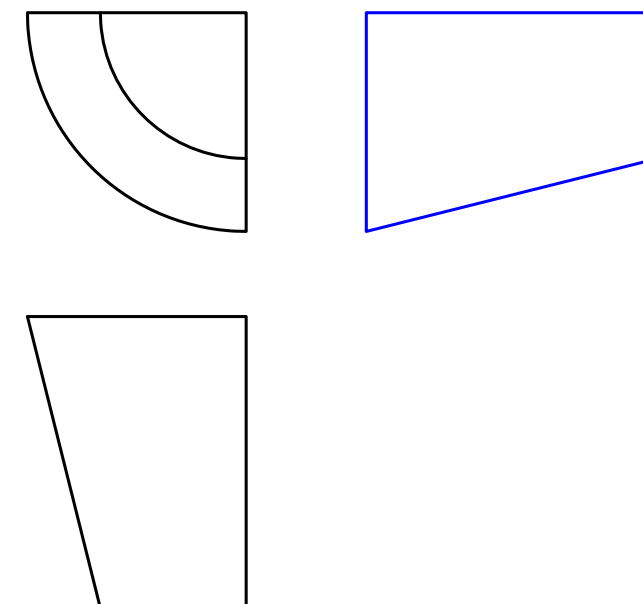
6. 完成1/2圆球的三面投影。



7. 完成1/4圆球的三面投影。



8. 完成1/4圆锥台的三面投影。



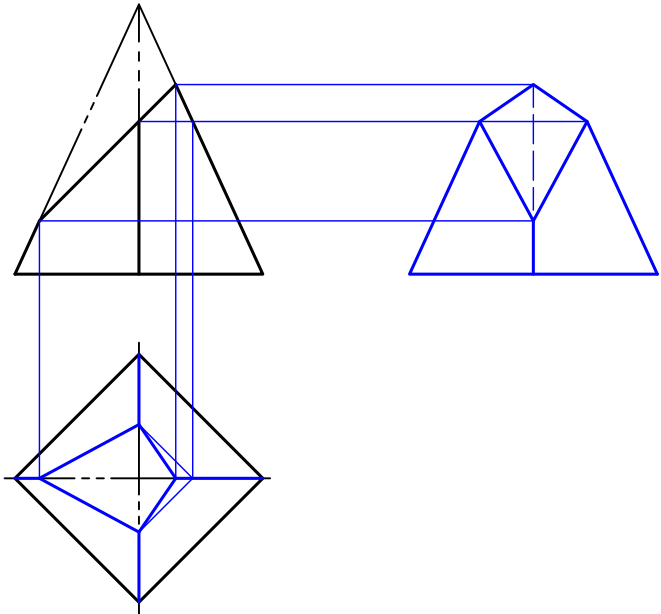
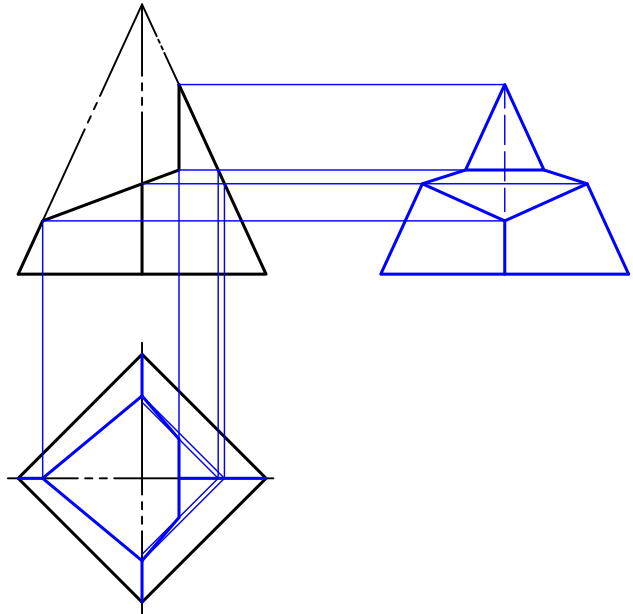
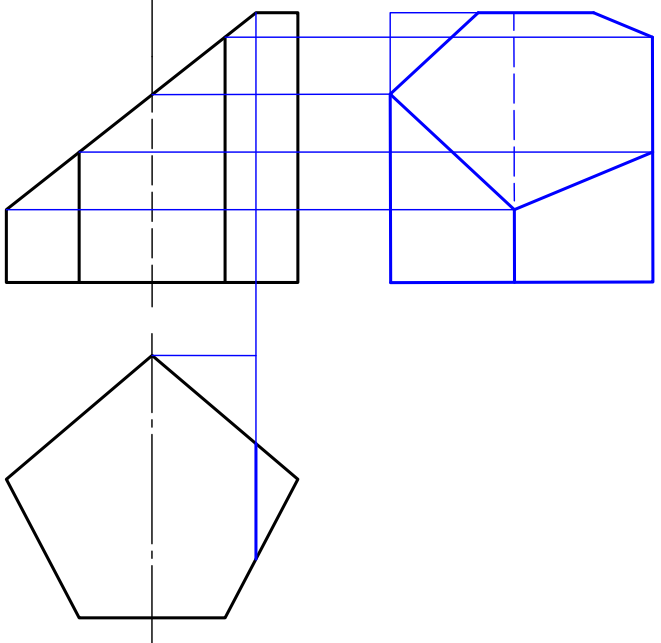
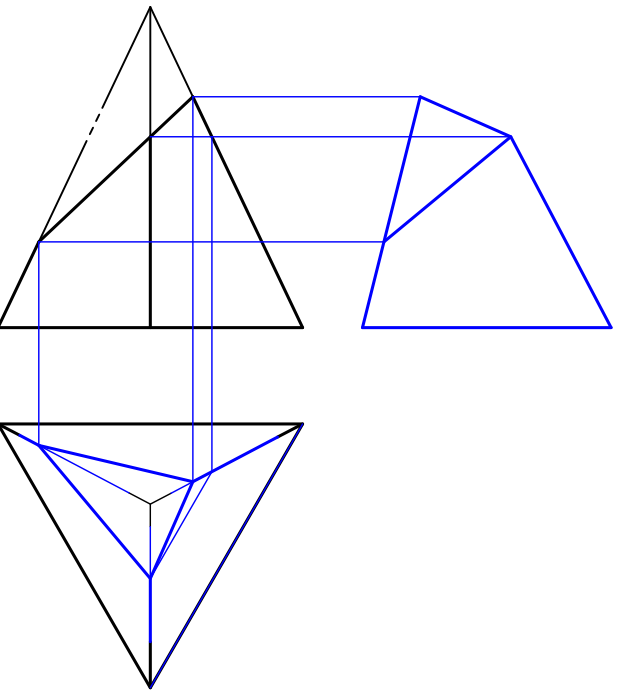
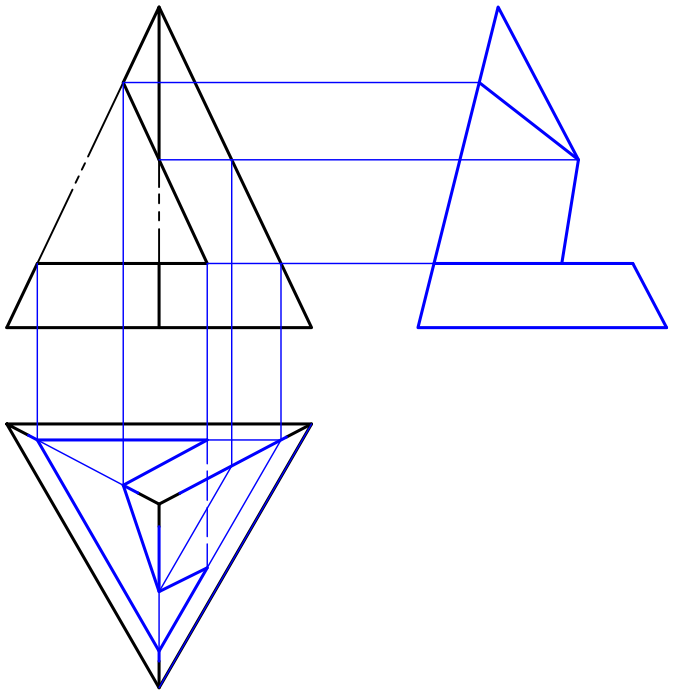
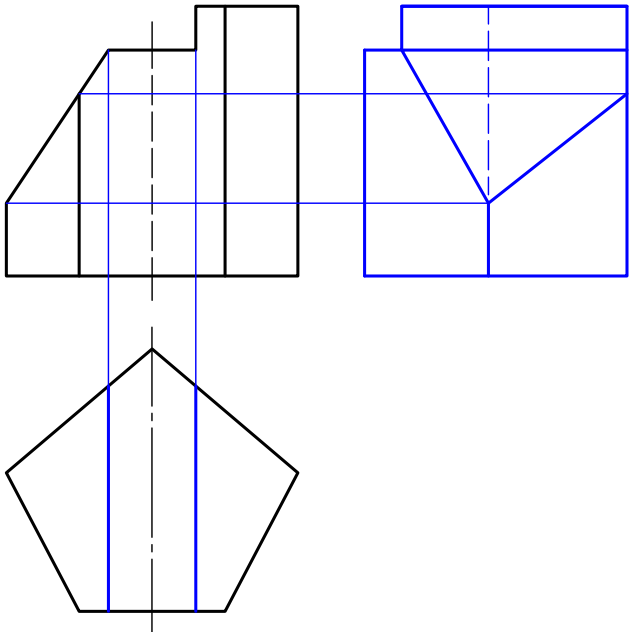
九、平面与立体相交(一)

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1.完成截切立体的三面投影。

(1) 	(2) 	(3) 
(4) 	(5) 	(6) 

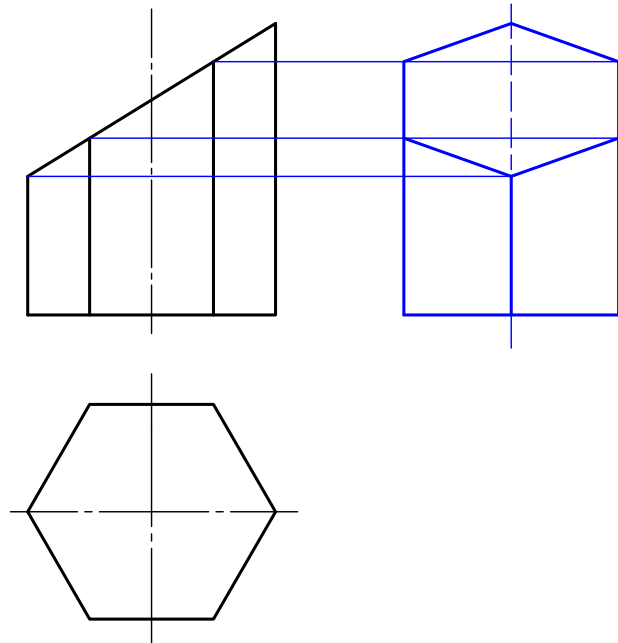
九、平面与立体相交(二)

班级

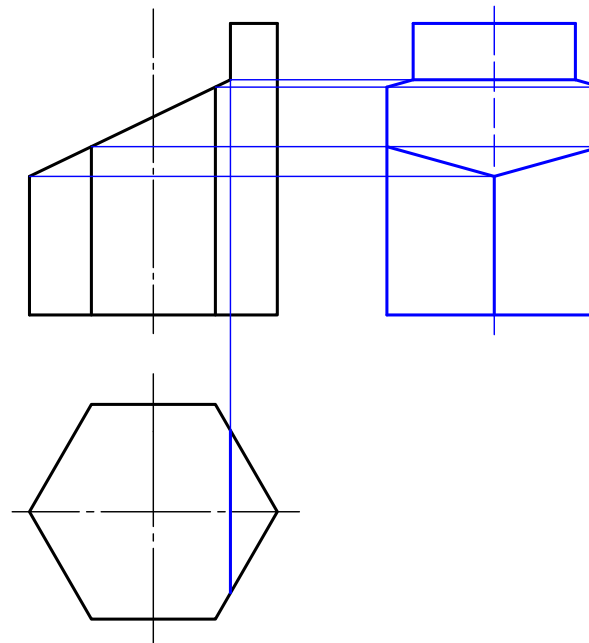
姓名

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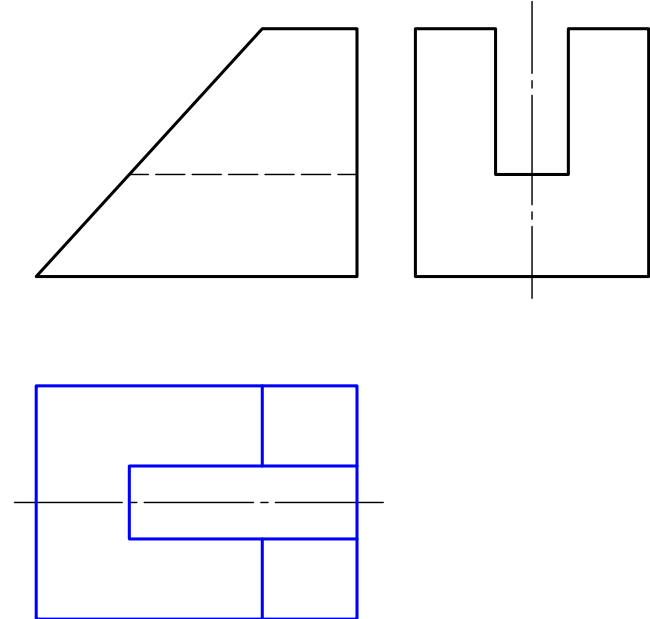
(7)



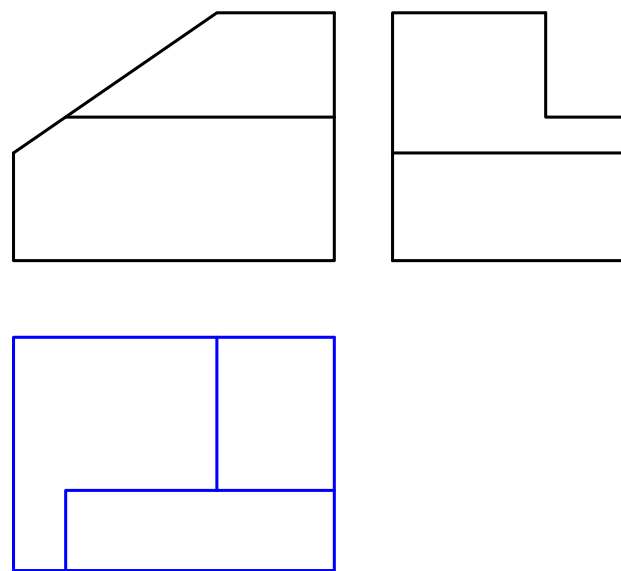
(8)



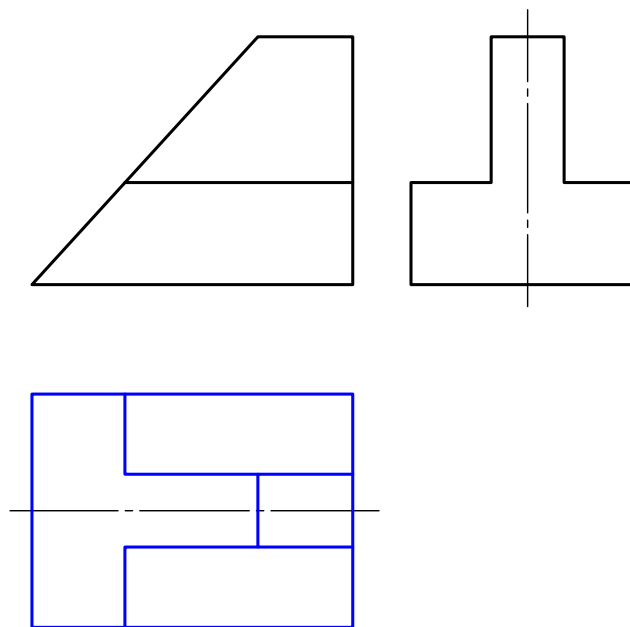
(9)



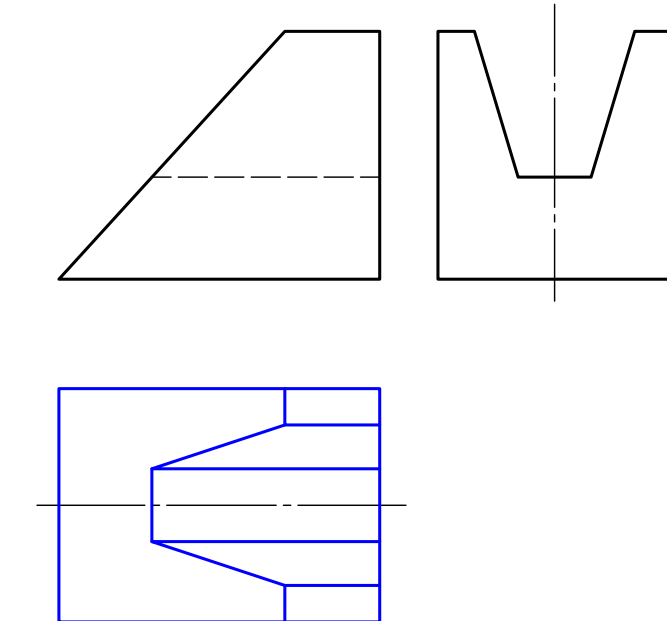
(10)



(11)



(12)



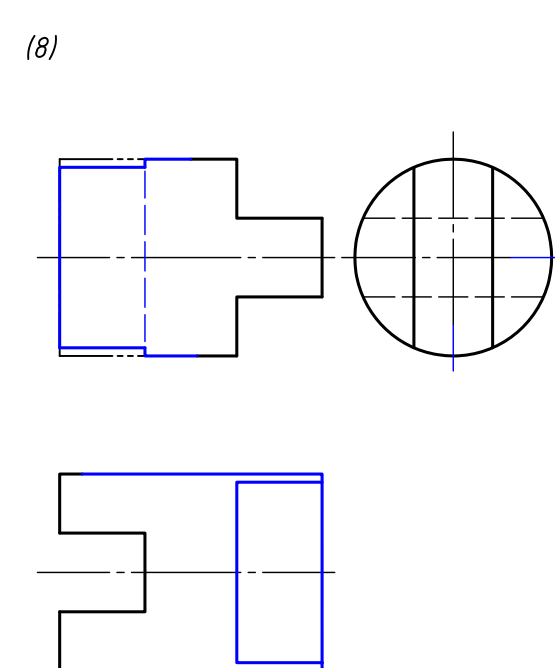
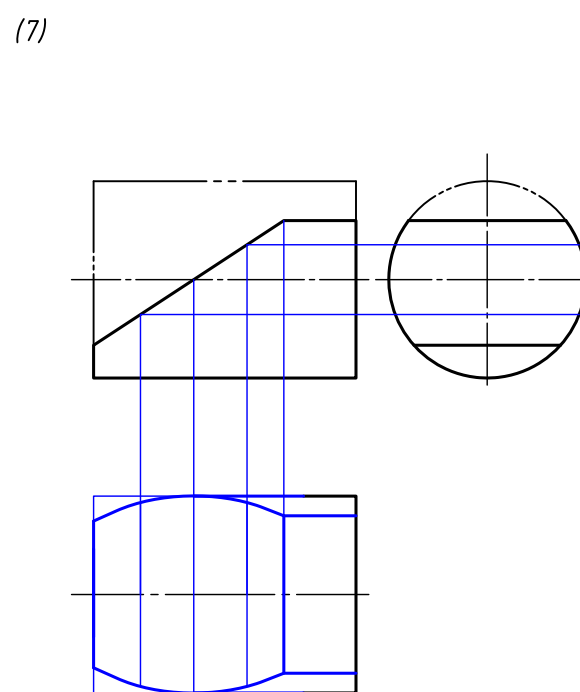
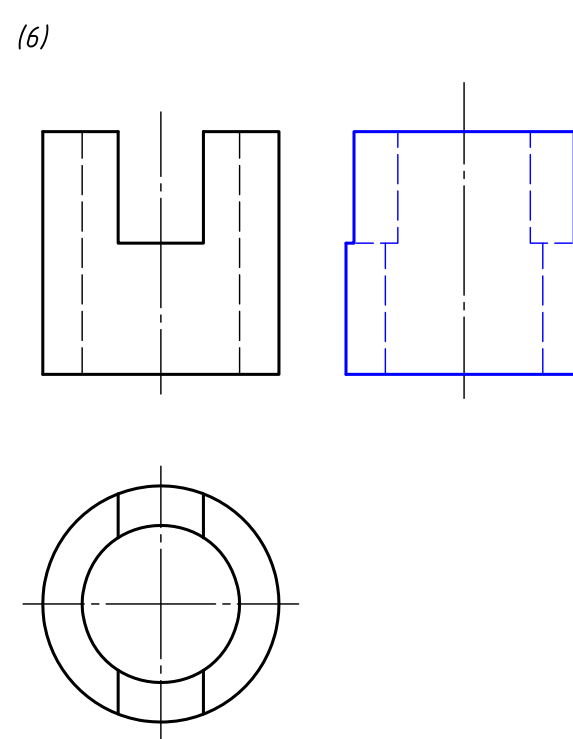
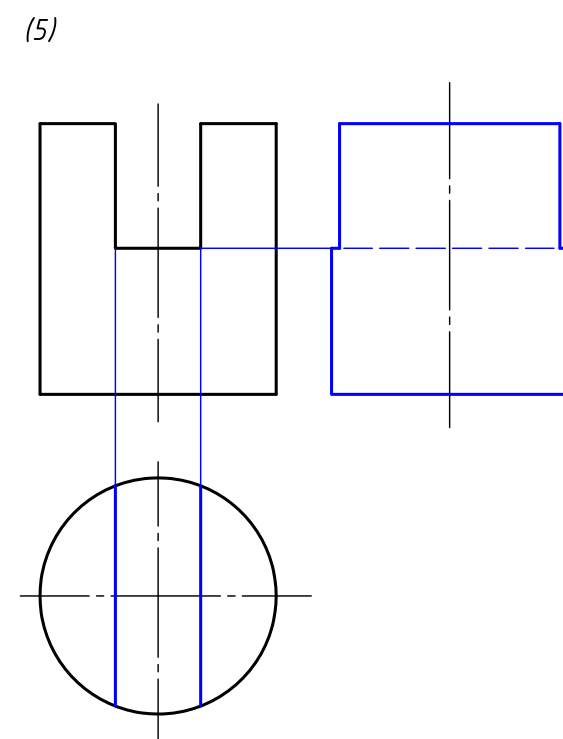
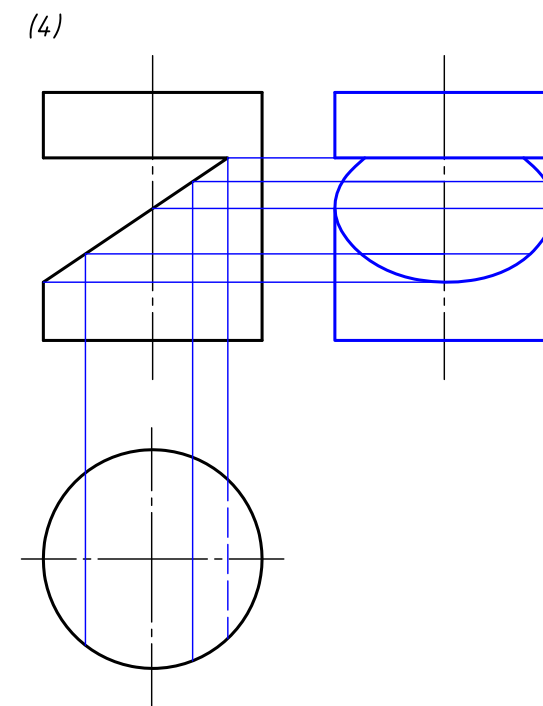
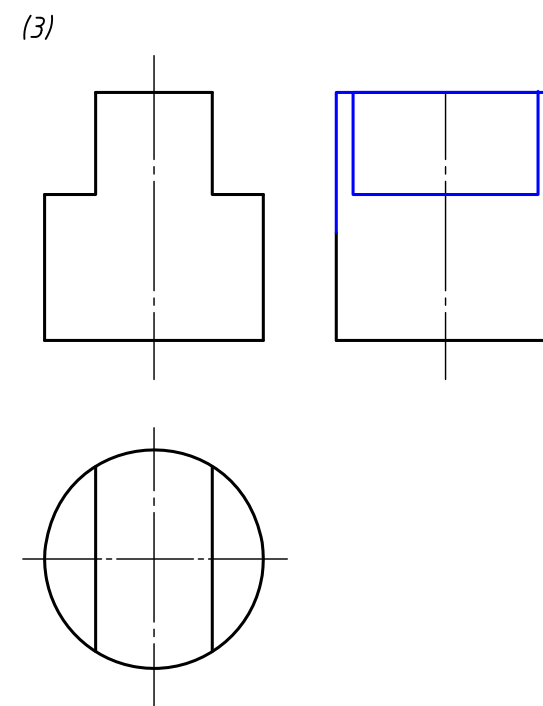
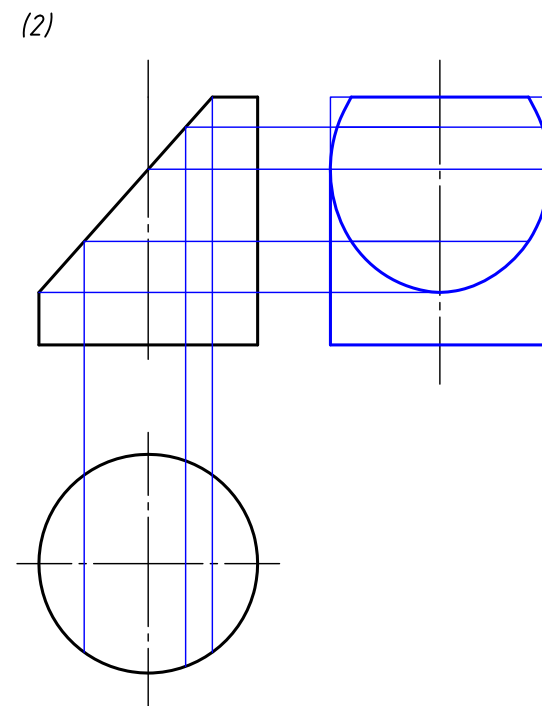
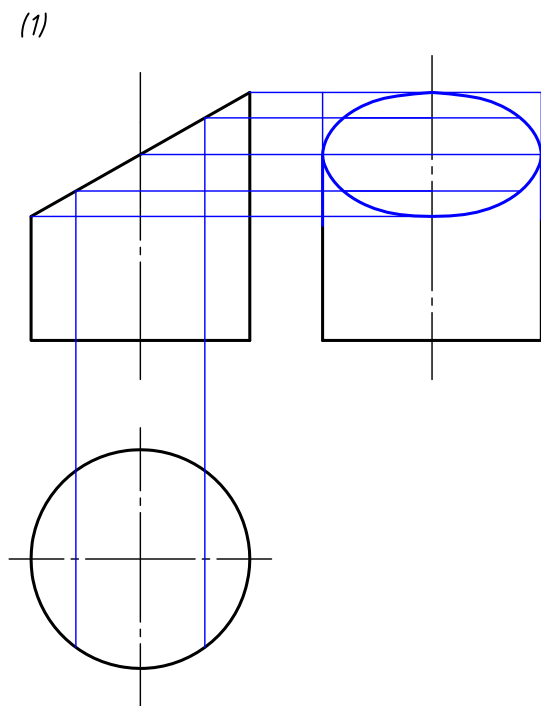
九、平面与立体相交(三)

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2. 分析下列圆柱被平面截切后所形成的截交线的形状和位置，并完成其三面投影。



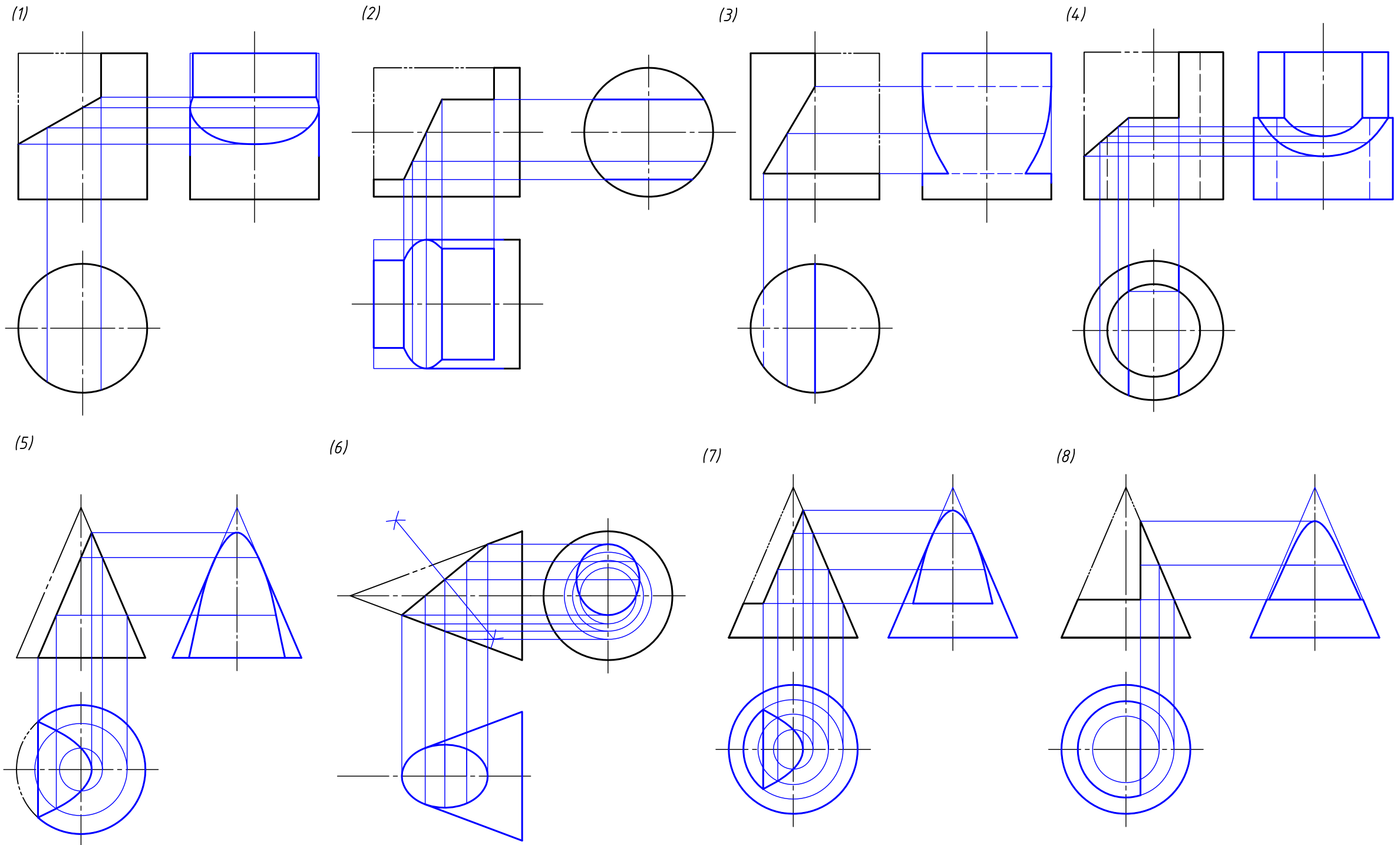
九、平面与立体相交(四)

班级

姓名

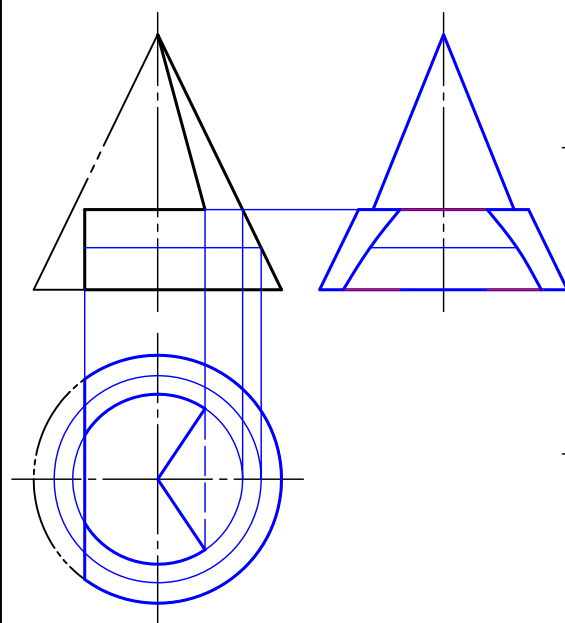
学号

3. 分析下列圆柱与圆锥被平面截切后所形成的截交线的形状和位置，并完成其三面投影。

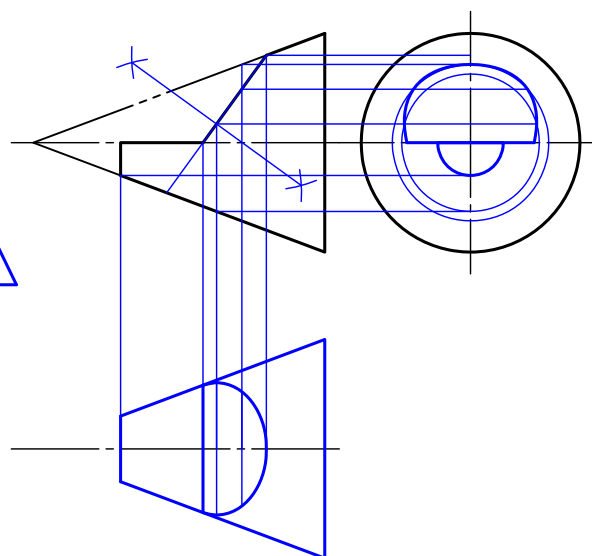


4. 分析下列回转体被平面截切后所形成的截交线的形状和位置，并完成它们的三面投影。

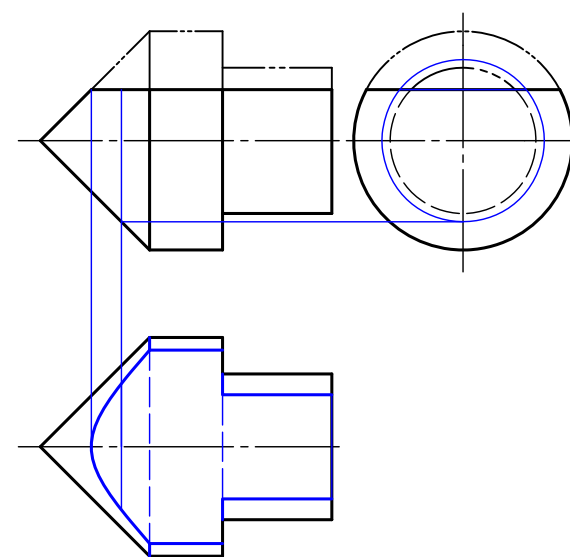
(1)



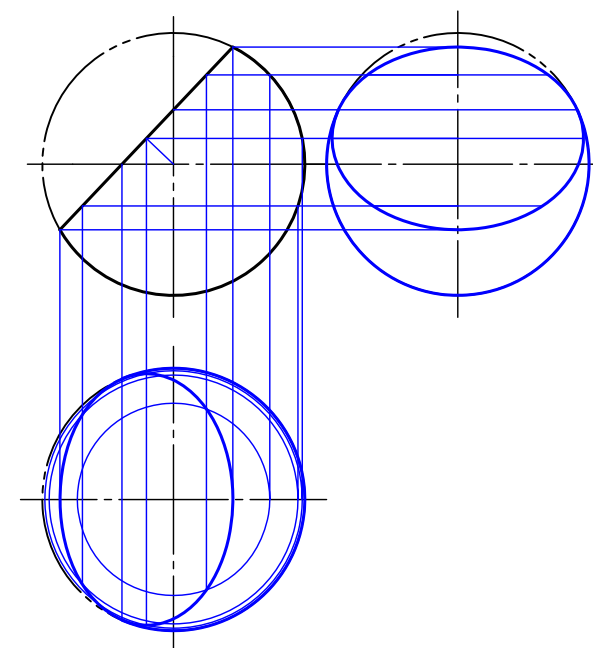
(2)



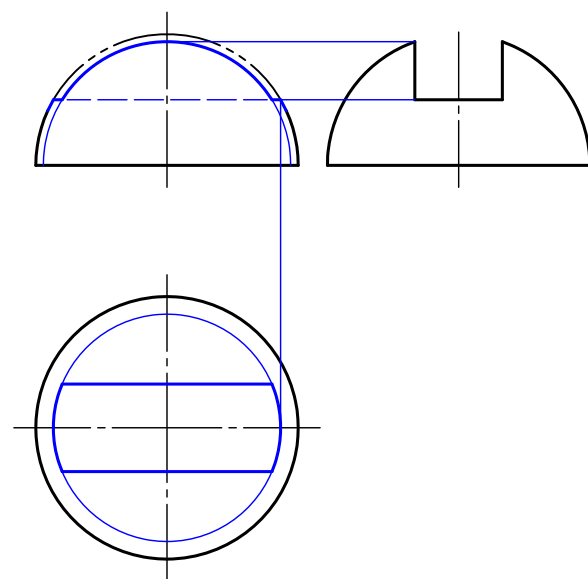
(3)



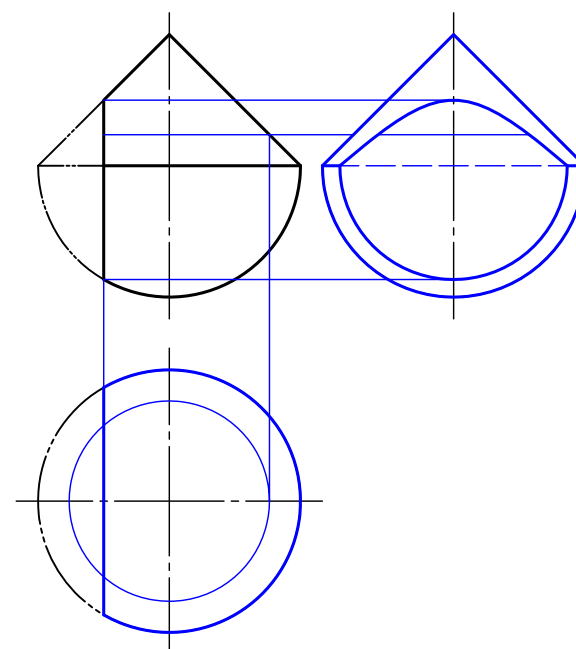
(4)



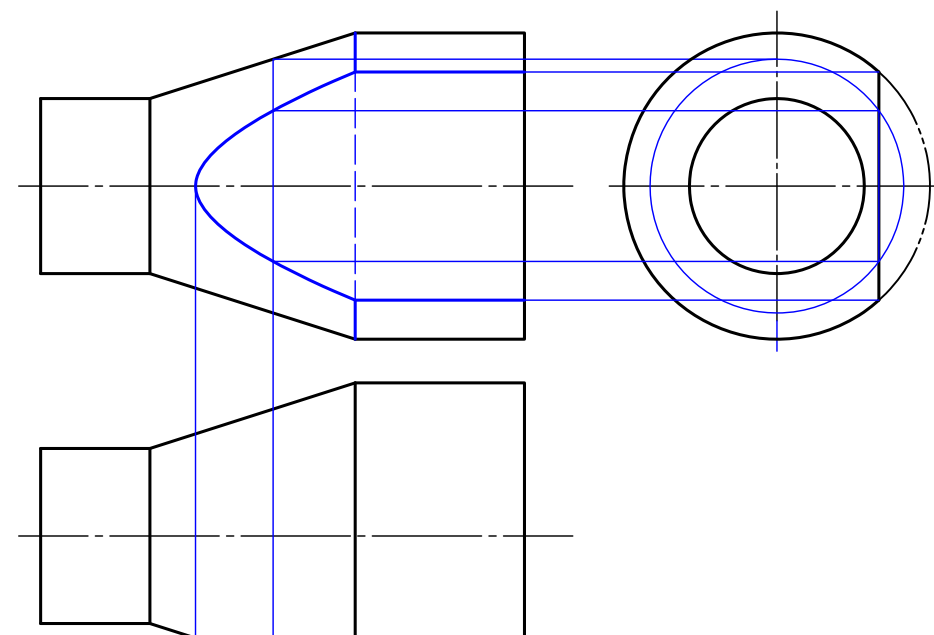
(5)



(6)



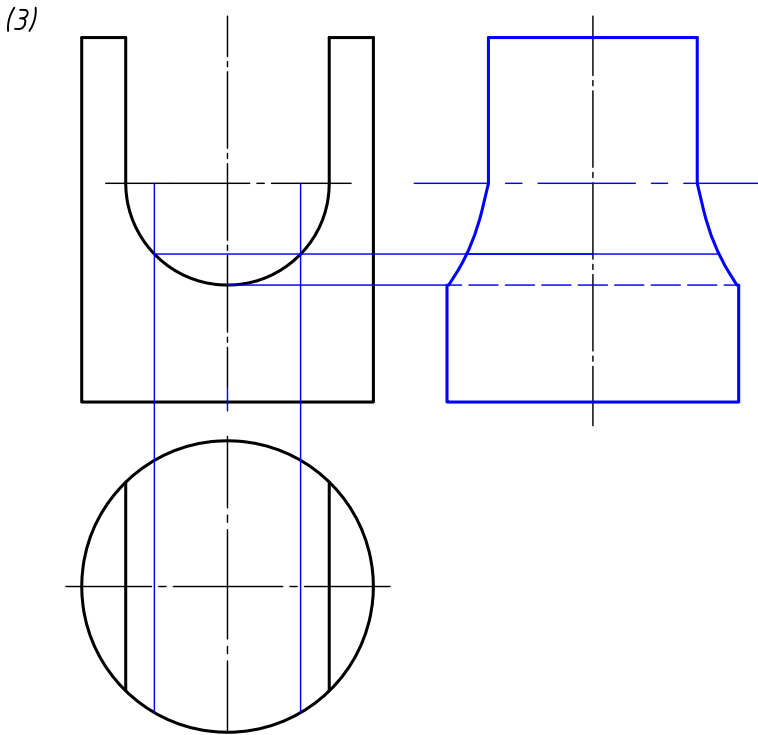
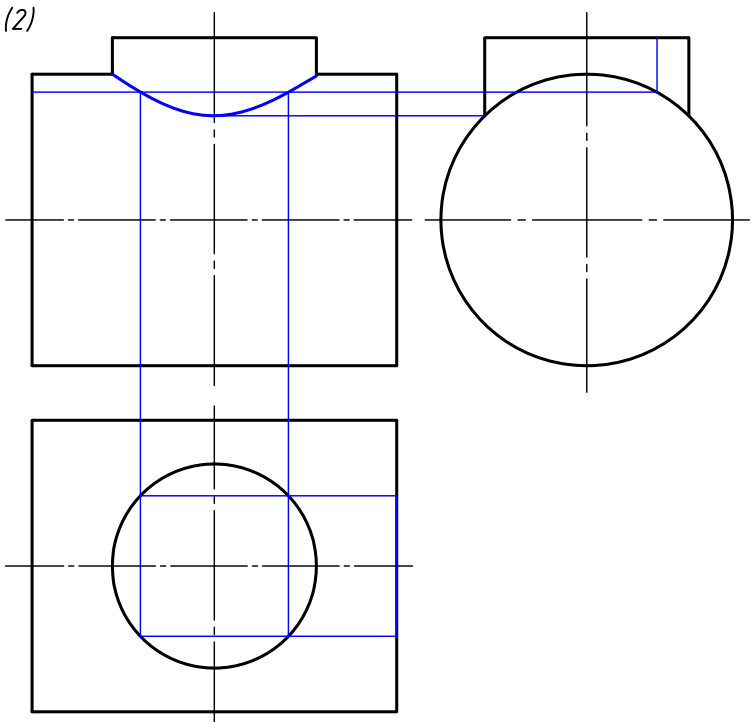
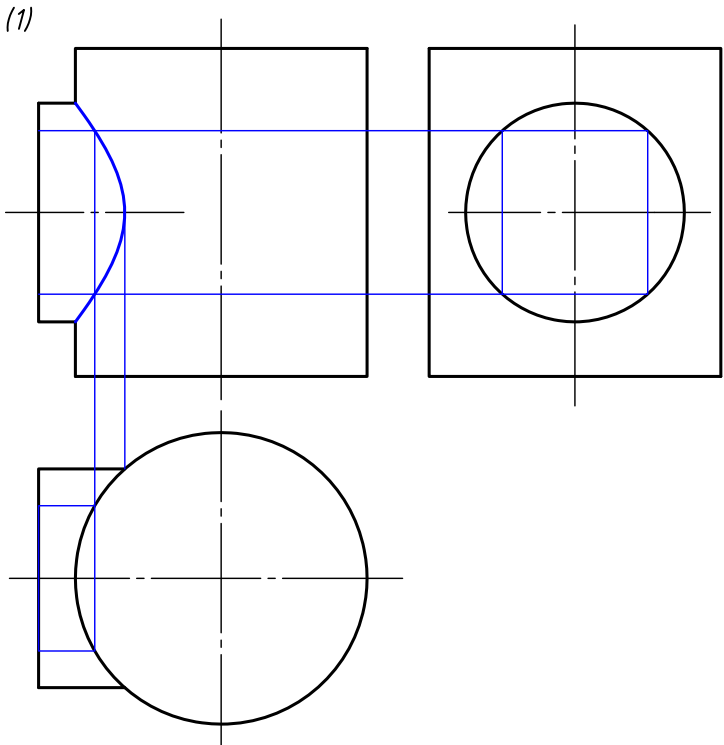
5. 完成截切体的三面投影。



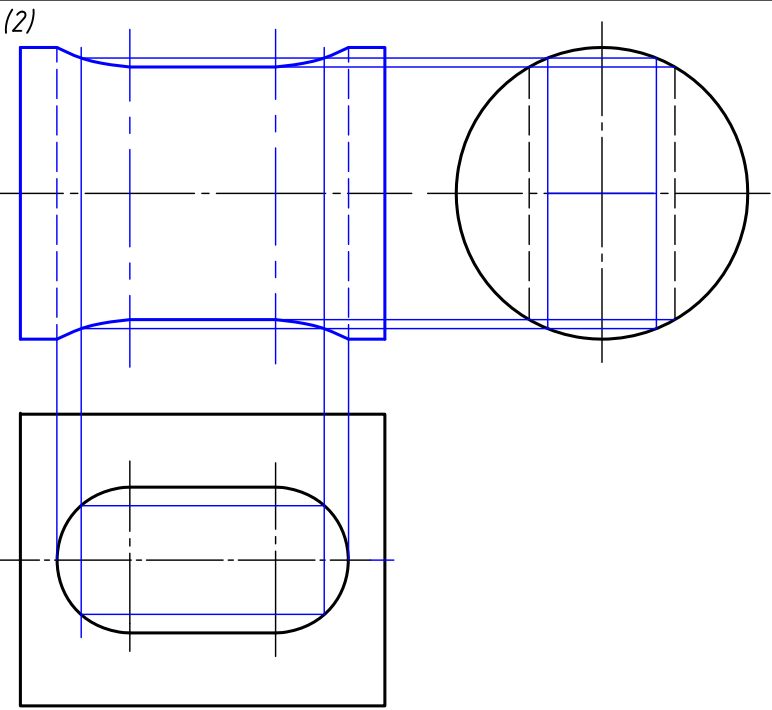
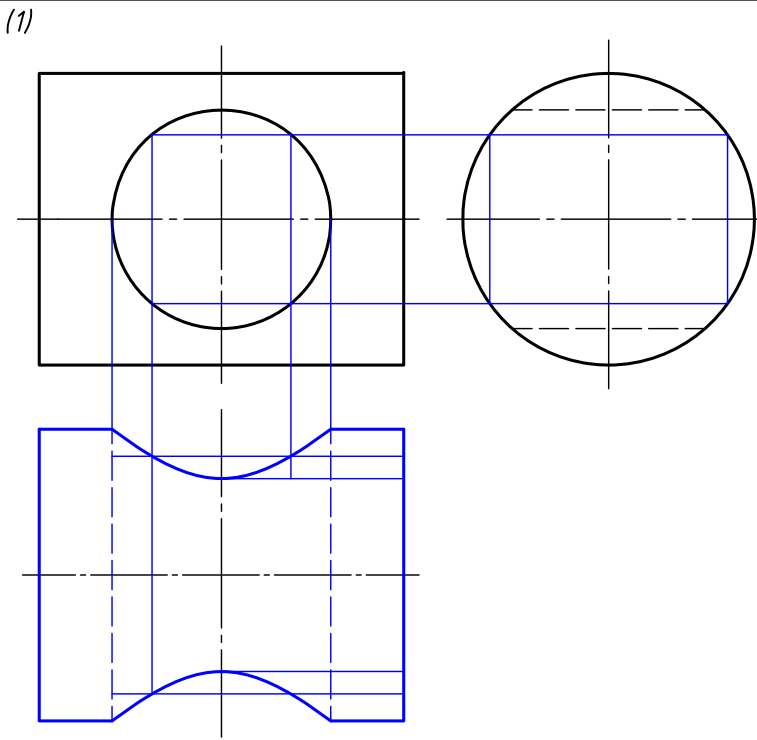
十、两曲面立体相交（一）

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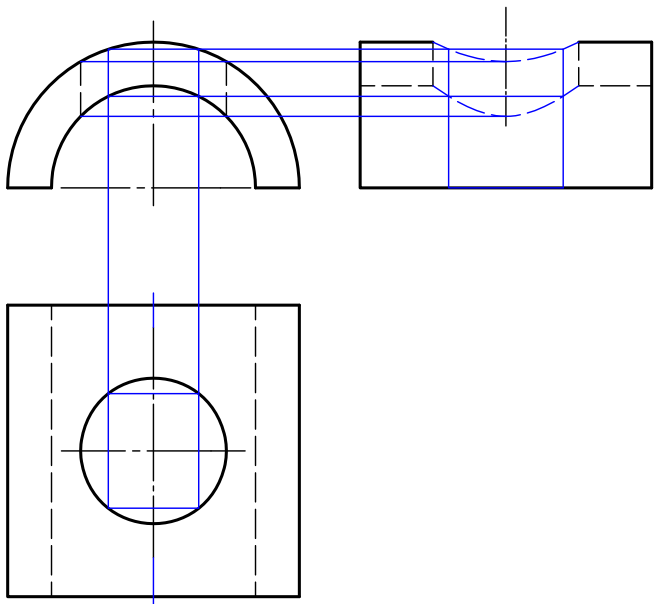
1. 求两圆柱的相贯线。



2. 求穿孔立体的第三面投影。



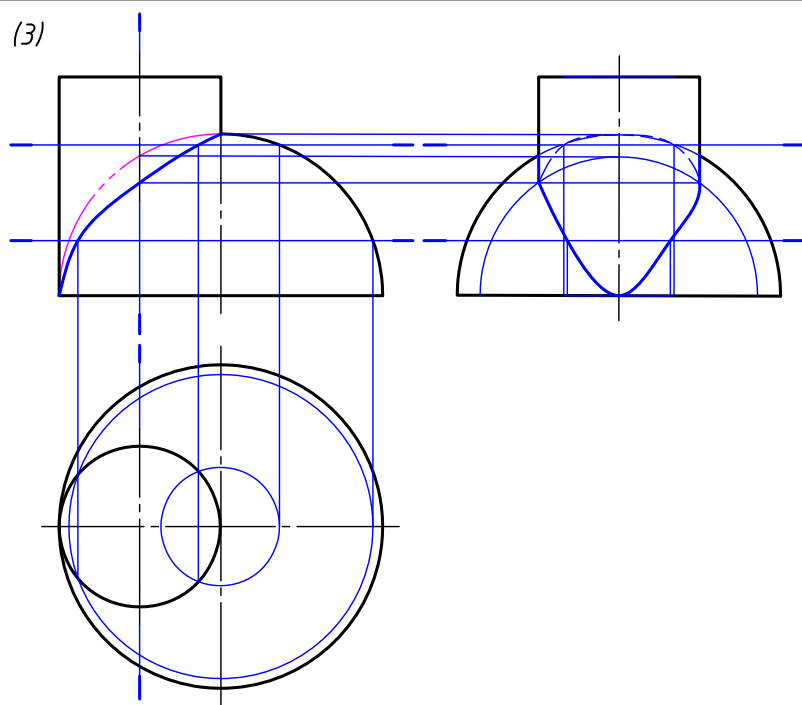
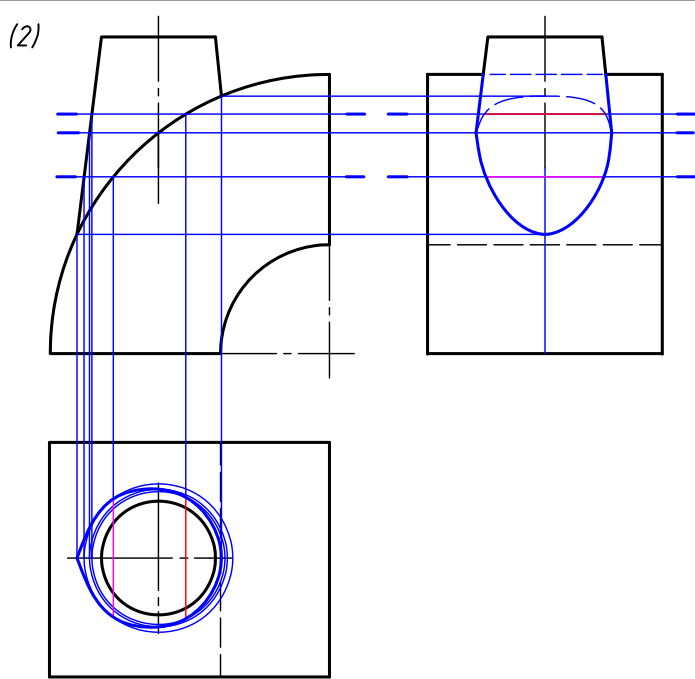
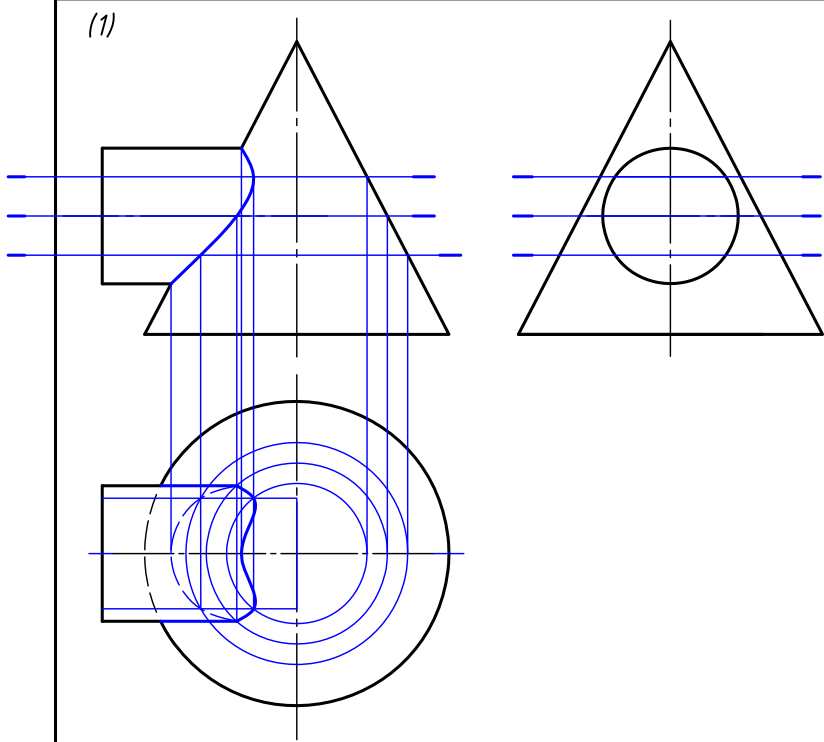
3. 补全相贯线的侧面投影。



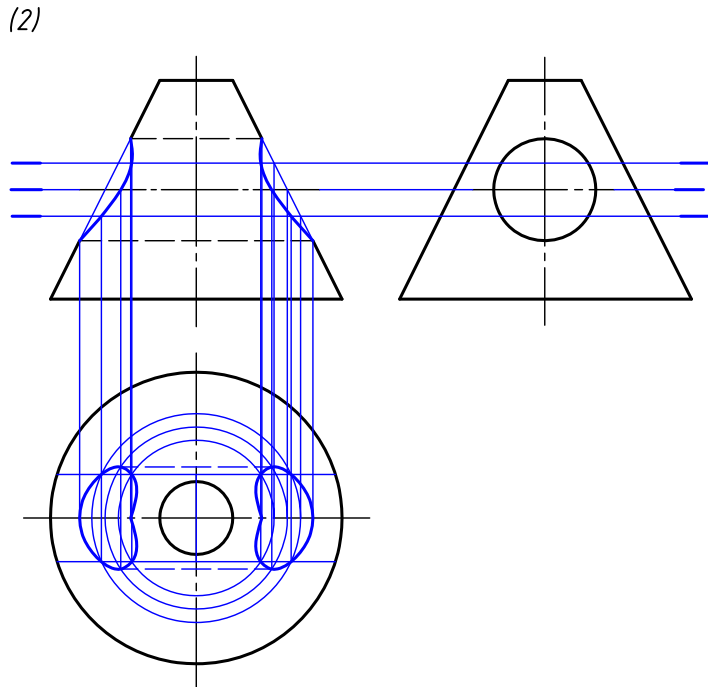
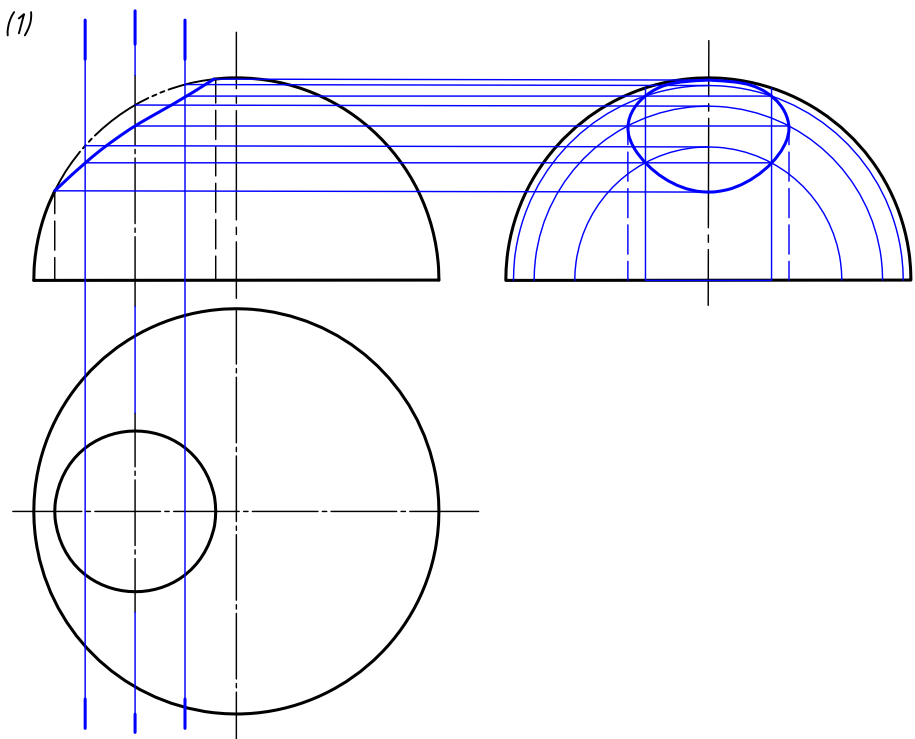
十、两曲面立体相交（二）

班级 姓名 学号

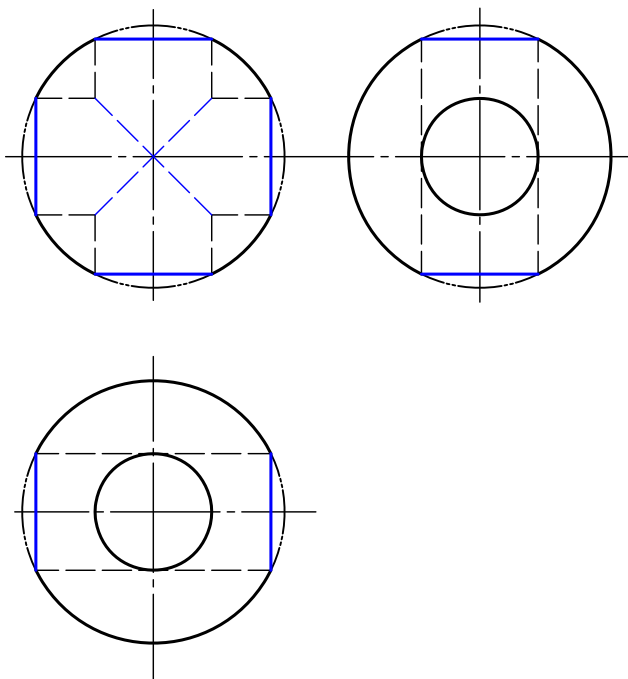
4. 求相贯体的三面投影。



5. 求穿孔立体的三面投影。



6. 求立体的相贯线。



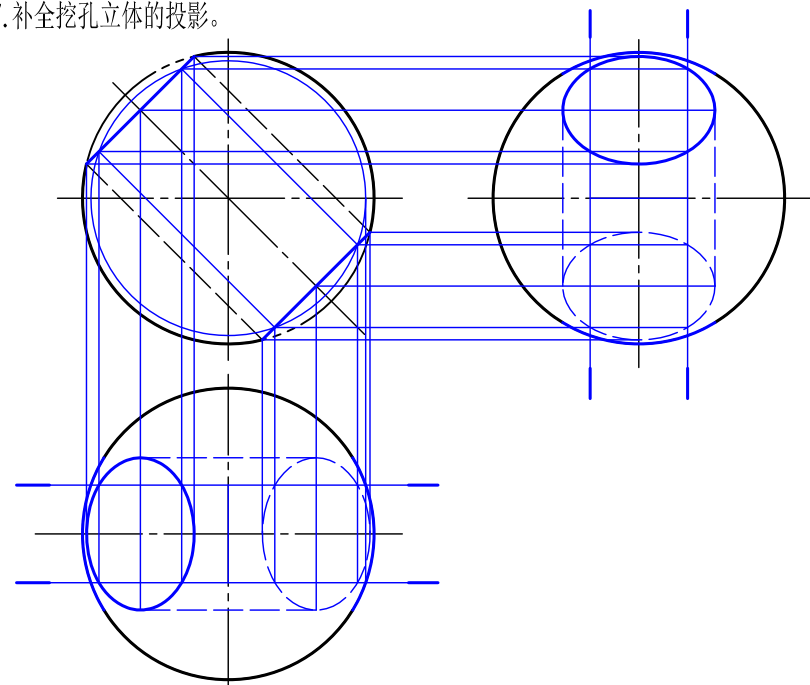
十.两曲面立体相交（三）

班级

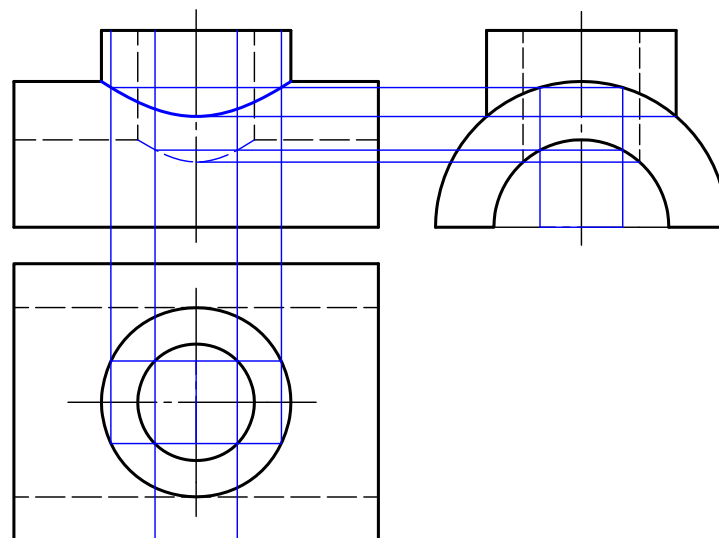
姓名

学号

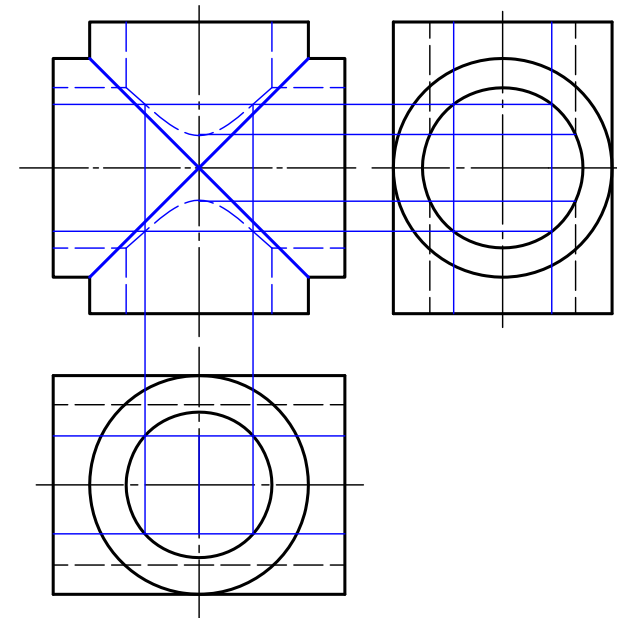
7. 补全挖孔立体的投影。



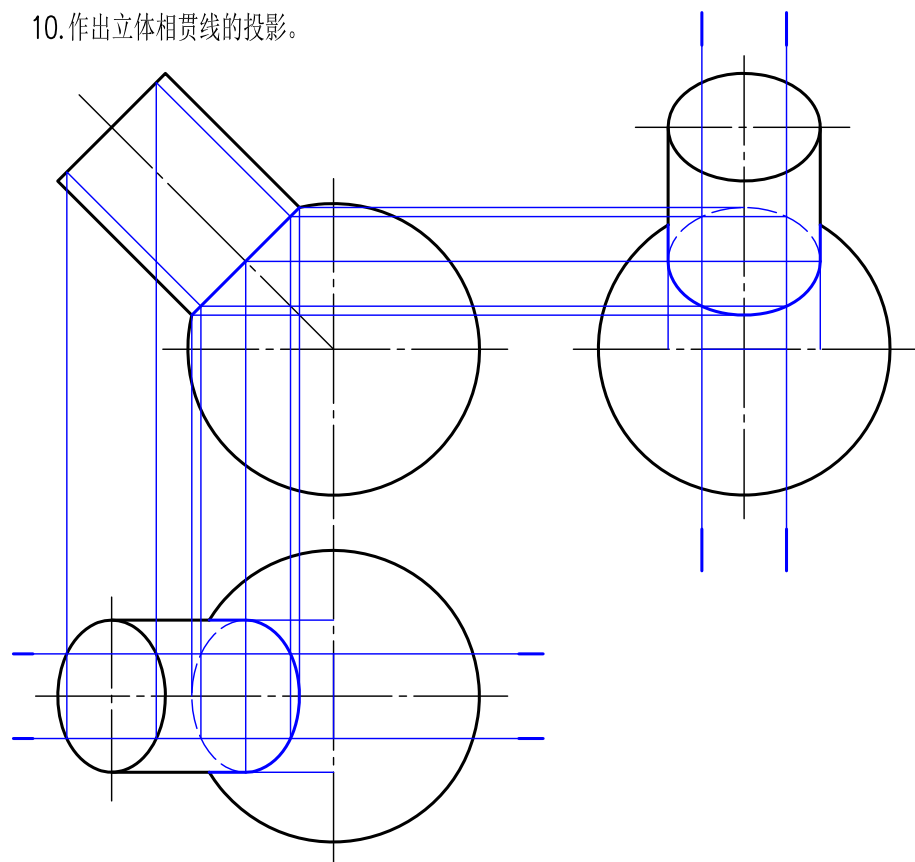
8. 完成立体的投影。



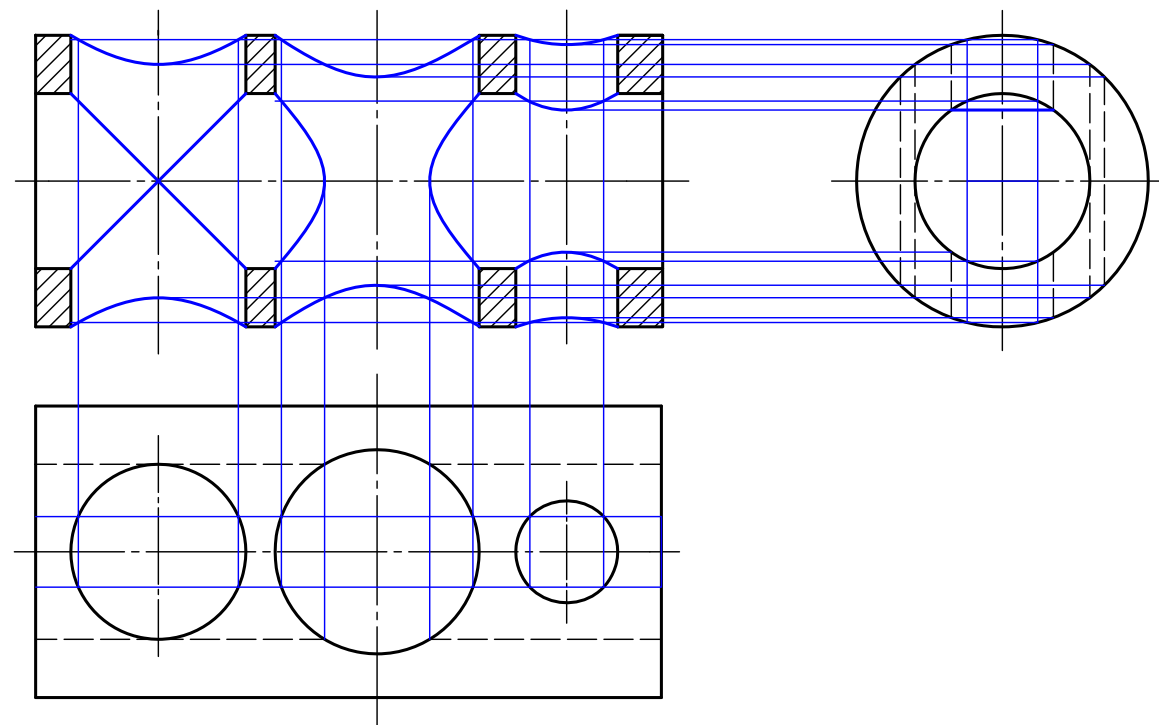
9. 求立体的正面投影。



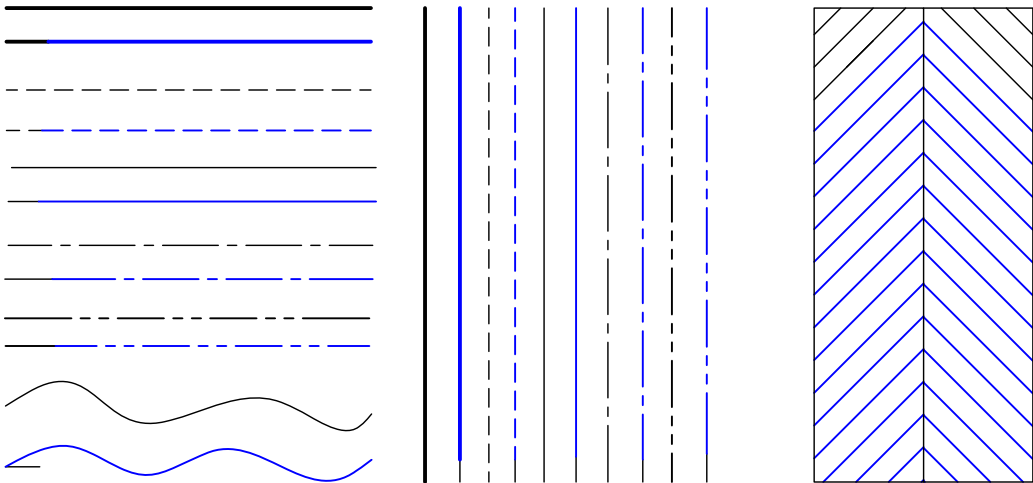
10. 作出立体相贯线的投影。



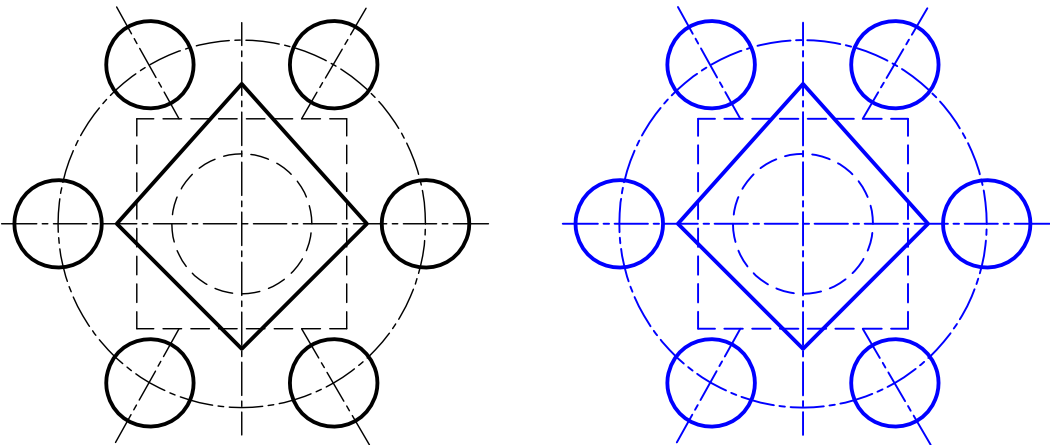
11. 在正面投影（剖视图）中，作出立体相贯线的投影。



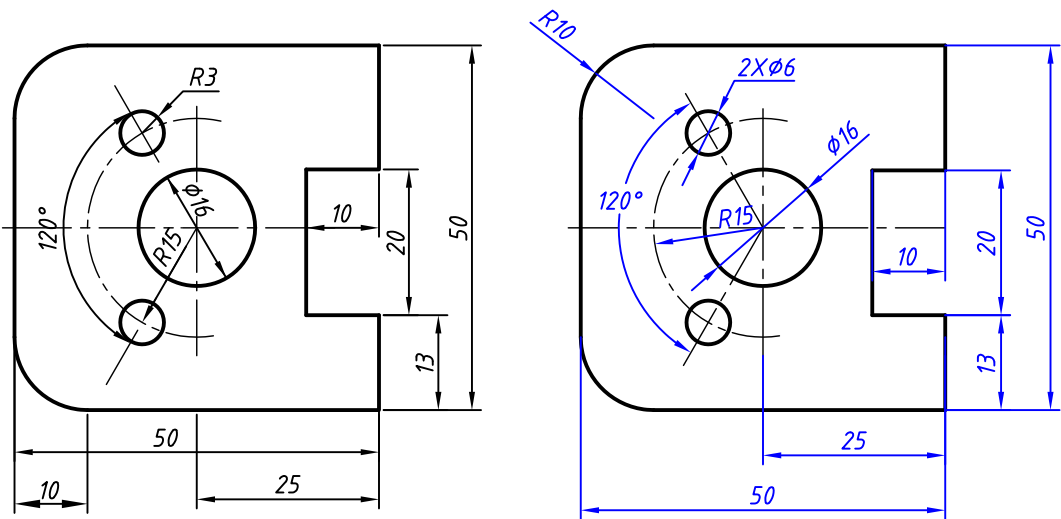
1. 参照给定图线在指定位置补画图线。



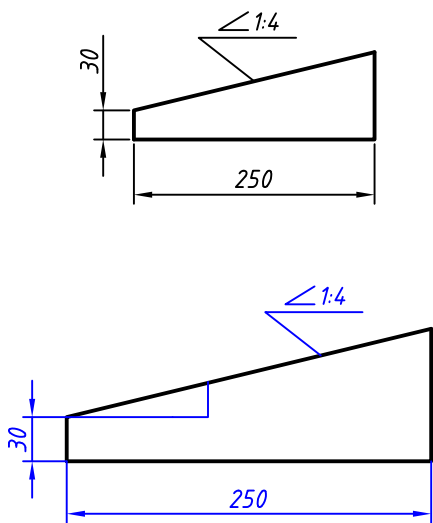
2. 在右侧照抄下面的图形。



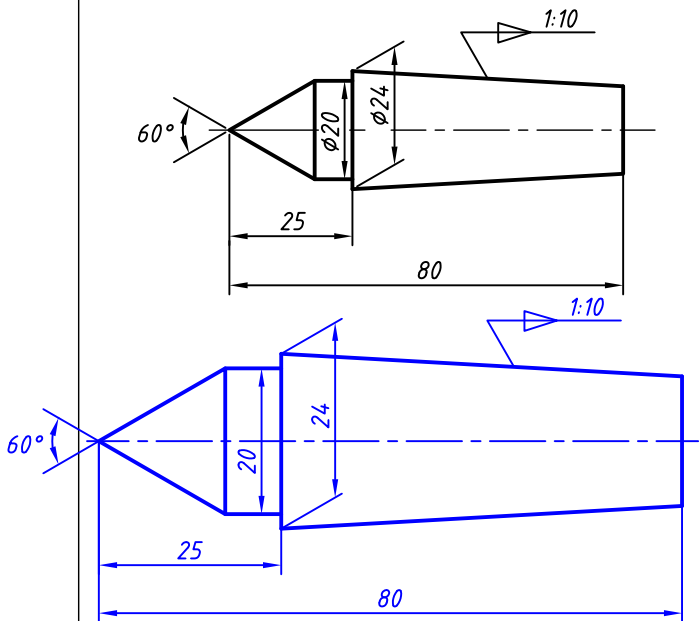
3. 指出尺寸注法的错误，并按正确的注法将尺寸标注在右边的图形上。



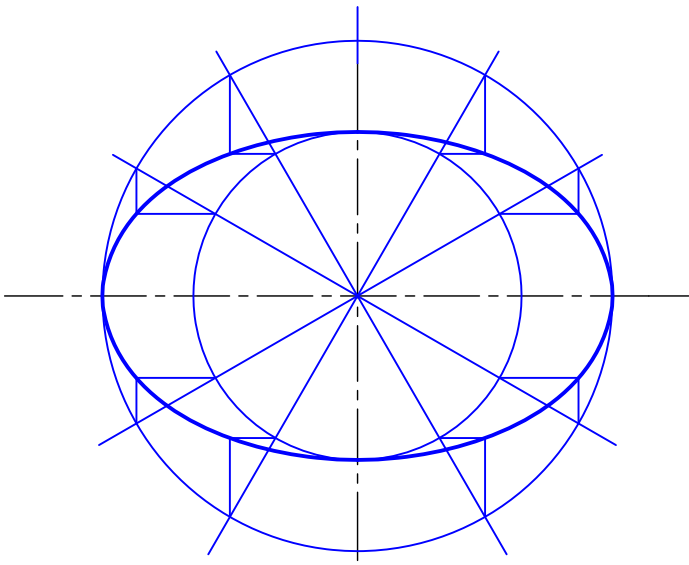
4. 参照所示图形，用1：5的比例画出图形，并标注尺寸。



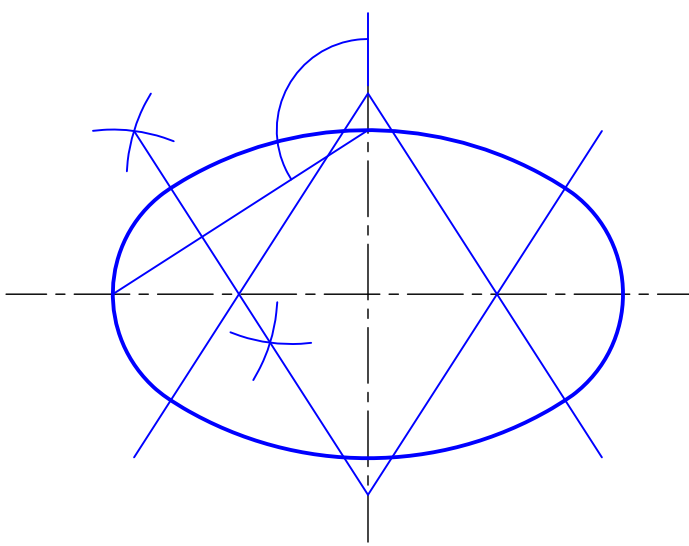
5. 参照所示图形，用1：1的比例画出图形，并标注尺寸。



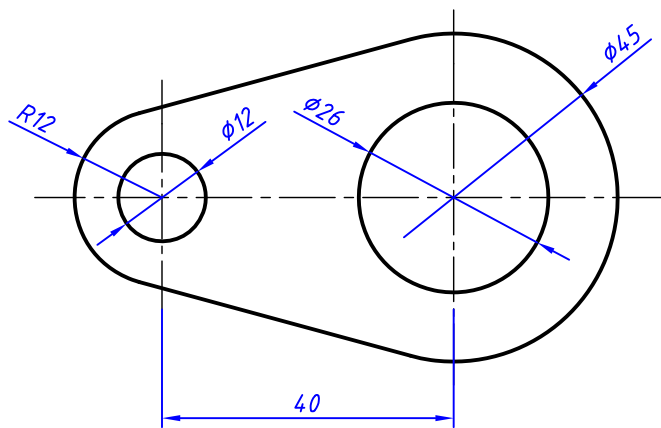
6. 已知椭圆长、短轴分别为70mm、45mm，用同心圆法画椭圆。



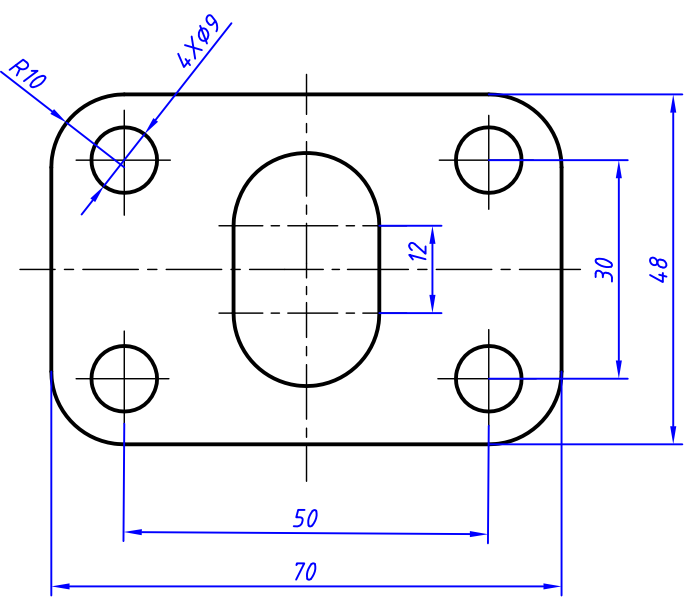
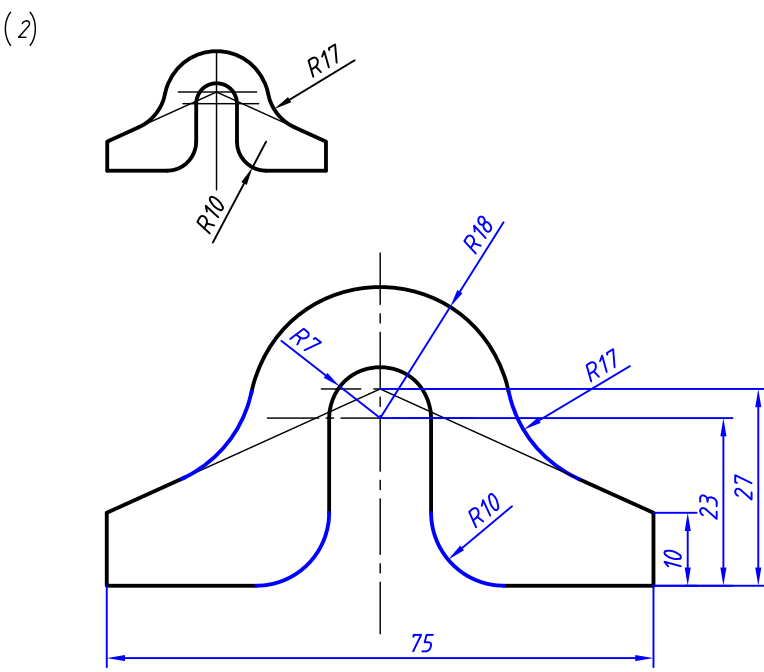
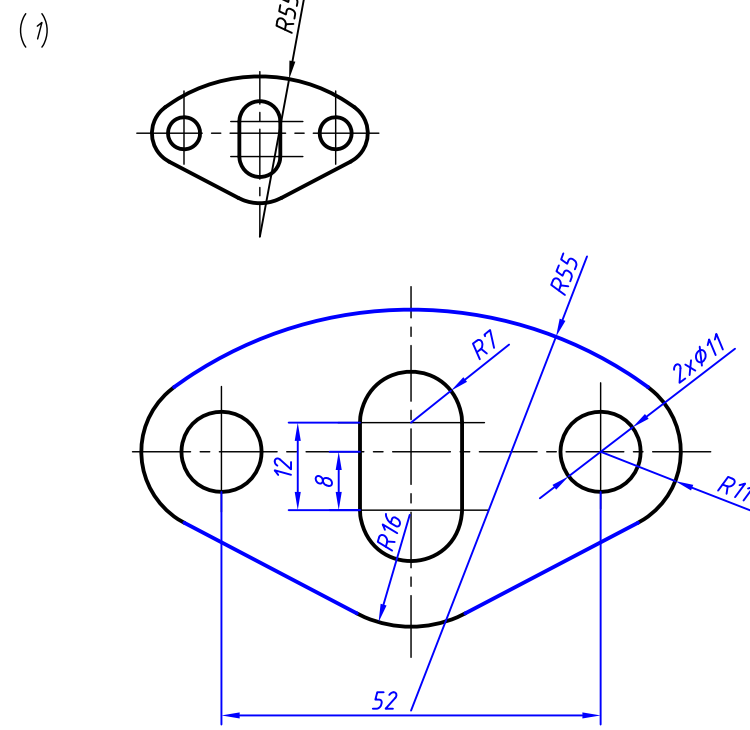
7. 已知椭圆长、短轴分别为70mm、45mm，用四心圆法画椭圆。



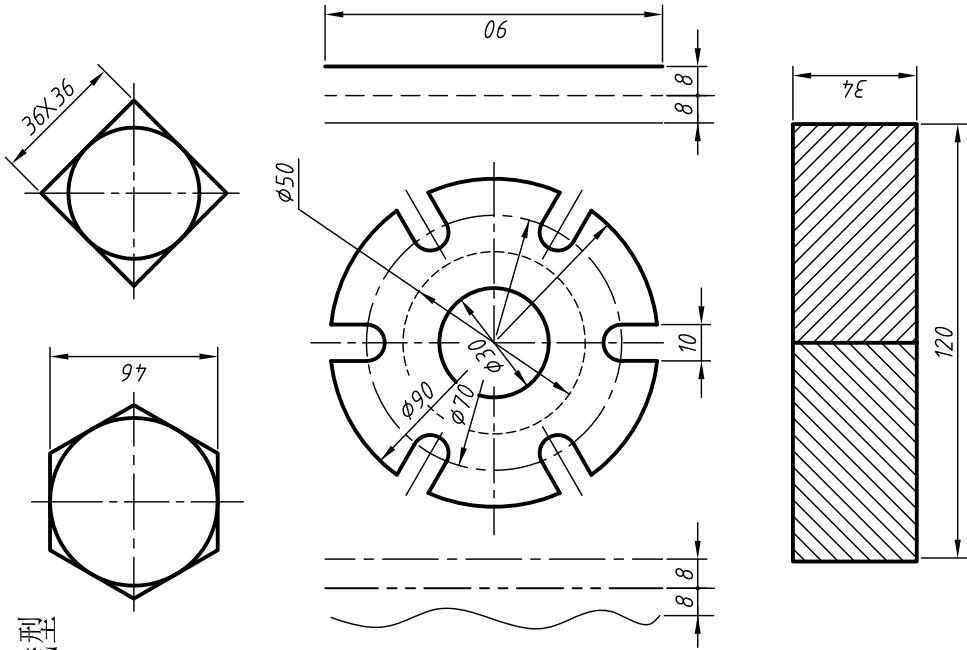
9. 在平面图形上用1:1的比例量后标注尺寸（取整数）。



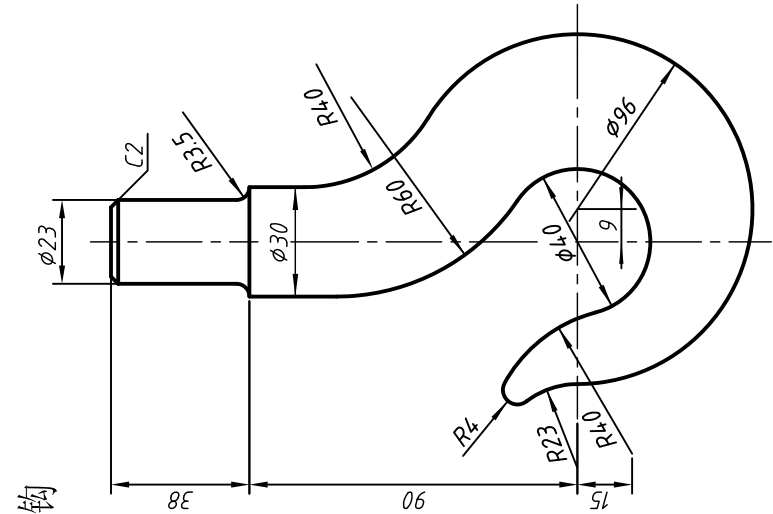
8. 按小图所给尺寸，在大图上作图，并注全尺寸。



线型



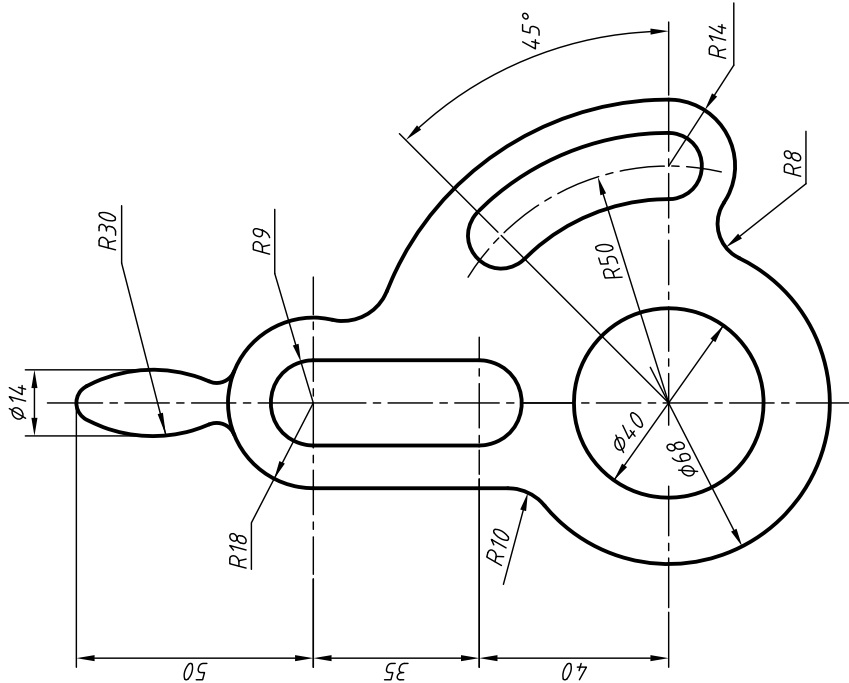
起重钩



基本练习		比例	1:1	材料	YT-01
制图	(姓名)	(日期)			
审核					

(校名、班级)

交换齿轮架



作业方法指示

- 一、目的、内容和要求
- 1.目的、内容
- 初步掌握国家标准《机械制图》的有关内容,掌握绘图仪器和工具的使用方法,抄画线型(不注尺寸),起重钩(或交换齿轮架)图形并标注尺寸。
- 2.要求
- 图形正确,布置适当,线型合格,字体工整,符合国标,连接光滑,图面整洁。
- 二、图名、图幅、比例
- 1.图名 基本练习
- 2.图幅 A3图纸
- 3.比例 1:1
- 三、注意事项
- 1.参照上图布置图形,先画出各图形的对称线、中心线等。
- 2.线型、箭头均按教材中的要求画出。
- 3.标题栏中汉字打格书写。图名、图号用10号字,校名用7号字,班级用5号字写在校名后面,姓名用5号字写在制图栏内。图中尺寸数字用3.5号字。
- 4.完成底稿后,经仔细校核方可加深。用铅笔加深时,圆规的铅芯应比直线的铅芯软一号。

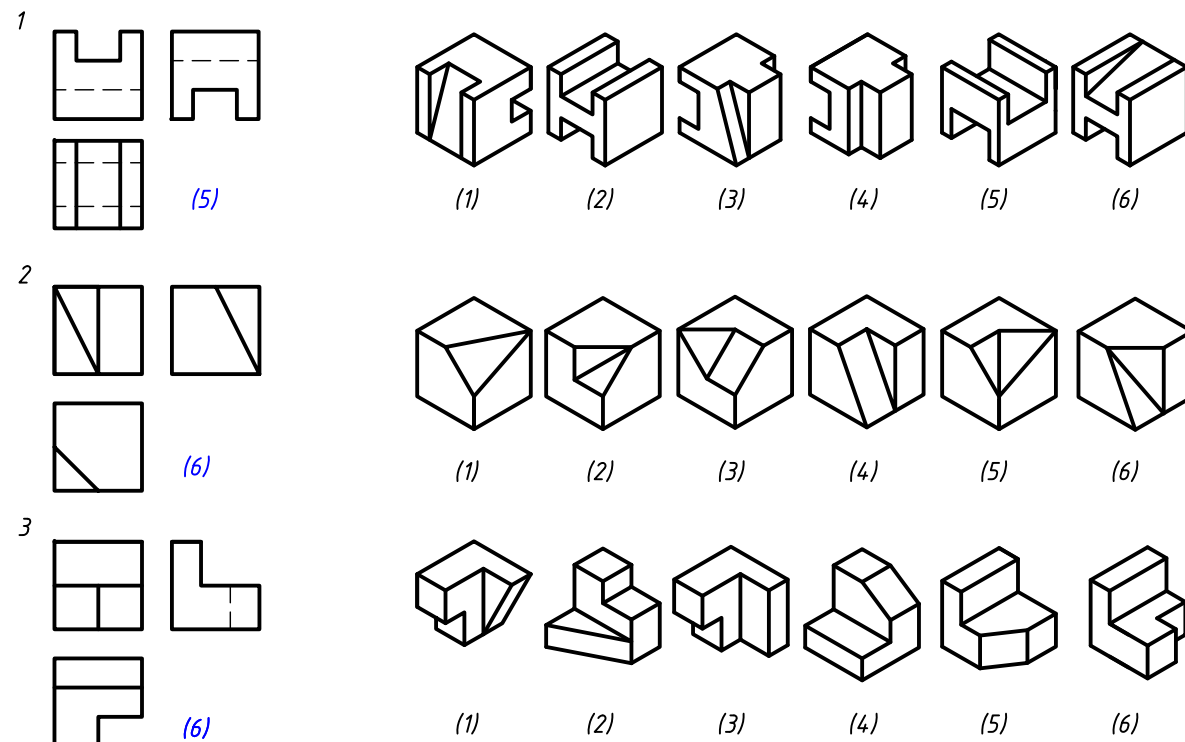
十三、画组合体视图（一）

班级

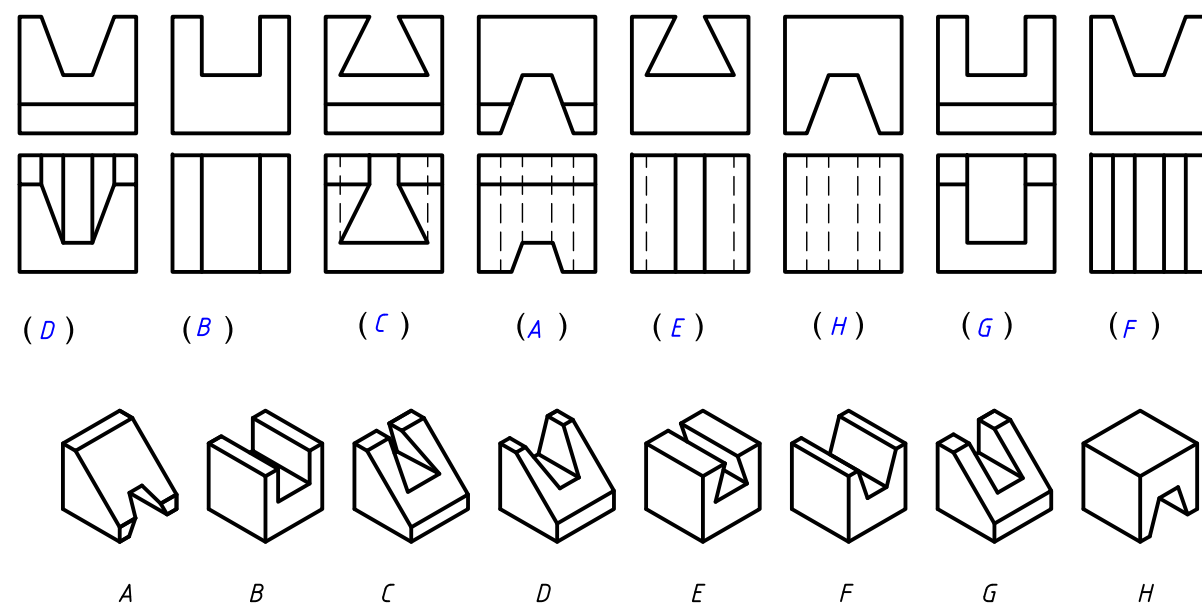
姓名

学号

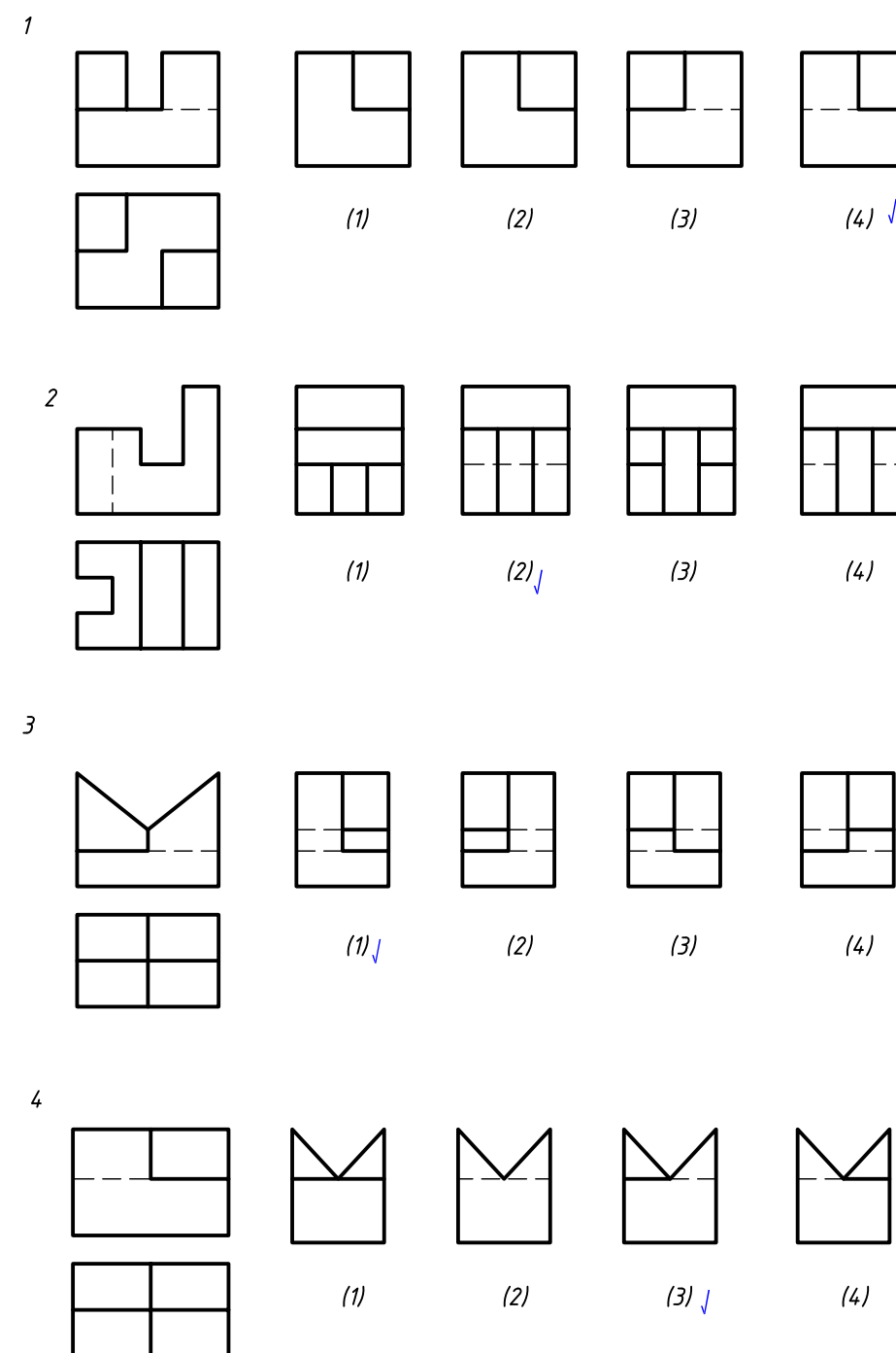
1. 选择与三视图对应的立体图，并将其编号填在三视图旁



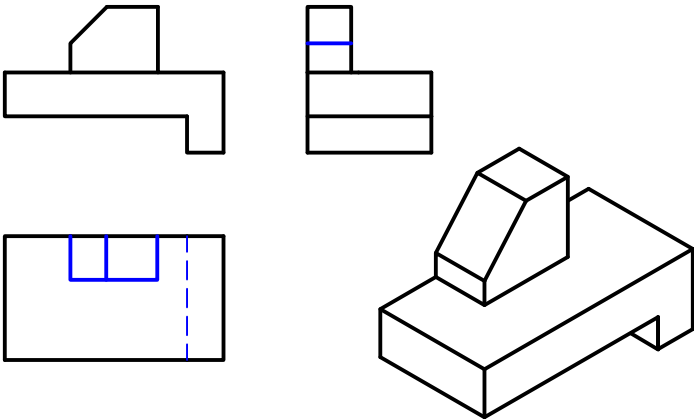
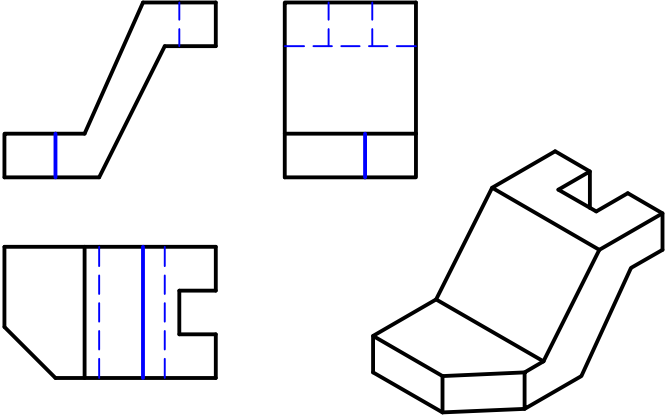
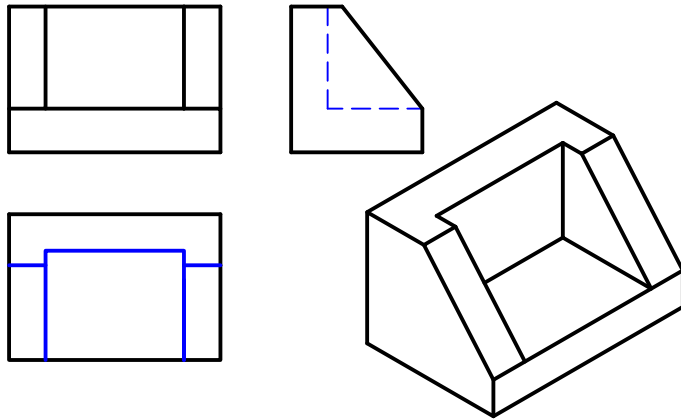
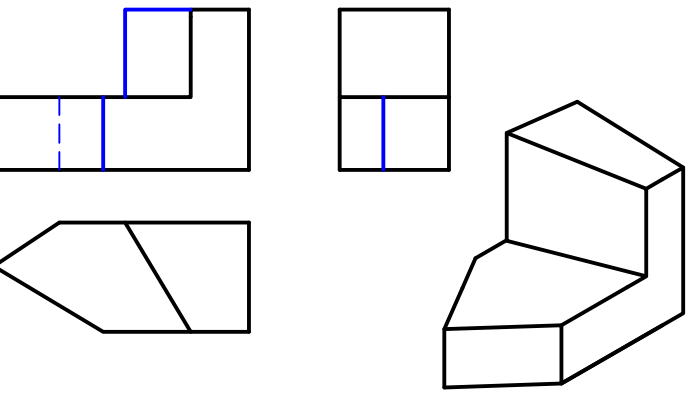
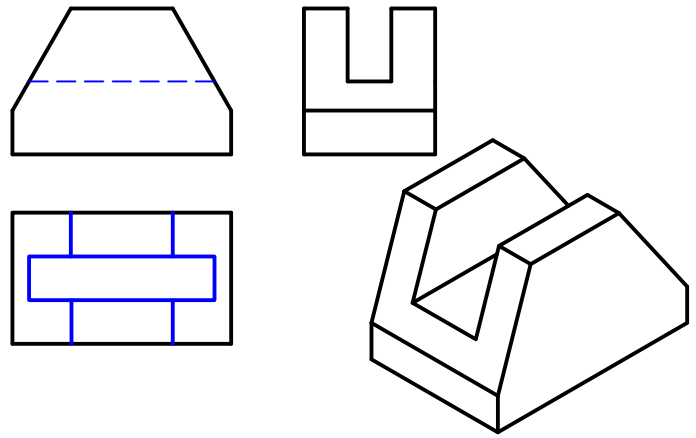
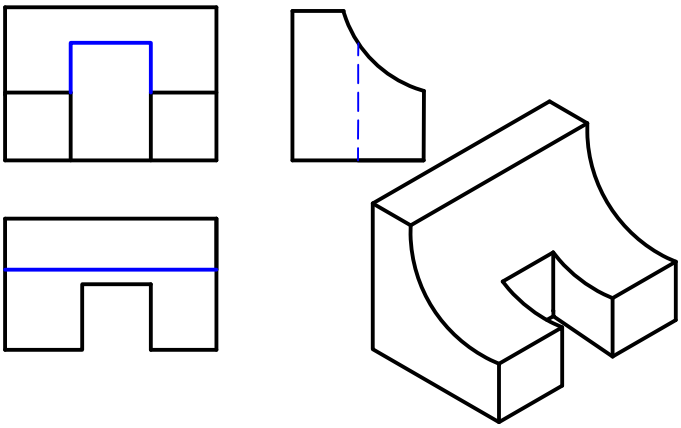
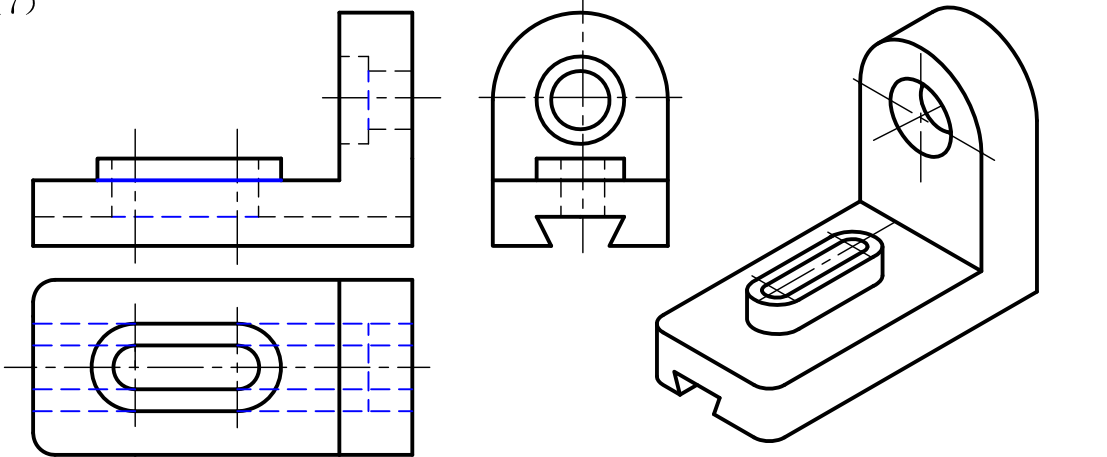
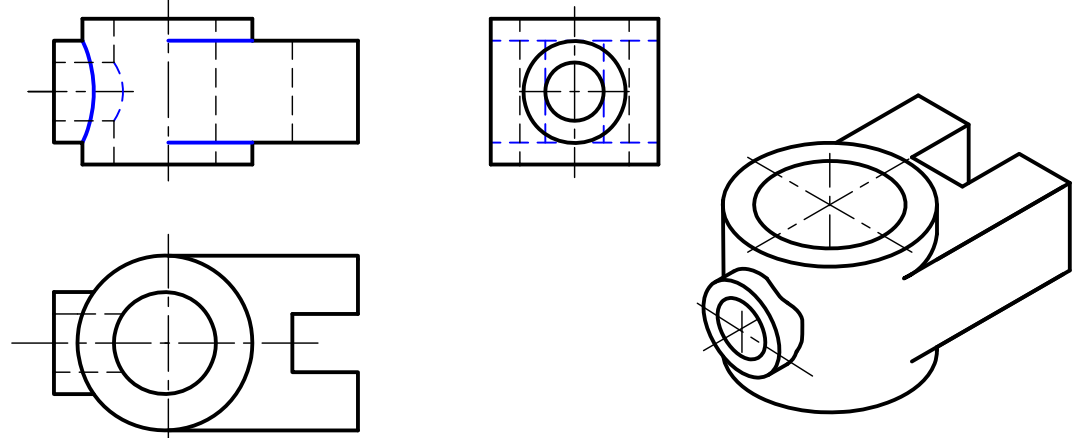
2. 选择与两视图对应的立体图，并将其字母填在视图下方



3. 已知主、俯两视图，找出正确的左视图（在正确的左视图编号上打√）。



4. 根据轴测图补画视图中所缺图线。

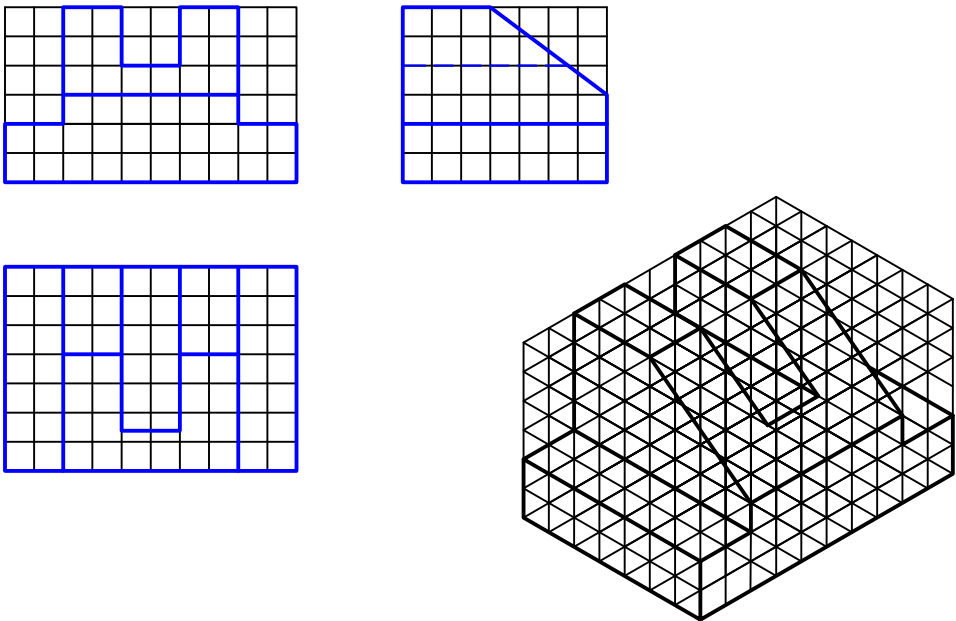
(1) 	(2) 	(3) 
(4) 	(5) 	(6) 
(7) 		(8) 

十三、画组合体视图（三）

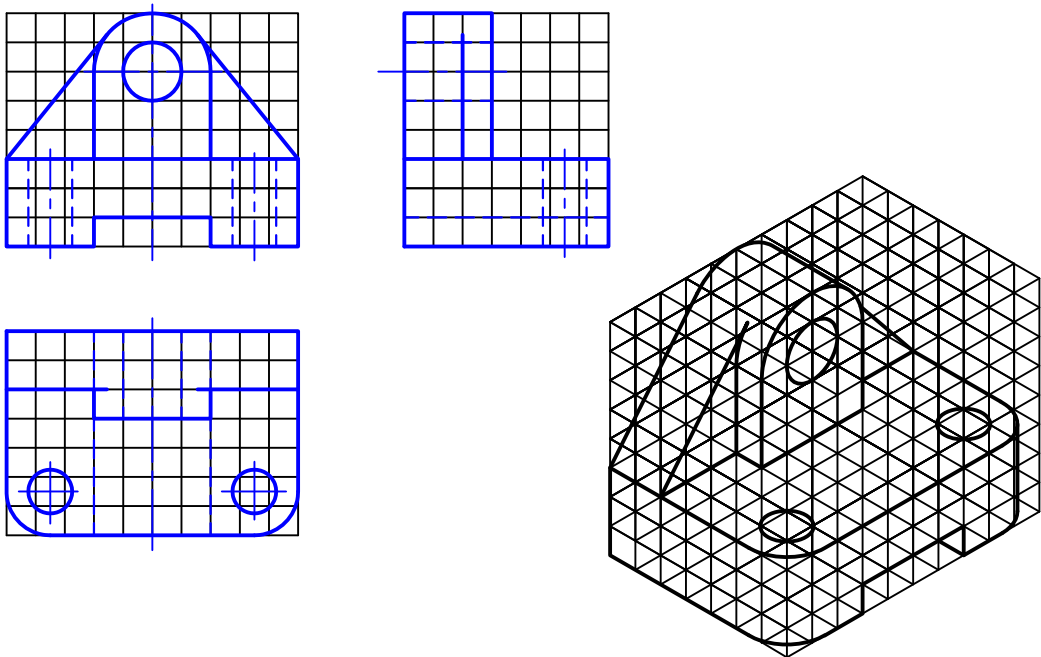
班级 姓名 学号

5. 根据立体图，利用方格徒手画出组合体的三视图（凡是图中未表明深度的孔均为通孔，槽为通槽）。

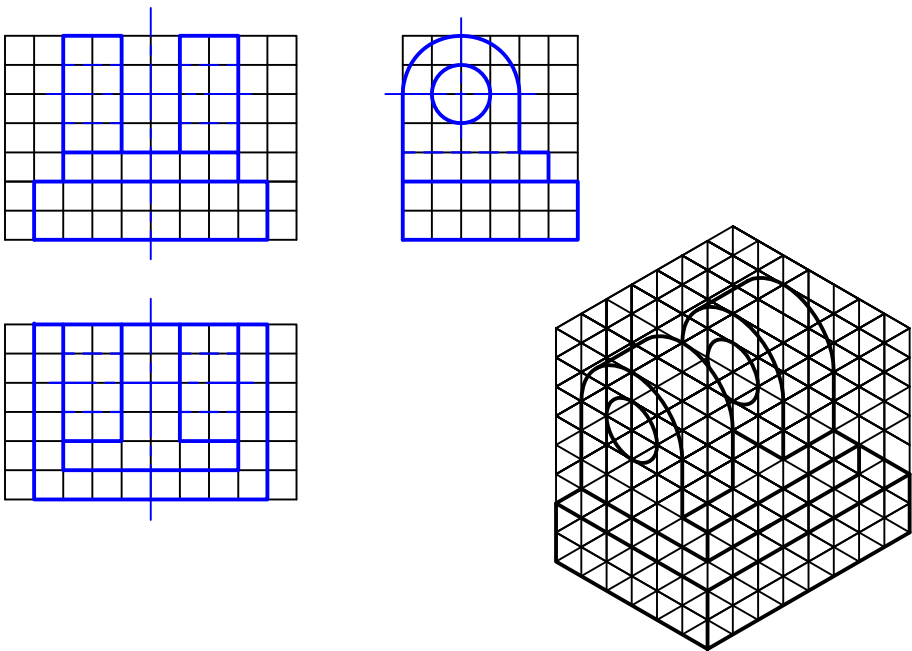
(1)



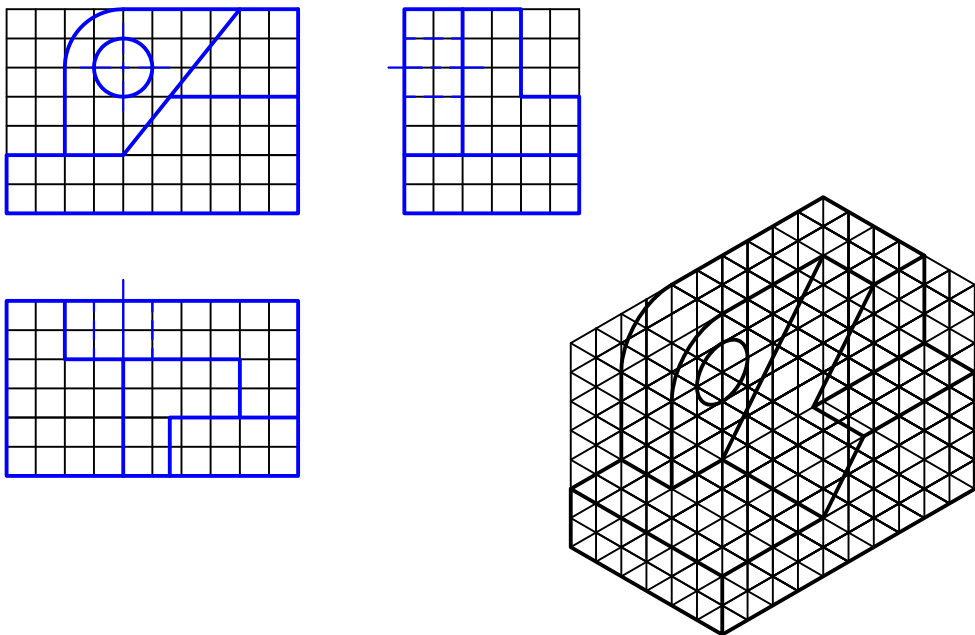
(2)



(3)



(4)

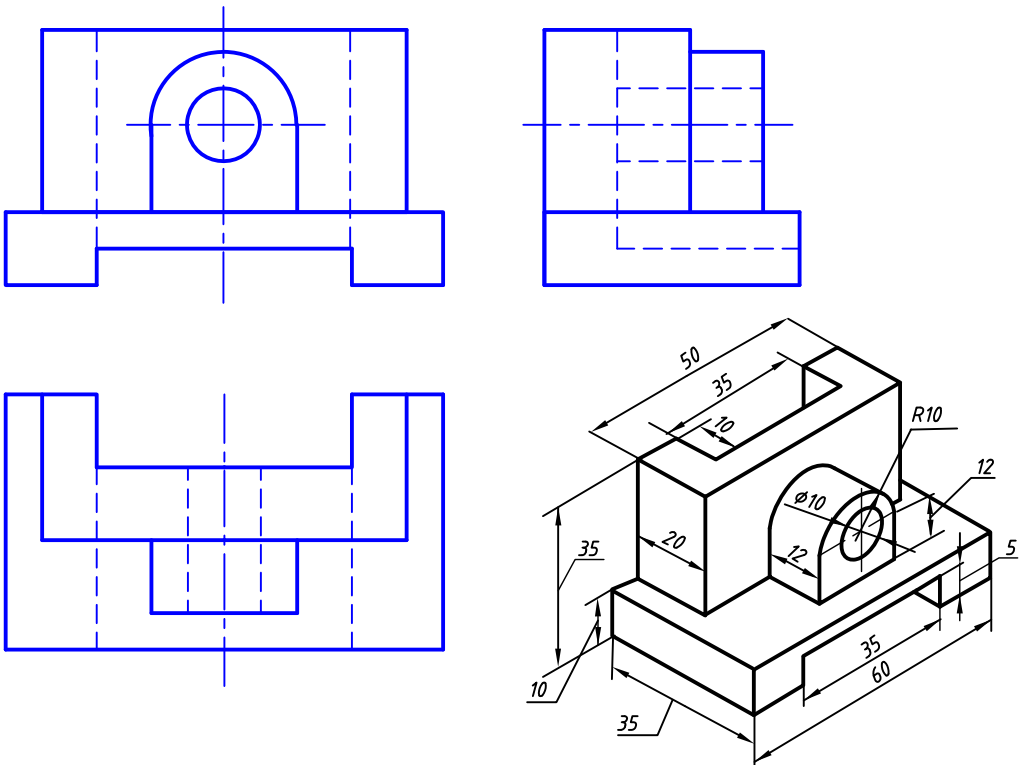


十三、画组合体视图（四）

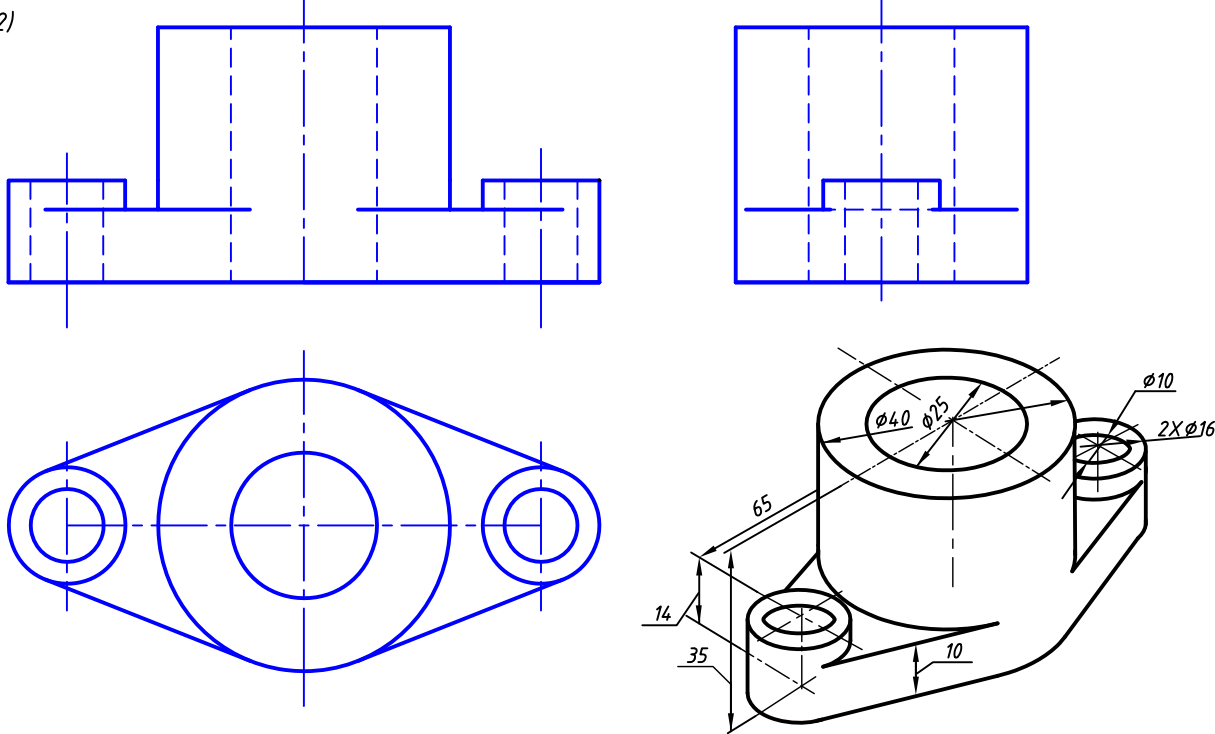
班级 姓名 学号

6. 根据立体图上所注尺寸，用1：1的比例画出组合体的三视图（凡是图中未注明深度的孔均为通孔，槽为通槽）。

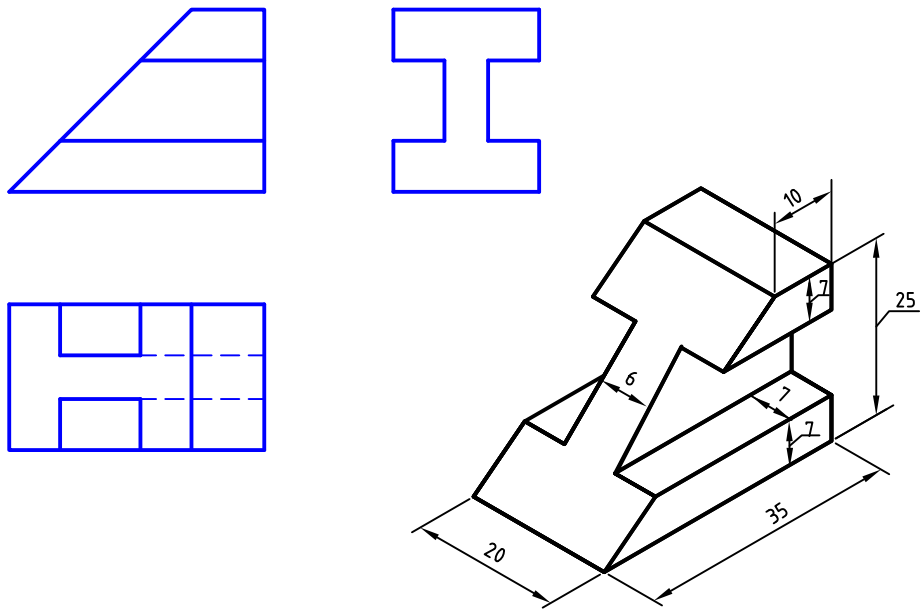
(1)



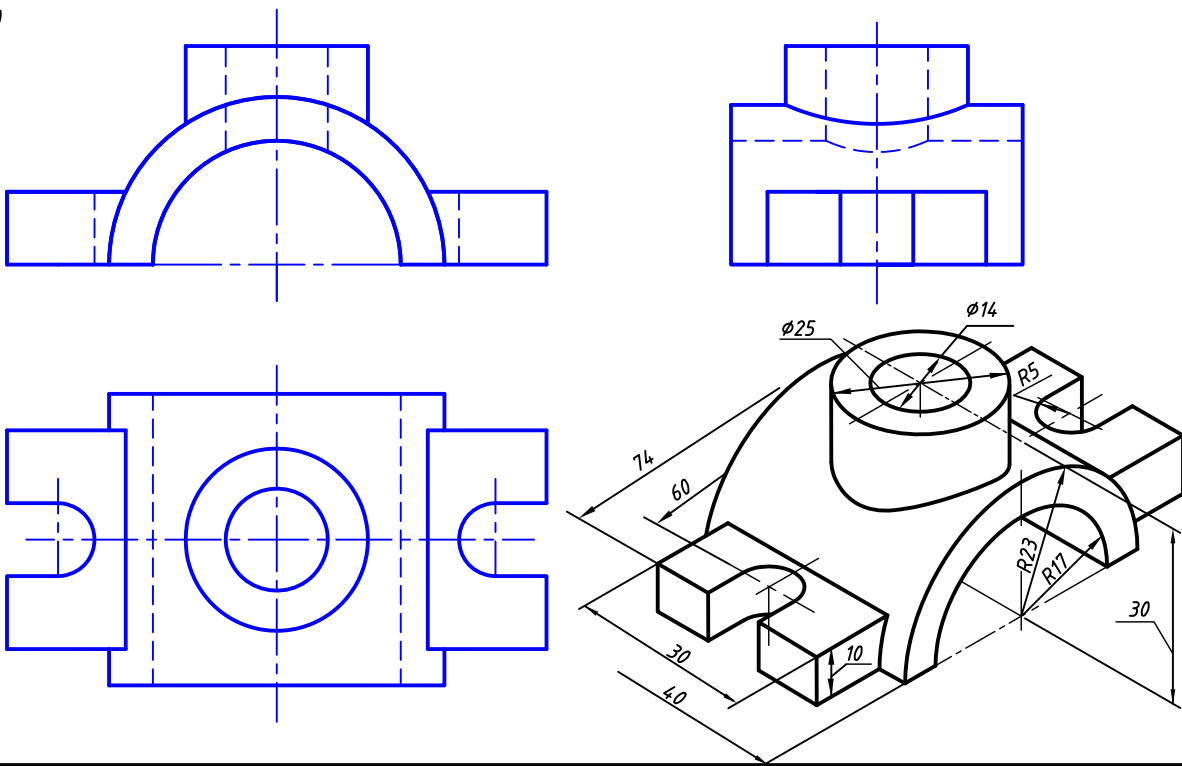
(2)



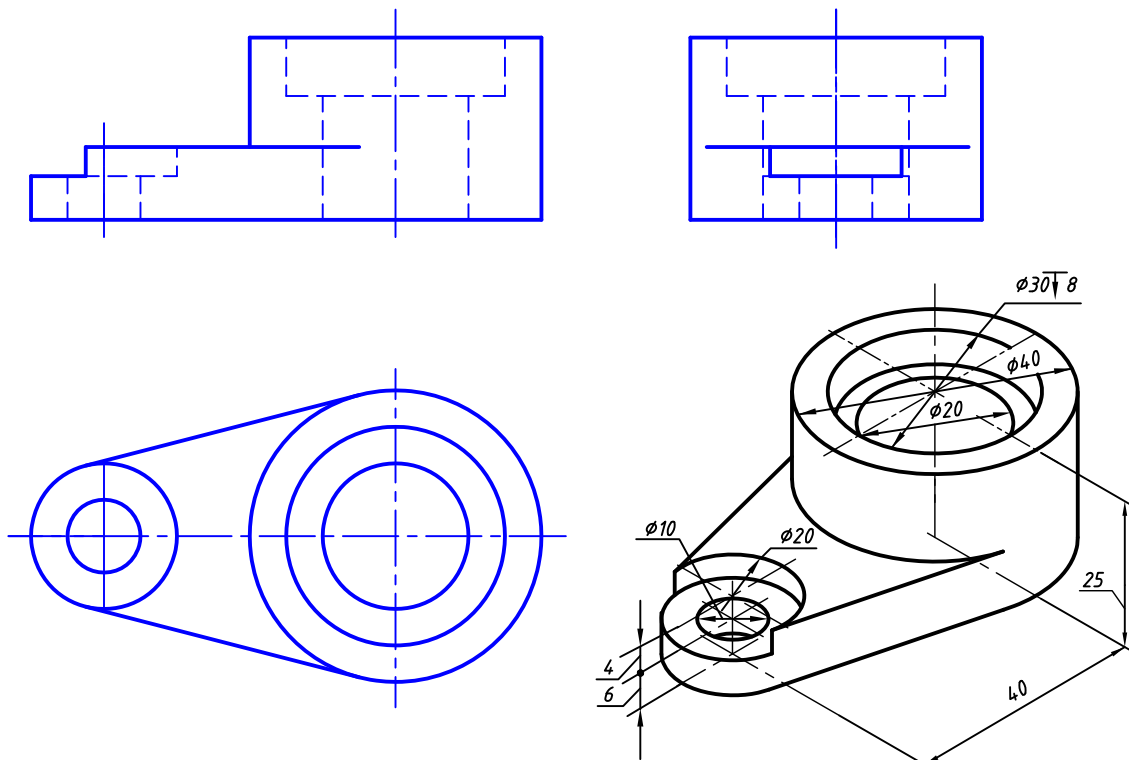
(3)



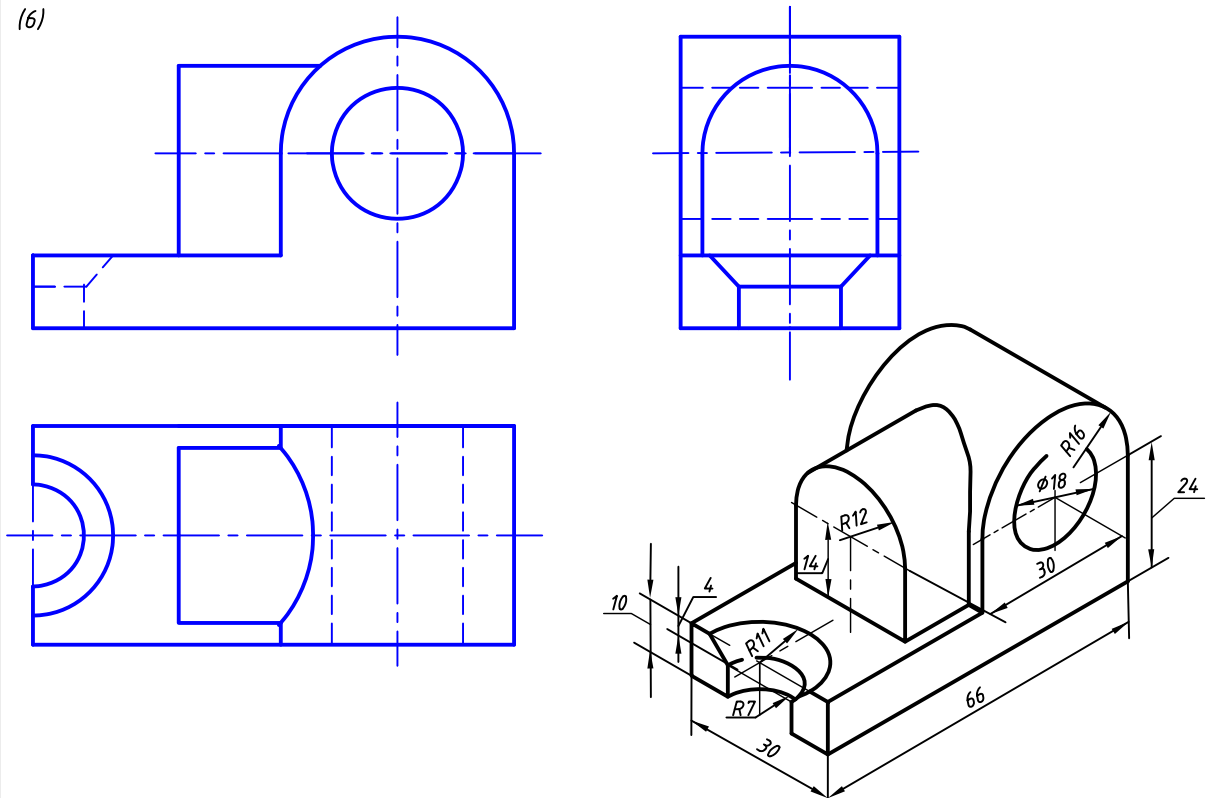
(4)



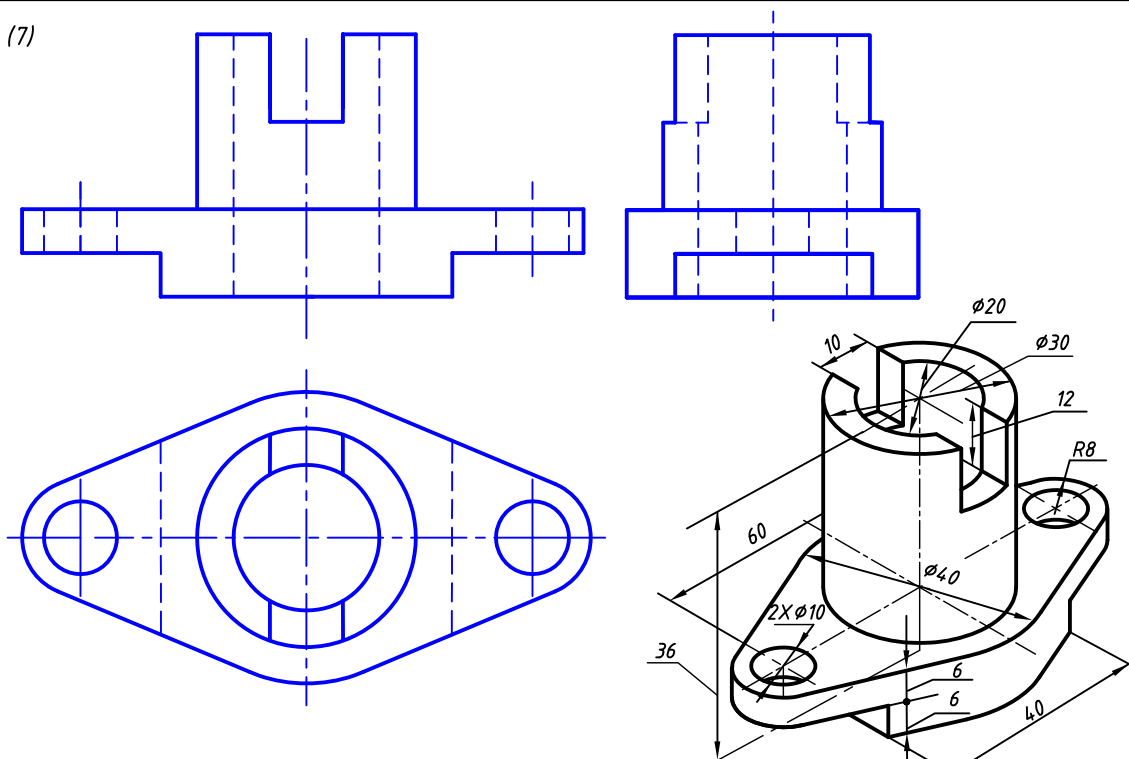
(5)



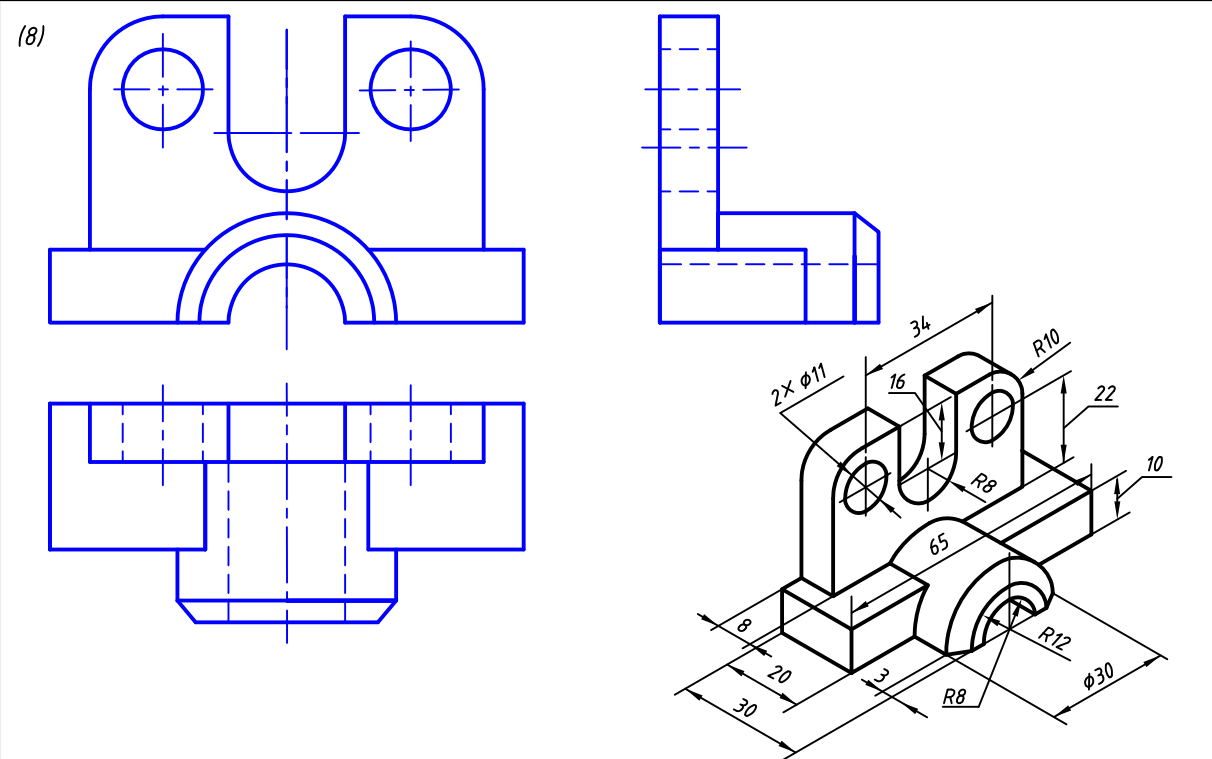
(6)



(7)

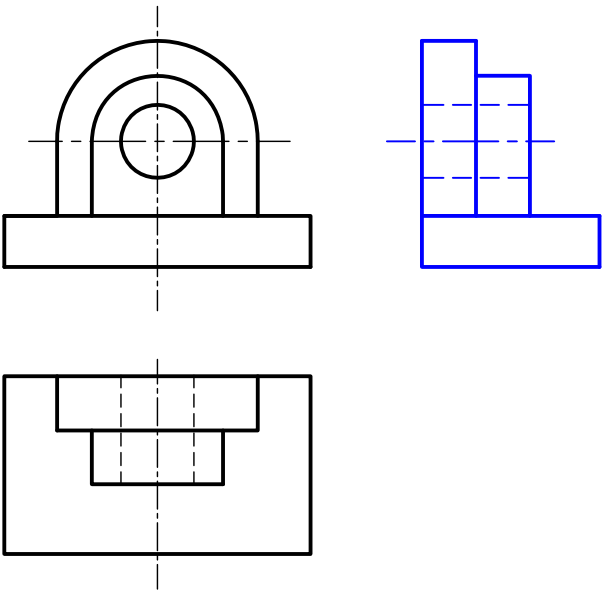


(8)

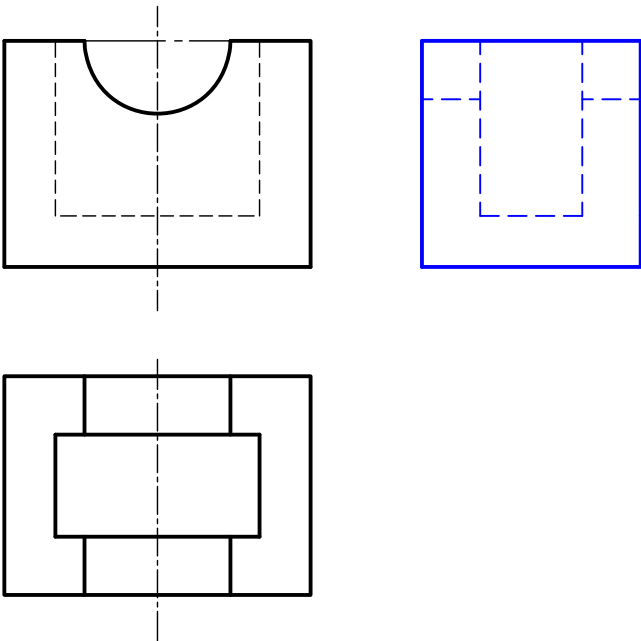


1. 读懂二视图后，补画第三视图。

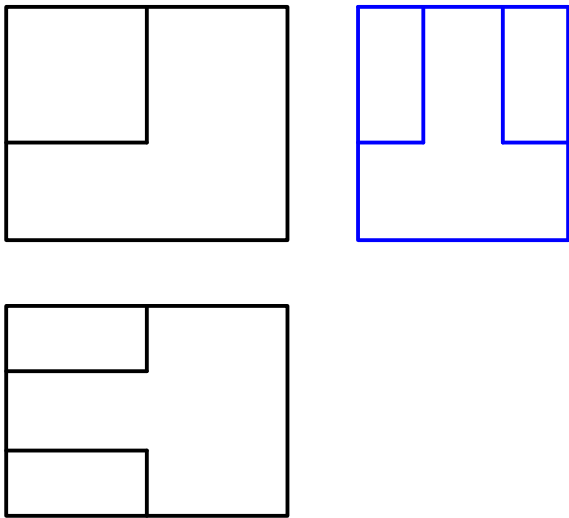
(1)



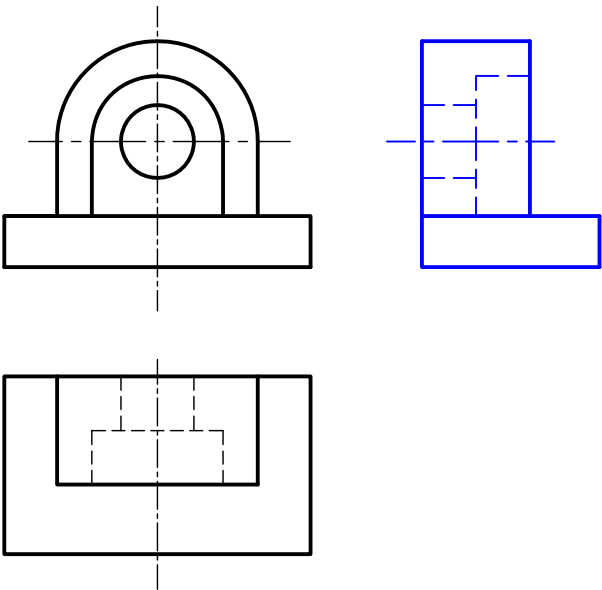
(2)



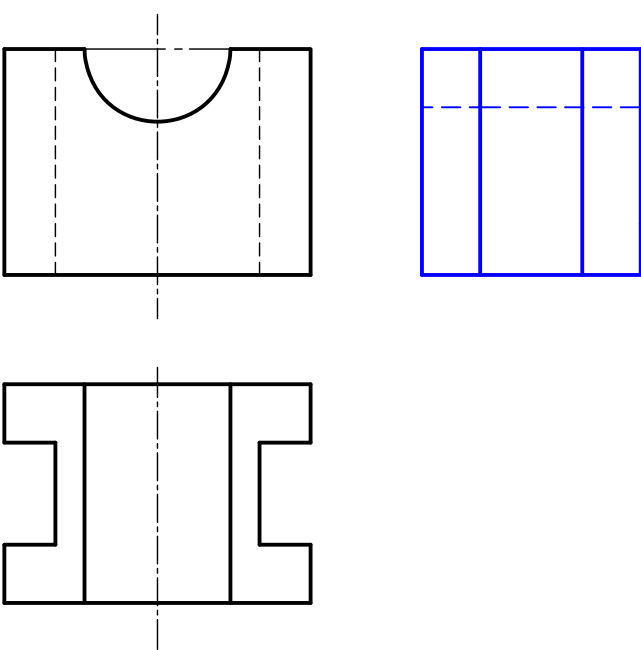
(3)



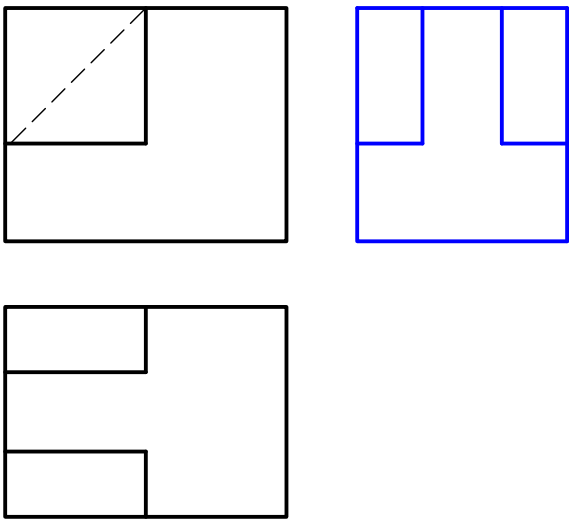
(4)



(5)

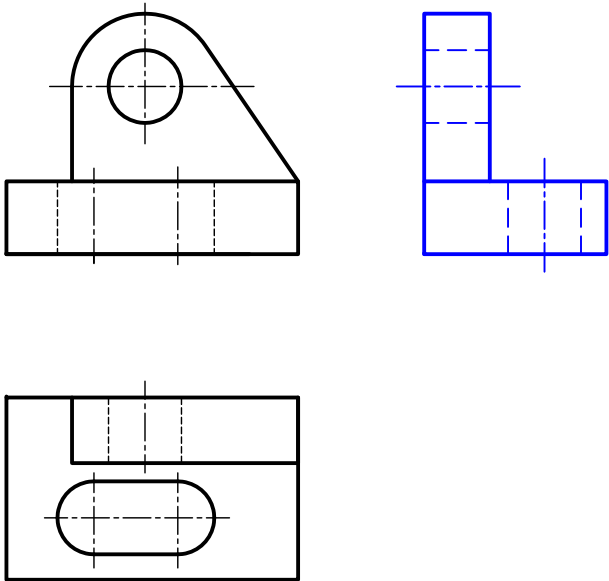


(6)

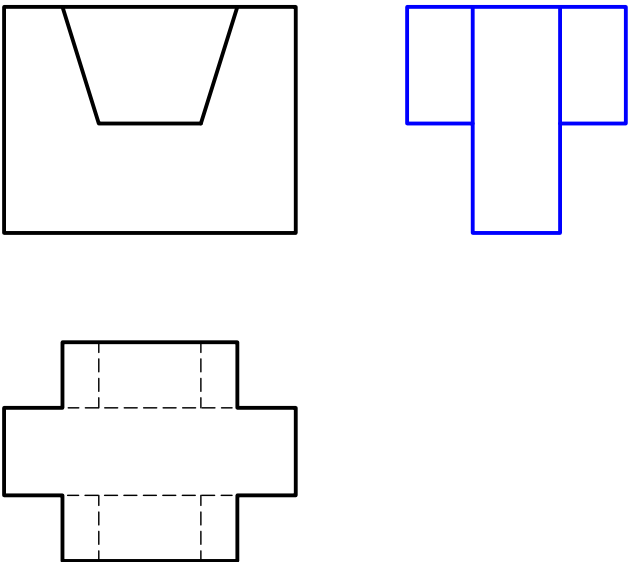


2. 读懂二视图后，补画第三视图。

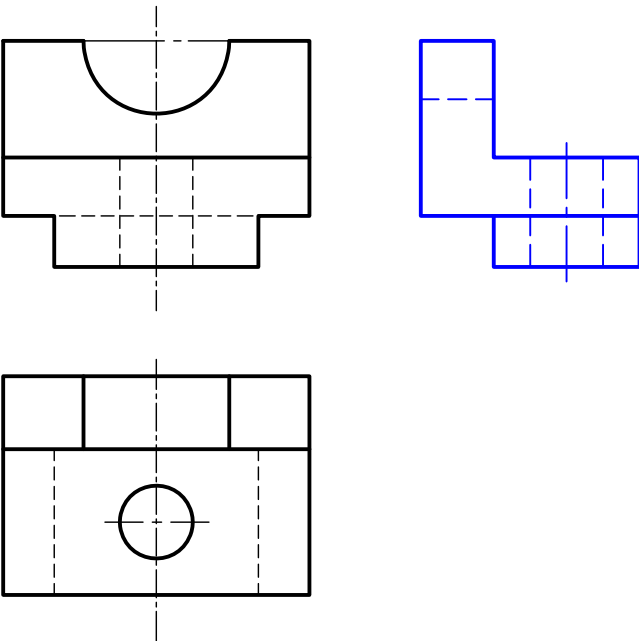
(7)



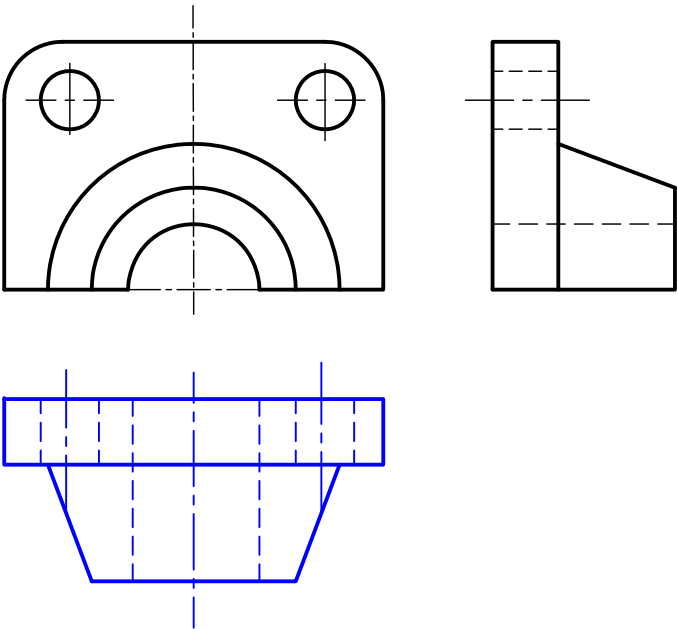
(8)



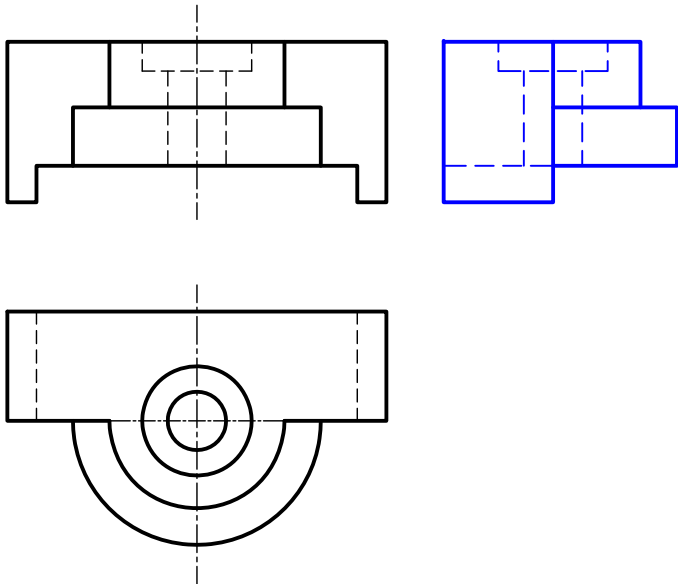
(9)



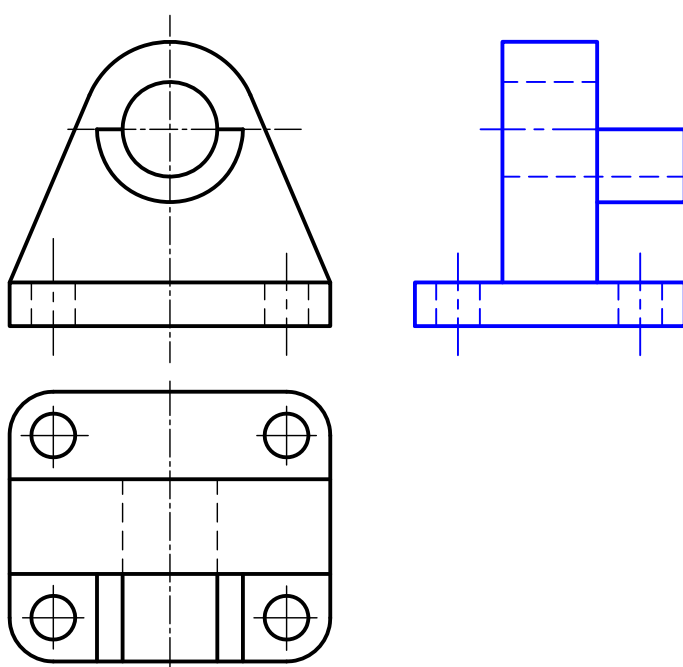
(10)



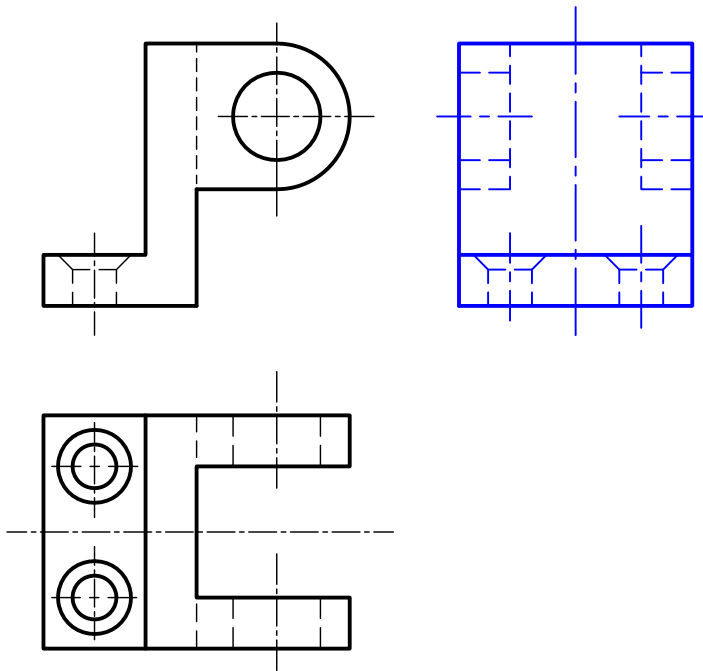
(11)



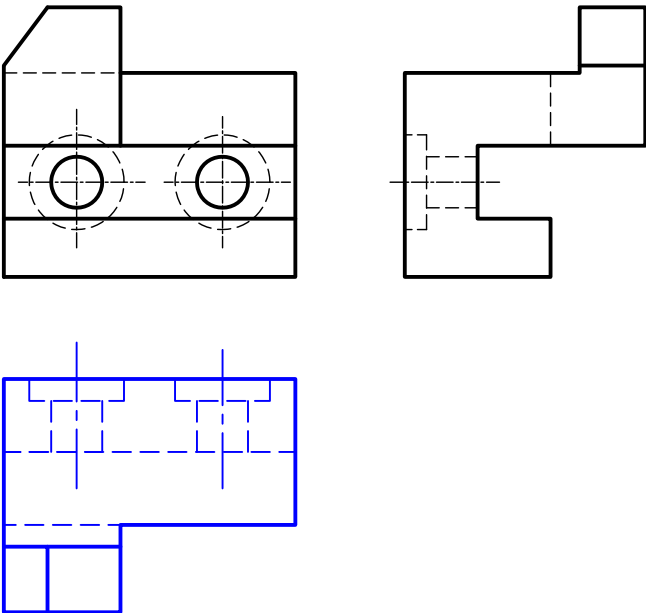
(12)



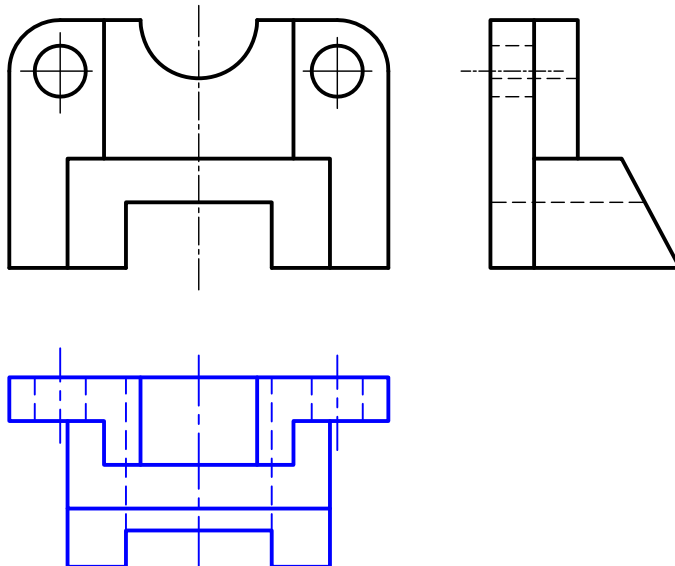
(13)



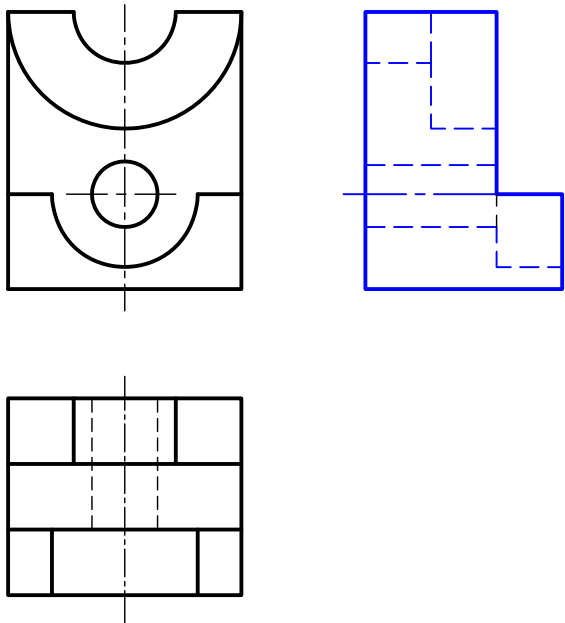
(14)



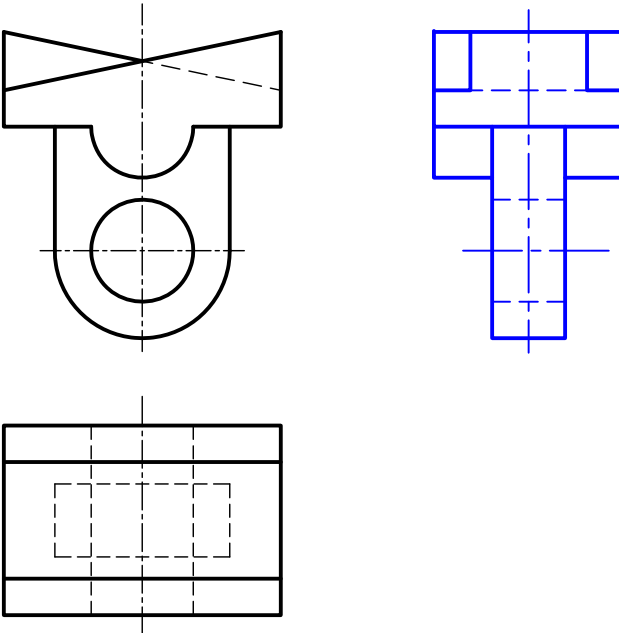
(15)



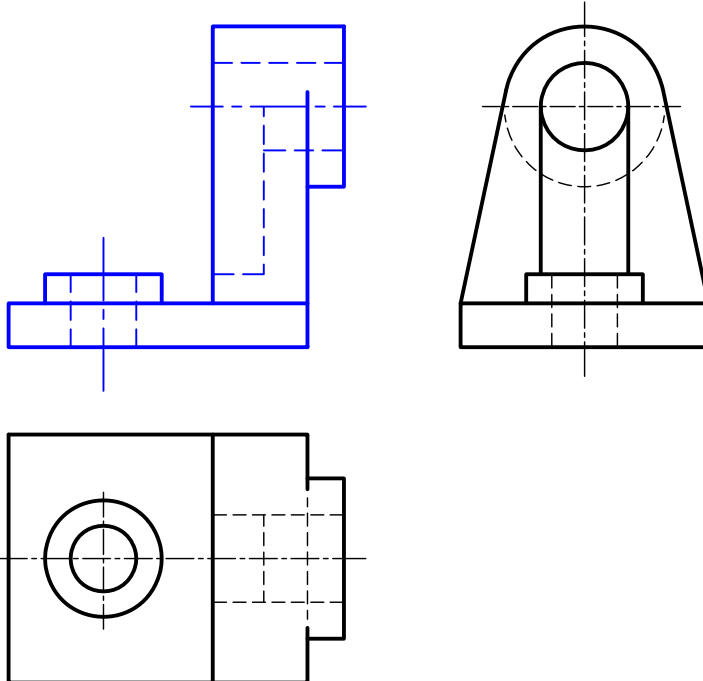
(16)



(17)

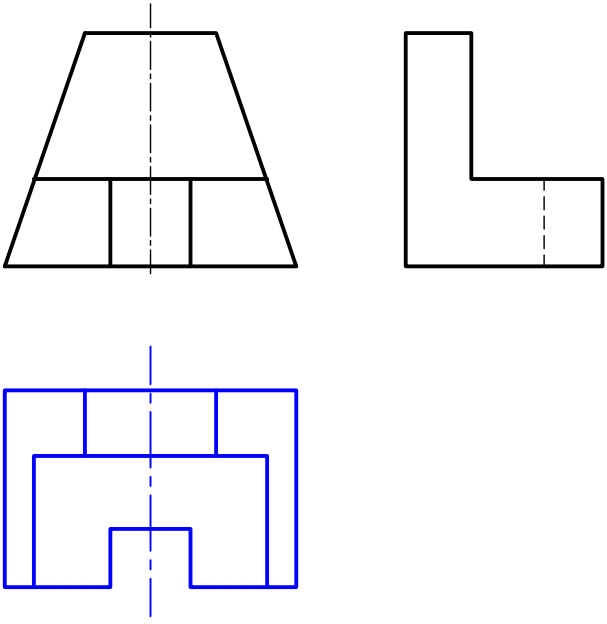
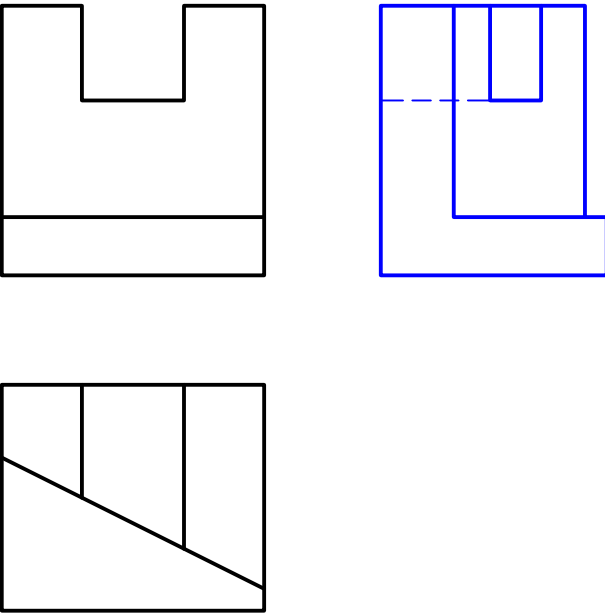
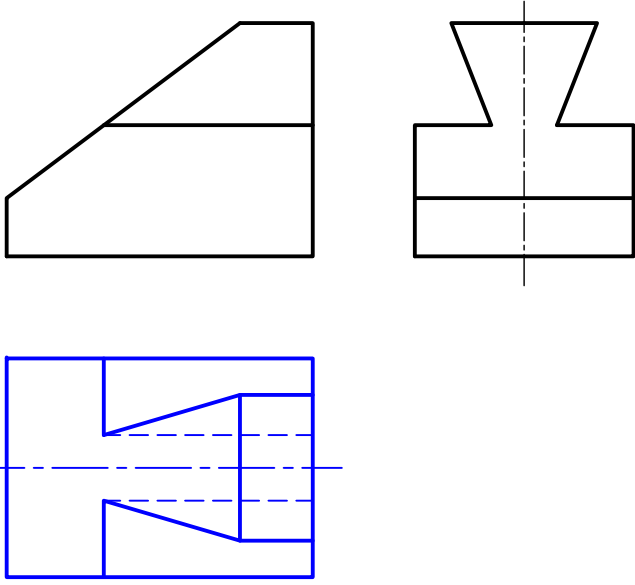
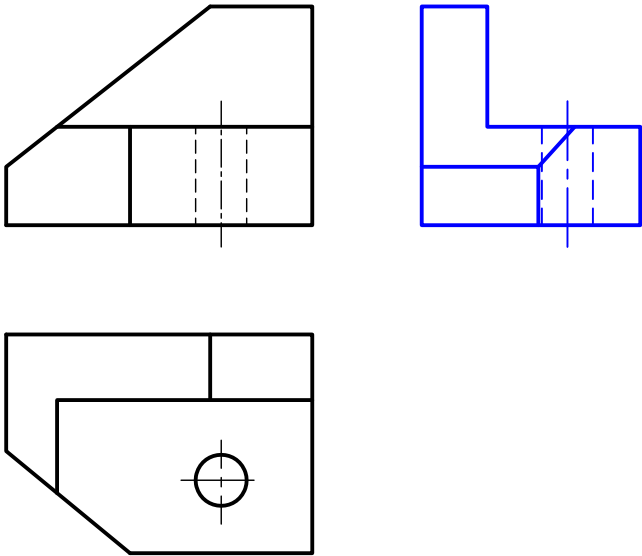
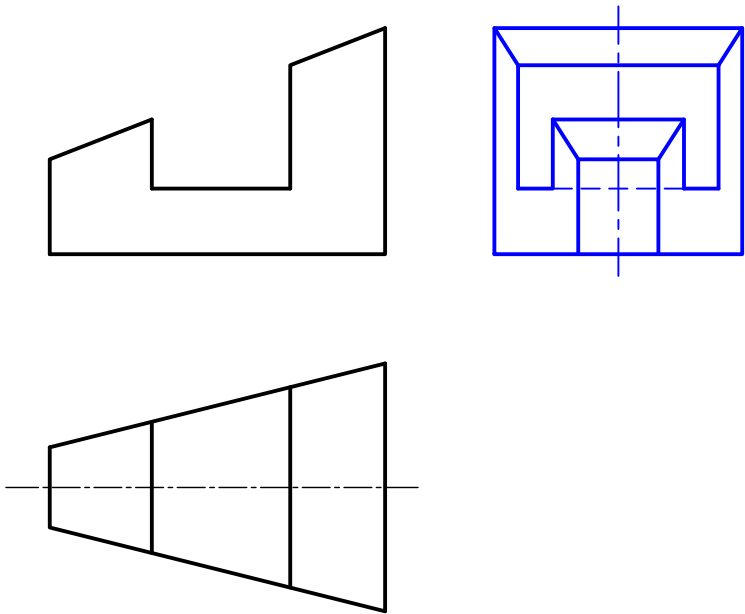
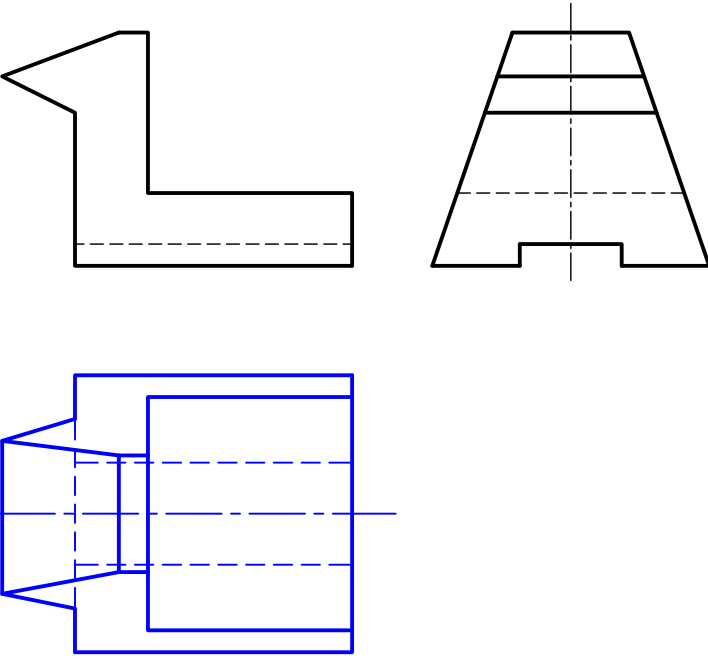


(18)



十四、读组合体视图（四）

班级 姓名 学号

(19) 	(20) 	(21) 
(22) 	(23) 	(24) 

十四、读组合体视图（五）

班级

姓名

学号

(25)

(26)

(27)

(28)

(29)

(30)

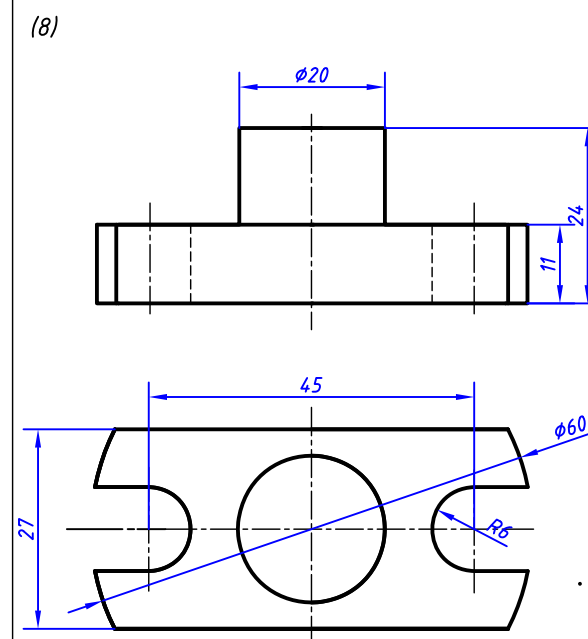
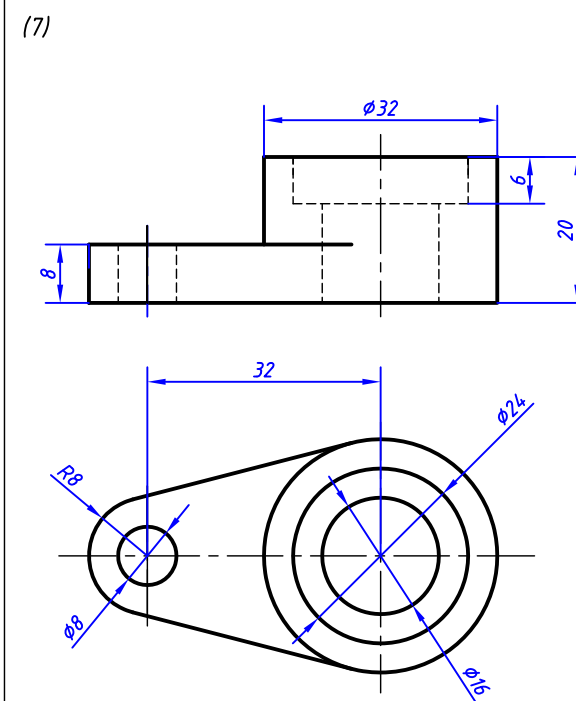
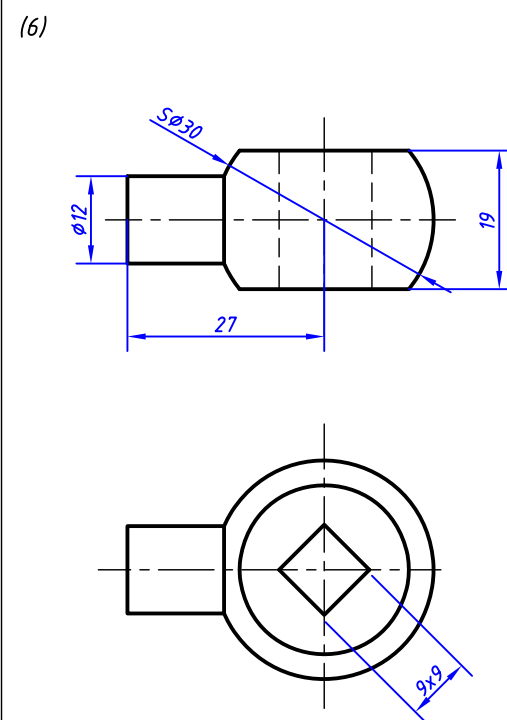
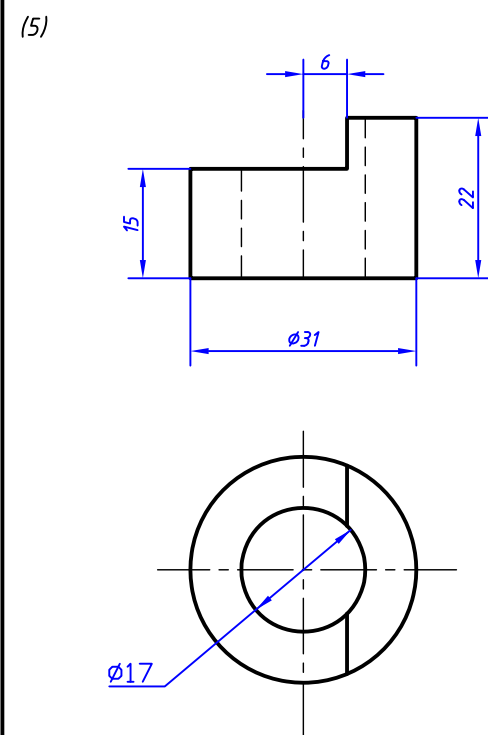
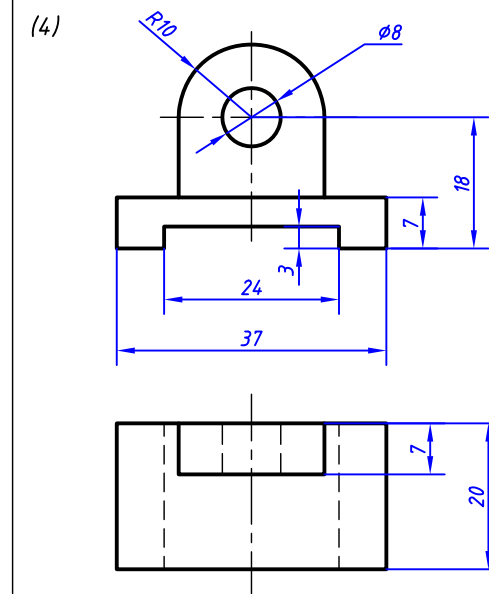
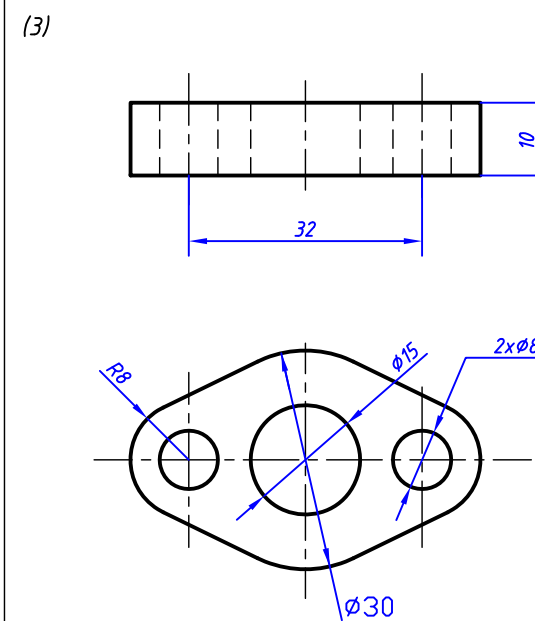
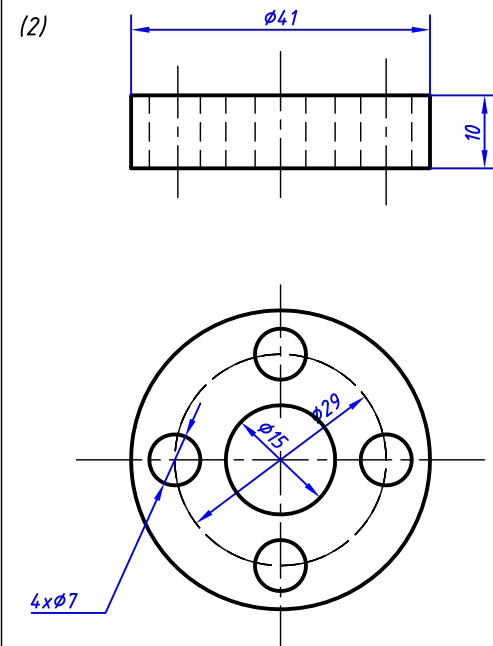
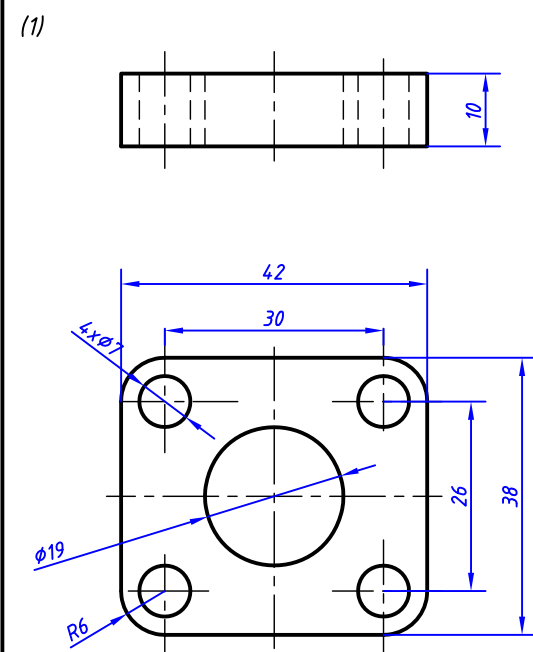
十五、组合体尺寸标注（一）

班级

姓名

学号

1. 标注下列各组合体的尺寸，尺寸数值按1:1的比例直接在图中量取，取整数。



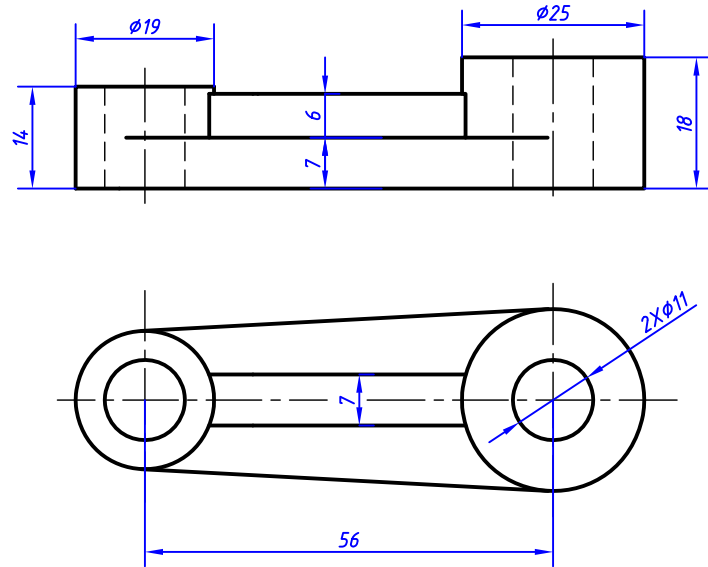
十六、组合体尺寸标注（二）

班级

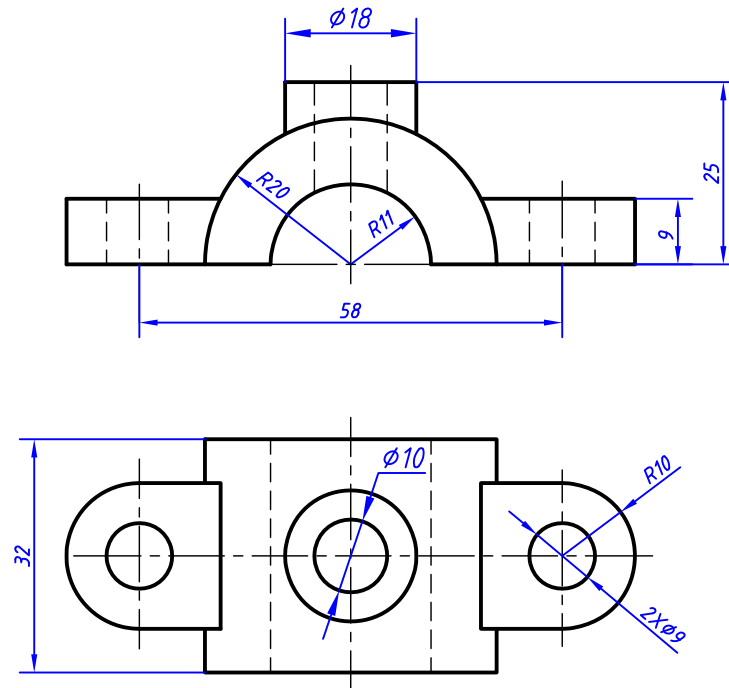
姓名

学号

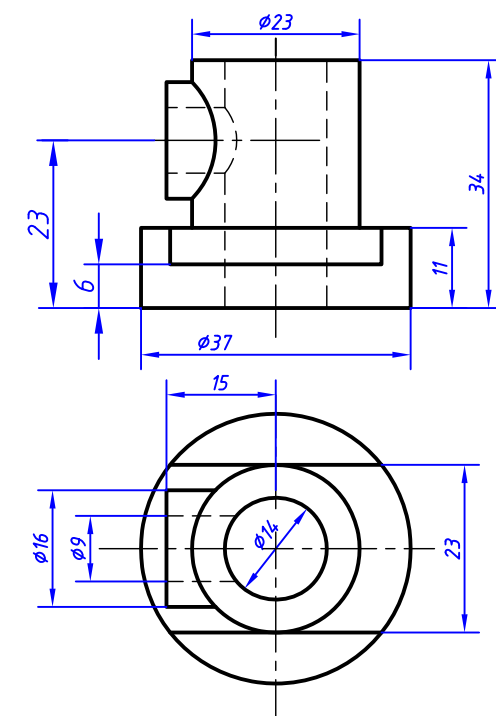
(9)



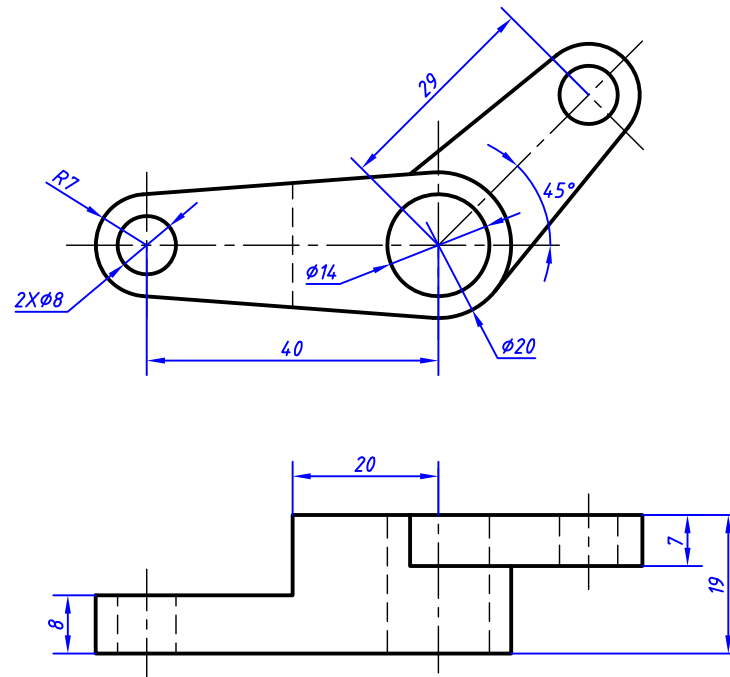
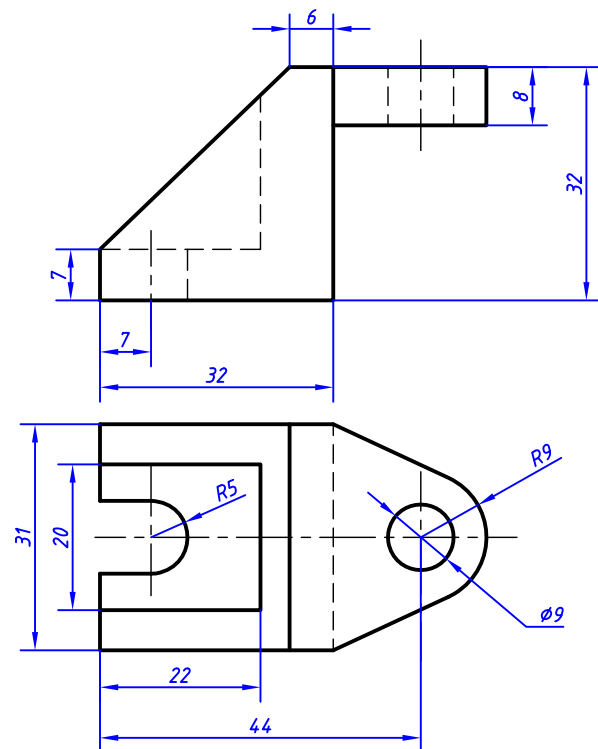
(10)



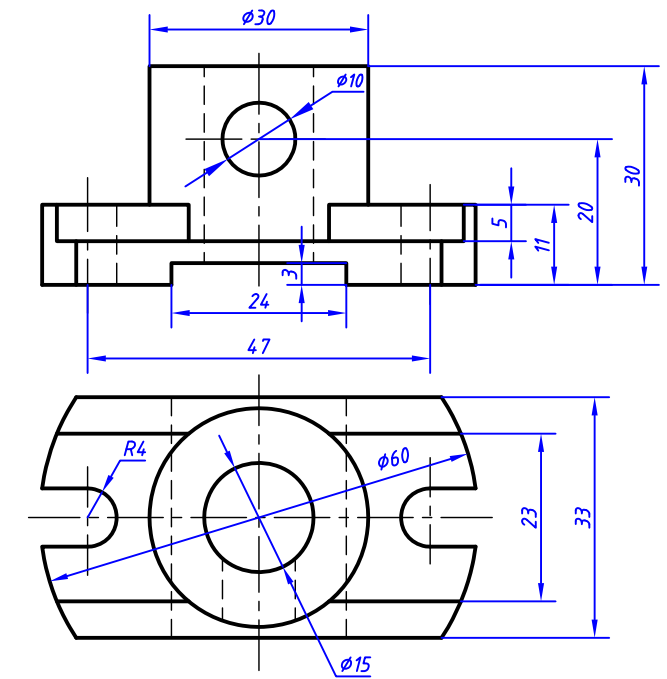
(11)



(12)



(14)



十六、组合体的构形设计

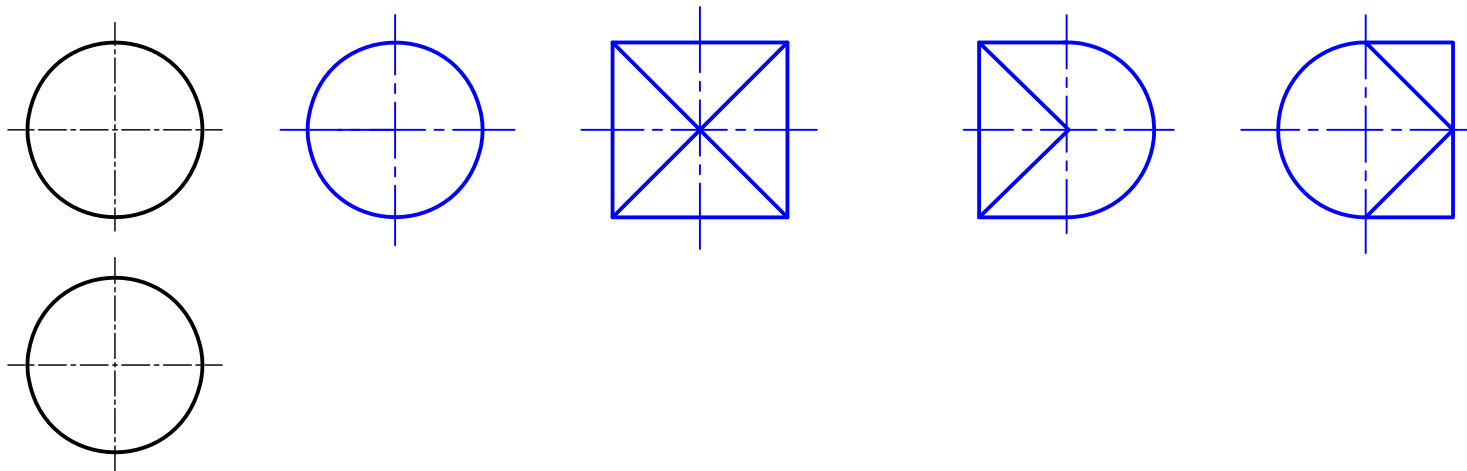
班级

姓名

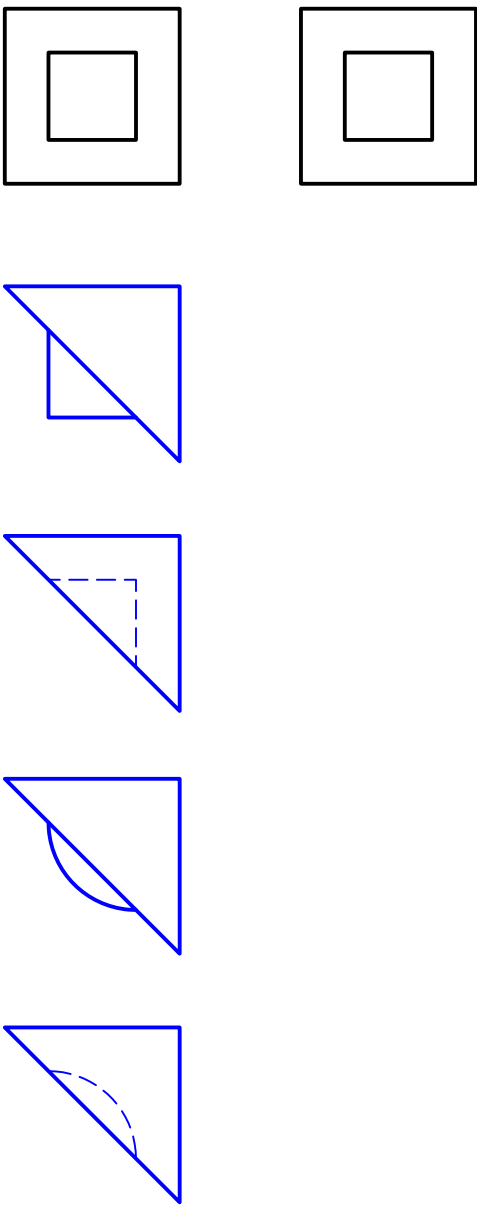
学号

根据给定两视图，想象并设计出其空间可能的形状（均有多解），然后补画各解的第三视图。

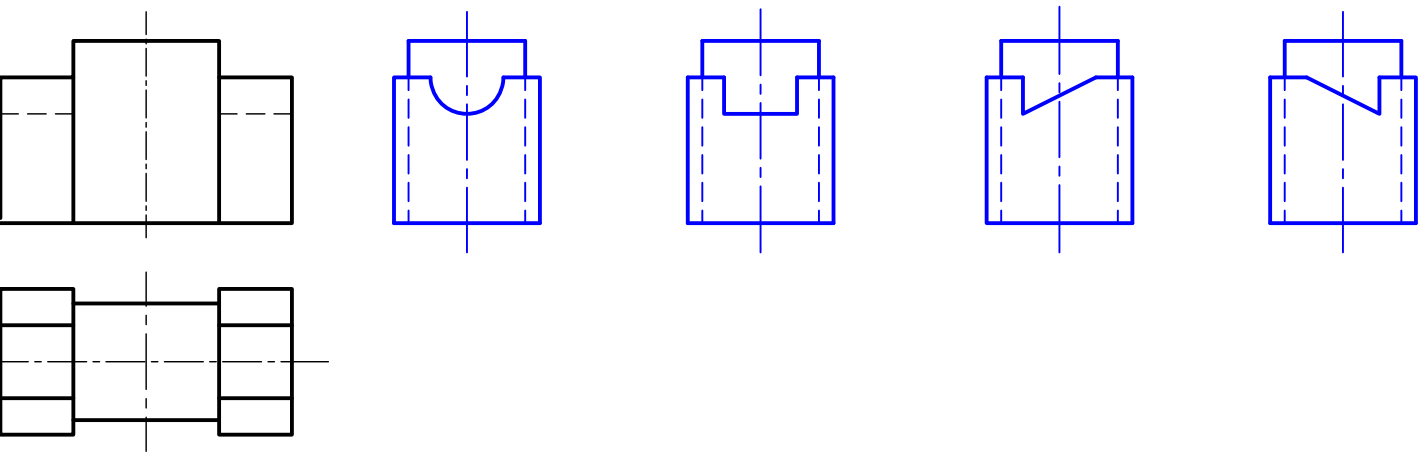
(1)



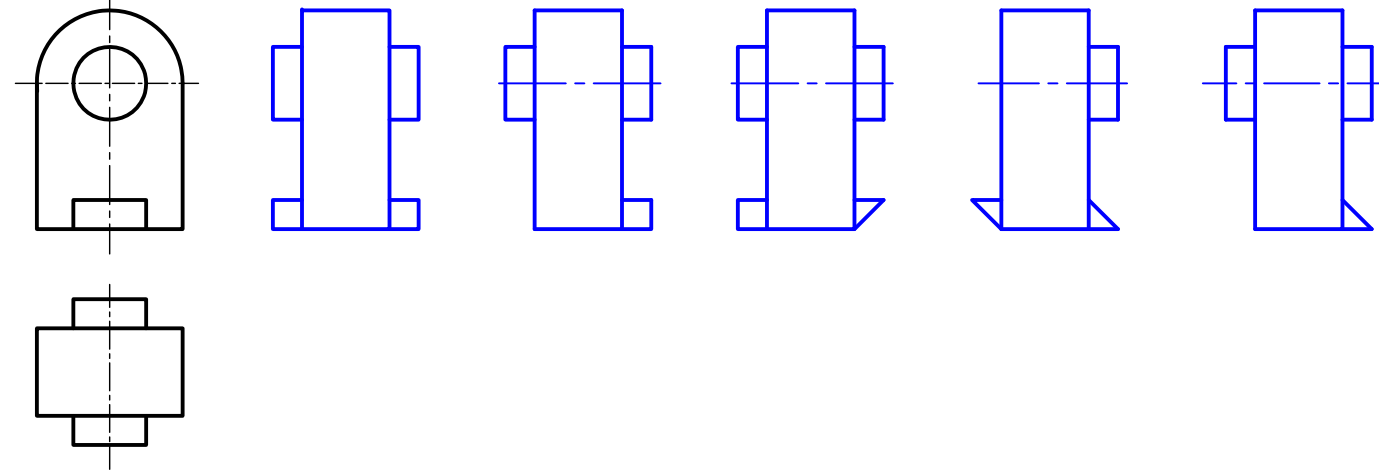
(4)



(2)



(3)

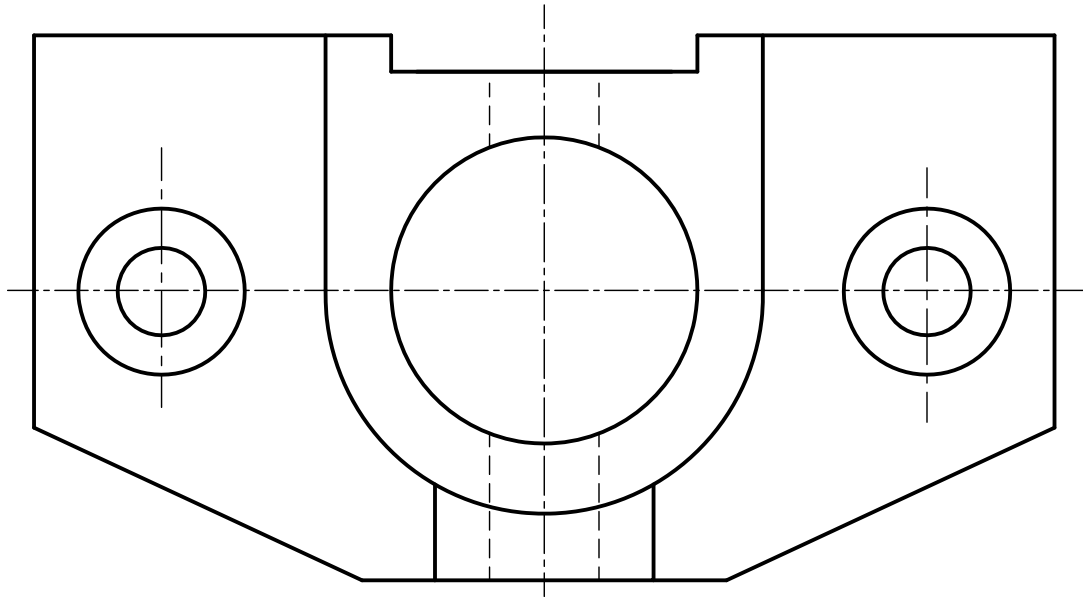
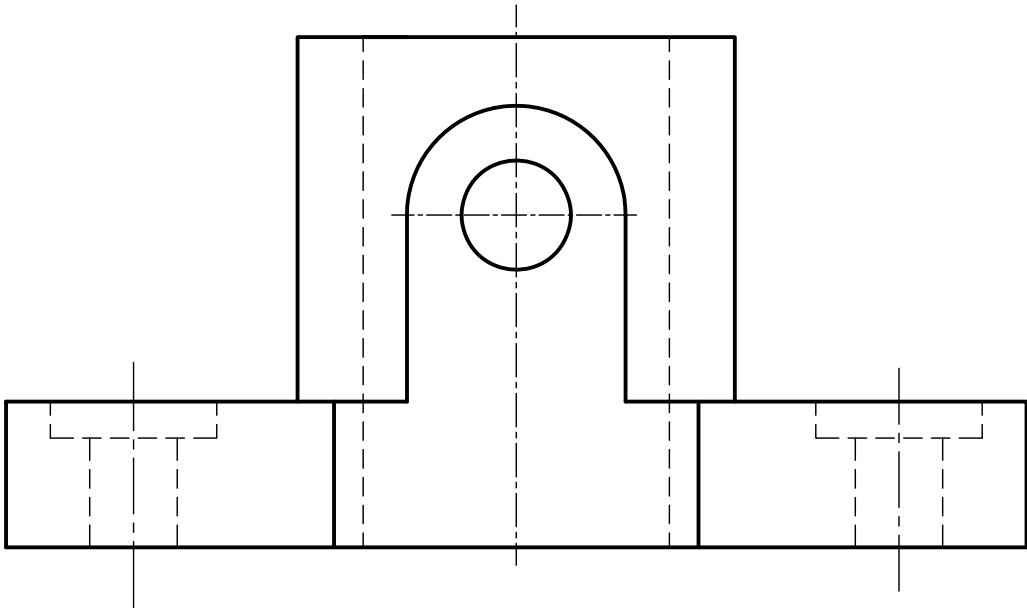


十七、组合体综合练习

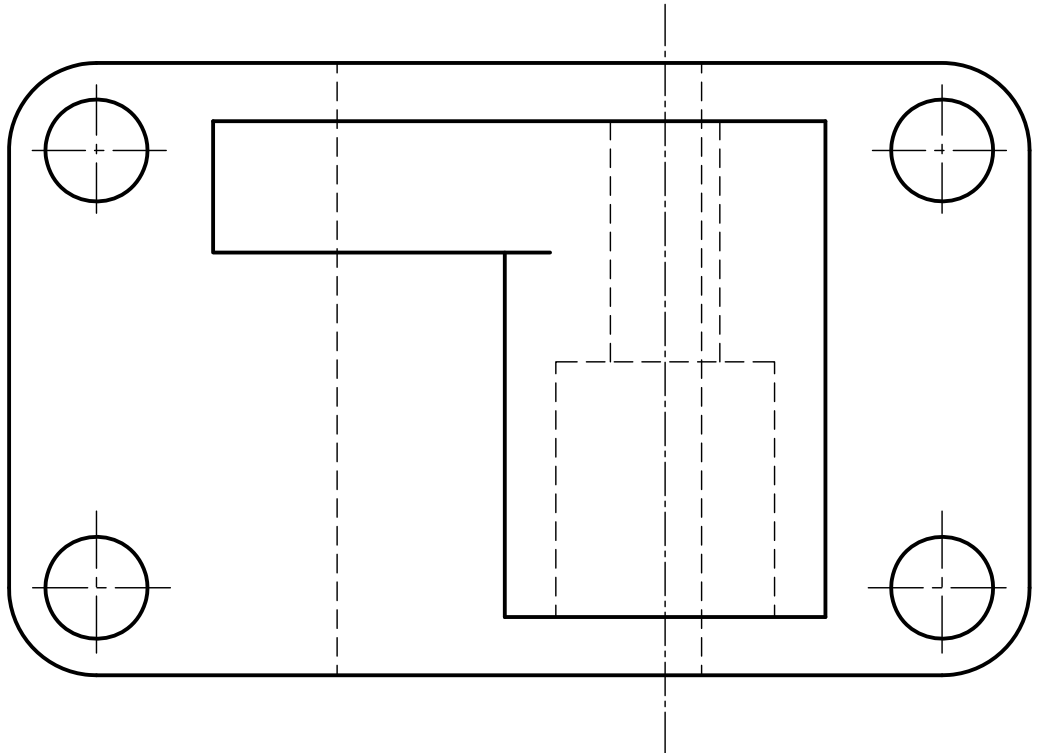
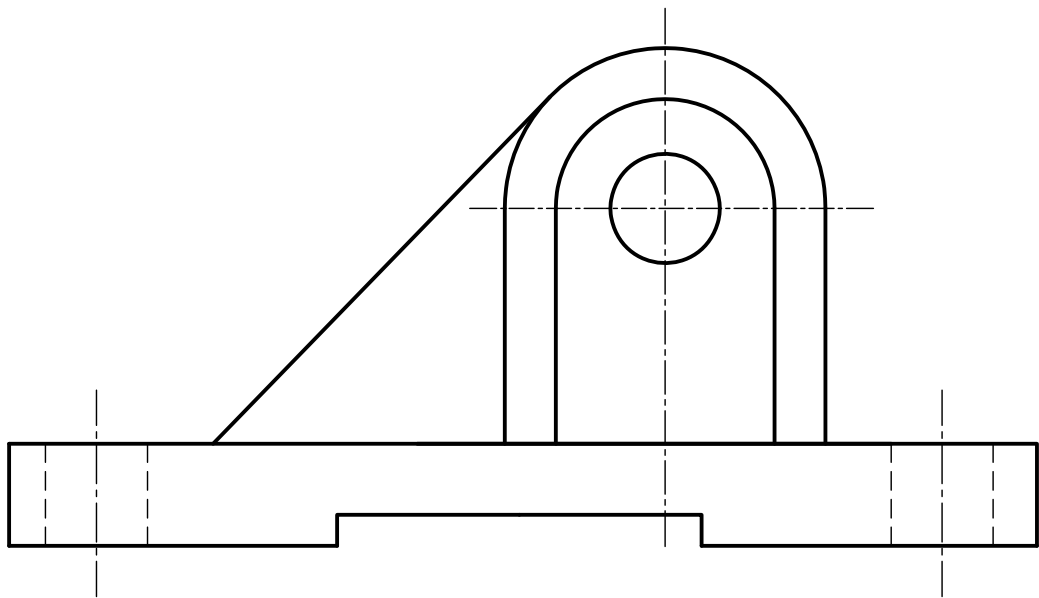
班级 姓名 学号

根据给定两视图，补画第三视图，并标注尺寸（用A3图纸，1：1比例）。

(1)



(2)

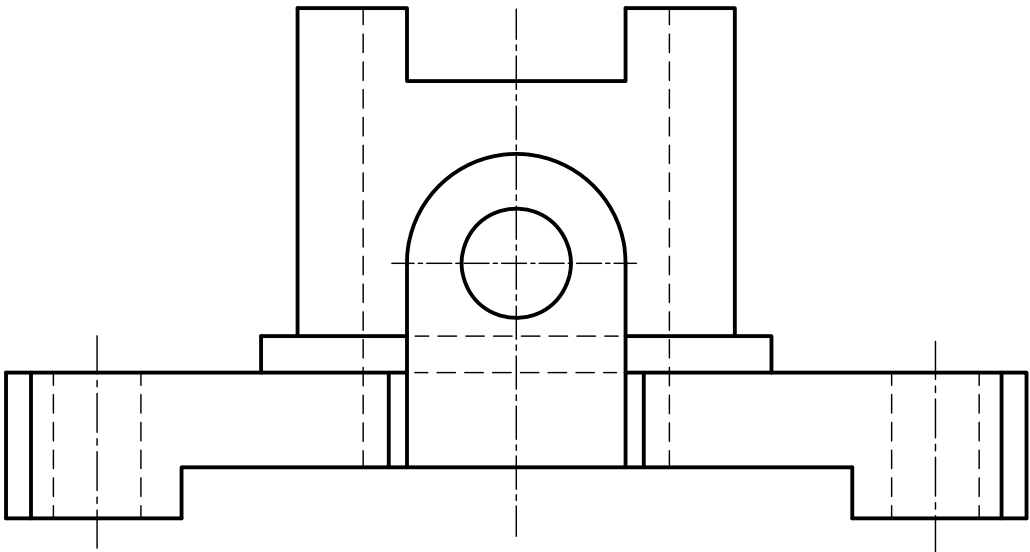


十七、组合体综合练习

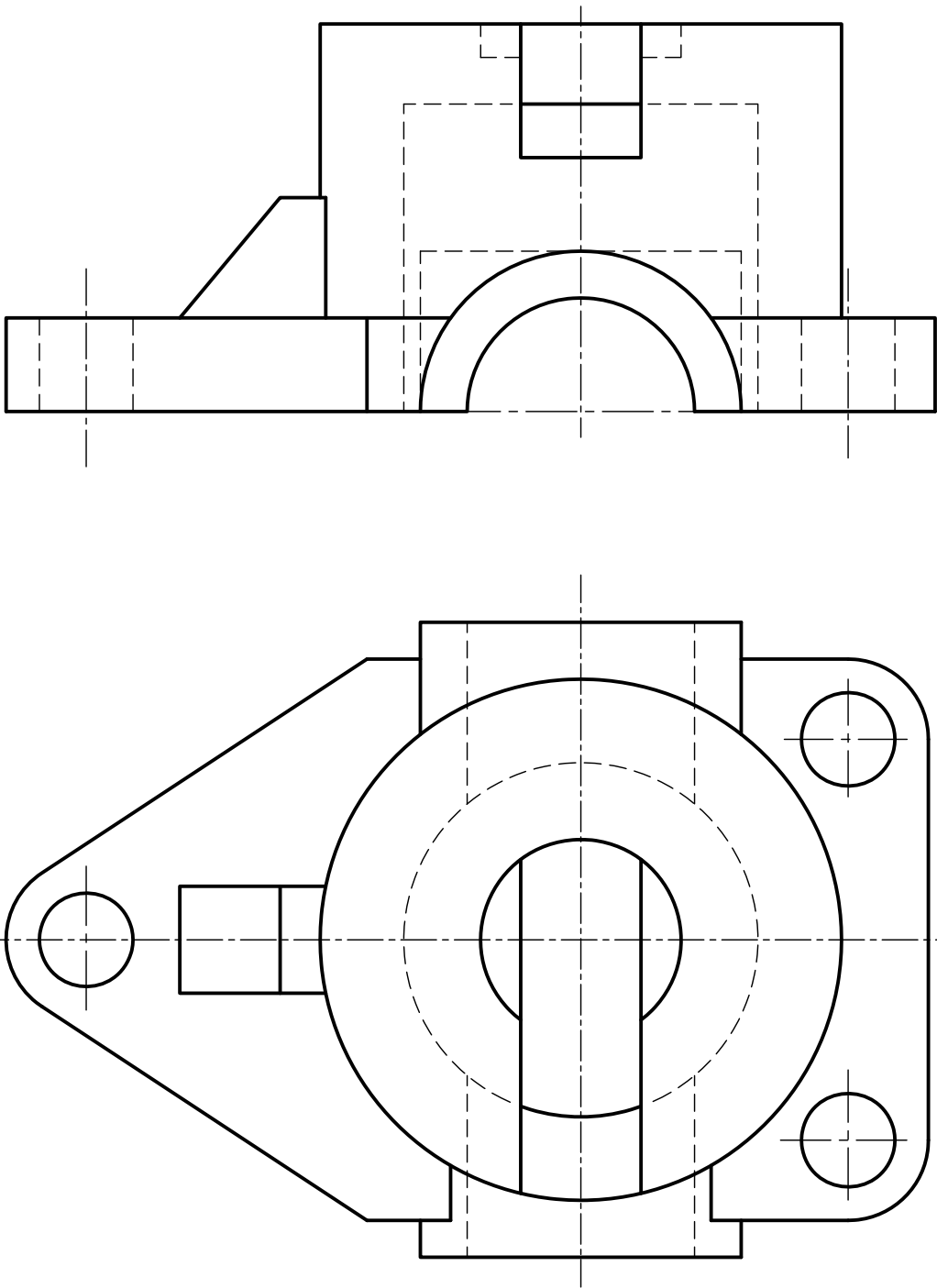
班级 姓名 学号

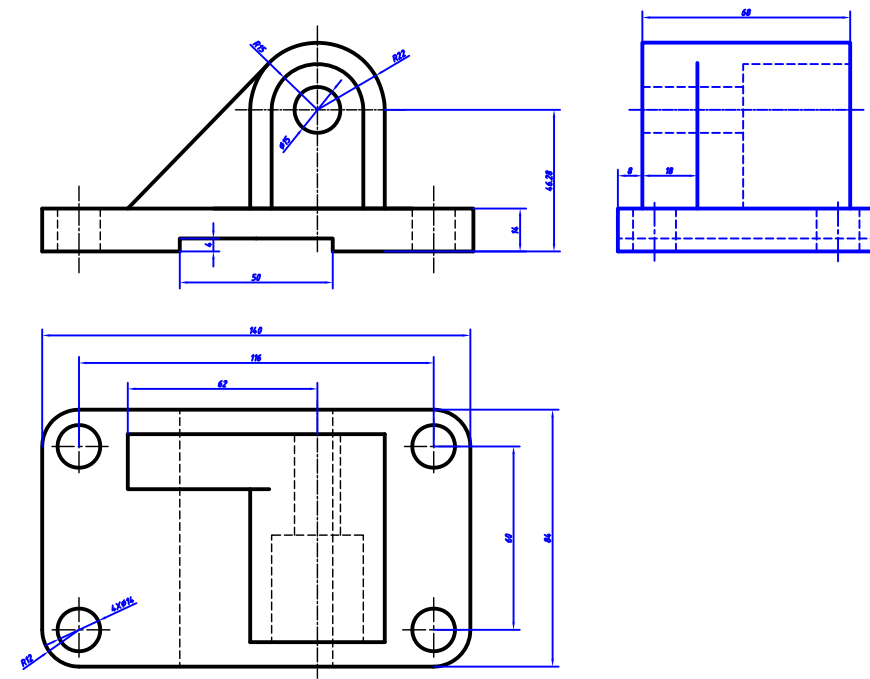
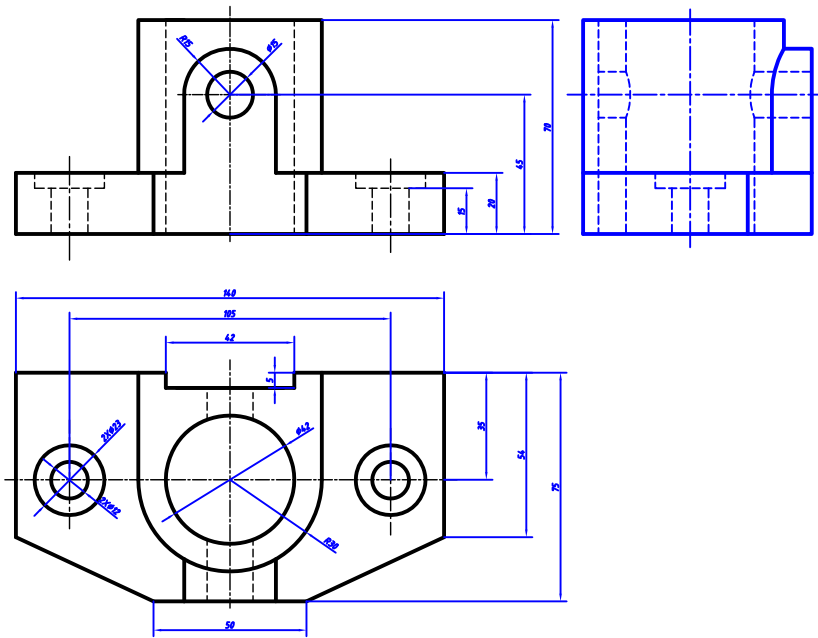
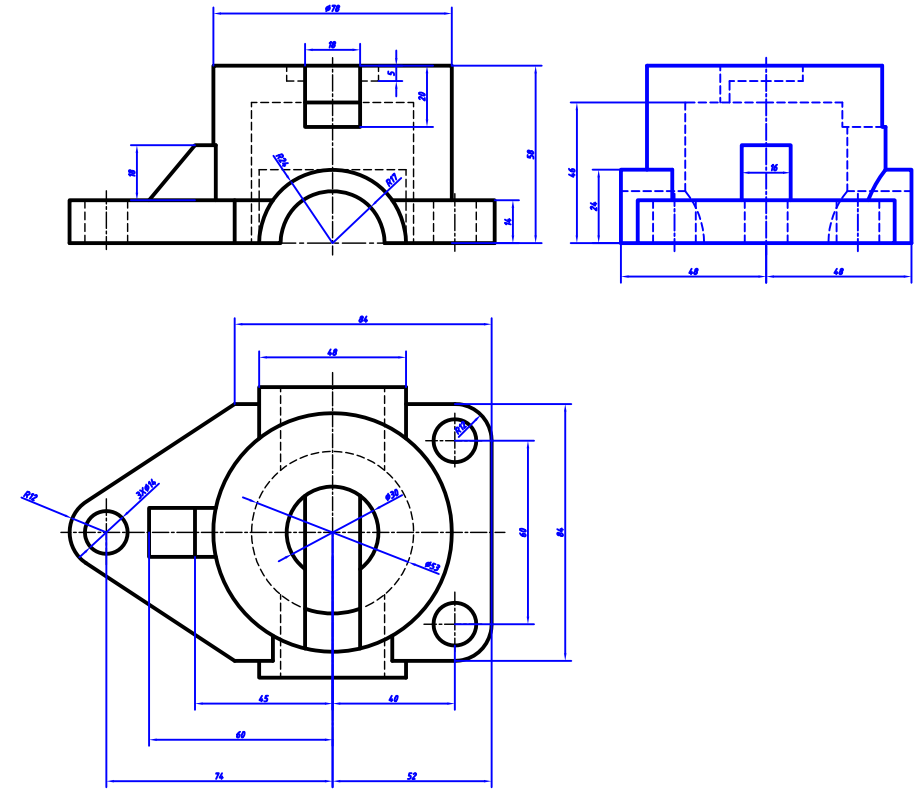
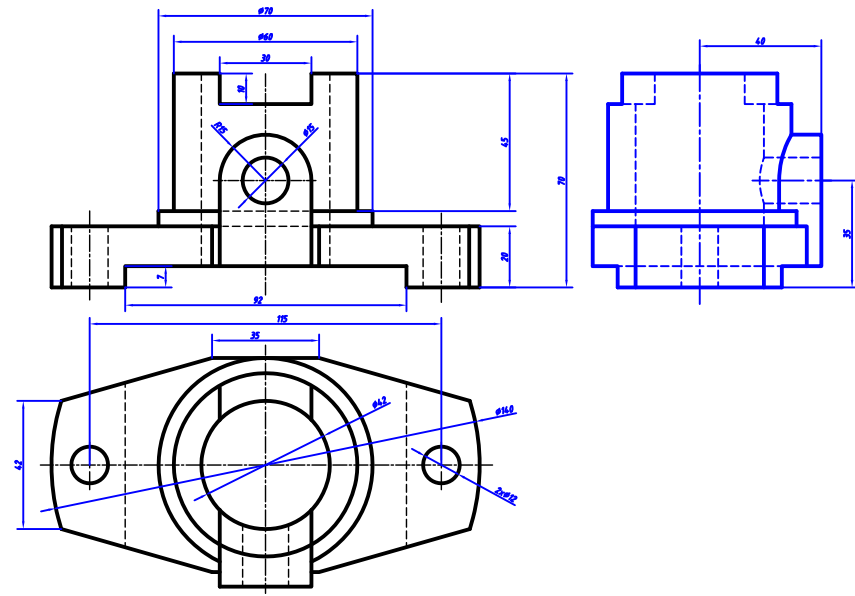
根据给定两视图，补画第三视图，并标注尺寸（用A3图纸，1：1比例）。

(1)



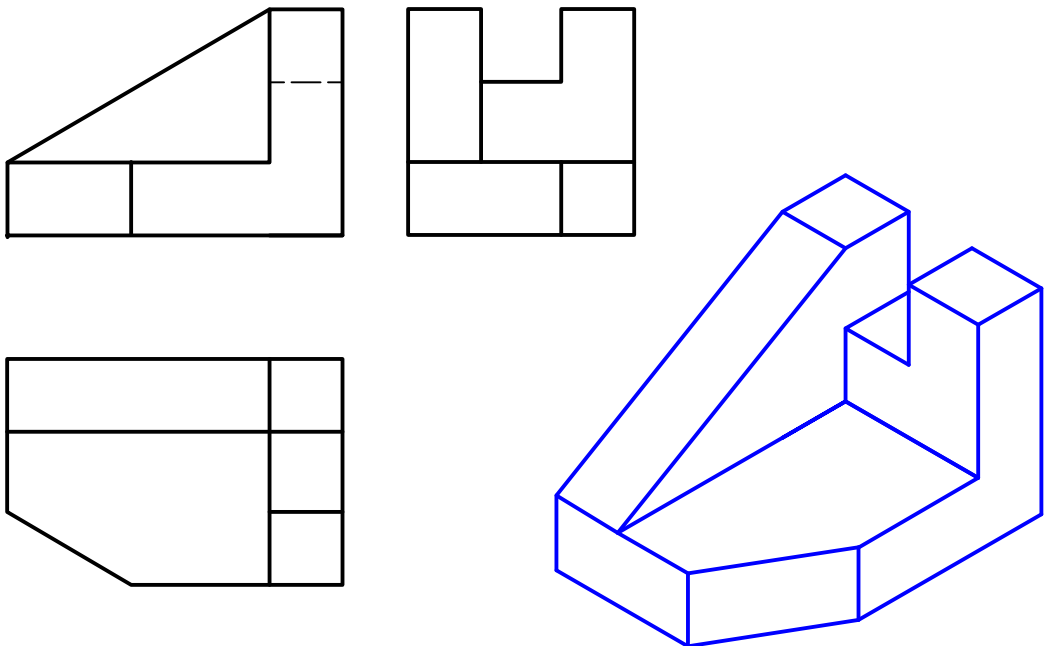
(2)





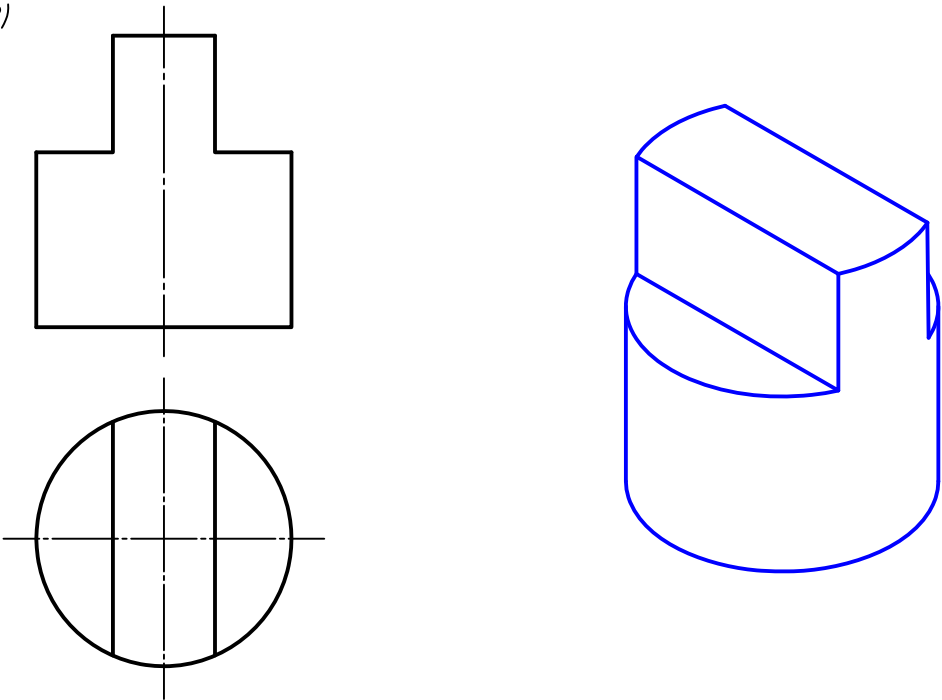
1. 画出立体的正等轴测图

(1)



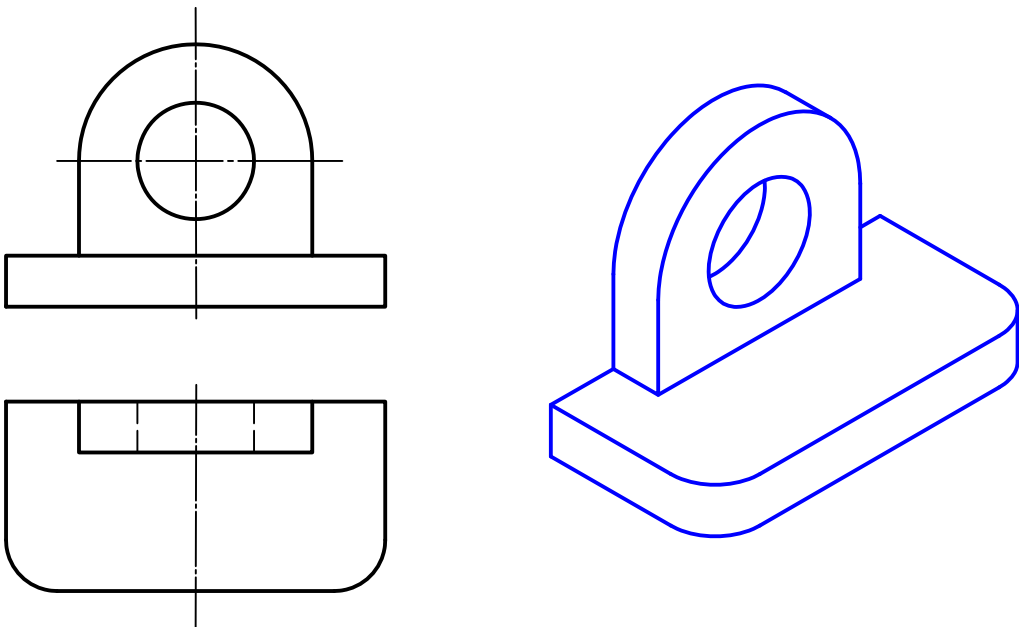
This block contains the orthographic projections and the isometric drawing of a 3D object. The object is a rectangular block with a triangular section removed from the top-left corner. The orthographic views include a front view showing the triangular cut, a top view showing the rectangular base, and a side view showing the stepped profile. The isometric drawing is shown in blue lines, illustrating the 3D form of the object.

(2)



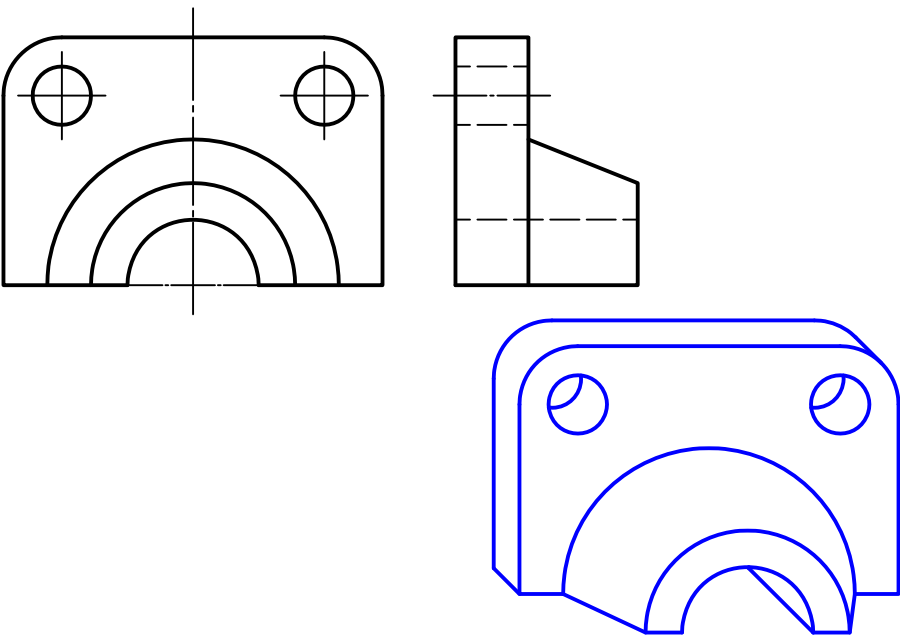
This block contains the orthographic projections and the isometric drawing of a 3D object. The object is a cylinder with a rectangular section removed from the top. The orthographic views include a front view showing the T-shaped profile, a top view showing the circular base with a rectangular cut, and a side view showing the cylindrical profile. The isometric drawing is shown in blue lines, illustrating the 3D form of the object.

(3)



This block contains the orthographic projections and the isometric drawing of a 3D object. The object is a U-shaped block with a semi-circular cutout on the top surface. The orthographic views include a front view showing the semi-circular cut, a top view showing the U-shaped base, and a side view showing the profile. The isometric drawing is shown in blue lines, illustrating the 3D form of the object.

2. 画出立体的斜二轴测图



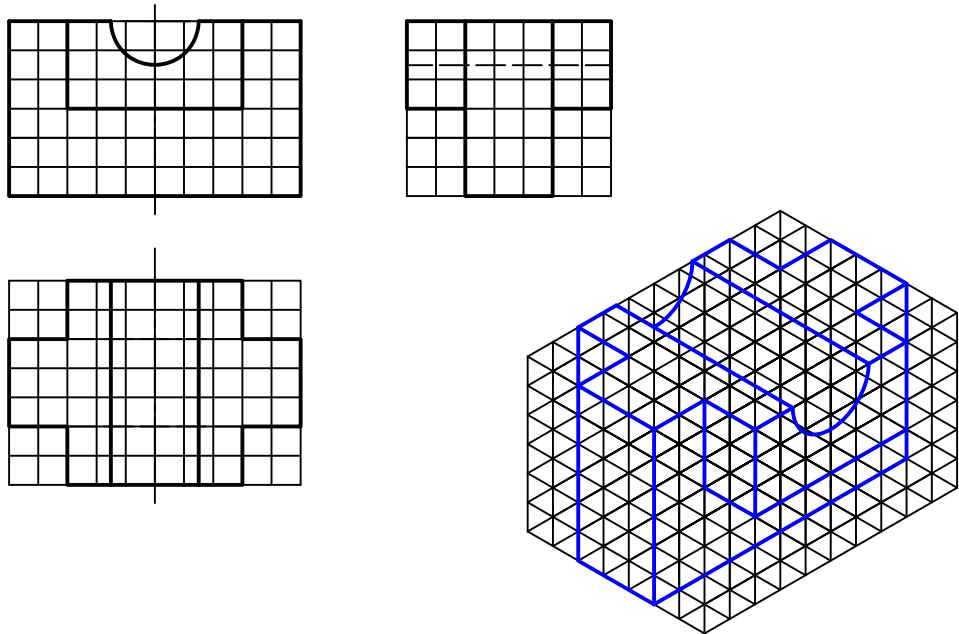
This block contains the orthographic projections and the isometric drawing of a 3D object. The object is a U-shaped block with a semi-circular cutout on the top surface. The orthographic views include a front view showing the semi-circular cut, a top view showing the U-shaped base, and a side view showing the profile. The isometric drawing is shown in blue lines, illustrating the 3D form of the object.

十八、轴测投影(二)

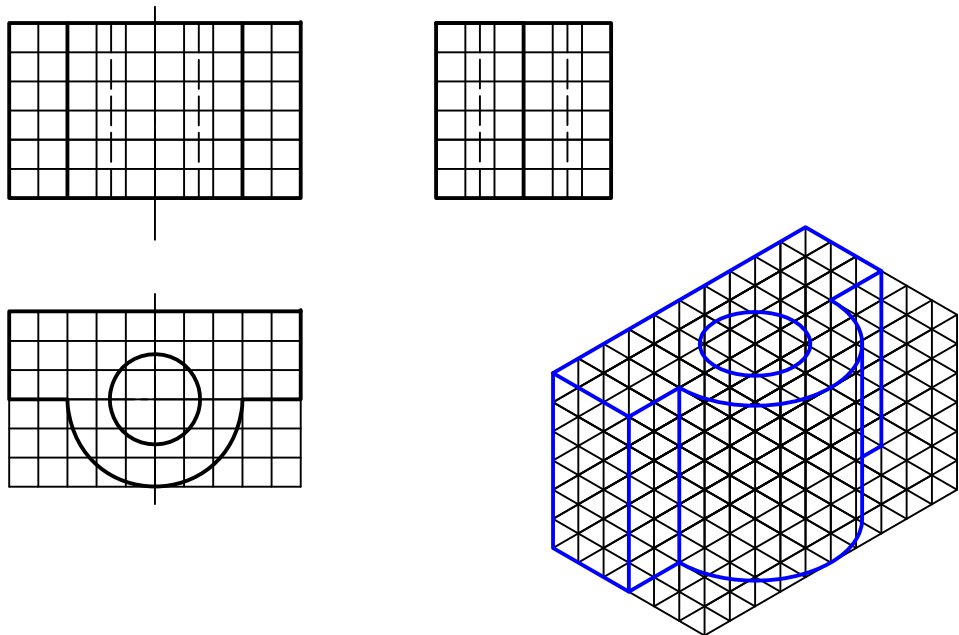
班级 姓名 学号

3. 徒手画立体的正等轴测图

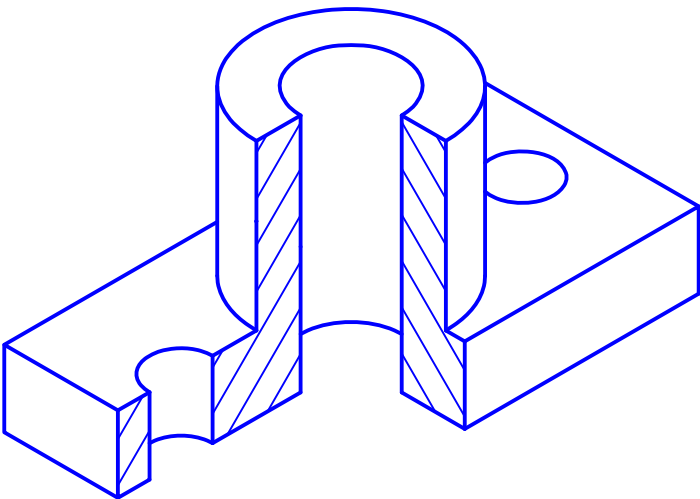
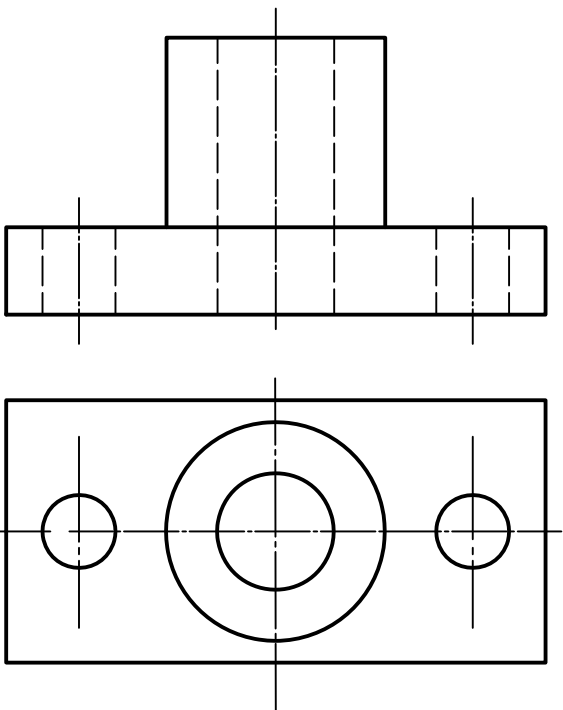
(1)



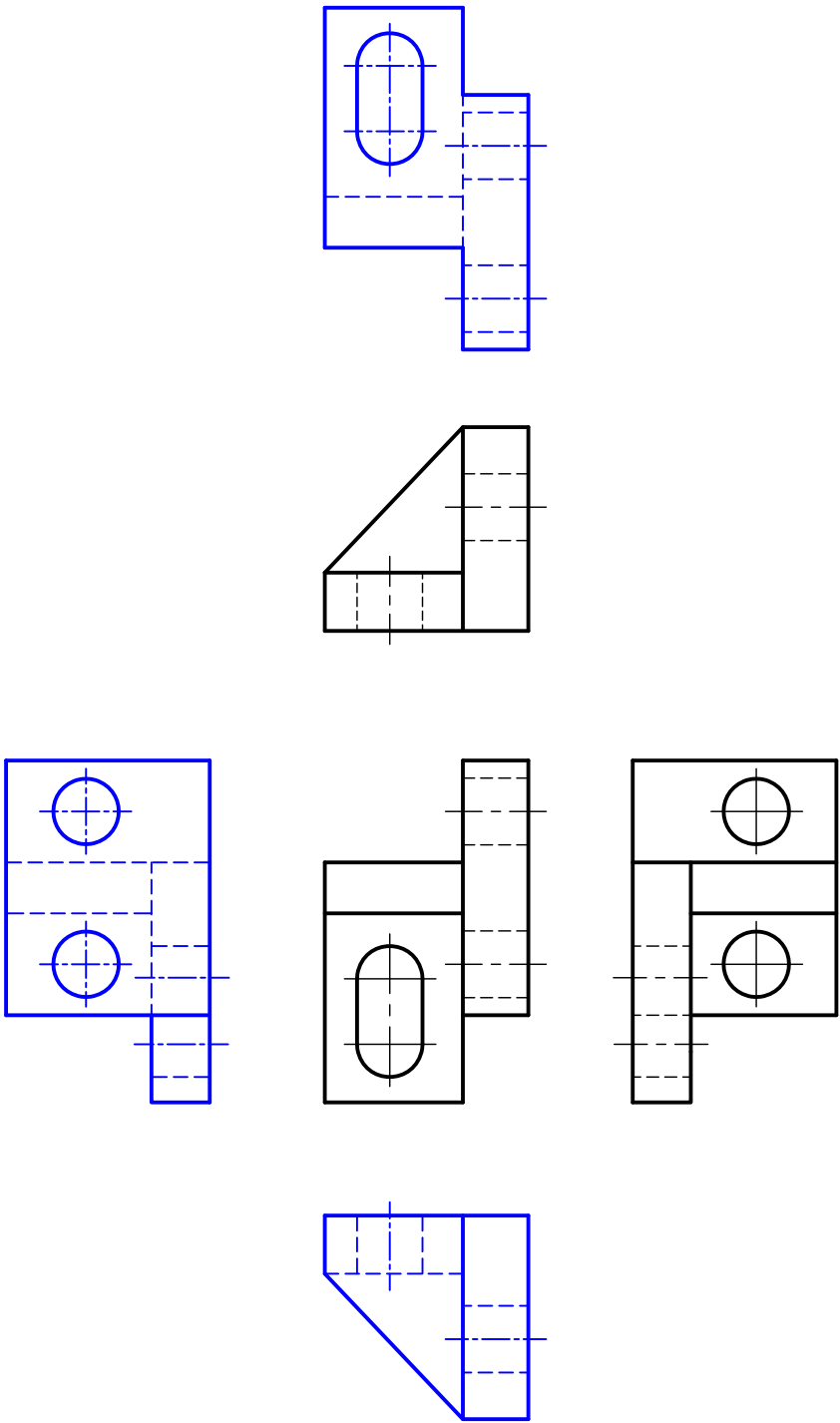
(2)



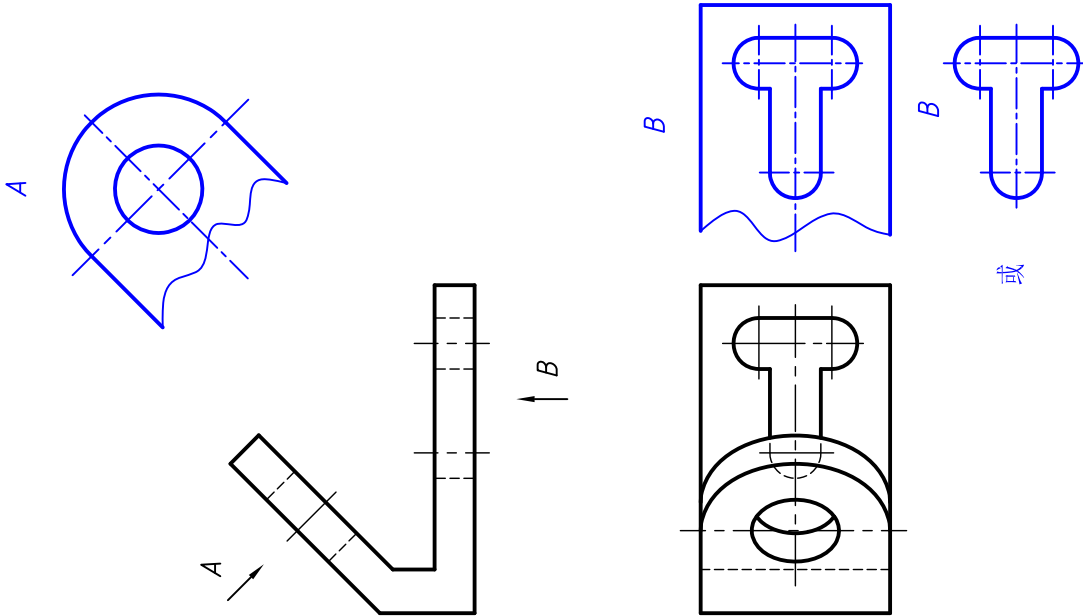
4. 画出立体的正等轴测剖视图



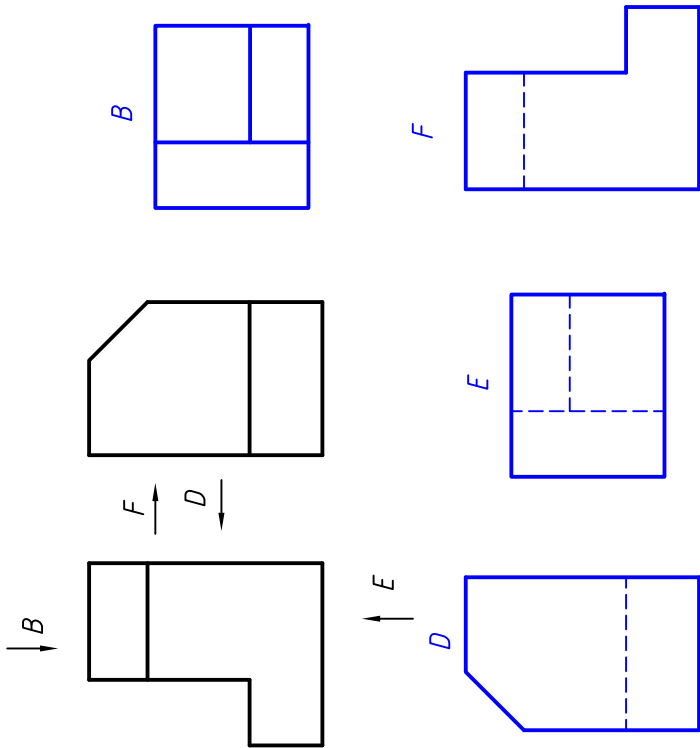
1. 画出机件的其余三个基本视图。（保留虚线）



2. 画出机件的斜视图、局部视图。



3. 按箭头所指的方向，在适当位置画出相应的向视图。（保留虚线）



1. 在指定位置将主视图画成全剖视图。

(1)

Exercise (1) drawing: Front view, half-section, and top view of a mechanical part. The front view shows a rectangular block with a vertical slot and a horizontal cutout. The half-section view shows the internal structure with hatching. The top view shows the circular base with a central hole and four small holes.

(2)

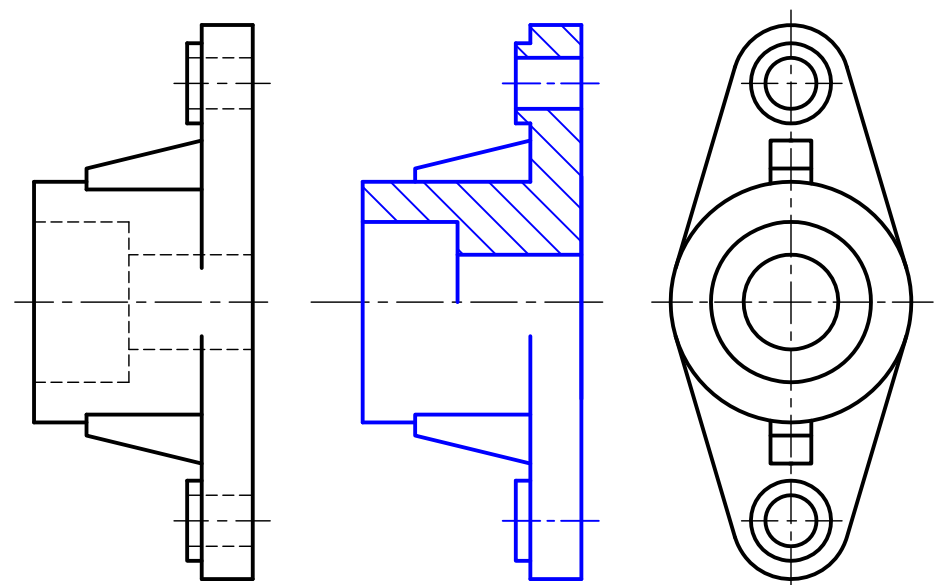
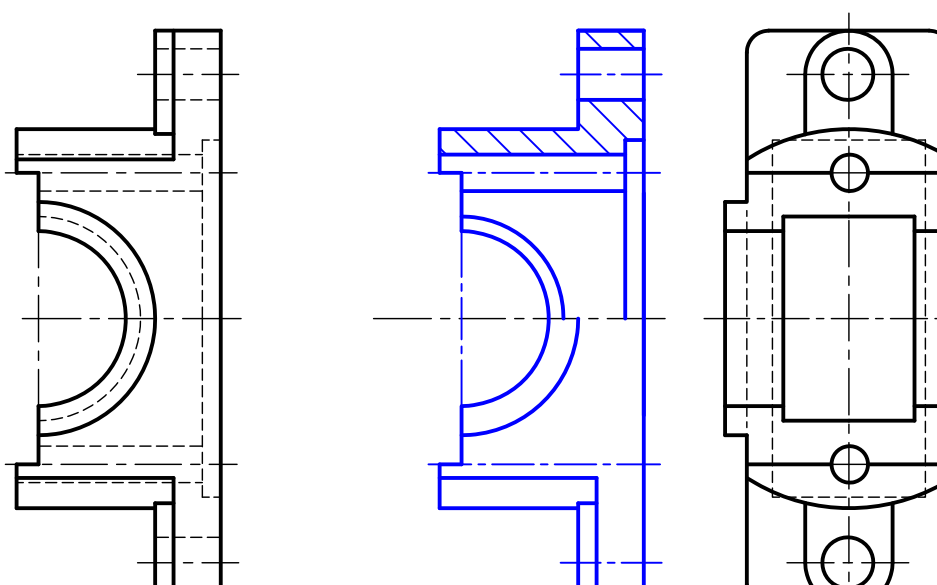
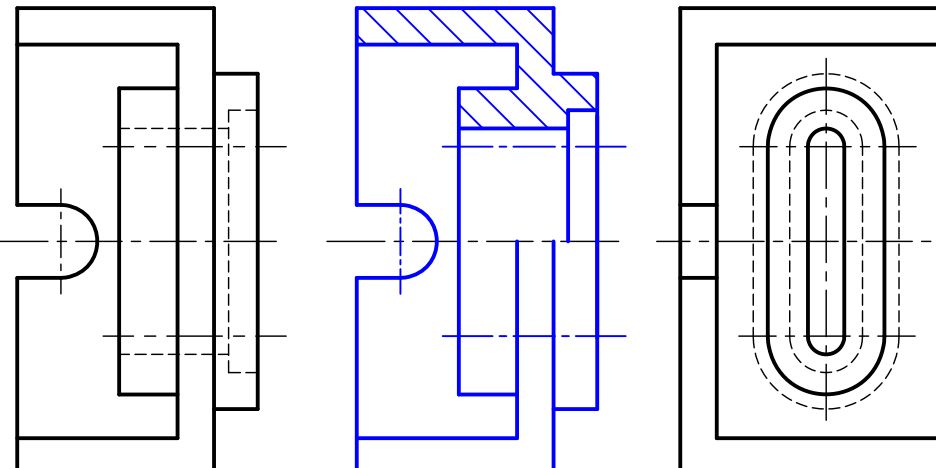
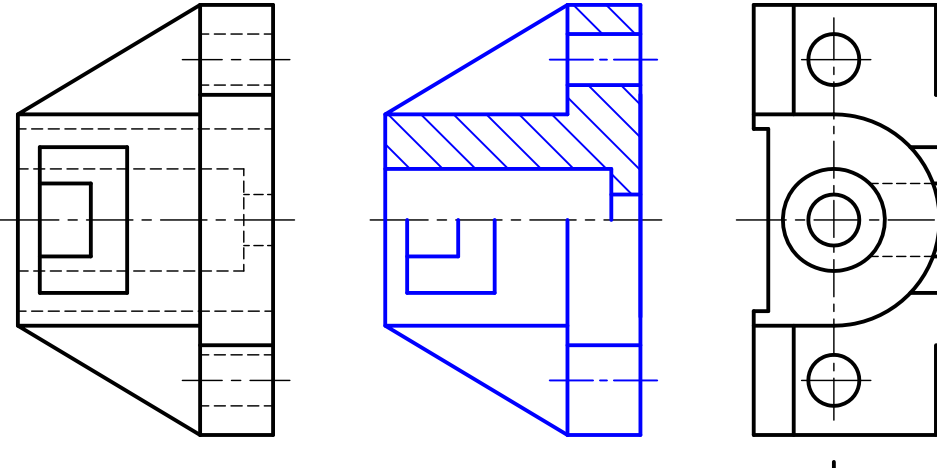
Exercise (2) drawing: Front view, half-section, and top view of a mechanical part. The front view shows a rectangular block with a vertical slot and a horizontal cutout. The half-section view shows the internal structure with hatching. The top view shows the circular base with a central hole and four small holes.

(3)

Exercise (3) drawing: Front view, half-section, and top view of a mechanical part. The front view shows a rectangular block with a vertical slot and a horizontal cutout. The half-section view shows the internal structure with hatching. The top view shows the circular base with a central hole and four small holes.

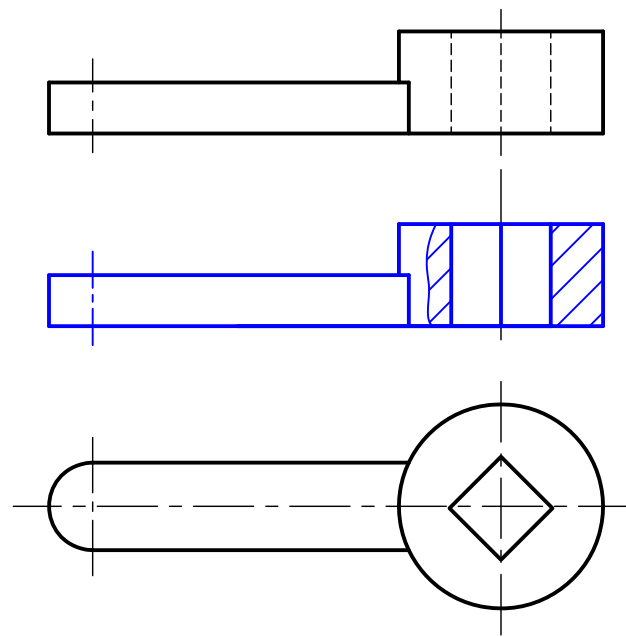
(4)

Exercise (4) drawing: Front view, half-section, and top view of a mechanical part. The front view shows a rectangular block with a vertical slot and a horizontal cutout. The half-section view shows the internal structure with hatching. The top view shows the circular base with a central hole and four small holes.

2. 在指定位置把主视图画成半剖视图。	(1) 	(3) 
	(2) 	(4) 

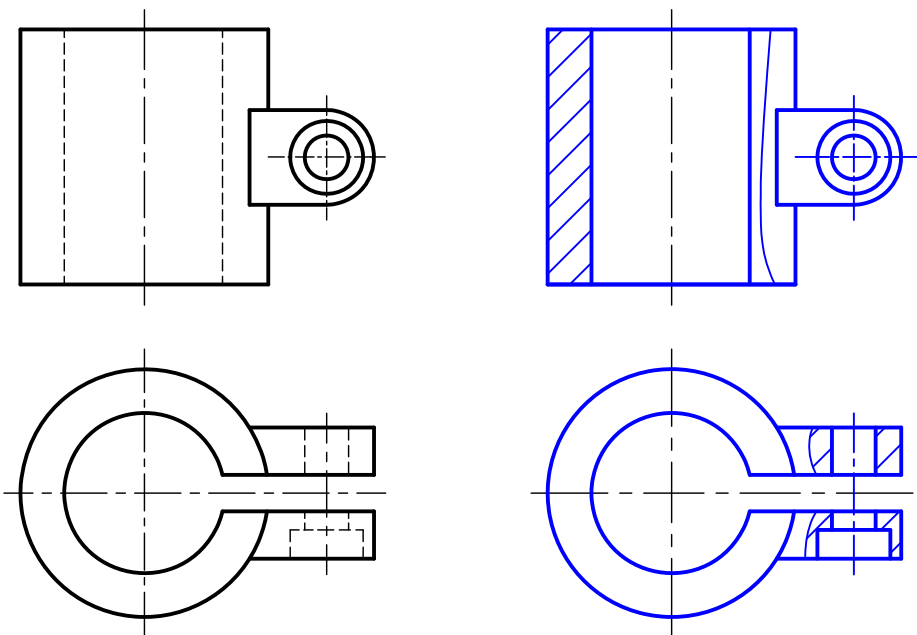
二十、剖视图（三）

3. 在指定位置把主视图画成局部剖视图。

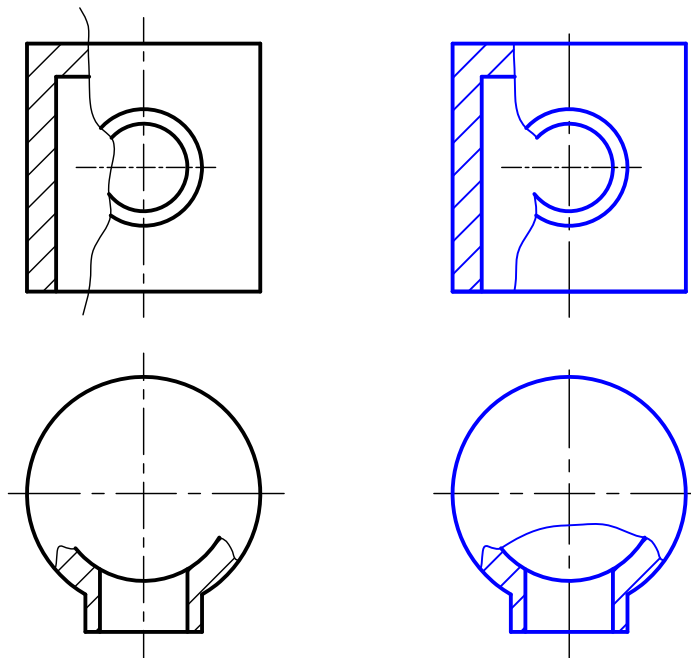


4. 重新选择表达方案表达机件。

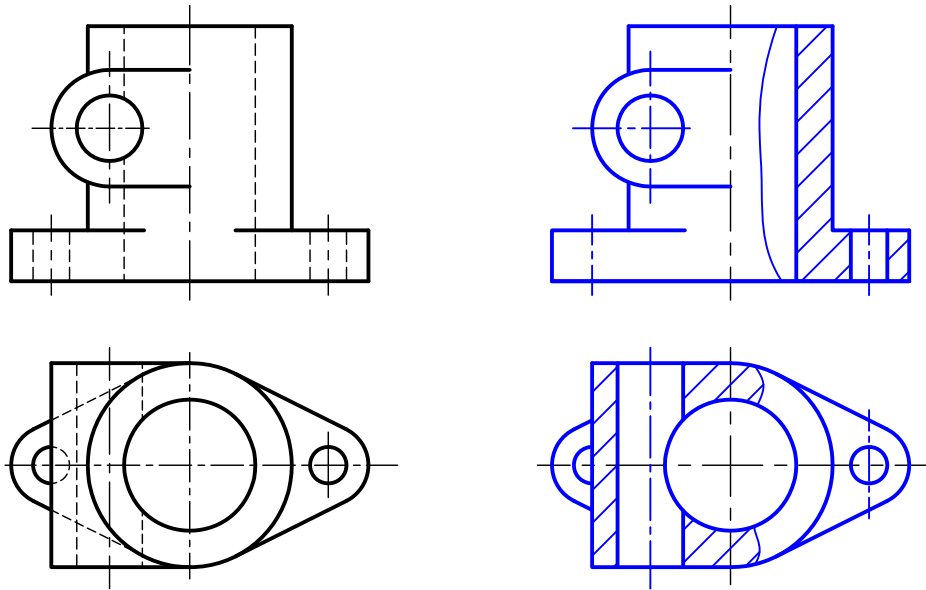
(1)

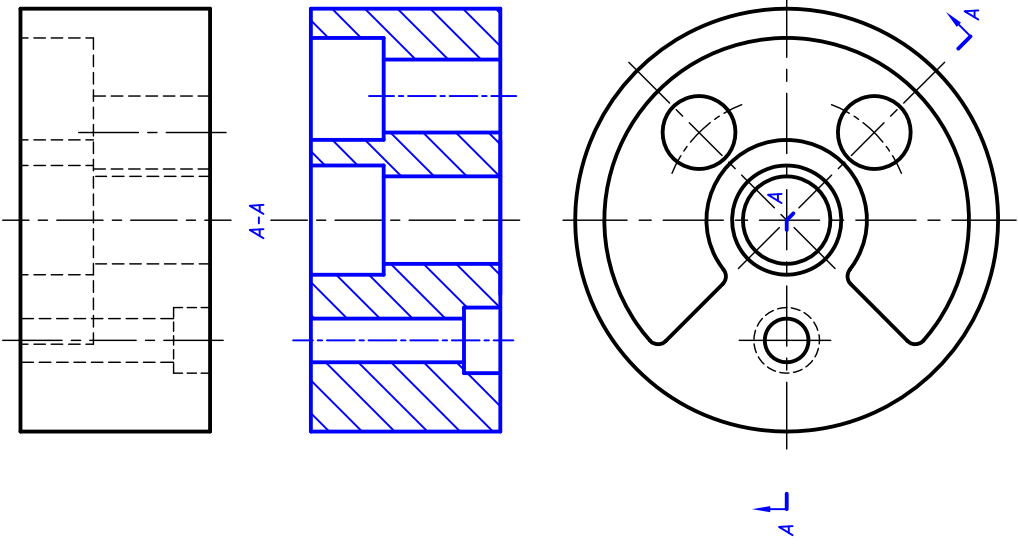
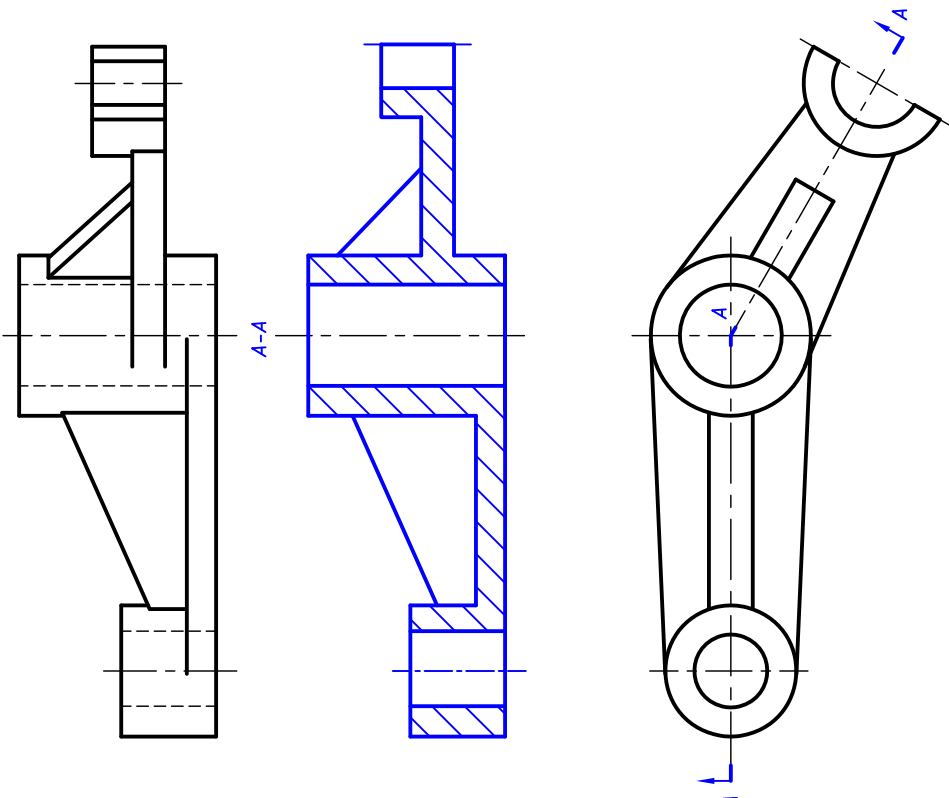
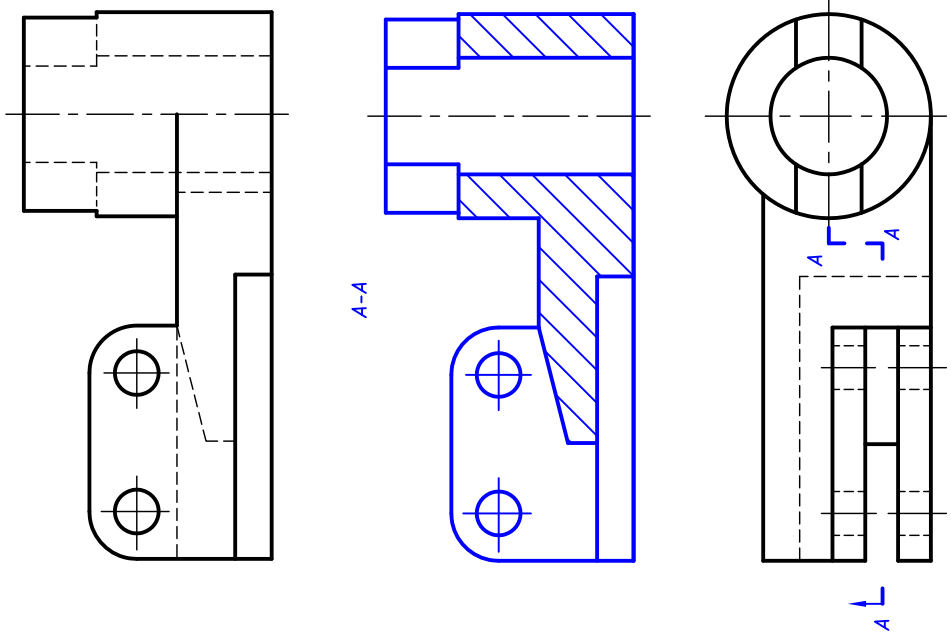
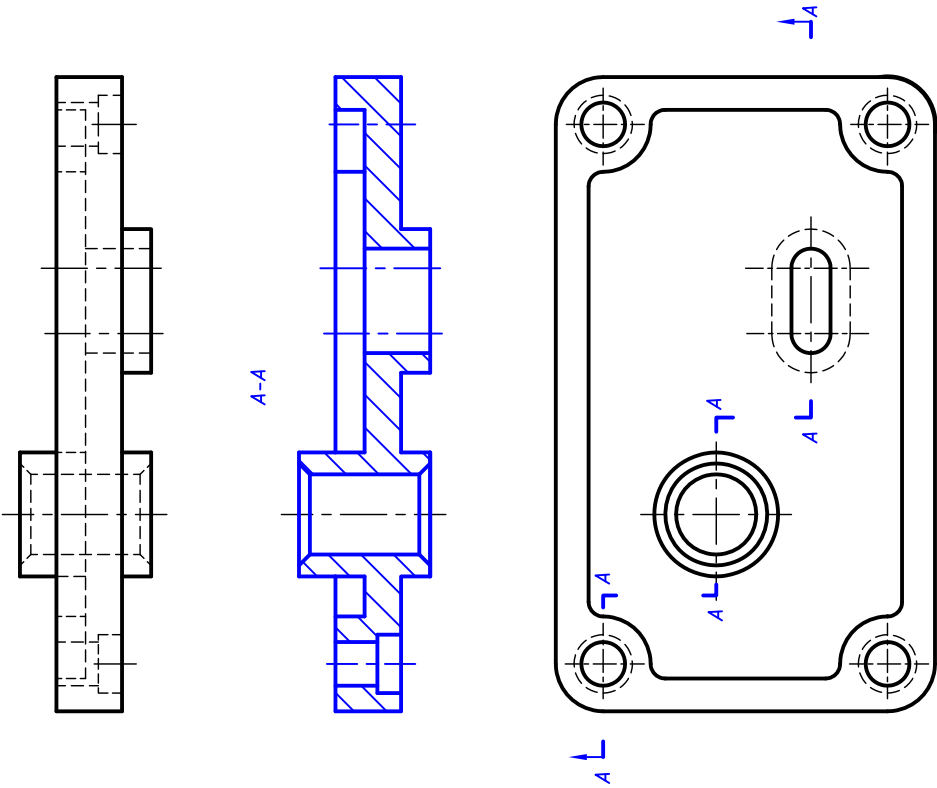


5. 分析视图中的错误，作出正确的剖视图。

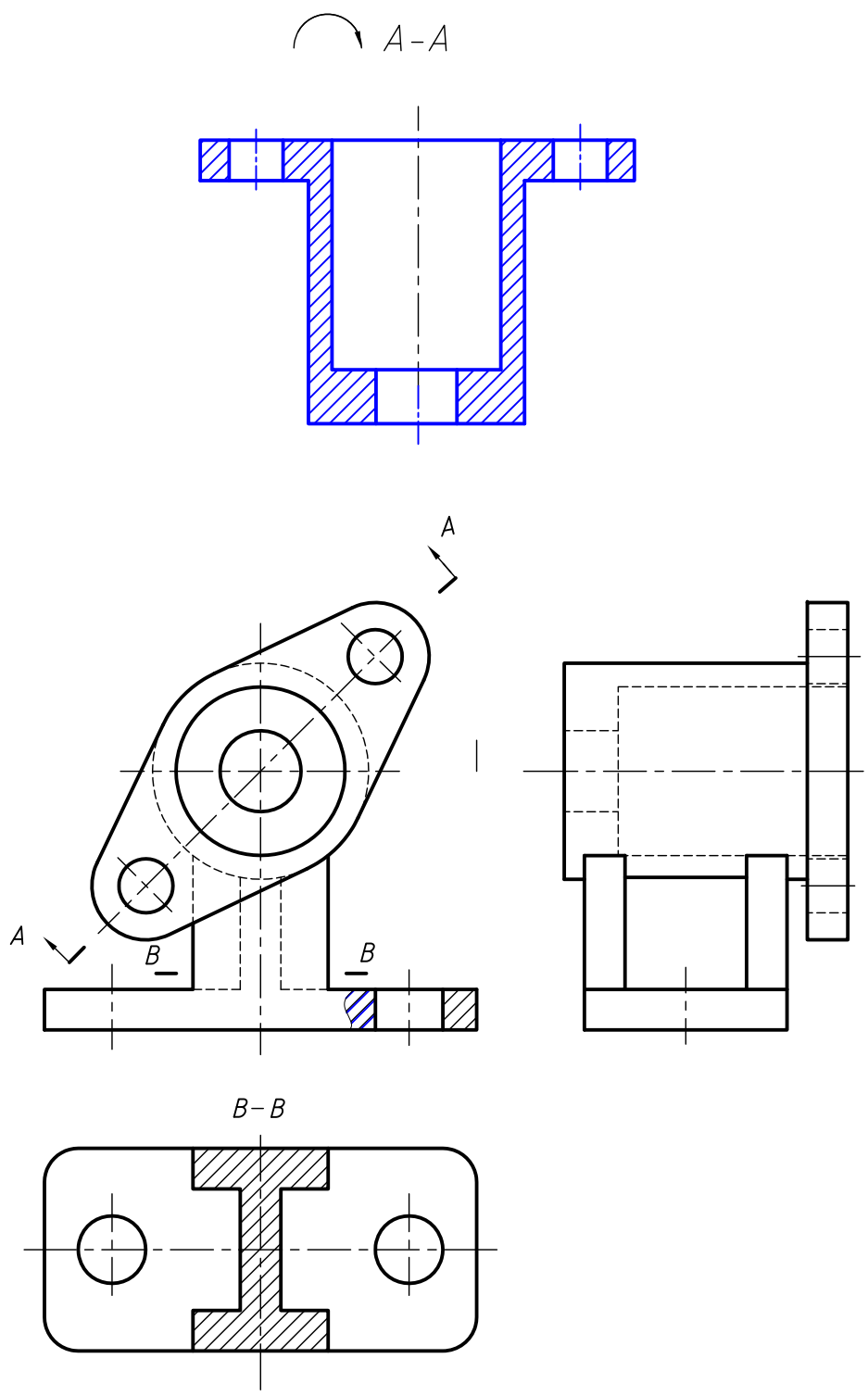


(2)

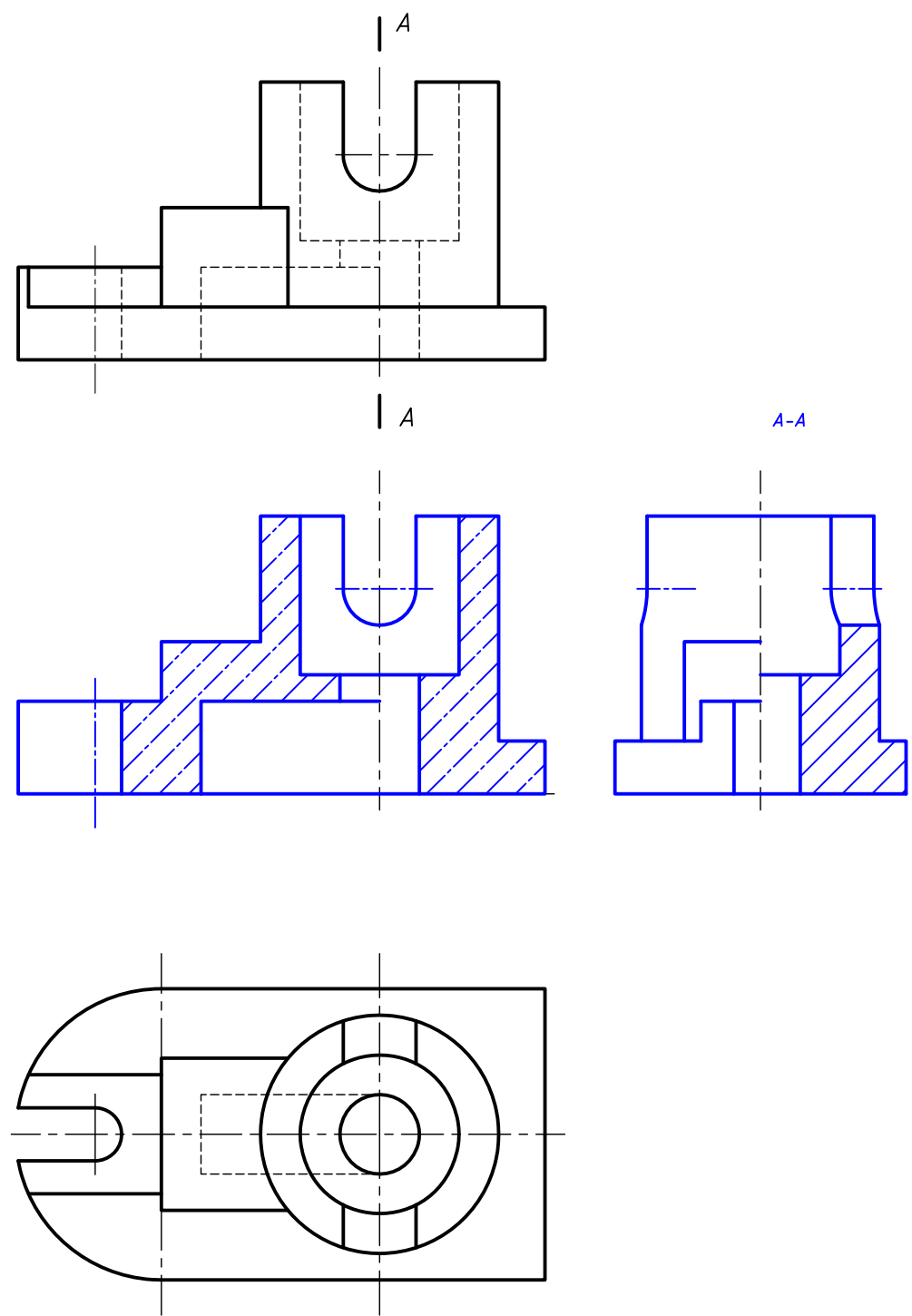


6. 在指定位置用相交剖面剖切的方法将主视图画成全剖视图。 (1) 	(2) 
7. 在指定位置用平行剖面剖切的方法将主视图画成全剖视图。 (1) 	(2) 

8. 在指定位置作A-A剖视图。

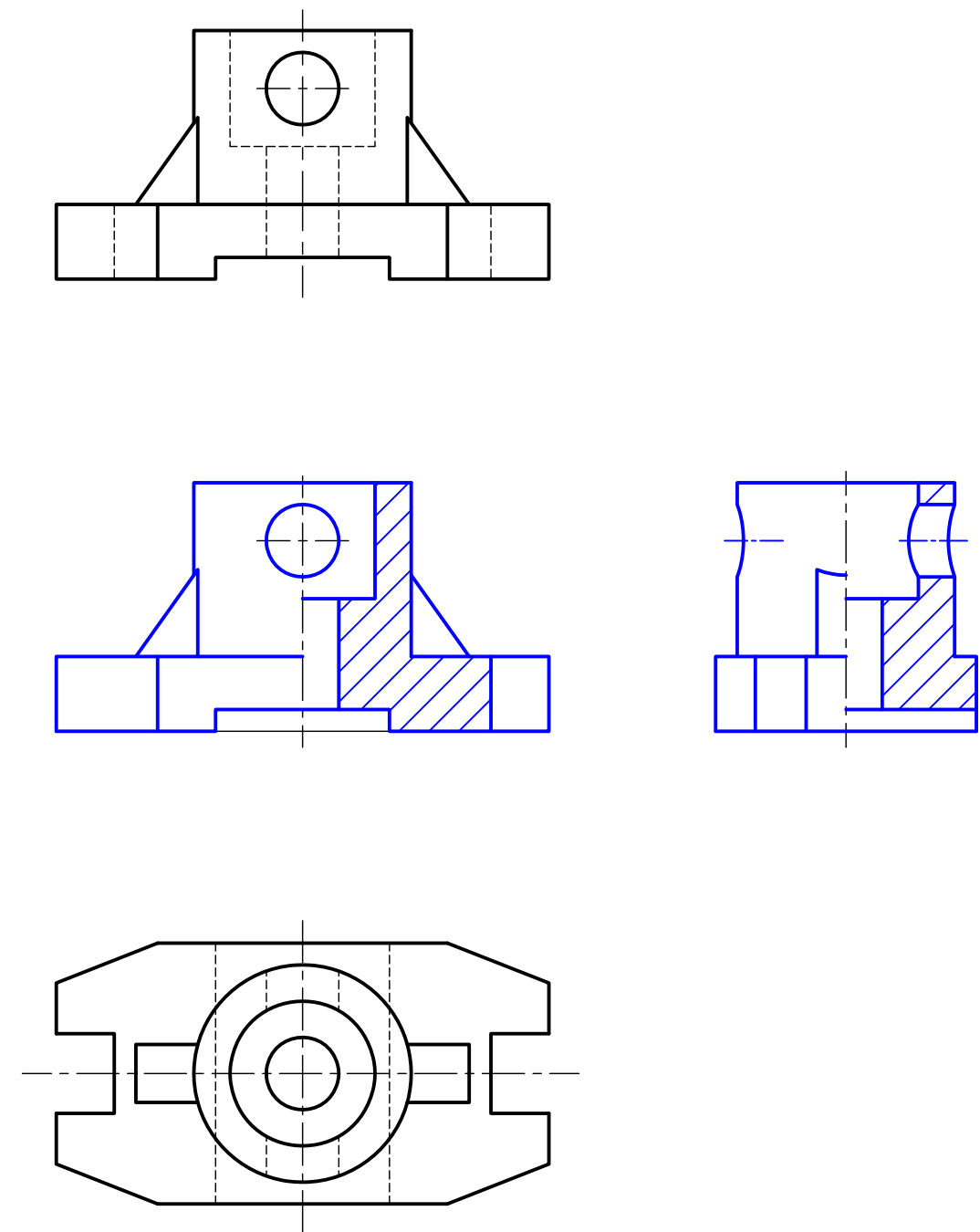


9. 在指定位置将主视图画成半剖视图，左视图画成A-A全剖视图。

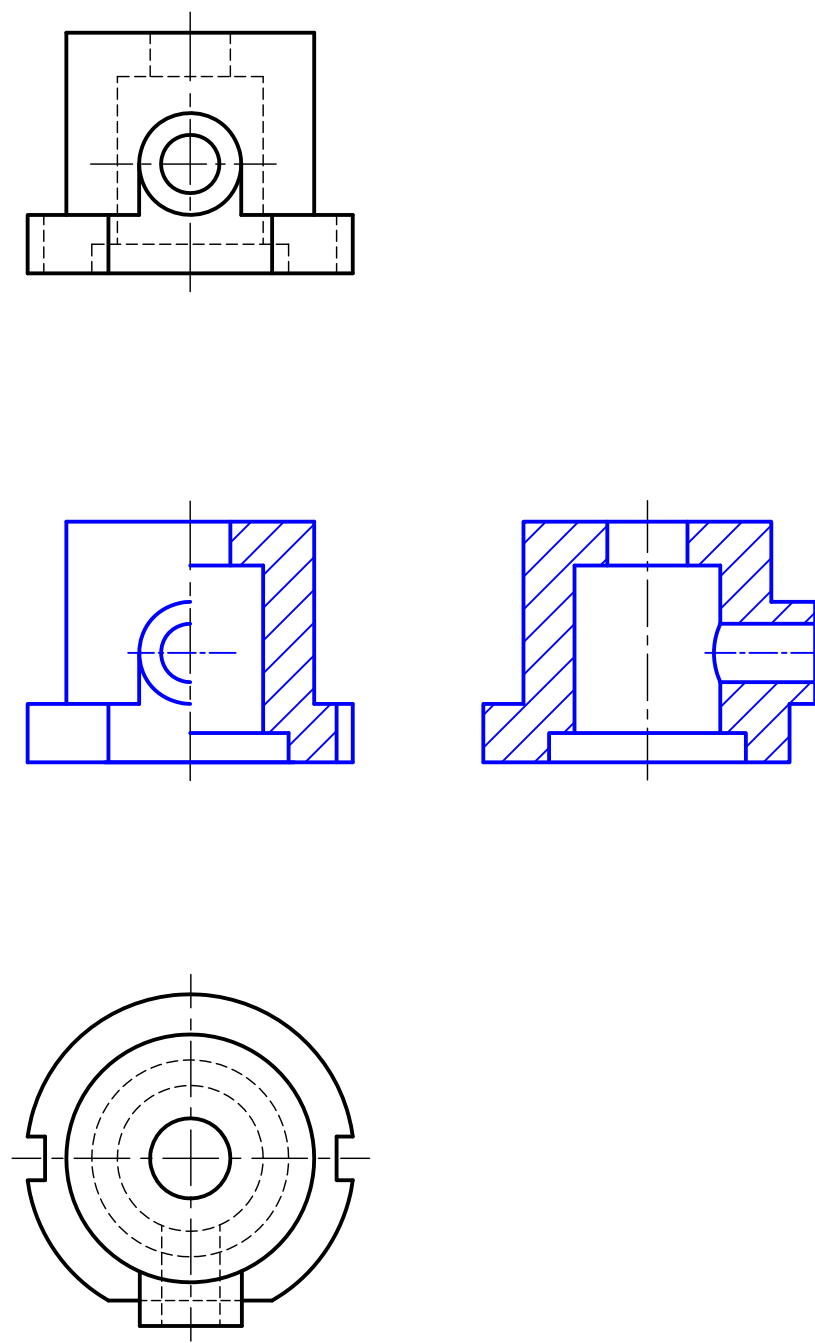


10. 在指定位置将主视图画成半剖视图，左视图画成全剖视图或半剖视图。

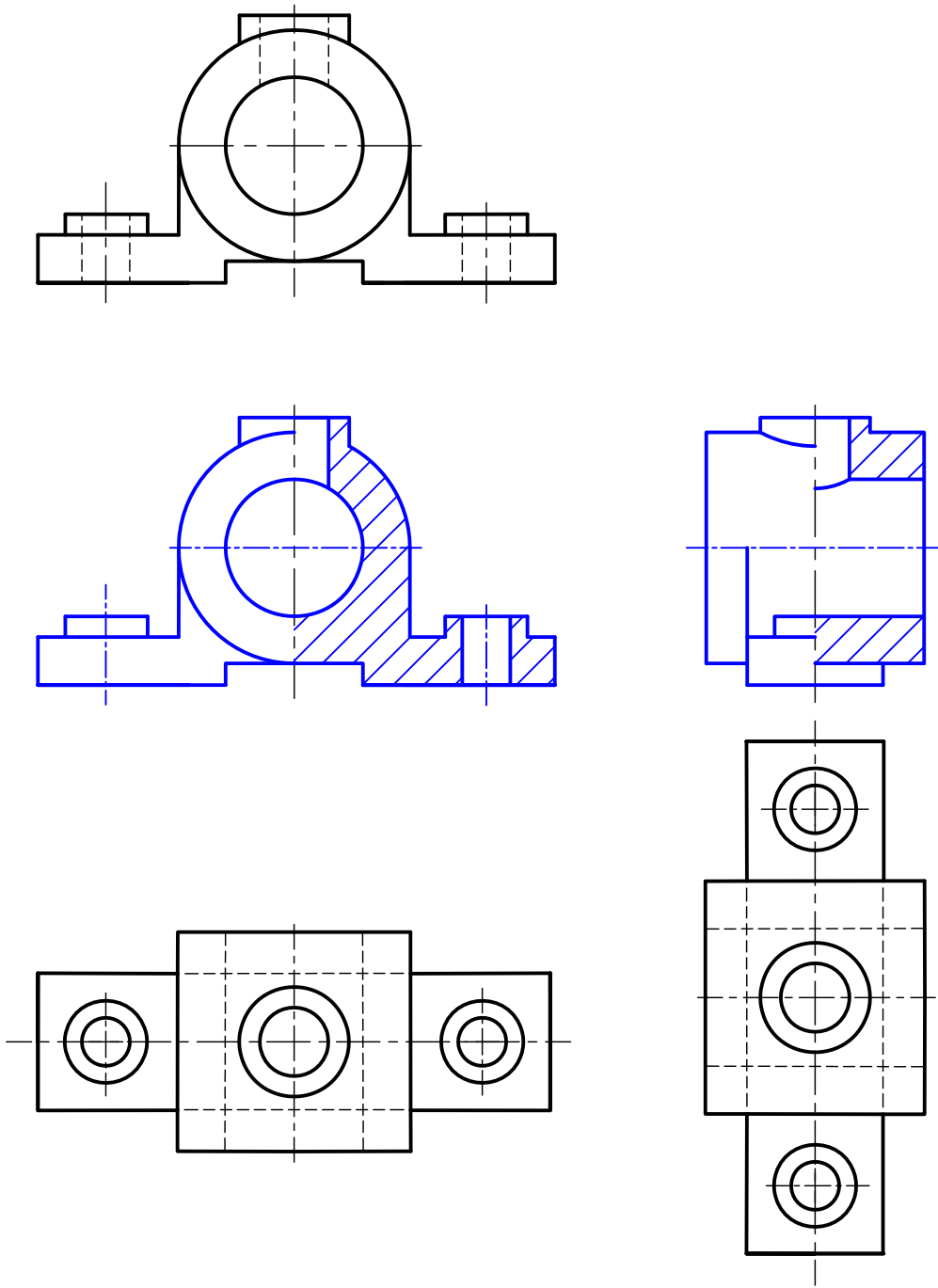
(1) 本题左视图画成半剖视图。



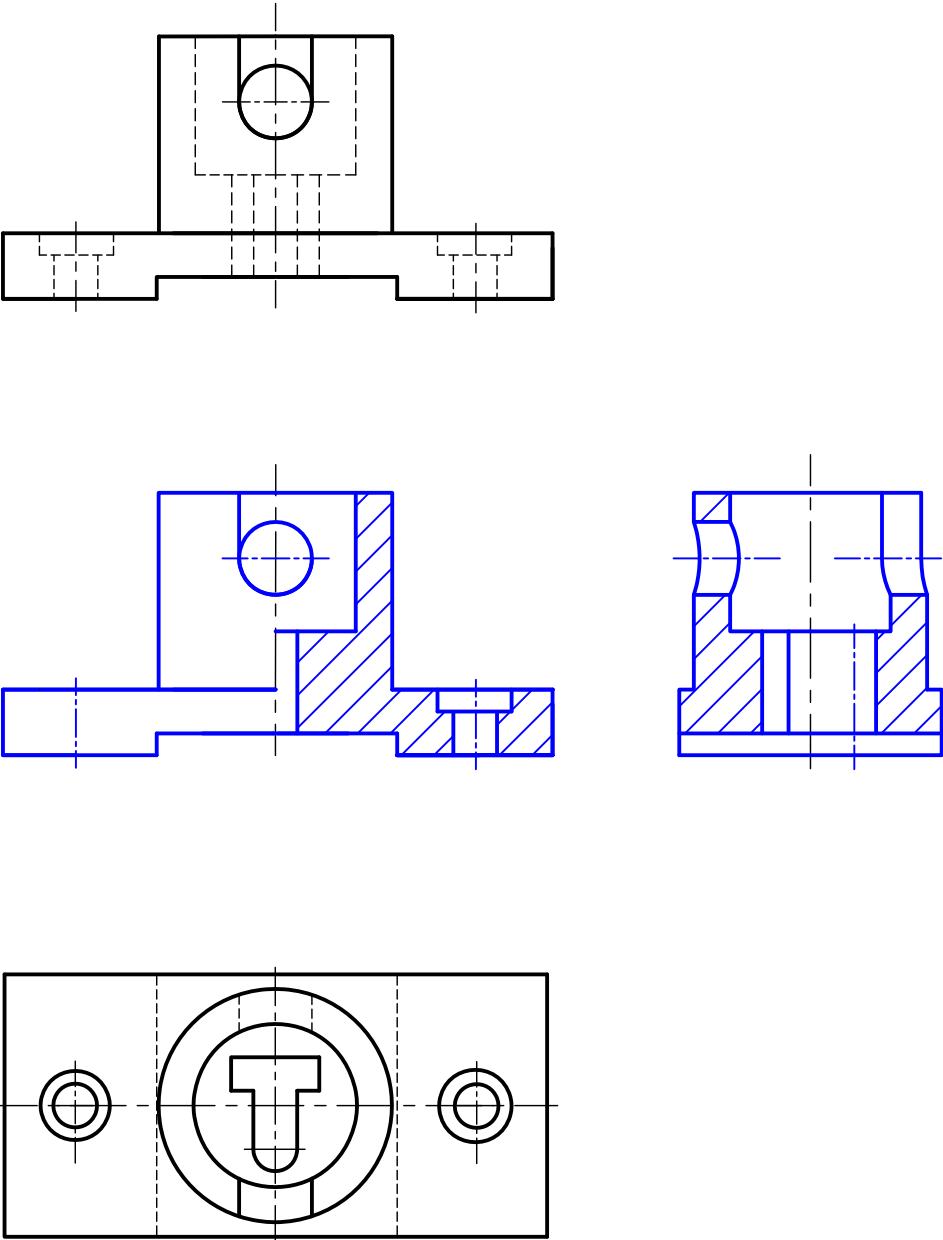
(2) 本题左视图画成全剖视图。



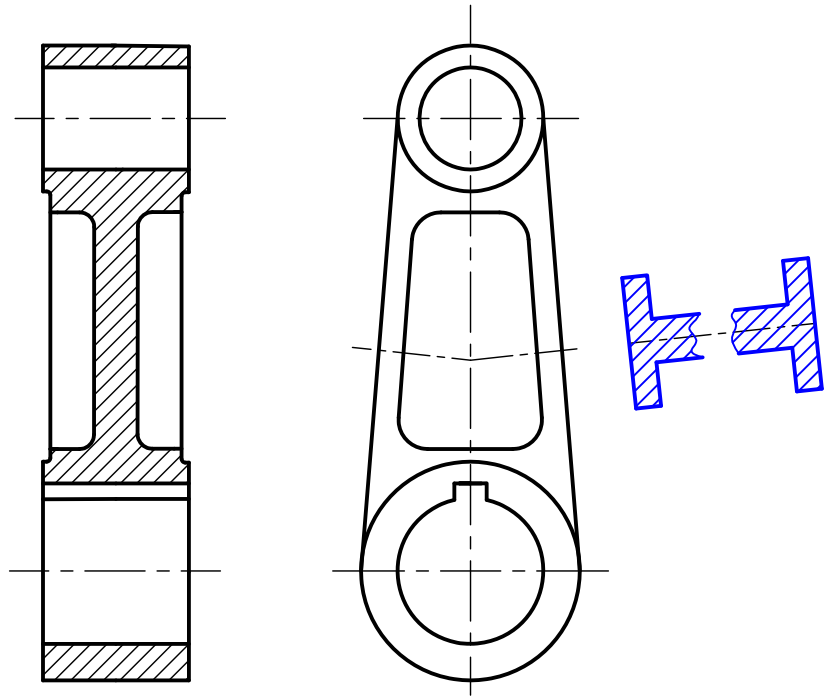
(3) 本题左视图画成半剖视图。



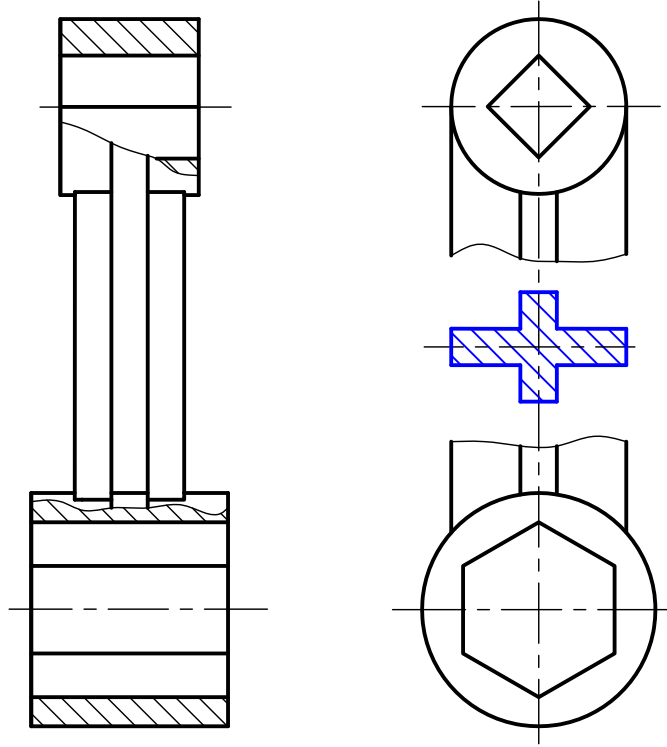
(4) 本题左视图画成全剖视图。



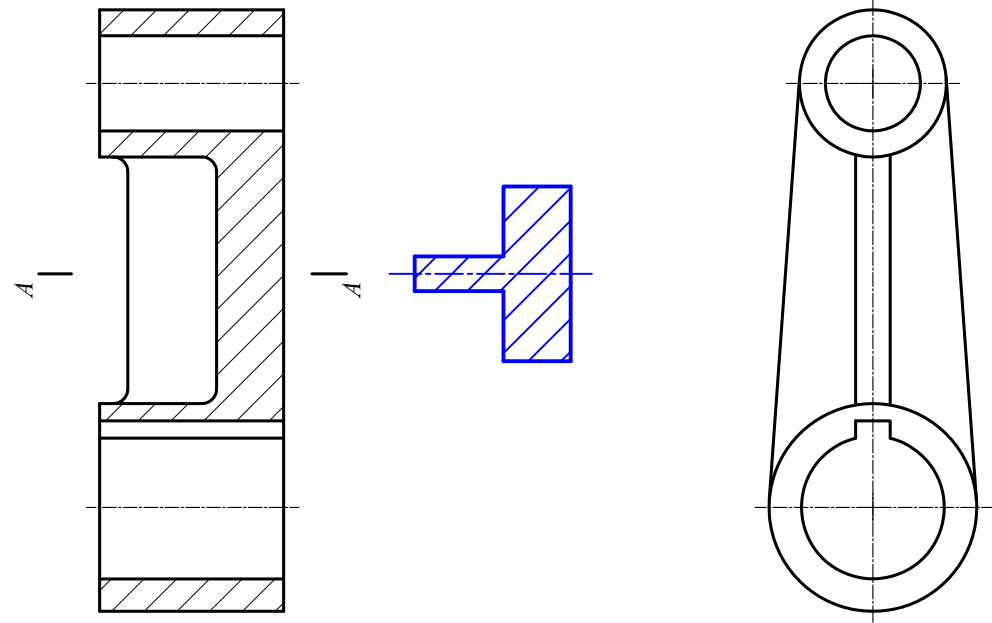
1. 在指定位置画出断面图。



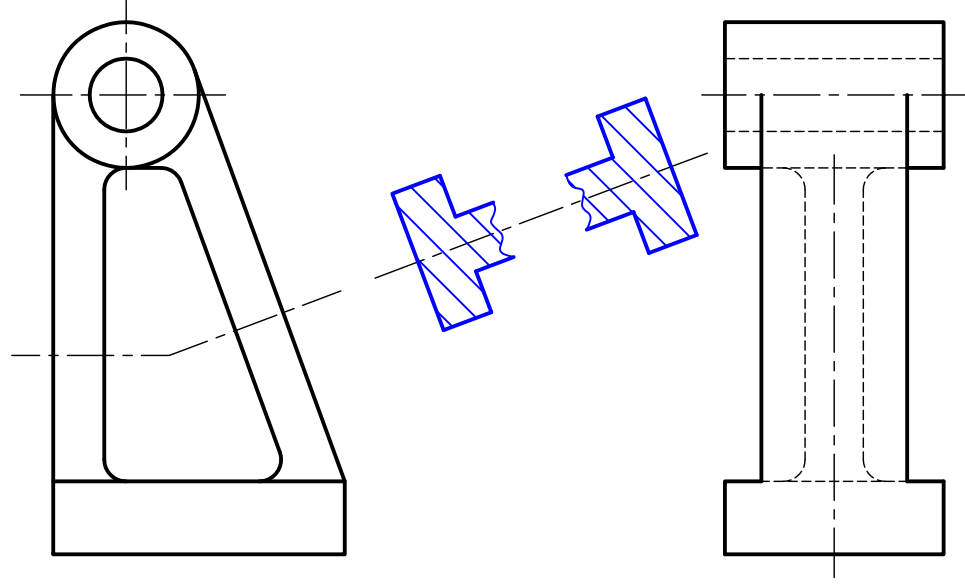
2. 在俯视图断处画出十字肋的断面图。



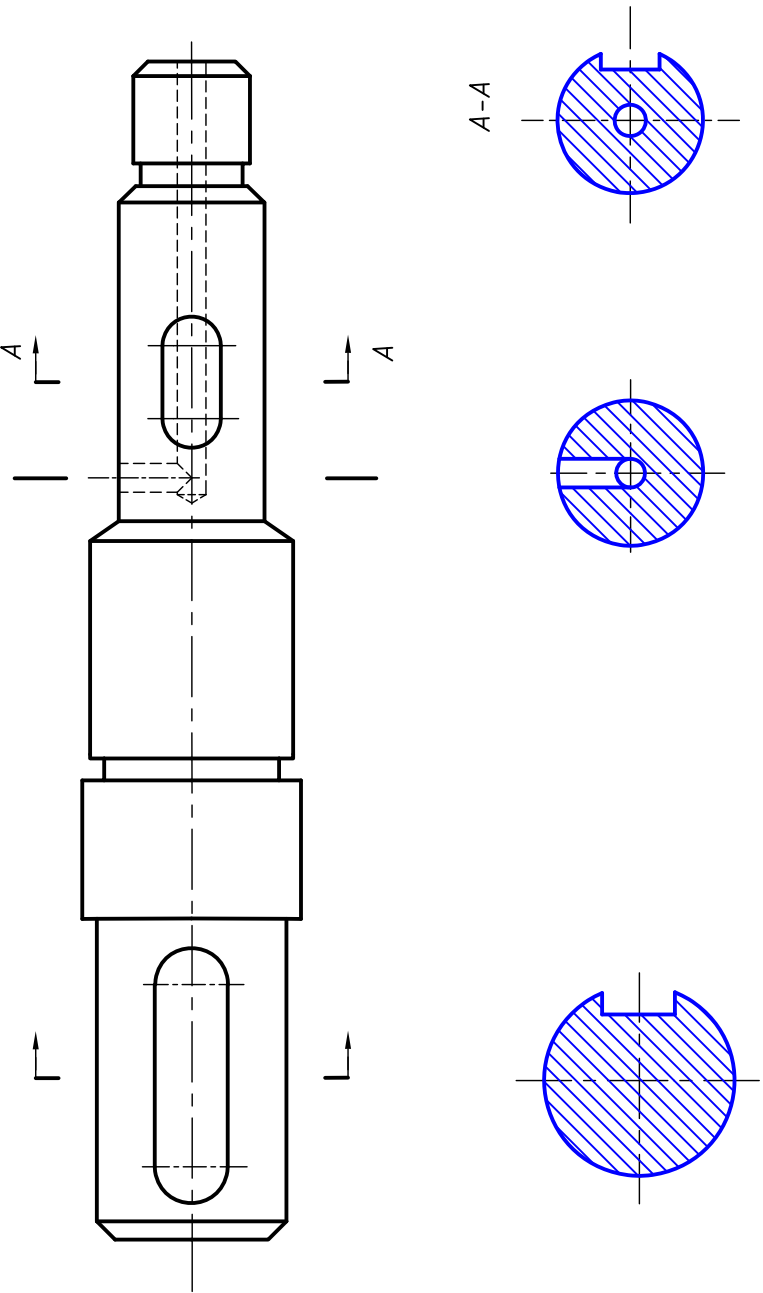
3. 画出指A-A断面图。



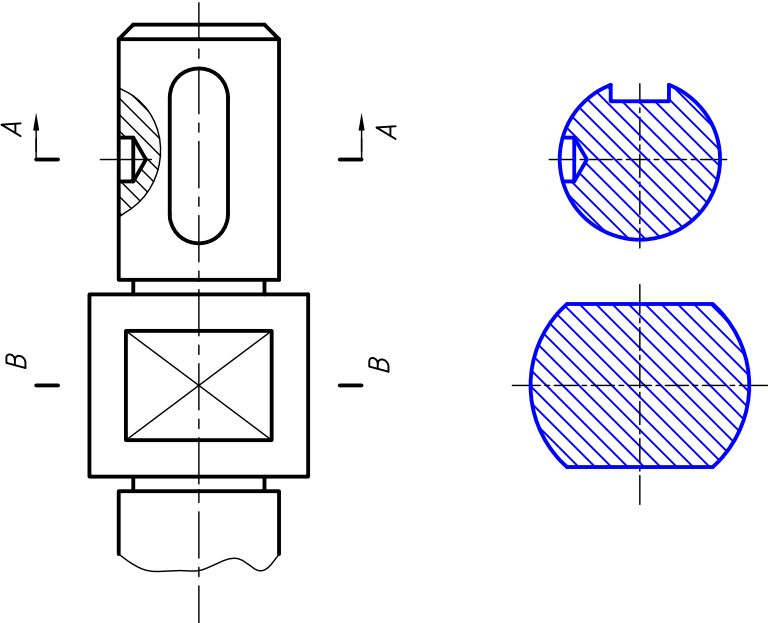
4. 画出工字型斜支承板的断面图。



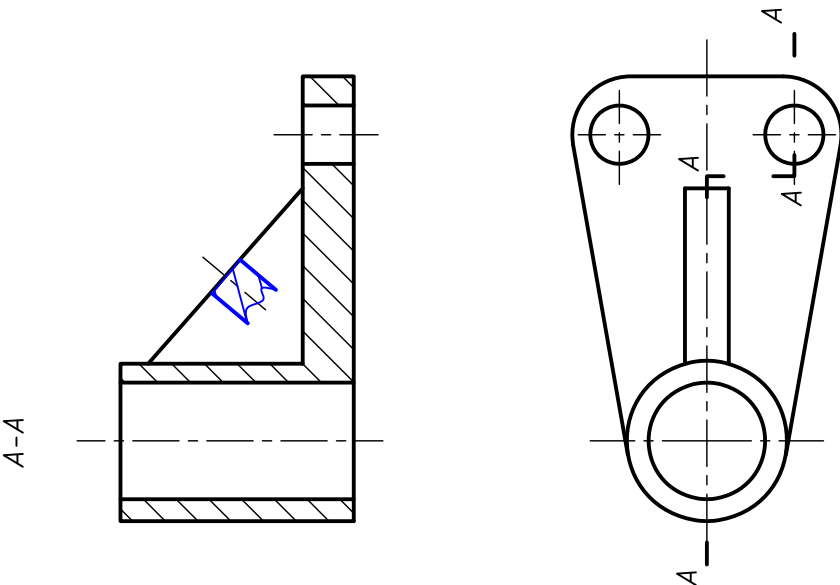
5. 画出指定位置的断面图（左侧键槽深4mm，右侧键槽深3mm）。



6. 画出轴上指定位置的断面图（键槽深3mm）。



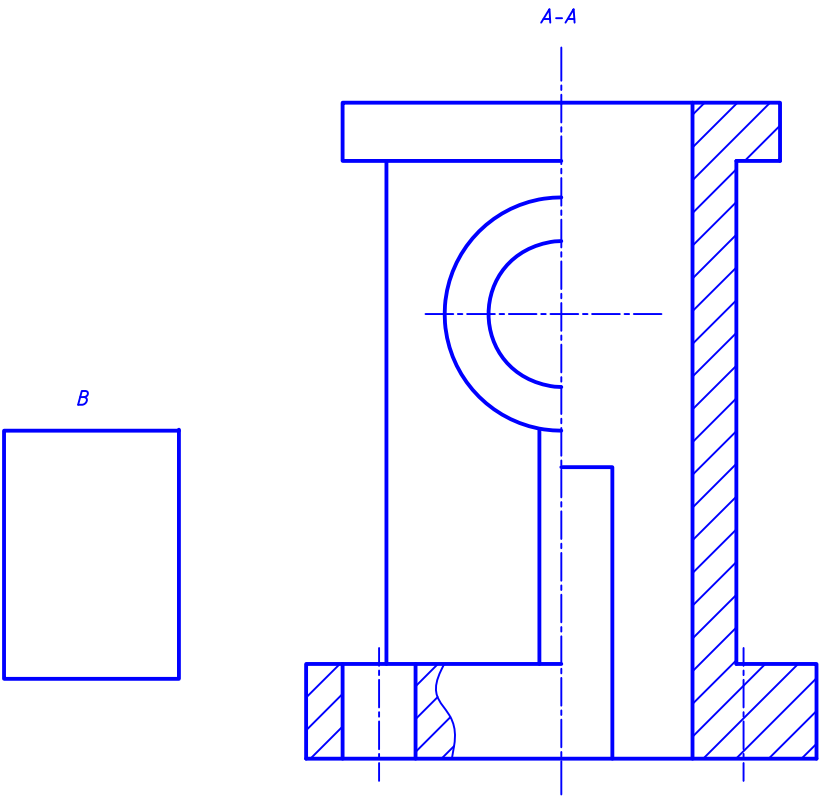
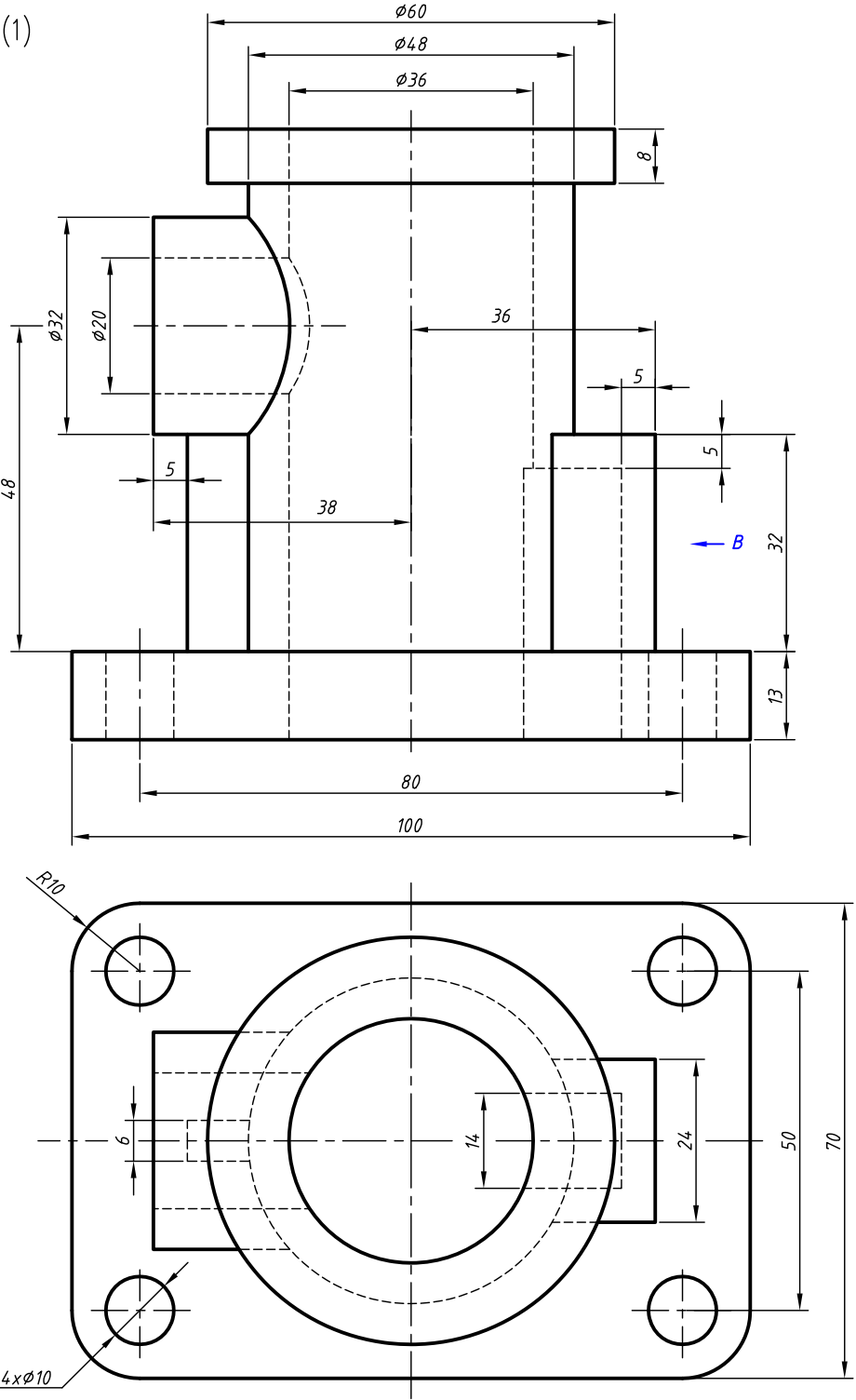
7. 画出肋板的重合断面图。



二十二、综合练习

班级 姓名 学号

1. 根据给出的视图想象机件的形状, 在形体分析的基础上选择合适的表达方案, 用适当的比例画在A3图纸上, 并标注尺寸。



二十三、螺纹（一）

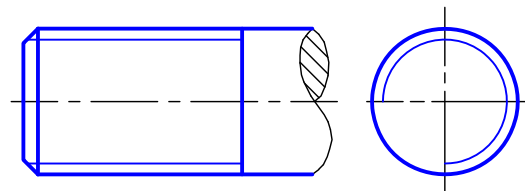
班级

姓名

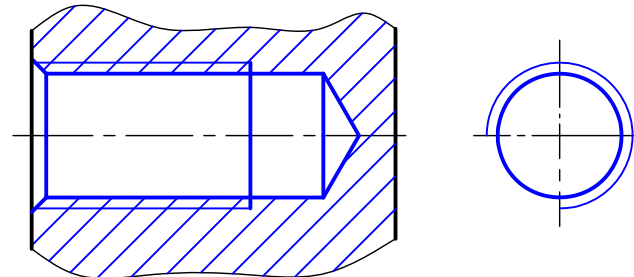
学号

1. 按给定的条件画出M20螺纹的主、左两视图。

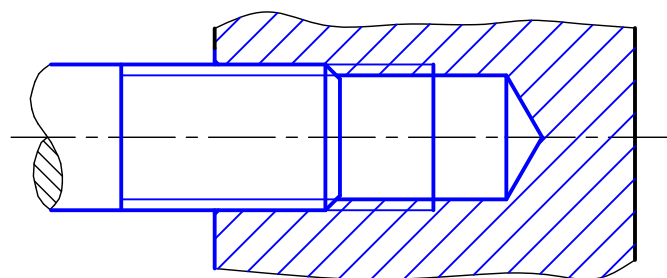
(1) 外螺纹：杆长40mm，螺纹长30mm（靠左端），螺纹倒角C2。



(2) 内螺纹：主视图作剖视，左端钻孔深40mm，螺纹深30mm，螺纹倒角C2。

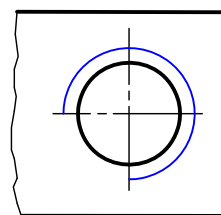
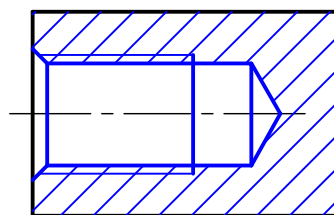
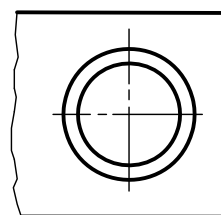
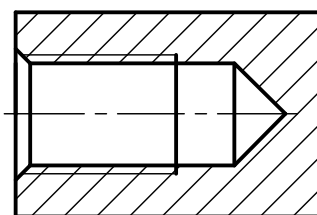


2. 将上题的螺杆调头，在指定位置画出内、外螺纹的联接图。

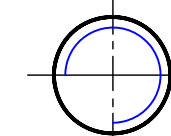
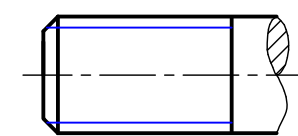
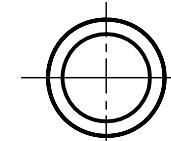
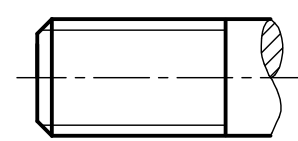


3. 分析下列各图的错误，将正确的图形画在下面。

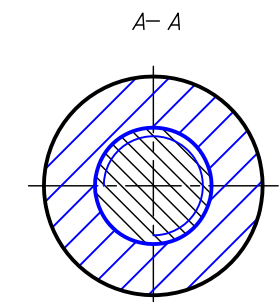
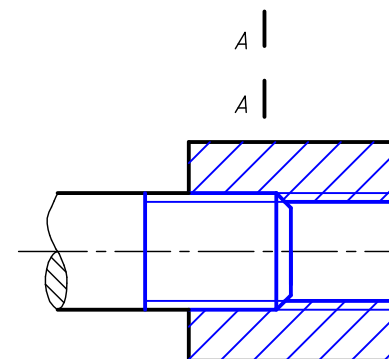
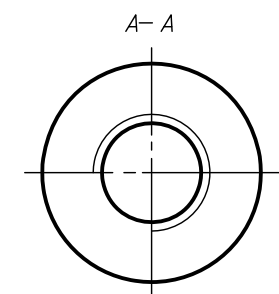
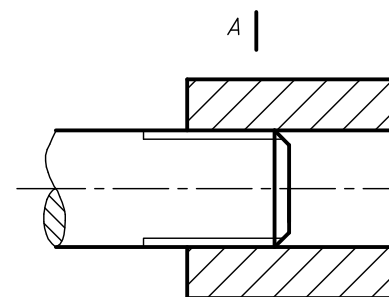
(1)



(2)



(3)

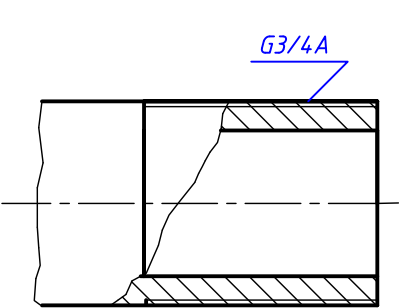


4. 识别螺纹的标记。

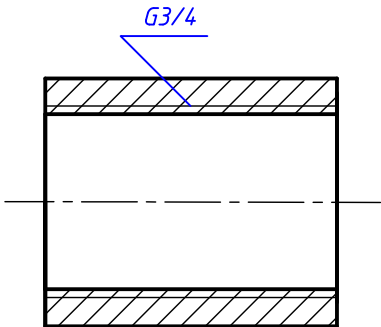
螺纹标记	螺纹种类	公称直径 /mm	螺距 /mm	导程 /mm	线数	旋向	中径公差带代号	大径公差带代号	旋合长度代号	内外螺纹
M10- 6H	普通粗牙	10	1.5	1.5	1	右	6H	6H	N	内
M10X 1- 5H- S	普通细牙	10	1	1	1	右	5H	5H	S	内
M20- 6g	普通粗牙	20	2.5	2.5	1	右	6g	6g	N	外
M16X 1.5- 6g7g- L	普通细牙	16	1.5	1.5	1	右	6g	7g	L	外
Tr32X 12(P6)LH- 8H- L	梯形	32	6	12	2	左	8H		L	内

螺纹标记	螺纹种类	尺寸代号	公差等级	内外螺纹	旋向	管子孔径 /mm	螺纹大径 /mm	螺纹小径 /mm
G1	55° 非密封管螺纹	1		内	右	25.4	33.249	30.291
G1/2A- LH	55° 非密封管螺纹	1/2	A	外	左	12.7	20.995	18.631

(3) 55°非密封管螺纹，尺寸代号为3/4，公差等级为A级。

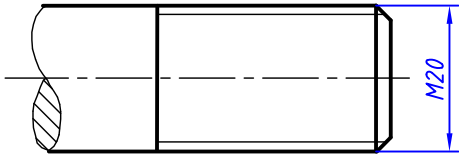


(4) 55°非密封管螺纹，尺寸代号为3/4。

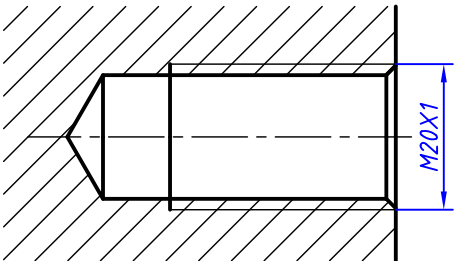


5. 按给定的螺纹要素在图中标注螺纹代号。

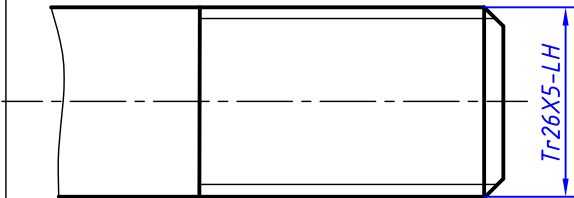
(1) 粗牙普通螺纹，大径20mm，螺距2.5mm，右旋。



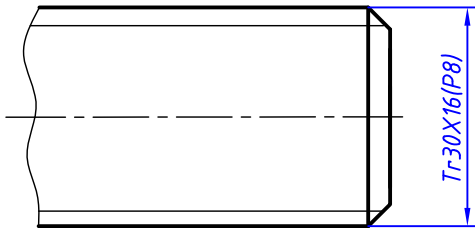
(2) 细牙普通螺纹，大径20mm，螺距1mm，左旋。



(5) 梯形螺纹，公称直径26mm，螺距5mm，左旋。

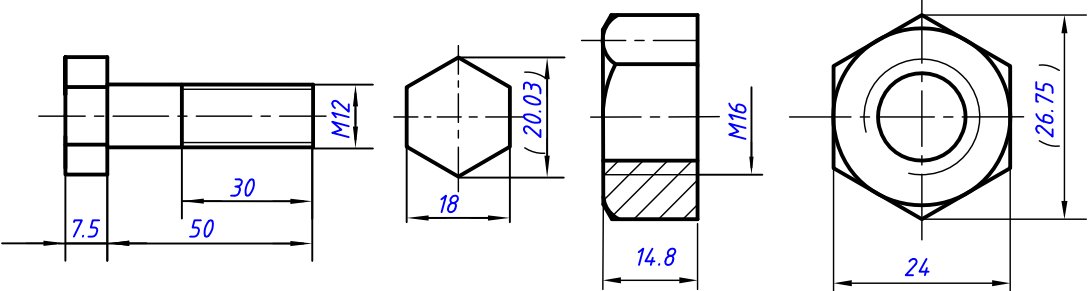
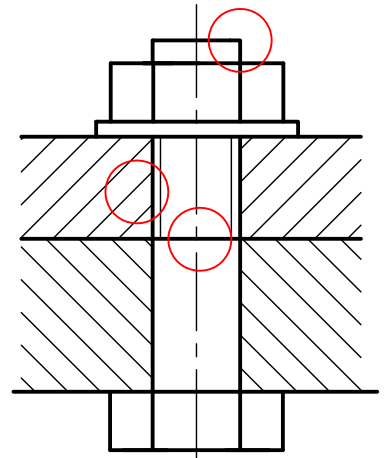
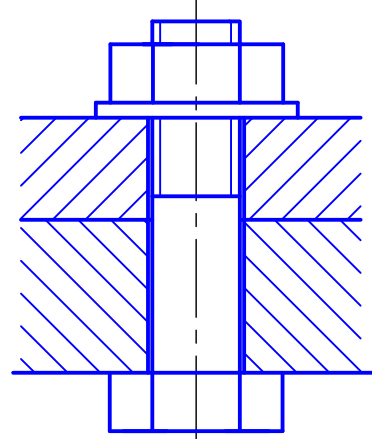
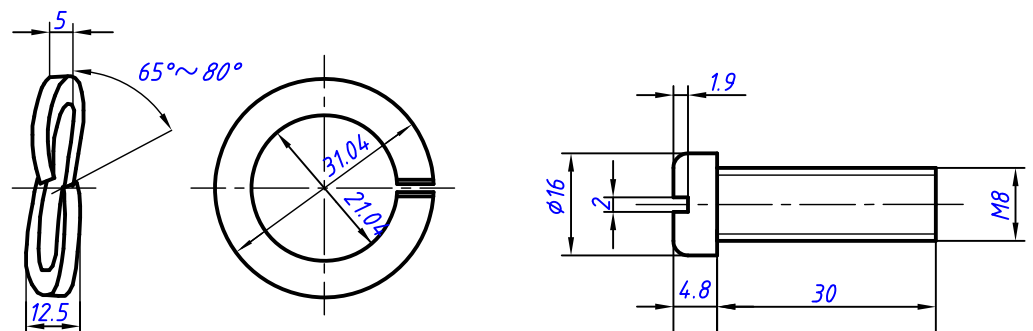
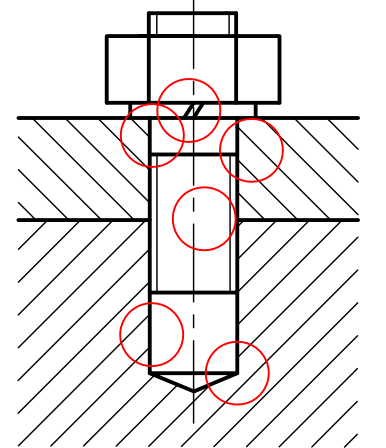
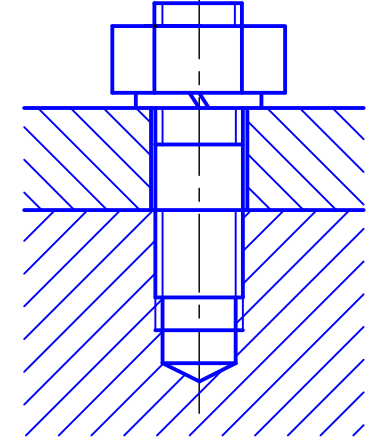
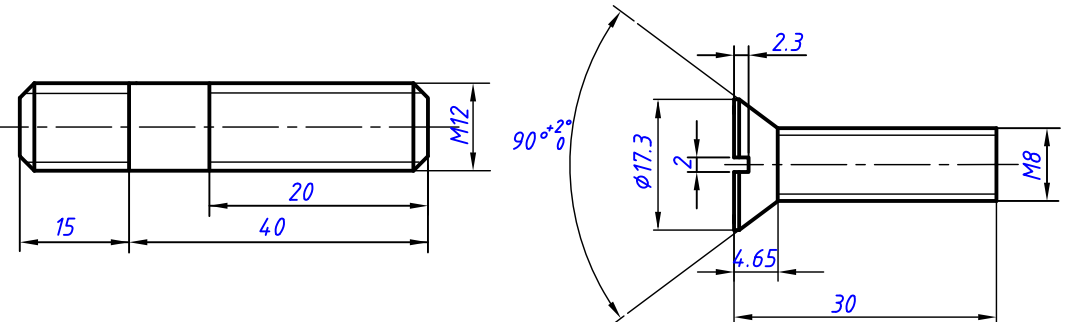
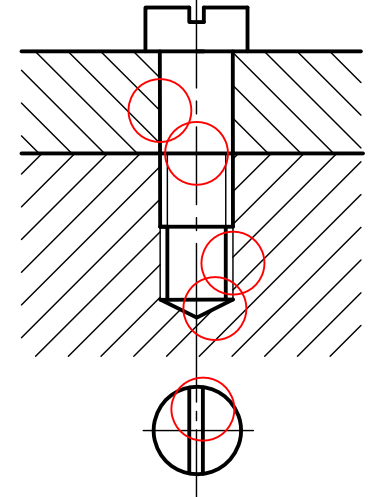
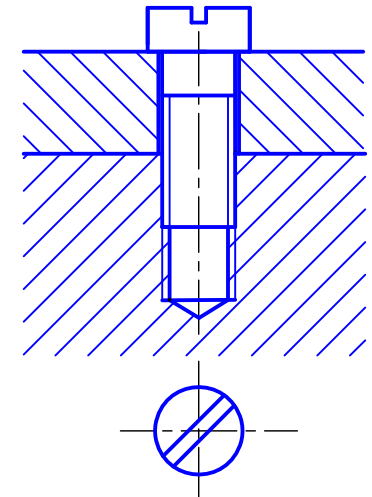


(6) 梯形螺纹，公称直径60mm，导程16mm，双线，右旋。
(本题绘图比例1:2)



二十四、螺纹联接件（一）

班级 姓名 学号

1. 查表后注出下列联接件的尺寸数值，并注写其规定标记。 (1) A级六角头螺栓：螺纹规格d=M12，公称长度 l=50mm。(2) A级 I 型六角螺母：螺纹规格D=M16。  规定标记： <u>螺栓GB/T5782 M12X50</u> 规定标记： <u>螺母GB/T6170 M16</u>	2. 找出下列各螺纹联接图中的错误，并在右侧画出正确的联接图。 (1)  
(3) 标准型弹簧垫圈：规格20mm。(4) 开槽盘头螺钉：螺纹规格d=M8，公称长度 l=30mm。  规定标记： <u>螺母GB/T93 20</u> 规定标记： <u>螺钉GB/T67 M8X30</u>	(2)  
(5) A型双头螺柱：螺纹规格d=M12，公称长度 l=40mm，bm=1.25d。(6) 开槽沉头螺钉：螺纹规格M8，公称长度 l=30mm。  规定标记： <u>螺柱GB/T898 M12X40</u> 规定标记： <u>螺钉GB/T68 M8X30</u>	(3)  

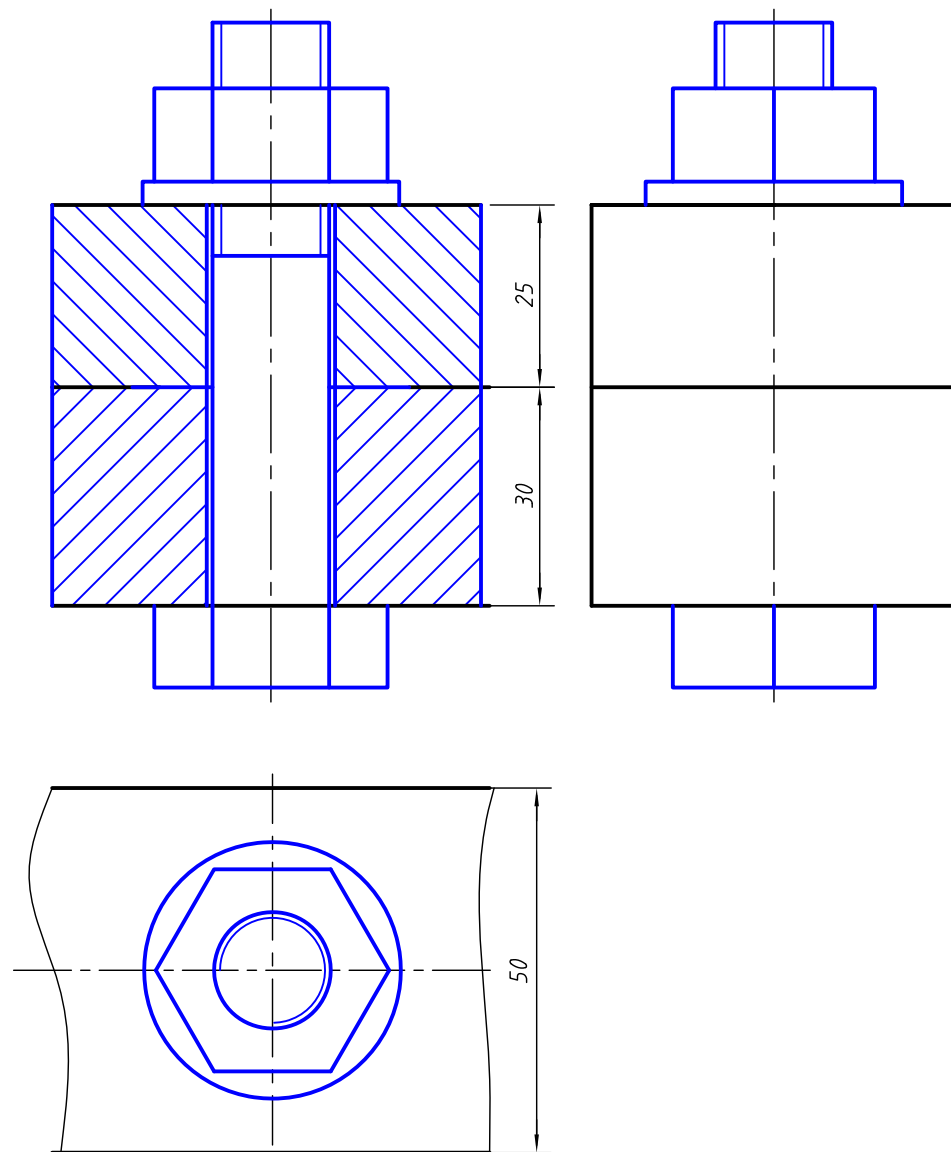
二十四、螺纹联接件（二）

班级 姓名 学号

3. 将A3图纸横放，画螺纹联接件的联接图。

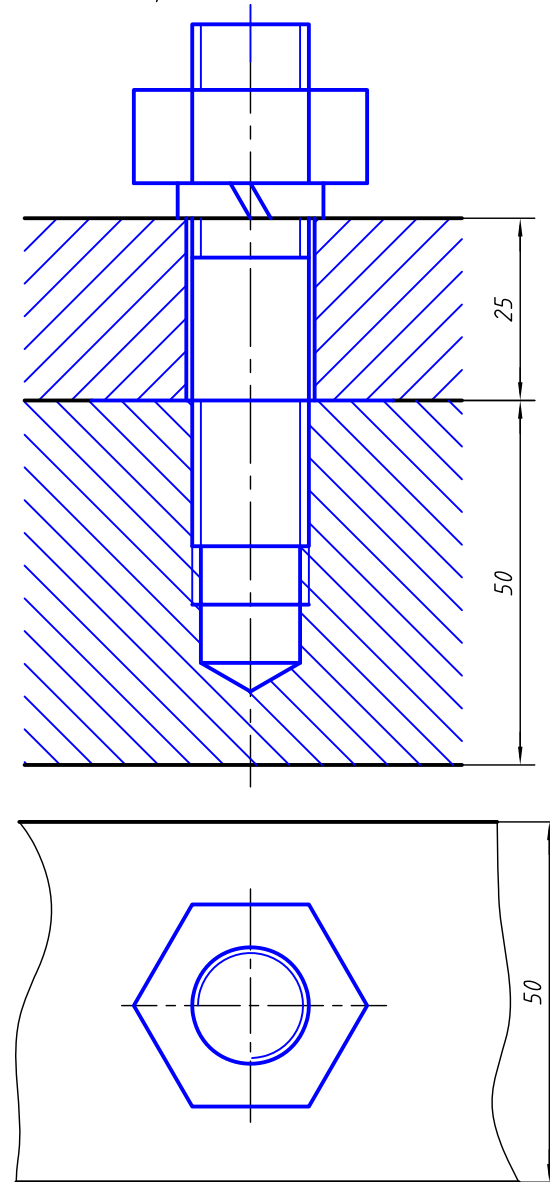
(1) 用比例画法画螺栓联接图（主视图取剖视），比例为1:1。

已知：螺栓GB/T5782 M16×80
螺母GB/T6170 M16 垫圈GB/T97.1 16



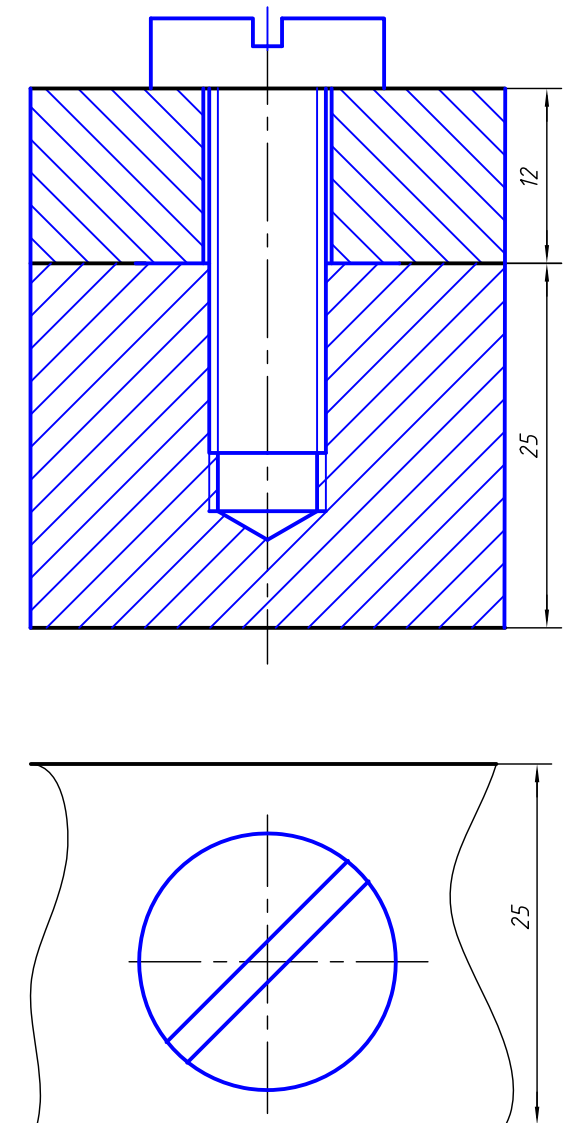
(2) 用比例画法画螺柱联接的两视图（主视图取剖视），比例为1:1。

已知：螺柱GB/T898 M16×50
螺母GB/T6170 M16
垫圈GB/T93 16

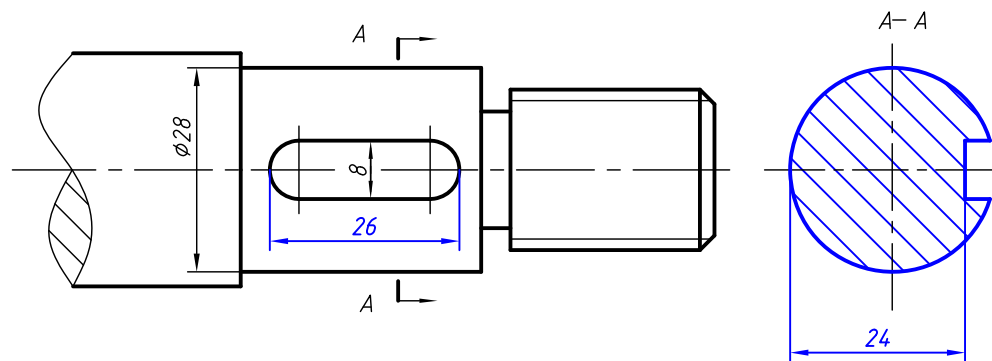


(3) 用查表法画螺钉联接的两视图（主视图取剖视），比例为2:1。

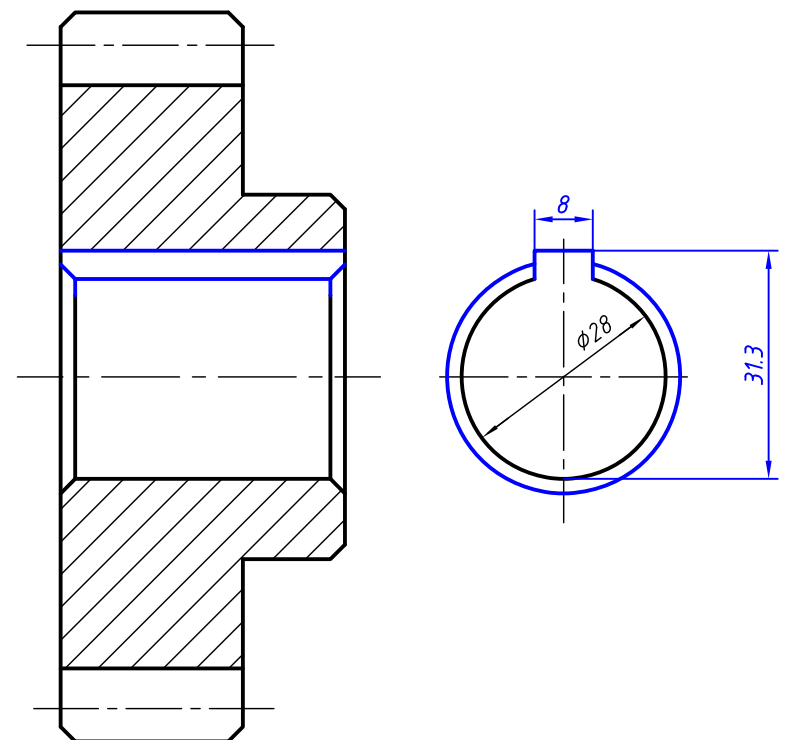
已知：螺钉GB/T67 M8×25



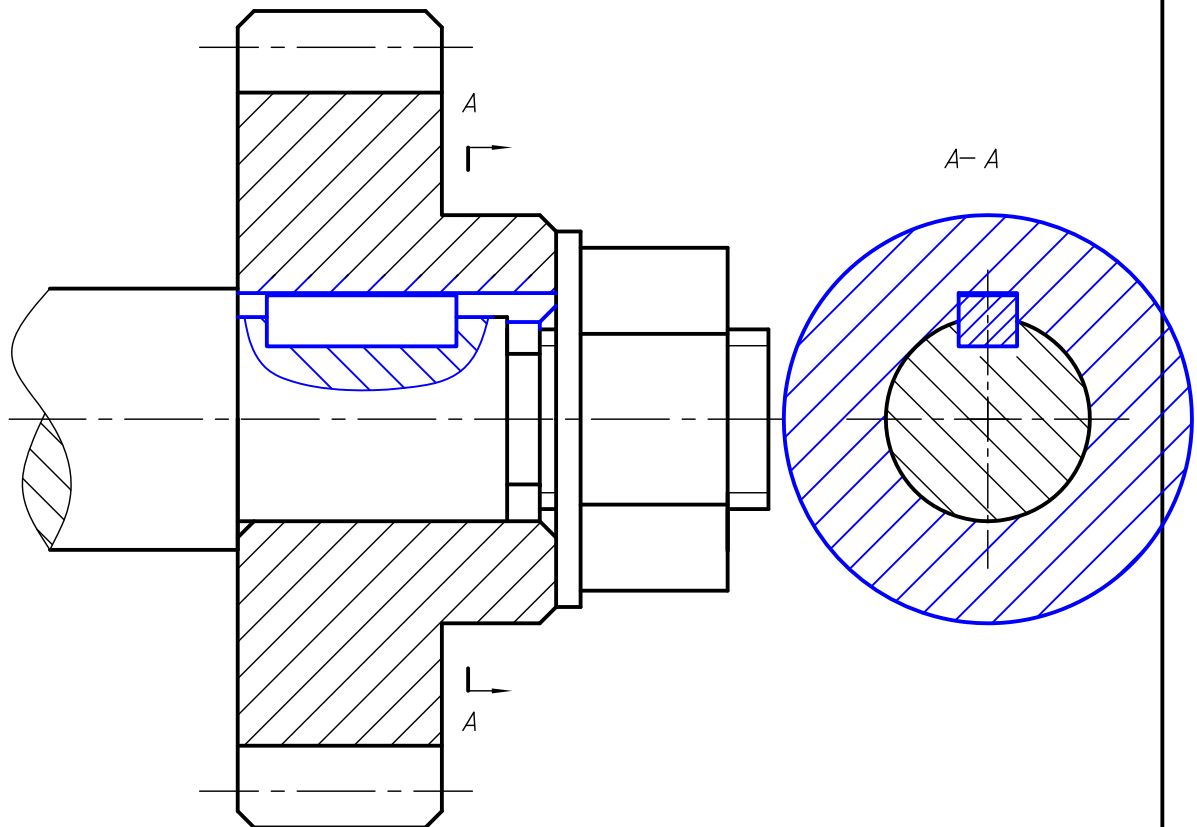
1. 画出A-A断面图，并注全键槽尺寸（键槽尺寸查表）。



2. 画出齿轮键槽的剖视图及其局部视图，并注全键槽尺寸(键槽宽8mm)。



3. 将题1、题2中的轴和齿轮装配在一起，试完成普通平键联接的装配图。



键的规定标记: GB/T1096 键 8X7X26

二十六、弹簧、销

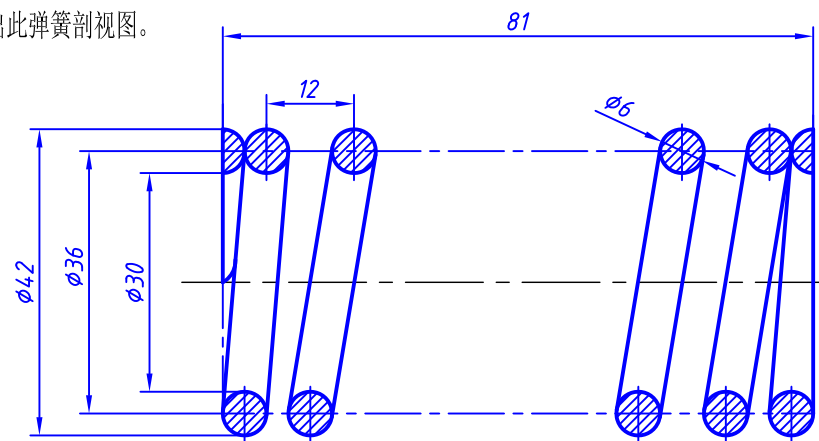
班级

姓名

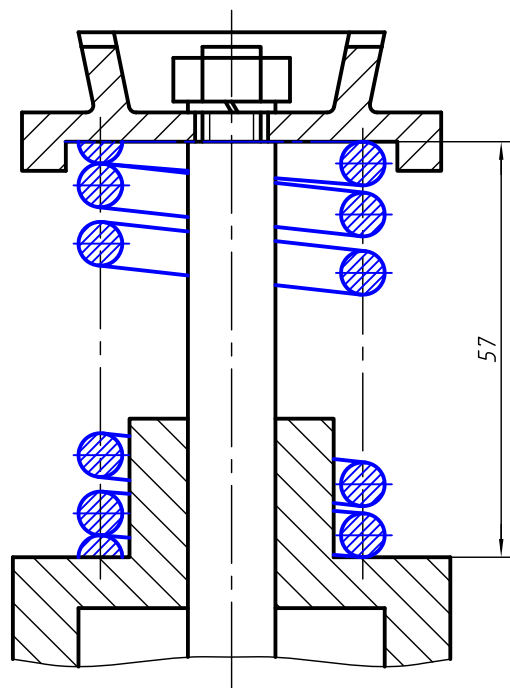
学号

1. 已知圆柱螺旋压缩弹簧外径 $D_2=42\text{mm}$ ，簧丝直径 $d=6\text{mm}$ ，节距 $t=12\text{mm}$ ，有效圈数 $n=6$ ，支承圈数 $n_z=2$ ，右旋。

画出此弹簧剖视图。

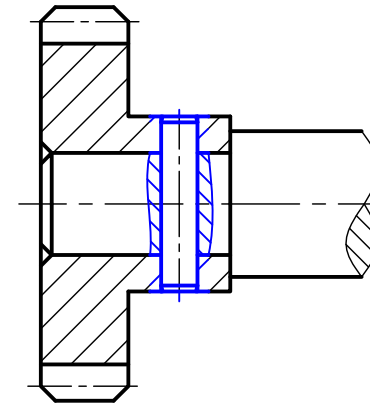


2. 将题1. 所画弹簧装入图示部位（将弹簧的自由高度 H_0 压缩为 57mm ）。

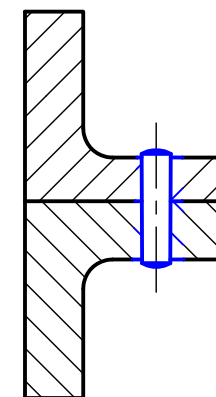


3. 销联接画法。

(1) 用“销 GB/T119.1 5×24”联接轴和齿轮，完成其装配图。

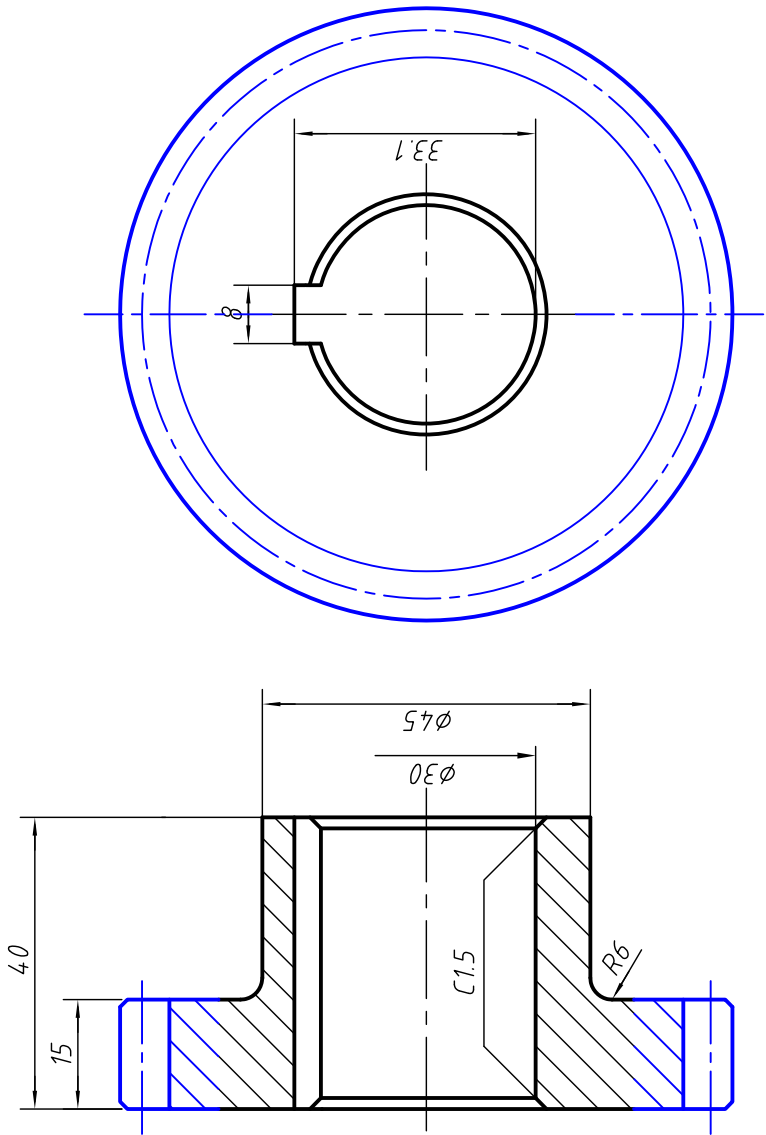


(2) 用“销 GB/T117 4×16”联接两零件，完成其装配图。



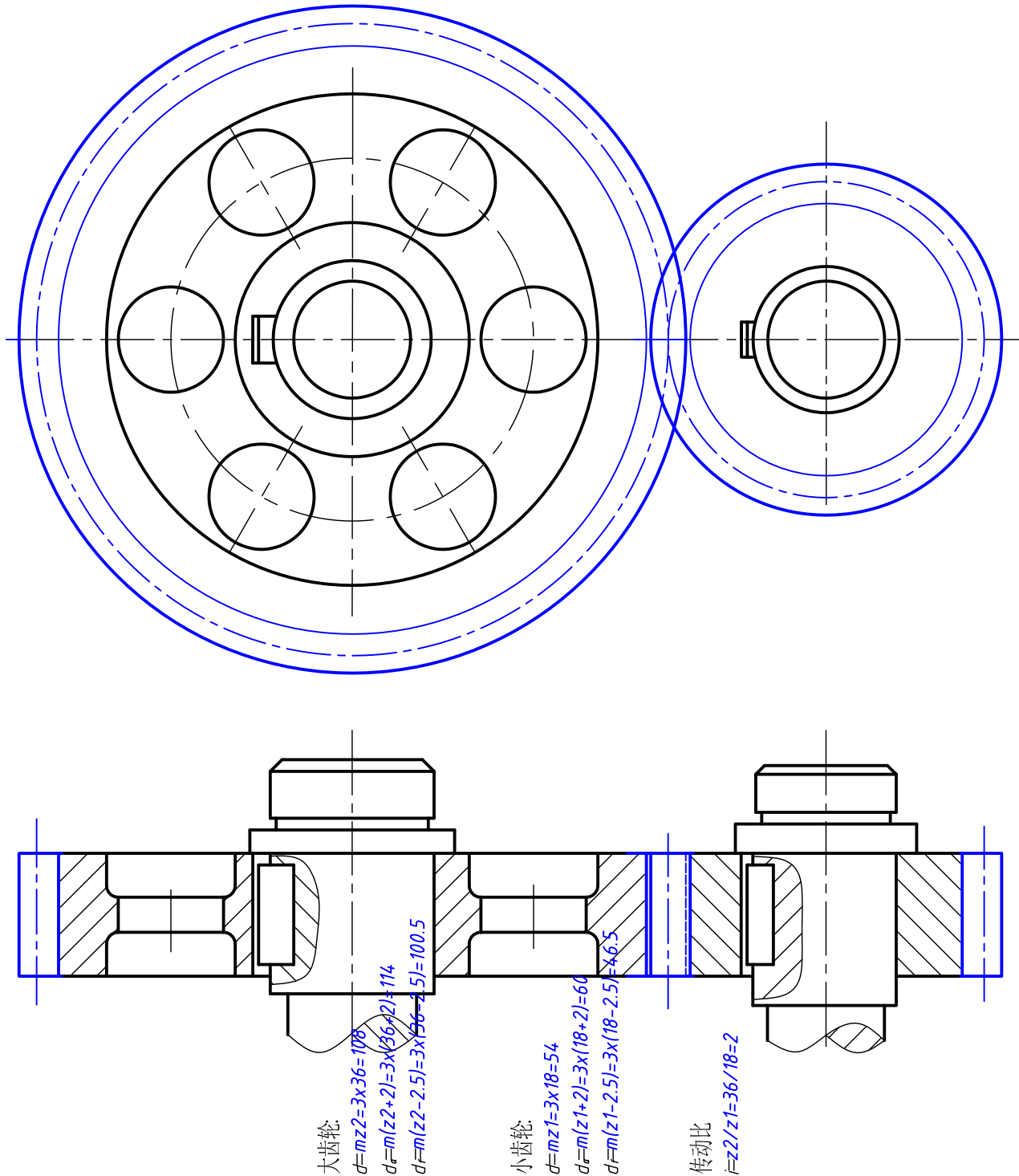
1. 已知 $m=3$ 、 $z=26$ 的直齿圆柱齿轮，计算其分度圆、齿顶圆和齿根圆直径，补全两视图，并注全尺寸。轮齿倒角 $C1$ 。

$d=mz=3 \times 26=78$ $d_a=m(z+2)=3 \times (26+2)=84$ $d_f=m(z-2.5)=3 \times (26-2.5)=70.5$

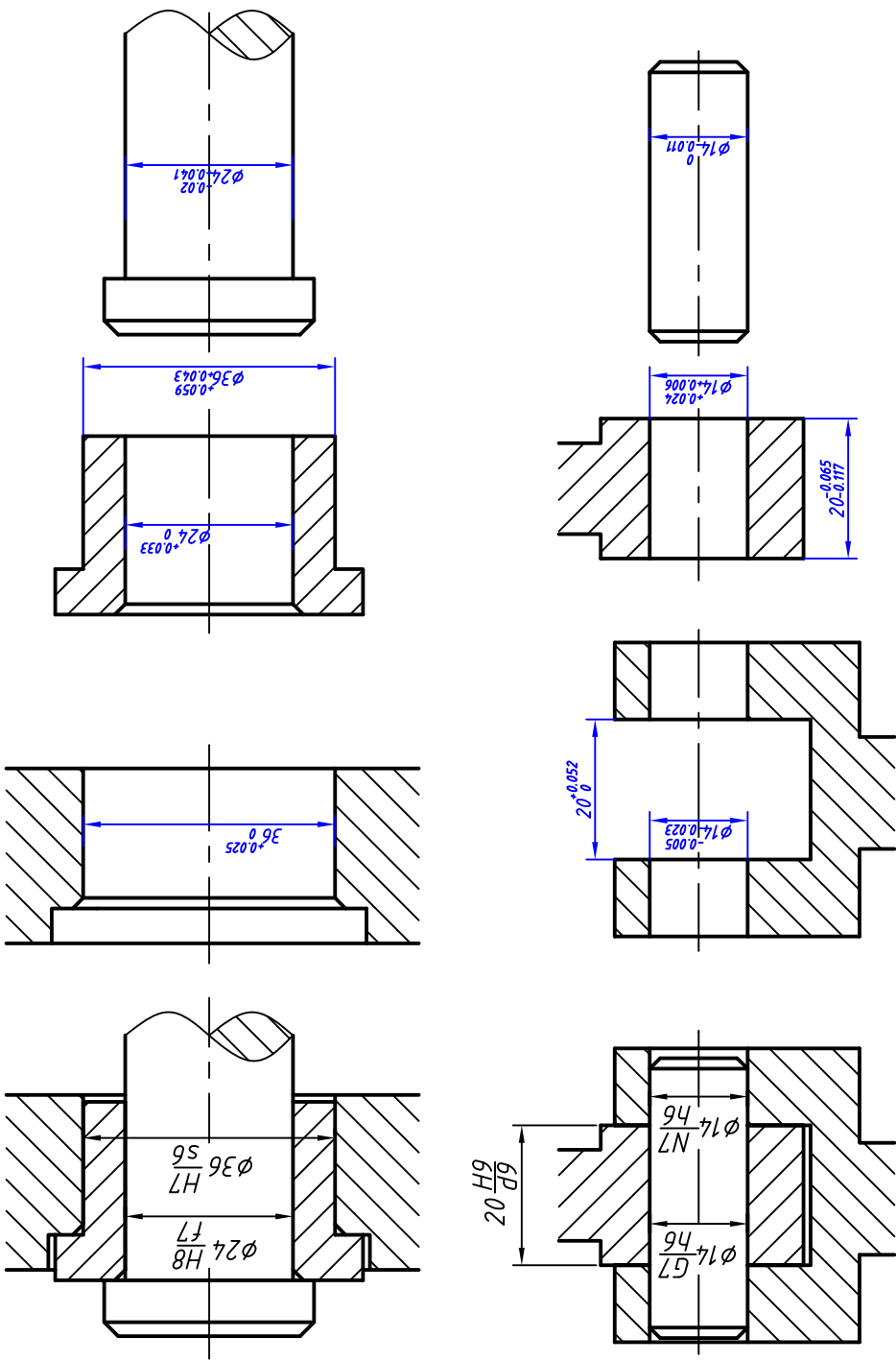


2. 一对直齿圆柱齿轮啮合，已知 $z_1=18$ ， $z_2=36$ ，中心距为 81mm 。试计算大、小齿轮的几何参数及传动比(其中小齿轮为主动齿轮)，并完成啮合图。

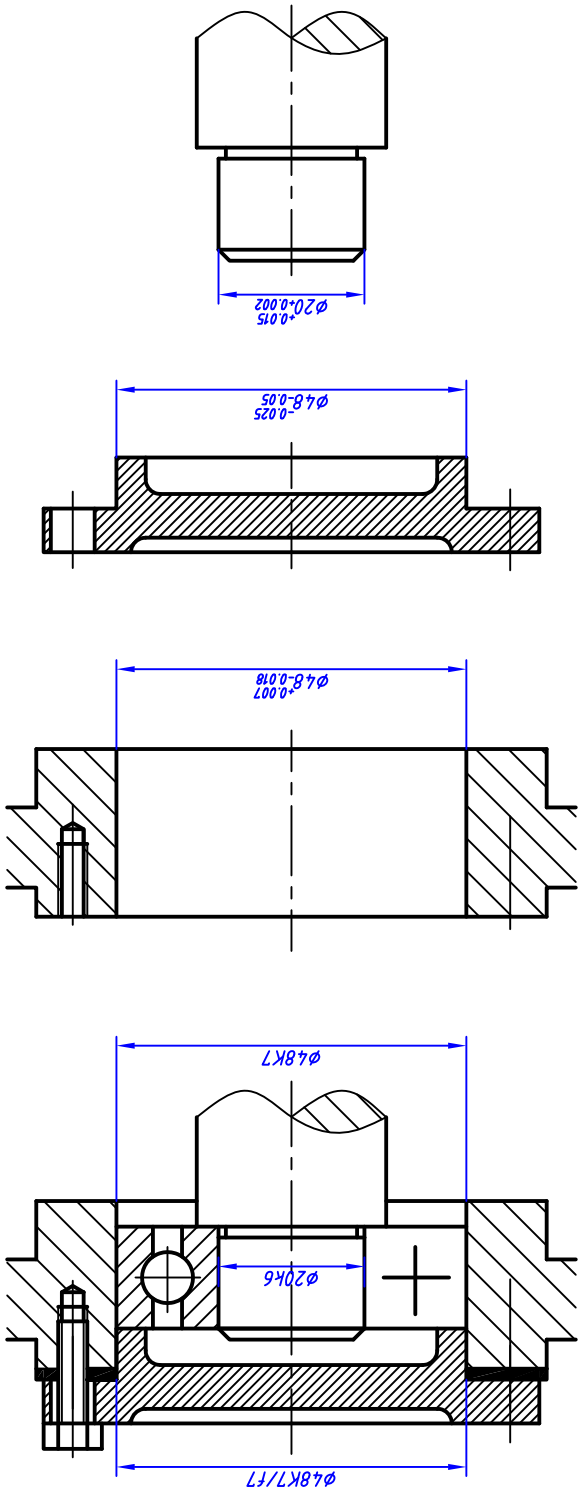
$m=2a/(z_1+z_2)=81 \times 2/(18+36)=3$



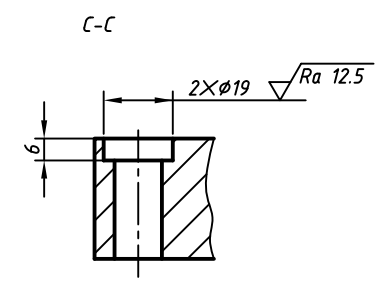
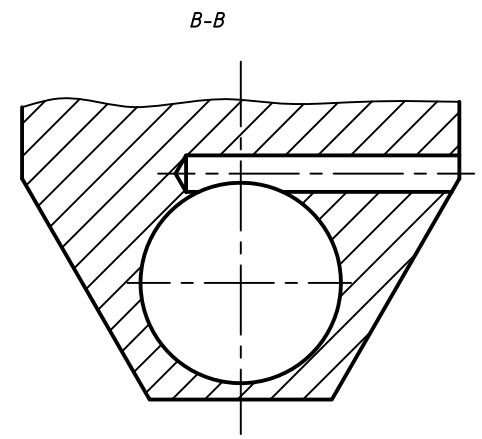
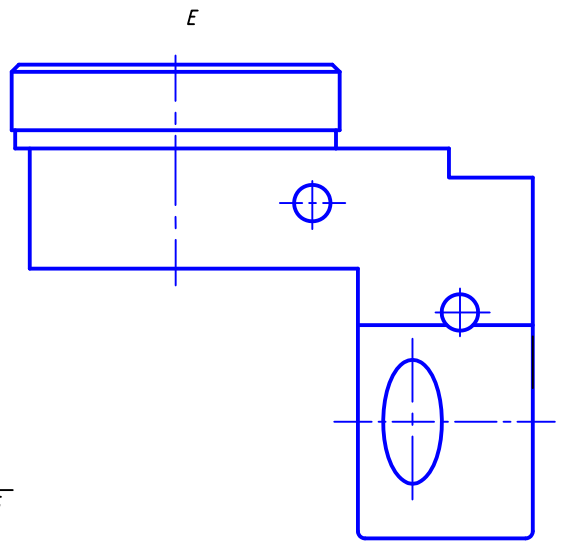
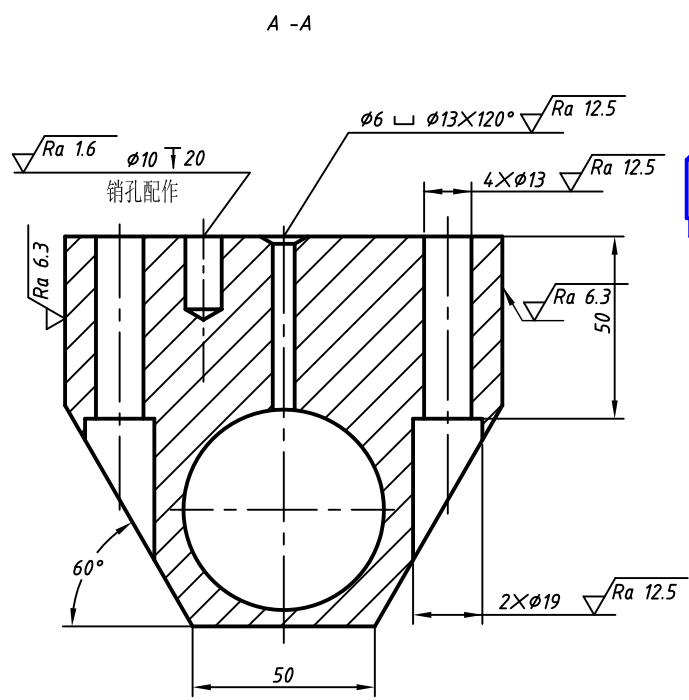
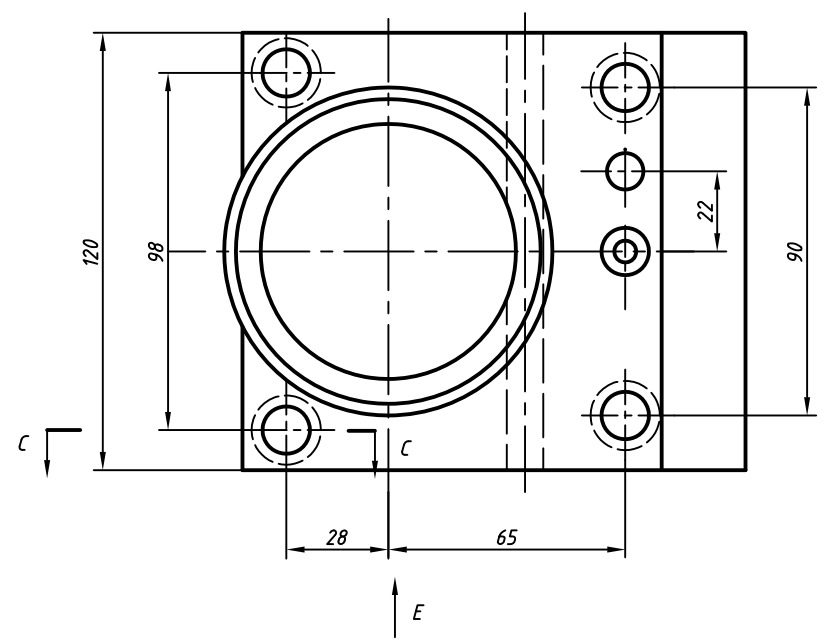
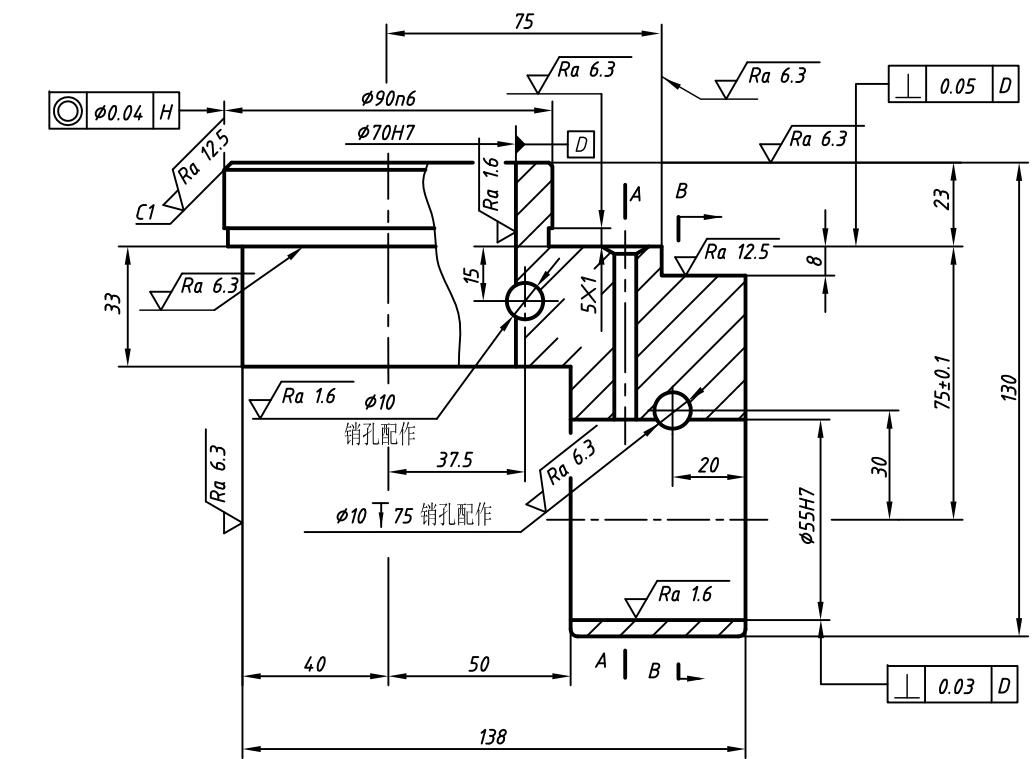
1.根据图中的配合代号，注出零件图中配合表面的尺寸偏差值（查表）。



2.下图为滚动轴承支座装配图。
已知: (1) 轴径与轴承内圈的配合为: 公称尺寸20mm, 基孔制, 轴的公差带代号为k6, 组成过渡配合。
(2) 减速箱壁孔与轴承外圈的配合为: 公称尺寸48mm, 基轴制, 孔的公差带代号为K7, 组成过渡配合。
(3) 减速箱壁孔与端盖的配合为: 公称尺寸48mm, 采用混合配合, 箱壁孔的公差带仍为K7, 端盖配合表面公差带代号为f7。
求: (1) 将各配合表面的配合代号注在装配图上。
(2) 在零件图中查表注出各配合表面的尺寸极限偏差值。



5. 看懂视图，补画E向外形视图。



思考题
B - B剖视图是否可以省略？为什么？

技术要求
未注圆角R2
✓ (✓)

托 架			比例	1:2	LD05	
			材料	HT150		
制图			(校名)			
审核						

7. 根据表中给定的表面结构代号，在视图中标注相应的表面结构。

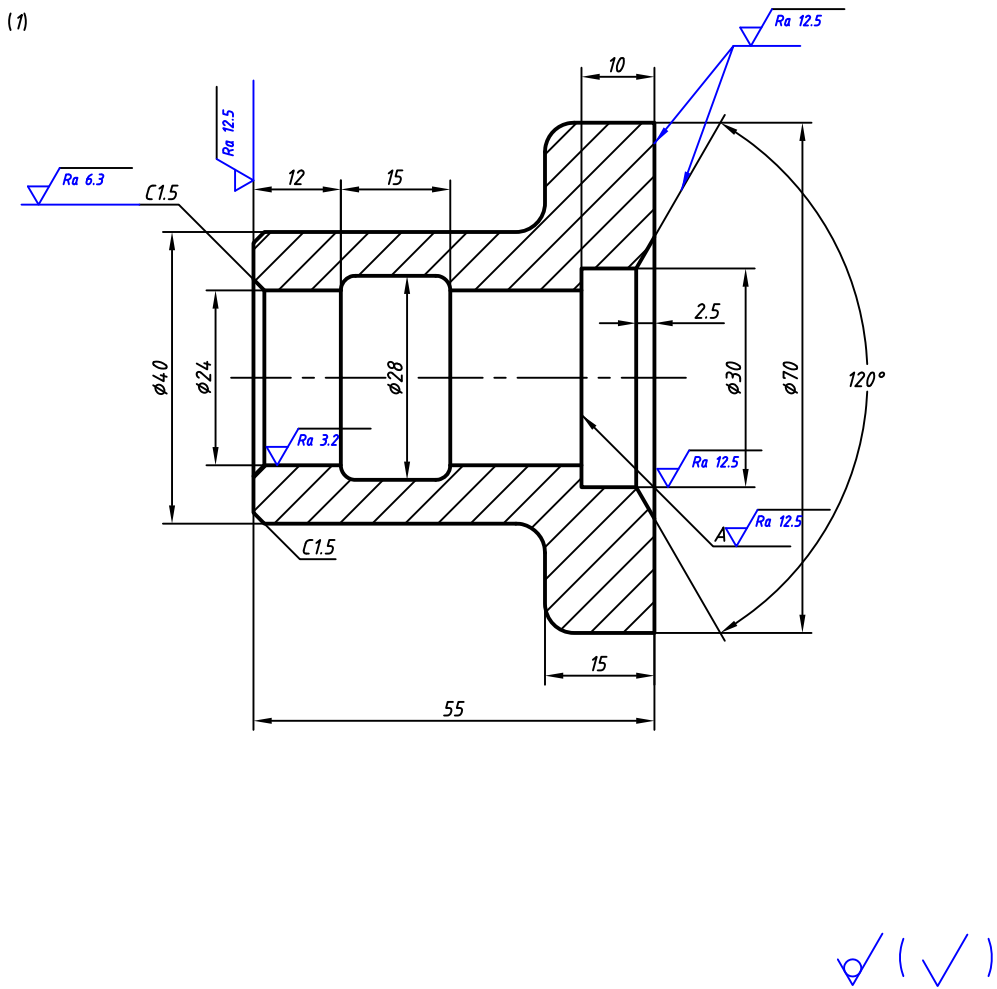


表 面	表面结构代号
左、右端面、120°圆锥面	$\sqrt{Ra\ 12.5}$
φ24圆柱面	$\sqrt{Ra\ 3.2}$
C1.5倒角	$\sqrt{Ra\ 6.3}$
φ30圆柱面、A面	$\sqrt{Ra\ 12.5}$
其余表面	✓

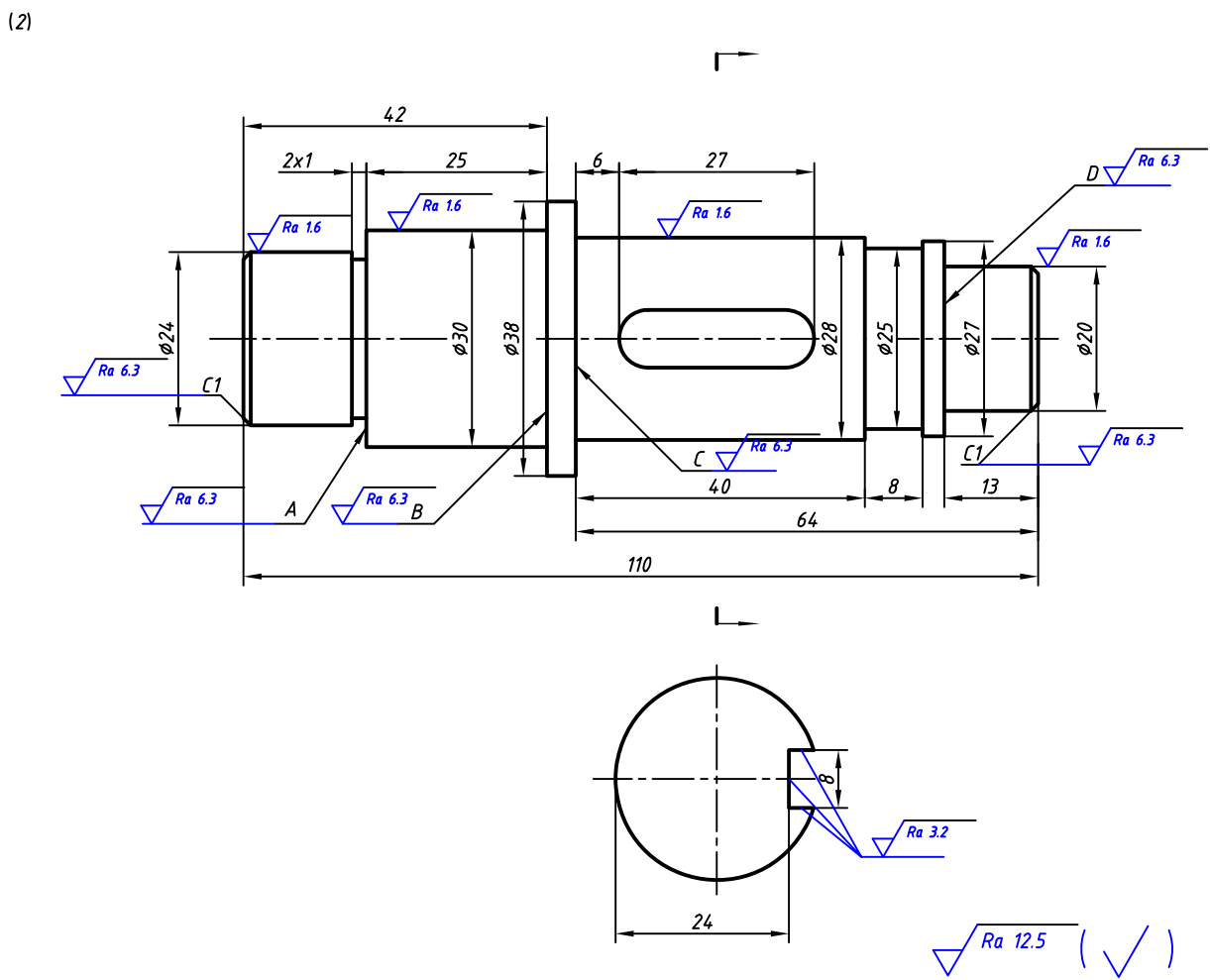
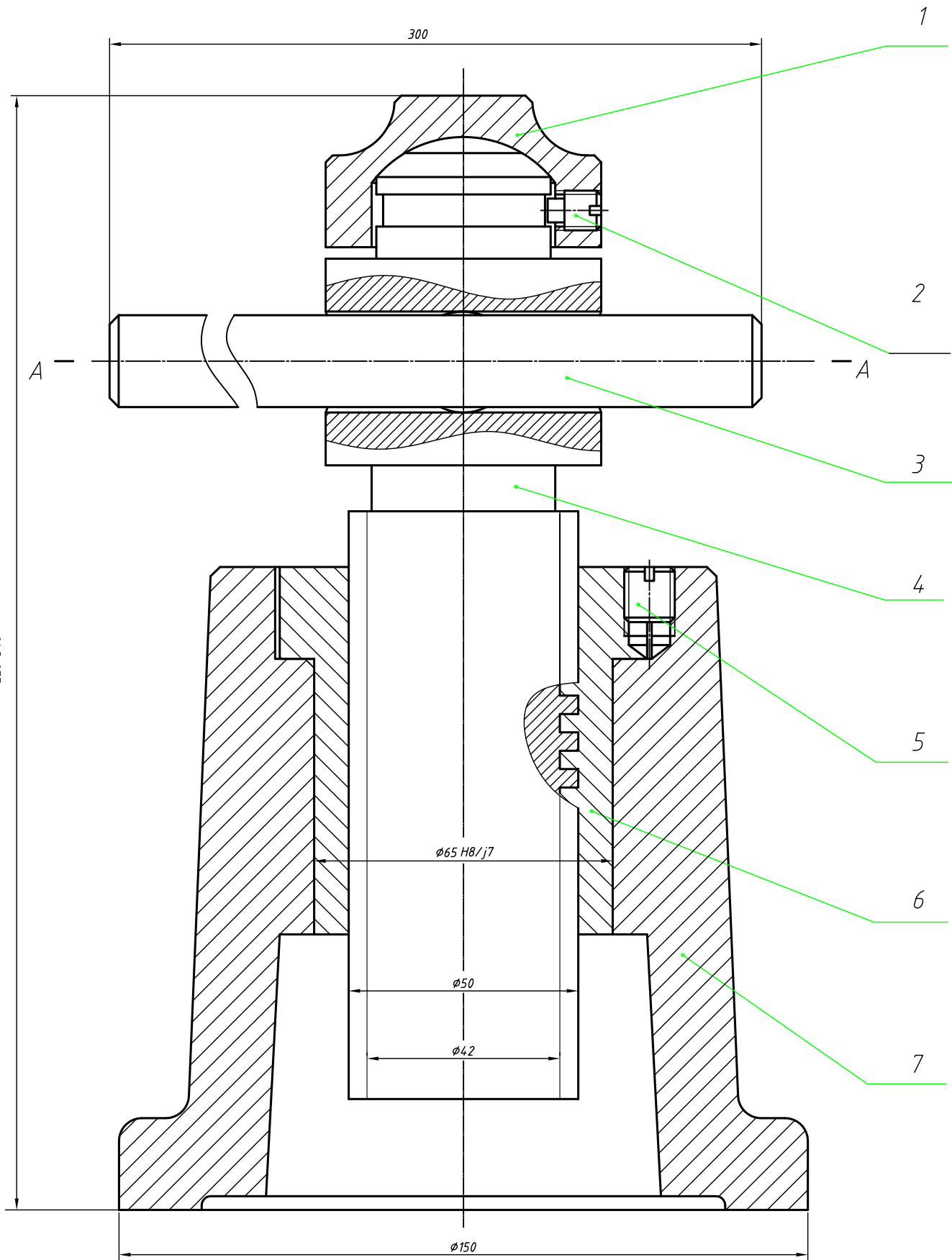
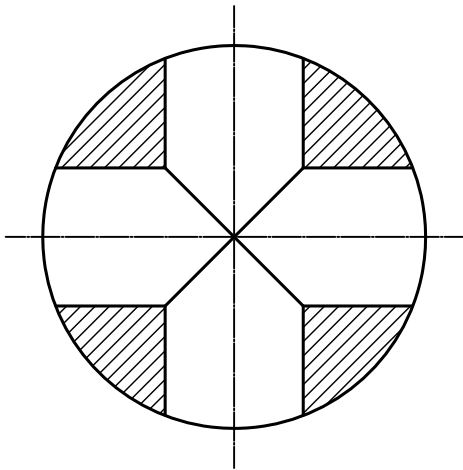


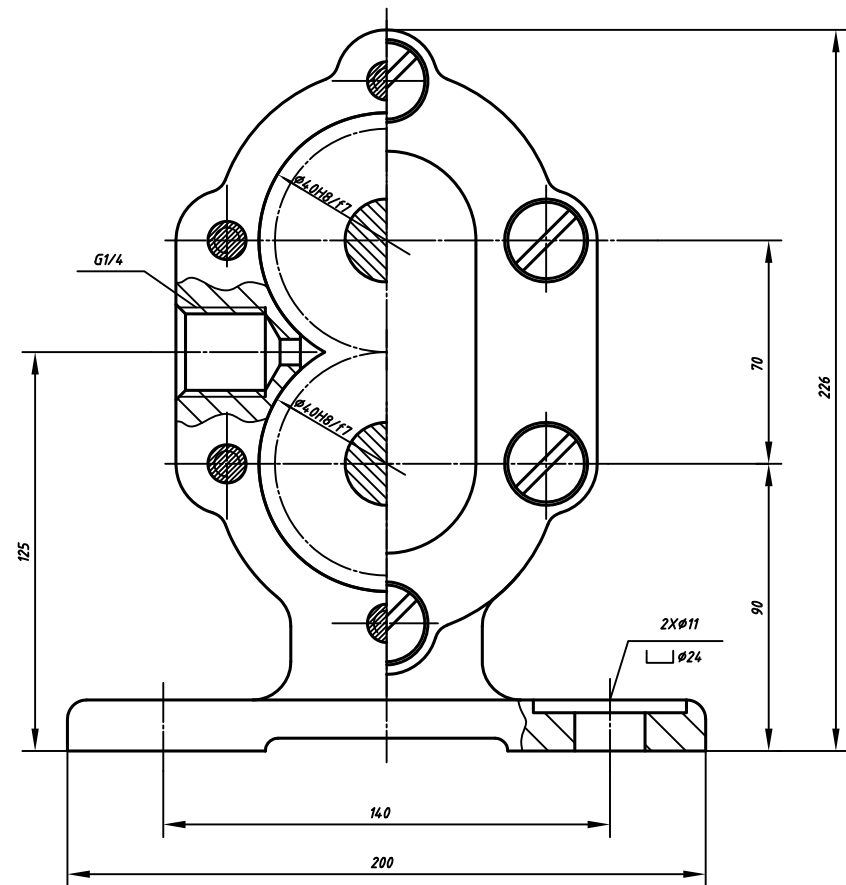
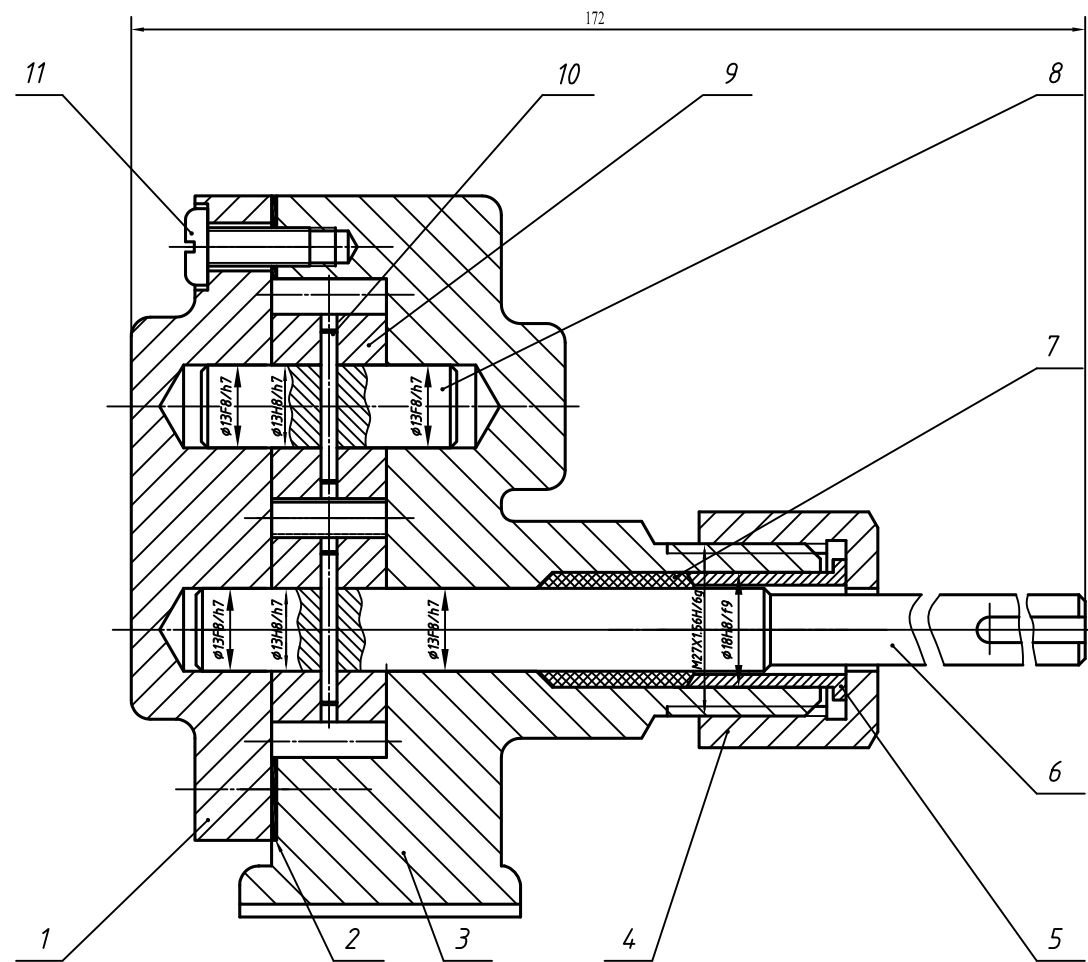
表 面	表面结构代号
φ24 φ30 φ28 φ20圆柱面	$\sqrt{Ra\ 1.6}$
A B C D面	$\sqrt{Ra\ 6.3}$
C1 倒角	$\sqrt{Ra\ 6.3}$
键槽侧面、底面	$\sqrt{Ra\ 3.2}$
其余表面	$\sqrt{Ra\ 12.5}$



零件4A-A



7	底座	1	HT200	
6	螺套	1	QA19-4	
5	螺钉M10X12	1		GB/T73-2000
4	螺旋杆	1	Q255	
3	绞杠	1	Q215	
2	螺钉M8X12	1		GB/T75-2000
1	顶垫	1	Q275	
序号	名称	件数	材料	备注
千斤顶		比例	1:1	YT-02
		材料		
制图	(姓名)	(学号)	校名（班级）	
审核				



					5	填料压盖		1	Q235A		
					4	压盖螺母		1	HT150		
					3	泵体		1	HT200		
					2	垫片		1	密封纸		
11	螺钉M6X16	6	A3	GB/T67	1	泵盖		1	HT200		
10	销M2.5X24	2	A3	GB/T119.1	序号	名称		件数	材料		备注
9	齿轮	2	45	$m=2.5, z=14$	齿轮泵			比例	2:1		YT-03
8	从动轴	1	45					材料			
7	密封填料	1	橡胶		制图	(姓名)		(学号)		校名(班级)	
6	主动轴	1	45		审核						